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The European Bathroom & Kitchen Product Markets

- 2021 Edition -



Germany

BRG
BUILDING SOLUTIONS

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Overview

BRG Building Solutions is a specialised market research consultancy serving the global building technologies industry. Combining unparalleled industry experience with expert research and analysis, our team delivers lasting results to hundreds of companies worldwide.

Our core expertise lies in our coverage of the bathroom, kitchen, HVAC, and other building products/technologies. Throughout a history spanning over 40 years, we have assisted industry players in identifying and realising growth opportunities through a combination of regular market monitoring and ad-hoc consultancy services.

BRG's unique Enterprise Database is at the centre of our offering. This is a comprehensive tool and very detailed source of historic, current and forecast market trends and market share data (by holding and brand) on a country comparable basis. The database provides an essential tool to our clients as a platform to support their strategic decision making.

BRG Building Solutions is a genuinely global company, staffed with individuals from varied backgrounds and nationalities, working collaboratively from four offices across four countries. Combining a high level of industry experience with expert research and analysis, our team delivers lasting results to hundreds of companies worldwide.

Our market coverage is also global, reflecting the nature of the industries we serve. Currently, 66 individual markets are covered annually in detail across the following key geographical regions (and number of countries):

- Africa (3)
- Australasia (2)
- East & Southeast Asia (6)
- Central America (4)
- Central Asia (1)
- Europe (31)
- South Asia (1)
- Middle East (7)
- North America (3)
- South America (5)
- Sub-Saharan Africa (3)

BRG has its own in-house dedicated modelling and forecasting unit. In cooperation with clients, we have since extended our offering to include market sizing for countries outside the traditional geographic reach of our main research

programme, via global market models

Product Coverage

Since the 1980s, BRG's Bathroom & Kitchen programme has provided industry players with insightful analysis and evidence-based information. In addition to detailed data on trends, values, end use, technical segmentations, market shares and distribution, our programme also includes perceptive text commentary, enabling our clients to easily identify and understand market characteristics and the latest industry developments.

BRG's programme has evolved steadily in terms of both geographic scope and product coverage and continues to do so each year. Each country report follows a uniform structure, to ensure that the data can be readily compared across regions and countries.

The product coverage has also expanded to cover virtually all bathroom, sanitary and kitchen related products, used in both residential and non-residential buildings, including:

Baths and Shower Trays

- Baths: cast iron, steel and synthetic
- Hydromassage baths
- Shower Trays: ceramic, steel/ cast iron and synthetic
- Shower systems (NAM): shower modules, integrated tub showers

Sanitary Ware

- WCs: wall hung, floor standing 1-piece and squat
- Washbasins: hand washbasins and countertops
- Other CSW: bidets, kitchen sinks, pedestals, shower trays, urinals and cisterns
- Plastic Cisterns: concealed and exposed.

Taps and Mixers ('Faucets'), both sanitary and kitchen:

- By application: basin/sink, bath, bidet, shower, kitchen sinks
- By type: pillar, two head, one head, thermostatic, self-closing, electronic
- By type (Americas): 4in centre set, 8in widespread, electronic, hand spray, pressure, single handle/one-head, self-closing, thermostatic, three handle.

Shower Wall

- Bath screens, conventional shower cubicles, hydrotherapy cubicles, bath & shower surrounds (NAM)

Bathroom Furniture

- Base units, mirror / mirror cabinets and separate cabinets.

Bathroom Accessories

- Soap holders and dispensers
- Towel holders, towel rings and towel bars
- Toilet paper holders and toilet brushes
- Towel and bathrobe hooks
- Toothbrush holders and tumblers
- Shelves, wastepaper bins, shower storage units and tissue holders.

Shower Accessories

- Shower heads, flexible hoses, shower supports, shower panels and shower pipes.

Kitchen Sinks

- Cast Iron, ceramic, enamelled steel, stainless steel and synthetic.

Definitions

Please note that all volumes given in the report refer to the number of units sold, where not indicated otherwise. In-depth product definitions are included in a separate document and available upon request.

All prices, both manufacturer and retail, exclude value-added-tax (VAT) and installation costs.

Methodology

The methodology behind the data remains true to our origins in that the industry participates in the research process, in many cases by providing BRG Building Solutions with annualised volume sales by product category per country and contacts for detailed expert interviews in all countries covered.

Our approach follows a multi-pronged strategy, gathering data inputs and information from many different sources. A further USP of BRG is the fact that we live and breathe the markets in which we specialise (bathroom, kitchen, plumbing & HVAC) and are in constant contact with manufacturers, distributors and other key influencers all of the time during our repeat annual research programme. The markets are built via a bottom-up approach. Extensive primary and secondary research complement the picture and provide all necessary insight for many

additional segmentations shown in the reports, like e.g. price segmentations, technical and end-use segmentations.

The reports are constantly evolving and refined based on industry feedback, and always reflect the latest developments in the industries.

All research is conducted in-house with a team of around 30 individuals who speak 20 different languages between them. As a result, we can transfer that knowledge and experience into services designed to assist stakeholders in understanding the opportunities available to them, at a fraction of that charged by the larger and less nimble global consultancies. BRG operates a policy of direct management of the team required to conduct the regular research work and avoids subcontracting its market studies to outside agencies.

The analysis has grown in complexity as the number of drivers affecting the markets grows, like e.g. economic uncertainty, construction cycles, new technologies, legislation. As a result, BRG Building Solutions considers all drivers through a holistic approach, recognising that products, end use segments and countries are all inter-linked.

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GERMANY

BACKGROUND

THE EUROPEAN BATHROOM & KITCHEN PRODUCT MARKETS



Geography and Demographics

Germany, officially the Federal Republic of Germany, is a federal parliamentary republic located in Western Europe. It is divided into 16 federal states, collectively referred to as *Bundesländer*, and each state has its own constitution and remains autonomous, with considerable power over certain policy areas and its own energy incentive schemes. Germany has 90 cities with over 100,000 inhabitants, 11 metropolitan areas, and each state has its own capital. Despite its large population, Germany has only a few large cities, and only four have a population of over one million: Berlin, Hamburg, Munich and Cologne. Still, these cities are much larger when the metropolitan area is considered, and around 73% of the population lives in urban areas.

The most important regions in terms of population are the Rhine-Ruhr metropolitan region around Düsseldorf and Cologne, the metropolitan regions around Berlin, Frankfurt/Rhine-Ruhr (the Rhine-Main region), and the Swabian industrial belt around Stuttgart. Other smaller concentrations exist around the cities of Bremen, Hamburg, Hannover, Nuremberg and Munich. Around 80% of the population lives in the *Alte Bundesländer* (ABL), the former West Germany, compared to just 20% of the population being in the *Neue Bundesländer* (NBL), the former GDR (the German Democratic Republic), including the capital Berlin. The population in the NBL has been constantly declining due to internal migration to the ABL, mainly for economic reasons.

German is the official, and predominantly spoken, language of the Republic and it is also one of the 24 official working languages of the European Union. German is the most widely spoken language within the European Union, with some 100 million native speakers. Minority languages within Germany are Danish, Low German, Sorbian, Romany and Frisian, while Turkish, Kurdish, Polish, the Balkan languages and Russian are the most used immigrant languages. According to the latest data released by EUROCONSTRUCT, Germany comprised about 83.167 million inhabitants in 2020. As Europe's largest economy and second-most populous nation (after Russia), Germany is a key member of the continent's economic, political and defence organisations.

Of all the European Union (EU) states, Germany has the highest percentage of immigrants in its population. Most immigrants come from other European countries, particularly Turkey, Russia, Poland and Italy. The country is the second-most popular destination for immigrants in the world after the United States. The increase of immigration flows with refugees arriving from the Middle East (including refugees from Syria) and Africa, forced to flee war and personal persecution, has sustained the growing path of the German population, compensating for the extremely low and still-declining birth rate of around 1.6 children per year. Germany has one of the lowest birth rates in the world. As a result, it has an ageing society: 21% of the population is over 65 years old (compared to 15.8% in 1997), whereas only about 13% is under 15 years old. The number of refugees entering Germany

has dropped significantly and stabilised in recent years, after a sudden influx that started in 2013. The years 2015 and 2016 had an exceptionally high increase of refugees, with Germany welcoming people who were forced to flee war and personal persecution. The ongoing inflow phenomenon was initially having a clearly positive impact on domestic consumption, since arrivals had to be supplied with essential goods and accommodation.

Macroeconomic Overview

The German economy is the fourth-largest economy in the world according to nominal GDP, and the largest in Europe, based on data from the International Monetary Fund. After four years with extremely strong growth rates (2014-2017), the German economy started to experience a strong downturn in 2018 and even more so in 2019. According to EUROCONSTRUCT, Germany's industry is in a recession and its production has been declining for the past couple of years, which is the main reason for the economic weakness.

With the entire manufacturing sector slowing down already, the impact of the COVID-19 pandemic contributed to a stronger recession in 2020. The global downturn of the economy added to the effect as Germany greatly relies on exports.

Unlike other industries, the construction industry remained at a high level in 2020. However, fewer completions were carried out as a result of the pandemic and the overall slowdown of economic activity. Generally speaking, the impact on the construction industry was hardly felt due to the huge backlog of pending projects and construction accumulated in the past. While some projects have been stopped and postponed due to the pandemic, there was still demand for other constructions from previous years. The need for housing is still a recent topic in Germany and ongoing demand is emphasised by the increase of dwelling permits in 2020 when compared to the previous year according to DESTATIS.

In 2020 the macro-economic indicators in Germany were negative and the GDP notably declined. Nevertheless, interest rates remained low and the motivation to invest into real estate is on a high level, despite lower disposable income and public consumption. The mood in the trade is mixed, with lower export and import growth rates in 2020 when compared to the previous year. The overall job market remains robust and a steep increase of unemployment was prevented by government measures such as short-time work programmes. The German government implemented the short-time working allowance until December 2021 as a means to secure jobs during the COVID-19 crisis. This is particularly helpful for employees working in businesses that have been strongly affected by the lockdown. The construction industry was less affected by the lockdown measures.

Germany has a clear problem of insufficient skilled labour, which hampers growth in the construction industry, among other industries. The government is aware of this problem and campaigns have been set in place to spread awareness about professions in the HVAC industry. Another approach to tackle the situation is by introducing a real immigration law. What started off as a debate in 2015 (as a result

of the strong influx of refugees), turned into a first draft that was passed by the new government at the end of 2018. With Germany's population growing older and the anticipated retirement of the baby boomer generation, the immigration law is primarily concerned with facilitating the process for qualified immigrants to work in Germany. In June 2019 the government passed the Immigration Act, which creates a legal framework for targeted immigration of qualified specialists from foreign countries to Germany. The law is no longer limited to certain industries but is open to all industries. It also offers foreigners the opportunity to come to Germany for a limited time to find a job. The law applied from January 2020.

The long-term forecast for the German economy remains promising with solid growth rates expected from private household spending, low interest rates and the need for new housing as well as government recovery programmes. In the short term, negative impacts of the COVID-19 crisis are expected to be overcome by significant government spending and recovery programmes to stimulate a positive economic development. According to EURCONSTRUCT new housing completions will slightly increase again from 2021 onwards, before turning negatively towards the end of the forecast period.

The unemployment rate is expected to recover in 2021 and to continue improving until the end of the forecast period. A similar development is expected for private consumption and the consumer price index. Both values are expected to plunge down in 2020, before rebounding the following year. These indicators support the positive trend of the construction industry, which is anticipated to grow again in the short term. The positive GDP value, that is expected for 2021, reflects the quick recovery after the COVID-19 crisis.

Residential Construction

Overview

German construction slightly declined in 2020 when compared to 2019 by about 13,000 dwellings. Overall decline was driven by fewer apartments as well as slightly less one-and two-family house completions in the year. Total construction activity of new housing and RMI in Germany, which includes both residential and non-residential construction, slightly declined in 2020 according to EUROCONSTRUCT. Although the booming growth rates from previous years are over, the construction industry is still on a high level. Domestic migration to top locations such as Berlin, Munich, Frankfurt, Hamburg, Cologne, Düsseldorf and Stuttgart puts great pressure for living spaces in certain regions. While the population in these top seven locations grew by 7-15% during that period (in total about one million more people in 2017 in comparison to 2007), only 286,000 new dwellings were built during the same period. The problem of available and affordable living space in popular areas is ongoing. Other contributing factors are low interest rates and rising real income levels, which motivate Germans to invest in their own property.

According to the Federal Statistical Office, DESTATIS, new housing construction permits slightly increased in 2020 when compared to 2019. This positive development is reflected in the marginal increase of apartment permits, and more so in the growth of one- and especially two-family house permits. In addition to more permits, there are still projects that are waiting in the pipeline for available labour. It is worthwhile to mention that the German construction industry is working at a high level capacity. More professional workers are needed to allow a stronger increase in refurbishments and new housings. Despite the COVID-19 pandemic, new housing development is expected to remain on a high level during the forecast period until 2023, before turning negative again.

New Housing

The German new residential construction market had a slight decrease in 2020, with total residential completions reaching around 280,000 dwellings. When taking a closer look, one- and two-family house completions marginally decreased by 3,100 dwellings when compared to 2019. The number of new dwellings in multi-family houses decreased by about 9,900 to reach a total of 180,000 completions in 2020. Despite the slowdown of the economy, the new housing market benefited from low interest rates, strong demand due to a prevailing deficit of accommodations, rising incomes, and investors searching for safe and stable forms of investment. Long term influencing factors that contribute to growth in new housing include the anticipated recovery of the German economy, decreasing unemployment, a growing population due to immigration and domestic migration to popular regions. A constraint to growth is a lack of professionals and a shortage of skilled workers.

The availability and construction of social housing in Germany is declining annually, despite the need for affordable living space. The main reason for this is the privatisation of social housing, whereas the construction of new social housing is not feasible for real estate companies, housing associations, pension funds, insurance companies and/or fund providers, as these all invest in the real estate sector because it is regarded to be a safer investment. The problem with social housing is its higher risk factor due to a low return on investment. This is not only caused by low rent prices, but also by so-called socially disadvantaged tenants since they are associated with higher maintenance and administration costs, as well as a worsening re-sale potential of the real estate. According to recent estimation, around 80,000 new social housing dwellings need to be constructed on a yearly basis to compensate for the overall decline. However, not even half of that is constructed on a yearly basis.

Demand for living space prevails, especially in popular destinations. Apartment completions are expected to increase from 2021 until 2023, whereas the completions of one- and two-family homes are set to decrease from 2022 onwards. The German government has announced a major housing construction programme to realise 1.5 million new dwellings – by allocating more than five billion euros until 2021 and introducing incentive programmes such as tax reduction for rented apartments, regulated rent prices, and submission of federal land to municipalities.

Its objective is to build more than 100,000 new social housing dwellings by 2021. With this kind of support, it can be expected that the construction industry is going to remain on a high output level. The COVID-19 pandemic has only slightly impacted the construction industry and it is believed that the negative outcome will become more evident towards the end of the forecast period. The main reasons are strongly increased prices of new constructions, which are linked to expensive labour costs, lack of construction land and an uncertain economic development of the pandemic.

Housing RMI

In 1990, the German reunification led to a massive increase in new construction. Between 1995 and 2009, annual housing completions fell from approximately 600,000 units to only 159,000 units per annum; this was mainly due to the overall population decline driven by a lower immigration rate, prolonged economic stagnation and high unemployment, and increasingly strict energy regulations for new build housing. According to DESTATIS, of Germany's building stock, 28% was constructed before 1948, 46% between 1949 and 1978, 13% between 1979 and 1990, and 13% was built after 1991. These dwellings that were constructed during the boom need renovation sometime soon.

Investments in the renovation of old buildings have risen only moderately in recent years, because renovation measures have been at an extremely high level for a decade now. Based on EUROCONSTRUCT data, around two thirds of total residential construction activity in terms of value is in existing buildings. In 2020, the value of residential RMI slightly decreased when compared to the previous year. A positive trend is expected for 2021, before turning slightly negative in 2022 and 2023. As long as new housing construction is growing and availability of professionals is limited, demand for renovations is likely to remain flat. An installer, for example, is more likely to choose a new housing project where higher margins can be gained. Instead, a renovation project is more likely to be postponed until no more lucrative projects are available. Most installers in Germany do both, the installation of heating and sanitary appliances. They can decide for the better project, e.g. a new bathroom installation, which have higher margins. Instead, the topic heating only becomes relevant when it breaks and less attention is paid to the heating system. This situation significantly changed in 2020 and the heating industry gained a stronger presence for several reasons. Homeowners spent more time at home and feeling safe and warm became more important. The contact with outsiders was reduced to a minimum, especially in a private zone such as a bathroom. Instead, heating systems can easily be upgraded in the basement and government incentives made it very beneficial to do so. The money that was saved by going on holidays was invested to improve the home.

At the same time, there are constraints on RMI activities in rural areas, where a high housing surplus exists. In such areas, low rent prices do not justify investments in the property and, therefore, renovations are not seen as a means to increase the rent prices of a dwelling. Large renovation projects are often subject to the same strict energy requirements as new builds, which makes renovation costly

while contributing to a small amount of energy savings, and it is, therefore, less appealing. Overall, it is expected that the RMI segment is likely to remain flat with a marginally declining value each year. Demand mainly comes from homeowners, who are keen to invest in their own properties while interest rates are low. Thanks to the lockdown consumers have had more available time and the ones who could afford it finally initiated renovation projects such as into a highly priced modern bathroom. Some demand for bathroom products is expected from properties, which are increasingly modernised to be age appropriate.

Non-Residential Construction

The German non-residential market is heavily dependent on overall economic developments. According to EUROCONSTRUCT, 80% of all construction output in the non-residential segment is carried out by commercial developers. Public sector investments only dominate in the areas of healthcare and education. In 2020, total non-residential construction output experienced a notable decline when compared to 2019. Demand was impacted by the pandemic and some new projects and anticipated renovations were put on hold. At the same time public budgets were impacted in a negative way. As for residential buildings, more than half of the total non-housing value is generated from RMI activities. For several years the German real estate market has been growing, and this attracts foreign investments. Germany offers low interest rates and a positive return on investment when compared to other European countries. However, during a global recession the slowdown for such investments was also felt in Germany. Commercial and retail buildings suffered from the lockdown and the shift towards online trade led to a significant decline in spending. In recent years, the government has increased funding for the renovation of schools, and the sector had available funds to grow in 2020. According to EUROCONSTRUCT, the volume spent for industrial, storage, office, commercial and miscellaneous buildings such as hotels declined in 2020 when compared to the previous year. The only exceptions were buildings for health and education, which underwent a marginal growth. The COVID-19 pandemic is expected to have a strong influence on the non-housing segment. Some facilities such as hotels, restaurants, trade fairs and events have been heavily impacted by lockdown measures. Businesses, which have been struggling before are likely to be in a challenging position, whereas previously successful businesses are expected to overcome the crisis.

According to EUROCONSTRUCT, there is a strong downturn of the non-residential new construction and renovation markets expected for 2021, followed by a small growth in 2022 and 2023. Some non-housing facilities are believed to benefit from the situation. Stricter hygiene requirements need to be followed and investments are needed to successfully implement these. The German government wants to provide significant subsidies to avoid further area-wide closings of kindergartens by investing into more washbasins for example. Overall, investments into buildings for education and health are expected to increase in 2021.

II.2 Key Figures

DE - DEMOGRAPHICS AND HOUSEHOLDS

Value	Year
Segmentation	Category
Demographics	Population (% Rural)
	Population (% Urban)
	Population ('000)
Households	Average Size (Persons h/h)
	Total ('000)
Persons per Household (% of Total)	1
	2
	3
	4
	5+

Source: BRG based on Country Statistics.

II.3 Macroeconomic Trends and Forecasts

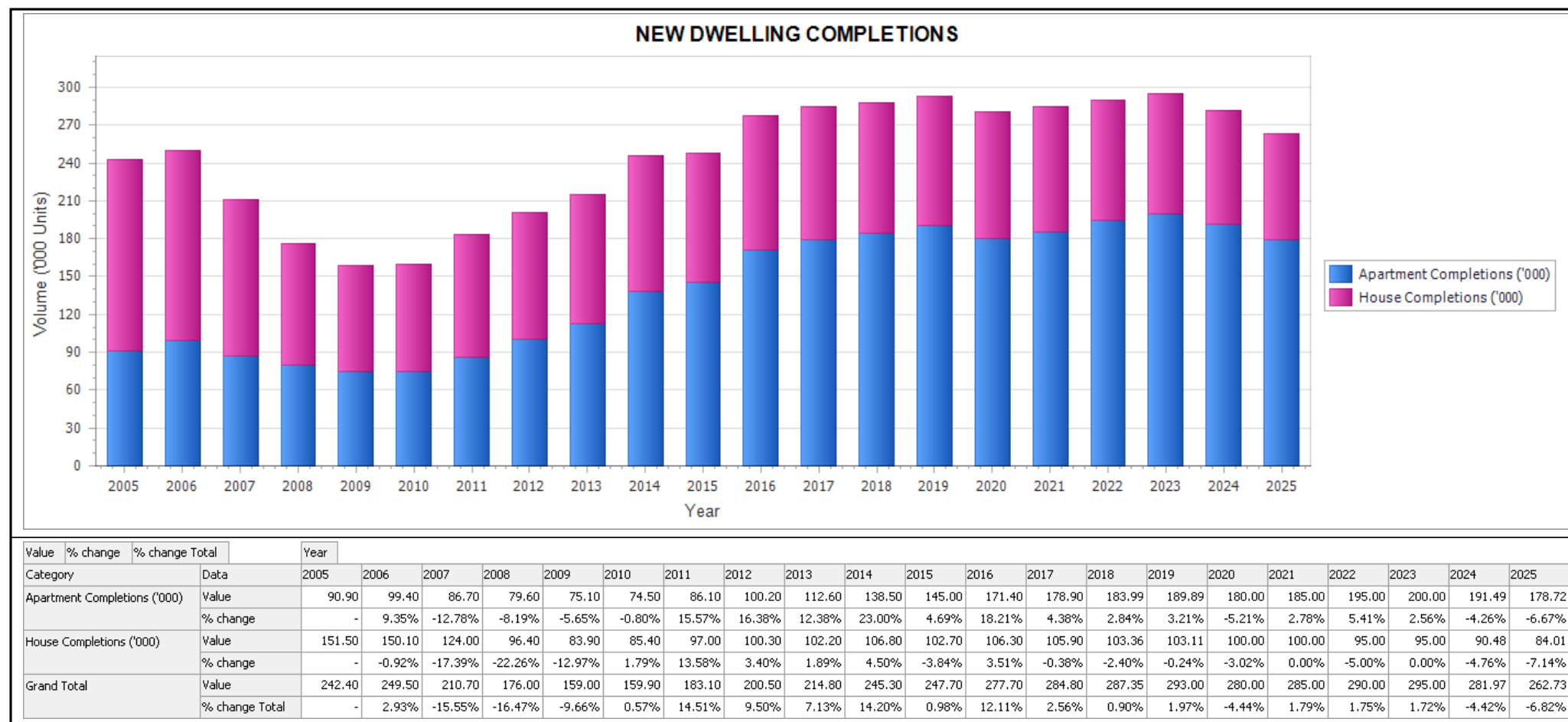
DE - MACROECONOMIC KEY INDICATORS

Value	Year																				
Category	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Consumer Price Index (% change)	2.13	1.39	3.08	1.00	0.88	1.85	2.24	2.19	1.33	0.20	0.20	1.71	1.48	1.75	1.53	- 0.66	3.07	0.99	1.95	1.76	1.96
GDP (% change)	0.73	3.81	2.98	0.96	- 5.70	4.19	3.91	0.43	0.43	2.22	1.49	2.23	2.60	1.27	0.56	- 4.90	3.60	3.42	1.57	1.46	1.15
Government Balance (% of GDP)	- 3.32	- 1.65	0.26	- 0.12	- 3.15	- 4.38	- 0.88	0.01	0.04	0.58	0.96	1.16	1.36	1.84	1.52	- 4.19	- 5.46	- 0.42	0.37	0.48	0.62
Household Saving Rate (% of Disposable Income)	10.62	10.56	10.64	10.70	10.91	10.37	10.31	10.02	9.65	9.29	9.84	10.06	10.20	10.59	10.93	10.87	16.63	0.00	0.00	0.00	0.00
Private Consumption (% change)	0.76	1.35	- 0.25	0.35	- 0.09	0.71	1.86	1.50	0.36	1.05	1.94	2.45	1.52	1.53	1.56	- 6.07	0.00	0.00	0.00	0.00	0.00
Unemployment Rate (%)	11.01	10.04	8.54	7.43	7.64	6.97	5.83	5.38	5.24	4.99	4.63	4.13	3.76	3.40	3.15	4.18	4.45	3.74	3.54	3.44	3.39

Source : IMF and OECD. N.B. Household Saving Rate and Private Consumption not Available for all Forecast Years

II.4 New Dwelling Construction

DE - NEW DWELLING COMPLETIONS ('000 UNITS)



Source: BRG based on Country Statistics.

II.5 Dwelling Stock by Type and Tenure
DE - DWELLING STOCK BY TYPE AND TENURE ('000 UNITS)

Value		Year
Segmentation	Category	2020
Dwelling Stock by Type	Apartments (%)	53.04
	Apartments ('000)	22 697.24
	Houses : Rural (%)	11.99
	Houses : Rural ('000)	5 131.00
	Houses : Urban (%)	34.97
	Houses : Urban ('000)	14 964.76
Dwelling Stock by Tenure	Primary Owned (%)	46.50
	Primary Owned ('000)	19 898.75
	Primary Rented : Private (%)	43.58
	Primary Rented : Private ('000)	18 649.19
	Primary Rented : Public (%)	2.22
	Primary Rented : Public ('000)	950.00
	Secondary (%)	4.45
	Secondary ('000)	1 904.29
	Vacant (%)	3.25
	Vacant ('000)	1 390.77

Source: BRG based on Country Statistics.

GERMANY

DISTRIBUTION

THE EUROPEAN BATHROOM & KITCHEN PRODUCT MARKETS



Overview

The distribution sector, typical for a Western European country, is dominated by wholesalers and large DIY chains. In Germany, even more so than in most other European countries, the wholesale channel occupies an extremely powerful, historically fast-grown, position on the bathroom and kitchen markets. Increased competition among wholesalers has led to a market concentration over the last 20 years, in which many smaller sized family-owned businesses vanished, since the wholesale sector over the last decades has increasingly become more concentrated. Originally consisting of hundreds of small '*mittelstand*' wholesale businesses, the German wholesale industry has undergone a significant consolidation, with the four largest wholesalers taking over many smaller, often family-owned, businesses. Due to the strong dominance of wholesalers, and DIY retailers, on the German bathroom and kitchen markets, the distribution primarily follows a so-called 'three-step' distribution chain, which comprises both sales from manufacturers/importers to specialist wholesalers, and sales from manufacturers/importers to wholesalers who then supply DIY stores. Traditionally, bathroom and heating products have been channelled through the same wholesalers. The main competitors for wholesalers are the retail distribution channels and e-commerce.

The German wholesale channel is very traditional by serving a loyal installer base. Installers prefer wholesalers as suppliers as they shoulder the credit risk while waiting for a customer's payment. The German wholesale industry is very competitive. It has rigid structures to remain powerful and to prevent distribution outside the traditional 'three-step' approach. Wholesalers, and their buying groups, offer private label products with competitive prices. In addition, they offer a wide range of products near installers. German wholesalers, as well as retailers to a lesser extent, have developed successful private label brands, that are often sourced from leading manufacturers. Installers play a key role in the distribution of sanitary ware products. They account for the biggest share of sales in terms of volume by linking wholesalers, and manufacturers to a lesser extent, with the end user.

German installers can be broadly divided into two groups based on a generational divide: (1) traditional installers, and (2) installers that are open to new products and changes in the distribution channels such as the use of smart phone 'apps' and 'e-commerce'. In the past years, an increasing number of installers have been purchasing online from wholesalers. Most wholesalers have sophisticated online services that allow installers to order in a quick and convenient way. They provide training and technology support to give installers more control over their installer base. According to a study by BAUINFOCONSULT from January 2018, around 40% of installers makes use of online purchases from wholesalers. This includes direct email purchases from trusted partners as well as website based and app purchases.

The DIY sector in Germany started to appear in the 1970s, when OBI opened its first store. Most bathroom products reach the DIY retailer via specialist distributors, so-called '*Sortimenter*', and through manufacturers specialising in DIY supply. DIY stores mainly supply customers with products that are part of the lower and economy price segments. The products from the economy segment are normally sourced from foreign manufacturers, or importers, and are divided into two categories: Economy I, which tends to be supplied by imports from Western European countries such as France, Italy, Portugal and Spain and Economy II, which is supplied mostly by imports from Eastern European countries such as the Czech Republic, Hungary and Poland. A significant share is supplied from Asia with China being the main manufacturing country. DIY stores no longer just sell individual products but, instead, they promote complete product lines and designed bath solutions. All the leading suppliers such as GEBERIT (former KERAMAG), VILLEROY & BOCH, DURAVIT, IDEAL STANDARD, LAUFEN and VITRA have a subsidiary or a sister company abroad to source products for the lower end of the market. An increasing number of major manufacturers is focusing on DIY segments and e-commerce. Following a multi-channel distribution strategy is becoming more acceptable. For example, major manufacturers are directly selling their products to REUTER or DIY stores. In the past, such distribution channels would have been supplied by wholesalers or *DIY-Sortimenter*, which in Germany are known for supplying DIY stores.

There are two main types of bathroom customers, who can be identified in Germany. On the one hand, customers are looking for a single piece of furniture such as a mixer or a flexible shower hose to replace a broken one. These customers need a product quickly to be able to use the broken facility again. In most cases, these end users are more price conscious and they increasingly turn towards DIY stores, online and furniture retailers out of convenience. It is unlikely that an end user would wait more than one month for a broken kitchen tap to be replaced by an installer. On the other hand, there are end users looking for a complete bathroom or kitchen solution for a new or refurbished dwelling. These end users seek more than just a product and they are willing to pay a premium price to receive a full service.

Wholesale Level

The German wholesale trade is powerful with more than 150,000 small to large sized wholesalers and nearly 1.5 million employees. Due to Germany's strong reliance on exports, German wholesalers are internationally oriented. Whereas historically wholesalers were solely the link between manufacturing and retail, nowadays, they have evolved into becoming a 'supply management provider'. Since 2012, German wholesalers remained stable in terms of value with slight fluctuations only. Although Germany's economy is performing well, the wholesale segment is not undergoing any significant growth. It is believed that supply is near its maximum capacity because there are not enough professional installers to cover the surplus work. According to industry feedback, manufacturers could have sold more if there had been a greater number of professional installers available to carry

out works.

The German wholesale trade can be characterised into two main types. On the one hand, there are single or smaller groups of regional and local independent retail businesses and wholesalers, most of which are members of buying groups to generate economies of scale and to benefit from outsourcing certain administrative tasks. On the other hand, there are wholesale groups, which have mostly grown through acquisitions and which act on a national level or in multiple regions. Moreover, some wholesalers such as REISSER and RICHTER + FRENZEL trade solely or primarily under a single trading branding, while others such as GC have regional companies trading under their own name. The German wholesale market has become increasingly concentrated in recent years and is now distinguished by only a handful of important national and multiregional wholesale chains/groups. These are independent from manufacturers and they combine various single or smaller groups of regional and local independent retail businesses and wholesalers, most of which are members of voluntary buying groups. Being members of buying groups allows these smaller businesses to benefit from economies of scale, not only in purchasing but also in management and marketing.

The four largest wholesalers in the industry are the GC GROUP, RICHTER + FRENZEL, REISSER, and PFEIFFER + MAY (in that order). Most wholesalers in Germany are privately owned and, hence, do not publish any financial figures. It is difficult to estimate the exact size and market power of these four groups. The larger wholesaler groups have their own central purchasing department and are not members of any buying group. The smaller and more local wholesalers typically join buying groups to generate economies of scale on the purchasing side. The biggest buying groups in Germany are E/D/E, HAGRO, IGH, ISG and GSH. Despite a trend towards more concentration in the market, the member numbers of the most important buying groups have been decreasing in the past few years. All buying groups, in recent years, have opened their doors to international members.

Retail Level

After the manufacturing and business sector, the German retail industry is the third largest sector. Close to three million employees work part and full-time for over 300,000 companies. Most revenues are generated by medium to small-sized companies.

DIY Retailers

The German 'Do-It-Yourself' (DIY) sector is strong and is the most mature market in Europe. It started to develop from the 1970s when OBI opened its first store in 1970. With the reunification of the *Neue Bundesländer* in the early 1990s, a surge in growth was attributed to the opening of many new stores. Although, in the more recent past the number of DIY stores has remained stable, there is a clear trend towards a DIY culture in Germany. DIY is increasingly associated as a lifestyle phenomenon. From the total DIY turnover, sanitary and heating products accounted for about 1.79 billion euro in 2019, which equals to 9.2% of the total

value according to the DIY association BHB (latest numbers available). Only construction materials have a higher turnover. For 2020, a strong increase in total DIY turnover is expected. In the first half of the year the total DIY turnover increased by 15.6% when compared to the same period in 2019.

OBI is the biggest DIY retailer in Germany in terms of value. In 2015, OBI confirmed its dominant position by taking over and rebranding 70 branches from the Austrian competitor BAUMAX. Even though the top three DIY players in Germany compete closely in volume, OBI is the clear value leader, followed by BAUHAUS and TOOM BAUMARKT. There is a mix of owned chains, franchises and buying groups on the German DIY market. The chains originate from three main sources: (1) the food retailing sector such as OBI, EXTRA & TOPBAUCENTER /METRO, TOOM /REWE, GLOBUS, and HELLWEG, (2) building materials suppliers such as HAGEBAU and INTERPARES/MOBAU, and (3) specialist DIY retailers such as BAUHAUS and HORNBACH. Several wholesalers have used the rapid development of the DIY stores by establishing specialised distribution of plumbing related products to DIY chains. The wholesalers operating this service for the DIY chains in Germany are referred to as '*DIY-Sortimenter*' or specialist distributors. These distributors source their products partly from German manufacturers, either directly or through traditional wholesalers for some branded products, and partly from abroad. They buy the products, package them, brand them and sell them on to the DIY stores, together with a marketing concept. The existence of these specialist distributors serves two main purposes. On the one hand, it enables German manufacturers with a 'wholesale only' policy to gain access to DIY stores without abandoning their loyalty to the wholesaler. On the other hand, specialist distributors provide DIY chains, which usually do not have central warehousing, except for some larger groups like BAUHAUS and HORNBACH, with a logistical infrastructure for bathroom products.

Furthermore, specialist distributors are an easy entry route for low-cost imports, especially for products such as taps and mixers, ceramic sanitary ware, and synthetic baths. There is a dozen established specialist distributors, of which the largest are CONMETALL, SANITOP-WINGENROTH and LEYSSER. While CONMETALL and LEYSSER offer a wide range of products covering metal goods, tools, gardening, and bathroom products, SANITOP-WINGENROTH specialises in bathroom products.

According to BHB, the turnover of DIY products purchased online is expected to more than quadruple in 2020 when compared to 2010. In 2010, the turnover of DIY products through e-commerce amounted to 1,065 million euro and increased to 4,000 million (selling prices including VAT) in 2019. This strong growth reflects the importance of the Internet as a distribution channel, with about 21% of the total DIY's turnover being purchased online. According to industry feedback, DIY retailers are increasingly positioning themselves as service providers. Advertisement campaigns on television and the radio are used to position the DIY as a one-stop solution for basic full bathroom renovations. The DIY targets end users who wish to carry out a private bathroom renovation for a reasonable price. These end users do not have high demand and try to get a better bargain by

avoiding the professional installers.

However, it might sound better than it is. The reason is the service providers, who work for the DIY are likely to charge higher hourly rates when compared to a professional installer. The reason is that the DIY installer does not make any money by selling the product and, instead, would have to increase his rates to compensate for lost margins. Despite the higher labour costs, the quality might be much lower, since a good installer would prefer to work for the professional distribution channel. Therefore, the only end users profiting from DIY stores are the ones who know how to install a product themselves. Instead, end users, who need someone to install it for them, might find it less attractive to have a DIY carry out their bathroom renovation, especially if quality and design are of importance to them.

Specialist/Other Retailers

Bathroom and kitchen retailers hold a strong position on the market for complete bathroom or kitchen solutions, but less so for single pieces. Specialist retailers usually provide tailor-made advice on the selection of all types of bathroom product. They make use of professional computer software to design 3D-bathroom plans and are responsible for the installation of all bathroom products. Bathroom retailers offer a complete package starting with the attractive display of products, individual and competent advice on the selection of all bathroom products, computerised bathroom planning, delivery and installation, realisation of all related masonry, tiling and electrical work, as well as the complete 'after-sales service'.

Some kitchen specialists are extending their portfolio by adding bathroom products to their existing kitchen portfolio to benefit from potential synergies. As a result, such specialists adjust their names by including the bathroom supply to their business, e.g. KUCHE UND BAD (kitchen and bathrooms). Like bathroom specialists, kitchen specialists provide complete services that range from exhibition, technical advice, sales, installation, to after sales-service at a reasonable price. According to industry feedback, demand for new kitchens has been increasing during the past years. This trend is fuelled not only by a favourable economic climate, but also by a growing importance of the kitchen as a status symbol among Germans. Demand for kitchens is increasing in the middle and upper price segments. General furniture specialist stores supply bathroom and kitchen products among all other types of furniture. The most popular example is IKEA, which supplies its own products, and which has rapidly gained popularity among German consumers. Another example is the Austrian XXXLUTZ, which currently has 48 stores in Germany. In 2013, XXXLUTZ launched its online shop in Germany to benefit from the digital consumer movement. POCO is a German furniture discounter that also has a limited supply of kitchen and bathroom products. The chain has more than 100 stores in Germany and targets the lower price segment.

In addition to these nation-wide chains, there are numerous regional and local furniture stores that might also offer bathroom and kitchen products, with the latter being more likely to be found in such a store. Other specialist retailers also include

food discounters such as ALDI, LIDL, PLUS, REAL and NETTO. On a promotional basis, they offer products such as taps and mixers, bathroom accessories and furniture for a limited amount of time. Due to their strong presence in the German market, a discounter can sell a considerable number of taps in just a week.

Internet Retailers

Sales via internet retail stores have been increasing in the last decade. Internet stores are a source of information and they make prices more transparent for consumers. Not all markets and products are suitable for online sales. Renovating homeowners and individuals who build their own houses are more likely to buy online than professional landlords and non-residential housing developers. Products with a perceived lower risk of buying online are more prone to be sold than products with high technical complexity, for example taps and mixers and shower heads compared to baths and shower trays.

Even if not all bathroom markets and products are believed to be suitable for online sales, purchases via internet retail stores have been increasing for the bathroom market over the last few years. It is expected that new internet retailers are emerging and that existing ones will increase their share of offered products over the coming years, as the Internet has established itself as an alternative distribution channel. In addition to online shops with a focus on sanitary products, there are also online shops that have some bathroom products only. According to a study by DEUTSCHLAND TEST, the following online shops are the most popular for bathroom products: MEGABAD, REUTER, NEUESBAD, BADMOEBELDIREKT, EMERO, ARCOMCENTER and MACH-DEIN-BAD. Moreover, there are online shops such as UNIDOMO and HAUSTECHNIKSHOP24 that supply both sanitary and heating products. According to ORBIS, REUTER's turnover is estimated to have been EUR 195 million in 2019, of which almost all was for sanitary products. AMAZON is believed to be significant, too. Online shops like AMAZON, ZALANDO, OTTO and TSCHIBO also include some bathroom products in their portfolio. The most sold products include taps and mixers, bathroom and shower accessories.

Some bathroom online shops opened physical showrooms to display products and to give advice to consumers. REUTER opened a physical store in the high street of Düsseldorf and MEGABAD is present near Bochum. The intention is to allow consumers to look at products before they can be purchased through the website only. Local installers generally refuse to install products bought by the end users on the Internet, due to warranty issues. For that reason, REUTER launched the website WERKPROFI, which provides a list with installers that do install products bought from REUTER. Most smaller industry-specific web shops are set up by installers, who receive their products from 'three-step' manufacturers and wholesalers.

Internet stores have a market edge in their price transparency approach by presenting prices to the end customer. Typical clients of online platforms are homeowners opting for renovations, as well as individuals building their own houses. Moreover, professional installers from smaller companies are likely to

make use of online shops, too. In contrast, non-residential project developers and large installer companies are unlikely to buy online. Generally speaking, 'simple-to-install' products are more likely to be purchased online than those with high technical complexity. In the past years there has been a trend towards more online sales of simple-to-install products. The reason is the lack of professional installers, who would not have time for smaller replacement jobs like broken taps and mixers, shower heads, or even WCs and washbasins. Major German manufacturers are aware of the growing position of internet retailers. Among other reasons, some major German manufacturers announced the reduction of their list prices by 20%. During the last few years, the list price, which is a fictive price suggested by the manufacturer to sell their product for, has increased significantly. This has led to a big gap between the list price and the actual selling price. By reducing the list significantly, manufacturers prevent online shops from communicating large discounts.

SUMMARY OF LEADING DISTRIBUTORS

DE - LEADING DISTRIBUTORS

Holding	Trading Name	Distributor Type	General Product Orientation Profile	Net Turnover		Home Market Coverage		Specific Product Orientation														Buying Group Affiliation
				Year End	€ million	No. of Depots	Geographical Coverage	Specialist						Specialist						Other		
								Total	Heat	Bath	Plumbing	Vent/ Aircon	Tiles	Total	Building Materials	Timber	Insulation	Roof	Other	Specify		
TENGELMANN	OBI Group Holding GmbH	DIY Retailer	DIY Assortment	2019	5,890.0	352	DE/International	xx	x	xx		x	x	x	x	x	x		xx			
BAUHAUS	BAUHAUS AG	DIY Retailer	DIY Assortment	2019	61.8*	126	DE/International		x	x	x	x	x		x	x	x	x	xxx	Kitchen, Electrics		
HAGEBAU	HAGEBAU KG	DIY Retailer	DIY Assortment	2019	6,600.0	477	DE/Europe	xx	x	xx		x	x	x	x	x	x		xx	Garden	HAGEBAU + E/D/E + EK	
GC GROUP	GC GROSSHANDELS CONTOR GmbH	Mixed merchant	Mixed merchant	2020	5,000.0	800	DE, Eastern Europe	xxx	xx	xx	xx	x	x	x			x	x				
REWE	TOOM BAUMARKT GmbH	DIY Retailer	DIY Assortment	2019	2,800.0	330	DE	xx	x	xx		x	x	x	x	x	x		xx		DIY Union	
STARK GROUP	STARK DEUTSCHLAND GmbH (former SAINT-GOBAIN Building Distribution Deutschland GmbH)	Mixed merchant	Mixed merchant	2019	2,019.1	220	DE	x			xx	xx	x	xxx	xx	xx	xx	xx				
HORNBACH	HORNBACH BAUMARKT AG	DIY Retailer	DIY Assortment	2019/20	4,700.0	161	DE/EU Countries	xx	xx	x		x	x	x	x	x	x		xx	Electrics, Kitchen, Garden		
GLOBUS	Globus Baumarkt Holding GmbH & CO K	DIY Retailer	DIY Assortment	2019/20	1,900.0	91	DE	xx	x	xx		x	x	x	x	x	x		xx			
BOSCH	BUDERUS Großhandel	Specialist Merchant	Heating, Vent/Aircon	2019/20	3,600.0	51	DE	xxx	xxx	xxx	xx	x										
RICHTER + FRENZEL	RICHTER + FRENZEL GmbH + Co. KG	Mixed merchant	Mixed merchant	2017	1,400.0	141	DE-South from Kassel	xxx	xxx	xxx	xx	x	x	x	x	x					ISG through COMIGRO	
HELLWEG	HELLWEG DIE PROFI-BAUMAERKTE GmbH & Co. KG	DIY Retailer	DIY Assortment	2018	758.0	75	DE, AT	xxx	xx	xx		x	x	xx	xxx	xx		x	x	Garden	DIY Union	
SANITÄR HEINZE	SANITÄR HEINZE GRUPPE	Specialist Merchant	Heating, Vent, A/C, Sanitary/Plumbing	2018	291.9	22	DE-SE (Bavaria, Saxony), AT, IT	xxx	xxx	xxx	xx	x									SANITÄR-UNION	
WIEDEMANN	WIEDEMANN DIENSTLEISTUNGS- UND VERWALTUNGSGESELLSCHAFT mbH	Mixed merchant	Mixed merchant	2016	117.0	75	DE, Poland	xx	xxx	xxx	xx	x	x	xx	x			xxx	Machinery, Building Automation	GSH		
RENSA GROUP	DEINZER + WEYLAND Vertriebs GmbH	Specialist Merchant	Heating, Vent, A/C, Sanitary/Plumbing	2018	204.0	13	DE	xxx	xxx	xxx	xx	xx									VGH	
ZANDER	J.W. ZANDER GmbH & Co. KG	Specialist Merchant	Heating, Vent, A/C, Sanitary/Plumbing	2019	250.0	80	DE	xxx	xxx	xxx	xx	x							xx	Electric	.	
MAINMETALL	MAINMETALL GmbH & Co. KG	Specialist Merchant	Heating, Vent, A/C, Sanitary/Plumbing	2019	161.0	20	DE-E	xxx	xxx	xxx	xx	xx		x			xx	xx			ISG	
HEINRICH SCHMIDT	MOSECKER GmbH & Co. KG	Specialist Merchant	Heating, Vent, A/C, Sanitary/Plumbing	2019	174.0	23	DE-W	xx	xxx	xxx	xx	xx		x					xx	Steel	ISG	
FRIELING	FRITZ FRIELING GmbH	Mixed merchant	Mixed merchant	2019	135.0	15	DE-NW	xx	xxx	xxx	xx	xx		x					xx	Steel	SANITÄR UNION	
CRH	ANDREAS PAULSEN GMBH & CO. KG	Specialist Merchant	Heating, Vent, A/C, Sanitary/Plumbing	2019	15.0	51	DE-N	xxx	xxx	xxx	xx	xx									ISG	
REINSHAGEN + SCHRÖDER	REINSHAGEN & SCHRÖDER GmbH & Co. KG	Specialist Merchant	Heating, Sanitary/Plumbing	2016	42.0	11	DE-W	xxx	xxx	xxx	xx	x									VGH	
PIETSCH	Dr. Pietsch Holding GmbH & Co. KG	Wholesaler group	Heating, Vent, A/C, Sanitary/Plumbing	2019	340.0	60	DE-W and Saxony	xxx	xxx	xxx	xx	x									SANITÄR-UNION	

Source: BRG based on company information and industry interviews

*Turnover in Germany only

SW = South West

N = North

E = East

W = West

SE = South East

x = marginal activity

xx = substantial activity

xxx = core activity

SUMMARY OF LEADING GERMAN BUYING GROUPS AND COOPERATIONS

DE - LEADING BUYER GROUPS AND COOPERATIONS

Trading Name	Company Name	Distributor Type	General Product Orientation Profile	Net Turnover*		Home Market Coverage		Private Label Brands	Members	Affiliated Organisations
				Year End	€ million	No. of Depots (DE/abroad)	Geographical Coverage			
E/D/E	EINKAUFBUERO DEUTSCHER EISENWARENHAENDLER GmbH	Buying Group	All building materials	2019	597.7	870/1386	DE/International	FORMAT, FORUM	1,000 Merchants (PVH) 376 Retailers (Subsidiary ZEUS)	WUPPER-RING
EDT	EUROPEAN COOPERATION OF PROFESSIONAL WHOLESALERS FOR SANITARY AND HEATING PRODUCTS	Buying Group	Heating, Sanitary/Plumbing	2016	5,510.0	215/1500	13 European countries		9 European buying groups with 250 wholesalers: ISG, CEALCO, ALGOREL, PTACEK, GEPESZCSOPORT, INSTAL KONSORCJUM, IDRO TEAM, LM, LUK	ZEUS
EUROBAUSTOFF	EUROBAUSTOFF HANDELSGESELLSCHAFT MBH & CO. KG	Buying Group	Building products	2019	5,990.0	1350	DE, AT, CH, FR, IT, LU, LI	PRIMA, OPUS, CERABELLA	474 medium sized companies	
HAGEBAU	Hagebau Handelsgesellschaft fuer Baustoffe mbH & Co. KG	DIY Cooperation	Building products	2019	5,610.0	>1000	DE, AT, CH		250 medium sized companies	ZEUS
HBG	HBG HANDELS BETEILIGUNGS GESELLSCHAFT mbH	Wholesaler Group	Heating, Sanitary/Plumbing	n/a	n/a	>180	DE		6 members (Elmer, Wiedemann, Börner, Pietsch, König, Jensen)	EDT and UHT
ISG	ISG SANITAER HANDELSGESELLSCHAFT mbH & Co. KG	Buying Group	Heating, Sanitary/Plumbing	2013	4.8	126/13	DE and Austria	CONCEPT, OPTIMA	28 members	
MAH	MITTELSTANDS-ALLIANZ HAUSTECHNIK	Buying Group	Heating, Sanitary/Plumbing	2019	4.3	n/a	n/a	TAKE	8 members (Bach, Bechem, Holtzmann, Reichwald, Richter & Röhrig, Vetter+Engels, Nerlich & Lesser, Wahl)	E/D/E
NORDWEST	NORDWEST HANDEL AG	Buying Group	All building materials	2020	512.0	732	DE/Europe	PROMAT, NOW, WELDING TEAM, DELPHIS	528 members (nordwest.com) (includes Nürnberger Bund)	
UNION HAUSTECHNIK	GSH GmbH & Co. KG	Buying Group	Sanitary/Plumbing	2019	120.0	n/a	n/a	SANIBEL, PREMIANO, MEDIANO, TOPARA, ARTIGO	UHT 31 members	
UNION HAUSTECHNIK	SANITAER-UNION GMBH & CO. KG	Buying Group	Sanitary/Plumbing	2019	3.8	127/6	DE/Austria	DIANA	UHT 31 members	
VGH	VGH VEREINIGUNG VON GROSSHÄNDLERN FÜR HAUSTECHNIK FÜR HAUSTECHNIK, MARKETING UND VERTRIEB VERWALTUNGS-GmbH	Buying Group	Heating, Sanitary/Plumbing	2012	1.4	31/135	DE, DK, NL, BE, RO, PL	KREOS, MIO	10 members (vgh-online.de)	E/D/E

Source: BRG based on company information and industry interviews

GERMANY

BATHS AND SHOWER TRAYS

THE EUROPEAN BATHROOM & KITCHEN PRODUCT MARKETS



Market Characteristics and Product Mix

Initially considered to be luxury products, baths were only present in upper-class multi-family houses, consisting of identically cut flats. The gradual shift in product material, which originally was expensive porcelain, towards the cheaper cast iron and, later on, enamel, broadly supported the increase of baths in residential dwellings. The post-war home building boom, that characterised Germany in the 1950s, influenced the bath market which, as a result, ultimately grew significantly. While it was not common to have a separate room as a designated bathroom in apartments, from this point on all dwellings were built with individual bathrooms, which normally included both a bath and a shower tray.

From the 1960s, new materials such as PVC and plastic were launched in the German market and gradually replaced porcelain and cast iron. Acrylic, which was introduced in the 1970s, swiftly gained popularity due to its moulding features, which permit various shape and colour designs. In addition to the mentioned perks, this new material denoted a higher comfort by giving a 'warm' feeling on contact with the skin. Sales of shower trays also benefited from the introduction of such a multifaceted material with increased sales. Nevertheless, the influence on shower trays was not as significant as it was in the bath market. Over the last decades, the everyday bathing and showering habits of Germans have changed, with a gradual move towards showering instead of bathing due to both economic and environmental reasons.

Contrary to the past, nowadays more residential buildings, especially apartments, are increasingly equipped with a shower only. This changing consumer preference has resulted in a long-term declining trend in bath sales in favour of the shower market. Furthermore, shower trays have been facing growing competition from the so called 'walk-in showers' (or 'wet shower rooms') since 2002. A walk-in shower, consisting of tile walls and base and using a shower drain, or shower channel, do not have a shower tray. To compete with walk-in showers, a trend towards flat, or even super flat shower trays, as well as larger sizes, has been seen now in the German market. Super flat shower trays represent the most fashionable type of showers that can be installed in a dwelling. The trend towards building larger and floor levelled showers is supported by the wide popularity among people of all ages, from young customers seeking a modern style to the elderly looking for more comfort and ease in the washing and cleaning process. More than half of new residential buildings is estimated to be equipped with drain products nowadays. In July 2017, the Abdichtungsnorm DIN 18534 was introduced to regulate the waterproofness of showers. Installers, planners and architects are now responsible for the level waterproofness of walk-in shower installations.

Another trend that is impacting the bath and shower tray markets is attributed to the revival of steel as a bath/shower tray material. This is mainly driven by the scratch resistance advantage that distinguishes steel to synthetic. Moreover, the margins in the synthetic market have been eroded over the years because of the high number

of competitors, which has impacted negatively both installers and producers. On the other hand, steel model producers ride the wave of circumstances by now offering a broader range of shapes and forms previously not available.

Recent Market Trends

The COVID-19 pandemic shaped the developments in the year 2020 and 'business as usual' was replaced by government measures. In Germany the official lockdown, which is better understood as a 'prohibition of contact', started at the end of March 2020. Bavaria was the first state to implement measures to contain the spread of the virus, which was followed by the remaining states towards the end of March. For a few weeks the German population was advised to work from home when possible, with most non-essential stores closing for a limited time only. The measures implemented varied by state (*Bundesländer*). Most stores, including DIY stores, were allowed to open again from the beginning of April 2020 onwards, with differences from state to state in the level of restrictions and measures. In certain regions, shopping in DIY stores was limited to professionals only.

During 2020, new housing construction continued without much interruption, with only large-scale project sites halted to implement protective measures. That being said, the overall production per employee is believed to have shrunk significantly, a trend that remains ongoing, since many employers introduced measures to protect the construction workers. The bathroom industry was declared part of the crisis infrastructure, but installers could still continue to work without many interruptions. Overall, there has been relatively little impact of the COVID-19 pandemic on the building related industry, including the bathroom market, in Germany. In some cases, end users may have postponed, or cancelled, bathroom related projects due to the uncertainty on the market, but overall demand remained relatively strong, as there was a large backlog of works from the past, benefiting sales of bathroom and kitchen equipment. At the same time, in certain instances, the COVID-19 pandemic may have contributed to additional demand. Since consumers were spending more time indoors, they had the opportunity to dedicate themselves to repair or upgrade works, whereas others might have decided to invest into their property by installing a new bathroom or kitchen due to (international) holidays being difficult because of the current pandemic.

In Germany in 2020, total demand for baths declined stronger when compared to 2019. The decline was caused by fewer sales to all three end-use segments. While sales of synthetic and steel baths declined, demand for cast iron and hydromassage baths slightly increased. Due to their low numbers, however, this growth did not compensate for the overall downward trend of the total bath market. Cast iron baths are a niche product and benefit from the trend towards free standing models. There are no known German manufacturers of cast iron baths and products are mainly imported from Italy, the UK, Portugal and Spain. Hydromassage baths experienced a small boost due to the pandemic and the desire to have more wellness at home. Sales are believed to be marginal when compared to total bath volumes. The main reason for the ongoing contraction of the market is lack of demand from bathroom renovations. When renovating an existing

bathroom, Germans tend to replace their old bath with a big shower and only if there is enough space will a new bath be installed. Baths are no longer a priority in a German bathroom. In new dwellings baths are rarely used for showering, the exception might be in social houses where lack of space might prevent the installation of both options.

The trend towards larger showers does not necessarily generate more demand for shower trays as the popularity of the walk-in shower continues to grow, particularly in upmarket bathrooms. Nevertheless, the shift away from shower trays towards tiled solutions seems to have reached its peak and it is believed that the share of 'walk-in-showers' is only marginally growing. In 2020, demand for shower trays developed stronger than in previous years. When taking a closer look, demand for synthetic shower trays increased and sales compensated for the small negative trend recorded for steel shower trays. Demand for ceramic shower trays remained positive, too, and the niche market continues to develop at strong growth rates.

Modern synthetic and steel shower trays with zero-threshold have the same visual appeal as a walk-in shower. Some end users opt for these types instead of a walk-in shower due to the lack of installers in the German market. It is easier and faster to install a shower tray than a walk-in shower. The German Sanitary Industry Association (Vereinigung Deutsche Sanitärwirtschaft – VDS) stated that, in 2017, of all German bathrooms 17% had a walk-in shower. In 2011, only 7% of all new build bathrooms in Germany had a walk-in shower. According to the 7th fundamental study of Germany's bathrooms by the German Sanitary Industry Association (VDS) from 2017, out of 46.2 million existing bathrooms, 76% had a bath and 53% had a shower with a shower enclosure. A significant share of these baths were also used for showering and are likely to be replaced by a statement shower during a refurbishment.

In terms of product mix, baths have continued to lose share in favour of showers (both shower trays and walk-in showers). This being said, the substitution trend has slowed down a little in the last two years as consumers have favoured converting the bathroom into a refuge for wellness and health and the need to lie in a bath to relax. Manufacturers have responded to the trend by trying to add more value to a bath, by equipping it with whirls, light, or even sound. Another example is the revival of free standing baths, which are believed to be modern and luxurious. Different designs and materials such as mineral cast and solid surface are used to create visually appealing free-standing baths. In many cases end users would place these free-standing baths in a corner close to the wall. In more luxurious dwellings freestanding baths can be placed in so-called wellness rooms or in the bedroom. While the share of freestanding baths is believed to be still small, there is a significant growth reported by the industry. While the overall bath market declined in terms of volume, the market value for synthetic baths increased compared to last year. The growing share of free-standing baths, which are usually made from mineral cast and solid surface materials, account for a higher turnover. Almost all manufacturers offer free-standing baths and the prices are significantly more when compared to a regular bath.

Ceramic shower trays remain a niche product, which has been gaining popularity in the upper price segment. It is unlikely that it will ever have a similar popularity as in France, for example. Demand for hydrotherapy baths marginally increased in sales volume. The main reason for this increase is the pandemic and lockdown measures, which prevented Germans from visiting wellness areas. With the growing trend for wellness and relaxation, the availability of having a relaxing hydromassage bath at home increased in popularity. A similar development has been seen for outside whirlpools, which are not captured in this report. According to manufacturers' feedback, demand has skyrocketed in 2020 and with end users not being able to go on holiday, they invested into more wellness products at home. Most bath manufacturers offer to equip their normal bath models with a hydromassage option. This can be in the form of clip-on jets or an integrated system. Some manufacturers started to observe a slight trend towards baths that are equipped with wellness features, such as whirls, light effect and music. Leading manufacturers, such as IDEAL STANDARD, GROHE and HANSGROHE, offer an upgrade to almost all of their different baths.

Demand for total shower trays slightly increased in 2020. The share of steel shower trays is strongest in the economy/lower price segment, which are common in social housing and rented dwellings, where prices and durability are priorities. The penetration of walk-in showers in this price segment is estimated to be around one third. In contrast, the upper/luxury market segments are believed to have almost only walk-in-showers, whereas in the middle segment it is estimated to be more than two thirds. If a hotel is renovated from scratch, most of its bathrooms will be equipped with a walk-in shower. Overall, the use of shower drains as opposed to shower trays has established itself on the German market. In fact, according to BRG research, the market size of shower drains is estimated to be around over half a million per year. BRG estimates a market size of around 805,000 shower trays including shower cubicles in 2020.

Prices and Market Values

Average prices, at manufacturer level, for most baths and shower trays continued to increase in Germany during 2020. This was in line with trends seen in previous years. Average MSPs for synthetic products were reported to have had the largest increase compared to the other materials. This development underlines the shift towards larger shower trays as well as more expensive materials.

Hydromassage baths overall underwent a marginal price increase. The growing share of the economy/lower price segment compensated for the small growth so that overall market value remained almost on the same level as in 2019. Hydromassage baths in Germany have a high share in the upper and luxury segments. Based on the quality of the hydromassage bath, prices can vary massively between different models. In general, products sold within DIY stores are significantly cheaper than those sold via direct sales and wholesale channels, which are a more exclusive product.

While hydromassage baths are shifting towards the lower price segments, the bath and shower tray markets are shifting into the upper-priced segments at the expense of the lower segments.

The MSP of cast iron baths is subject to strong fluctuation. All cast iron baths are imported from abroad and, depending on the country of origin, the MSP can differ a lot. For example, baths from the UK and Italy tend to be more expensive, while products manufactured in Spain and Portugal have a lower MSP. The total number of the niche product can vary significantly and demand is very volatile.

New Housing

The overall new build market slightly declined in 2020 mainly due to fewer multi-family dwelling constructions. All products experienced a decline in that segment. Steel baths and shower trays were influenced in a more negative way when compared to other materials as the steel models are common in project works in Germany. In general, shower trays continued to face competition from walk-in showers; however, the impact on shower tray sales volume was limited when compared to 2019. The growing penetration of walk-in showers among the new housing segment is believed to have almost stabilised in recent years. BRG estimates that around 61.4% of all showers in new build houses and around 42.7% of all showers in new apartments were equipped without a shower tray in 2020.

As a result of fewer new housing completions, demand for hydromassage baths and hydrotherapy shower cubicles in new construction was negative in 2020. Wellness in new bathrooms mainly comes from the upper price segment, where the owners have enough money to spend on such luxurious products.

There is a trend in new housing that two bathrooms are built in new dwellings, rather than one main bathroom and a guest toilet. Therefore, even a 100 m² apartment would have two bathrooms with two showers. The option of a second bath is given to buyers, and installation depends on their decision. Overall, the number of showers is increasing, whether it is a larger second bathroom or an en-suite bathroom in the master bedroom.

Housing RMI

Replacement demand for baths slightly weakened in 2020, whereas demand for shower trays underwent a solid growth. Sales of all types of baths, except for cast iron, underwent a decline in 2019. This can be explained by the trend towards showering and the consumer's reluctance to replace a bath, which can have a long life span. Instead, if an end user decides to replace a bath, it is more likely to be for design and wellness purposes. In this case a modern free-standing bath made of mineral cast or solid surface would be preferred.

For shower trays, ceramic and synthetic shower trays gained in volume in 2020 compared to the previous year. This can be explained by the trend towards upper

market products, where new types of materials and designs are getting popular for large and floor levelled shower solutions. The German replacement market is dominated by steel products, which are used to replace an existing bath or shower tray. End users, who want a product with a stronger focus on design, are more likely to choose more expensive materials such as mineral cast or ceramic. The growing demand of ceramic shower trays in the upper price segment continued in the RMI sector.

Sales of baths and shower trays used for RMI account for most of the market share. This replacement demand comes from private middle-aged homeowners, who are on the verge of retiring. These consumers built or bought their homes during the 1980s and 1990s and are motivated to modernise their bathrooms. The market quickly adapted to these clients, who are looking for higher comfort and more sophistication. Although most end users in that age do not want to talk about the needs of old age, some products such as step free entry models prove to be suitable when considering the demographic change. When it comes to declining bath sales, it is primarily these end users who are contributing to the long-term trend of replacing an old bath with a modern shower. This is leading to a considered slowdown of bath sales for RMI.

Non-Housing

In 2020, total non-residential construction output notably decreased as a consequence of the pandemic. The only exceptions, which are buildings for education and health, have developed slightly positive in 2020. All other sectors, such as industrial, hospitality, office and commercial building, had a strong decline as projects got postponed or cancelled due to the economic uncertainty. As a result, demand for all products notably declined in 2020 when compared to 2019. Most of the demand for non-housing bathrooms comes from the hospitality sector, as well as from healthcare buildings, offices, schools and sports clubs. It is interesting to mention, that many of these facilities are usually equipped with a walk-in-shower. Renovated hotels tend to have only walk-in-showers, while some older hotels might still replace their old showers with a modern walk-in-shower solution. There are exceptions where traditional showers are installed. Hospitals or homes for the elderly, for example, prefer steel shower trays to tiled floors because of hygienic reasons and wellness/luxury hotels are more likely to have a freestanding bath to enhance the comfort.

Supply Structure

In the steel bath and shower tray market two manufacturers, namely KALDEWEI and BETTE, have been dominating the market for a long time. KALDEWEI is the market leader and there is a continued competition between these two German manufacturers. Thanks to their dominating position, the two producers manage to keep the prices on a profitable level. Importers that are active in the market are the Spanish manufacturer ROCA, which is active with the ROCA and LAUFEN brands as well as a significant amount of OEM products to German distributors. KALDEWEI is believed to be the main supplier for private labels of DIY stores and

wholesalers. However, the volume is believed to be small when compared to total KALDWEI branded sales.

All cast iron baths are imported and mainly come from the neighbouring countries Poland, Portugal, the UK, Italy and Spain. Ceramic shower trays represent a niche product, which is supplied by few manufacturers only. This market was dominated by VILLEROY & BOCH in 2020 with its product called ViPrint. End users can choose among individual styles to be printed on their new ceramic shower trays to individualise their own bathroom design.

Supply of synthetic baths and shower trays is fragmented with a high number of domestic and foreign players competing in a saturated market. Companies such as VILLEROY & BOCH, IDEAL STANDARD and HOESCH play leading roles in the three-step distribution, which is the main selling process for both the steel and synthetic segments. While imports for steel products are low, due to the presence of domestic players, the overall synthetic market is characterised by a high share of imports, which are mostly sold via DIY, representing branded products from other EU countries. The largest non-EU exporters to Germany are Egypt and China, whose products are mainly sold in DIY chains. The portion of imported synthetic baths is significantly higher than the portion of imported synthetic shower trays.

Some domestic as well as foreign suppliers offer single shower tray lines made of a different material. One example is BREUER, which offers cast marble shower trays. In the case of wooden shower trays, they are sometimes supplied on a regional level or online by woodworking shops. Most major manufacturers offer cast mineral products to compete for higher profit margins. An example of a manufacturer with cast mineral baths is VILLEROY & BOCH, with its own material called quarly (which is a type of cast mineral). Additionally, DURAVIT, HOESCH, TREOS and PV HOLDING (GALATEA) offer mineral cast baths. While the number of manufacturers offering mineral cast or solid surface is limited to a handful of bath manufacturers, almost every shower tray manufacturer produces a mineral cast or solid surface shower tray. This includes manufacturers such as HOESCH, DURAVIT, HUEPPE, IDEAL STANDARD and VILLEROY & BOCH among others.

After having passed its growth cycle, the hydrotherapy market is no longer attractive and many large manufactures focus on other market segments instead. Only the traditional hydrotherapy suppliers, for example HOESCH, actively push sales in this segment. Some manufacturers have invested in the development of new wellness add-ons to their existing baths. The manufacturer OTTOFOND is active in the economy/lower price segment and sells *inter alia* through DIY. The high share of 'Others' in all hydrotherapy segments is a result of the decline of brand sales and the large number of small importers and suppliers.

Forecasts

The German bath and shower tray market is set to continue with a similar development as seen in 2020. A less negative sales trend is anticipated for baths, while shower trays are believed to gain in volume only marginally. New housing

completions are believed to increase in 2021 and a small boost is expected from more completions. In terms of renovations, consumers investing into their homes is likely to continue as long as Germans have money and are not allowed to go on holidays. Demand for baths and shower trays is likely to benefit from bathroom renovations in the upper middle and luxury segment. Consumers who can afford such products have more time available to carry out such projects. In contrast, households with families that are financially impacted by the pandemic are likely to halt and postpone renovations that are not necessary. This situation is likely to contribute to a growing overall value of the bath and shower trays market, while the actual volumes continue to stagnate.

Regarding demand from non-housing, the tourism and leisure industries are struggling to remain in business and, therefore, it is likely that demand for baths and shower trays is going to continue declining from that end-use segment. There are exceptions such as large hotel chains and successful business, which will make use of this low occupancy to upgrade their facilities.

The long-term outlook for the German bath market remains negative and sales are believed to decline on a yearly basis. Demand for shower trays on the other hand is likely to develop slightly positively from 2023 onwards. Due to ongoing weakening RMI demand for baths combined with the trend towards showering in the new housing market, overall sales are expected to stagnate. In terms of product mix synthetic products might gain share in the short term as long as the steel price remains high. Manufacturers reported steep price increases for 2021 due to growing steel prices as well as transportation cost.

There are signs that demand for walk-in showers has passed its peak growth rates and consumers are starting to prefer shower trays again. With the introduction of new materials, sizes and colours there is more competition for wet rooms. Many manufactures such as VILLEROY & BOCH already offer thin shower trays with printable designs similar to floor tiles. It is expected that demand for shower trays should stay positive during the forecast period, with the exception of the year 2022.

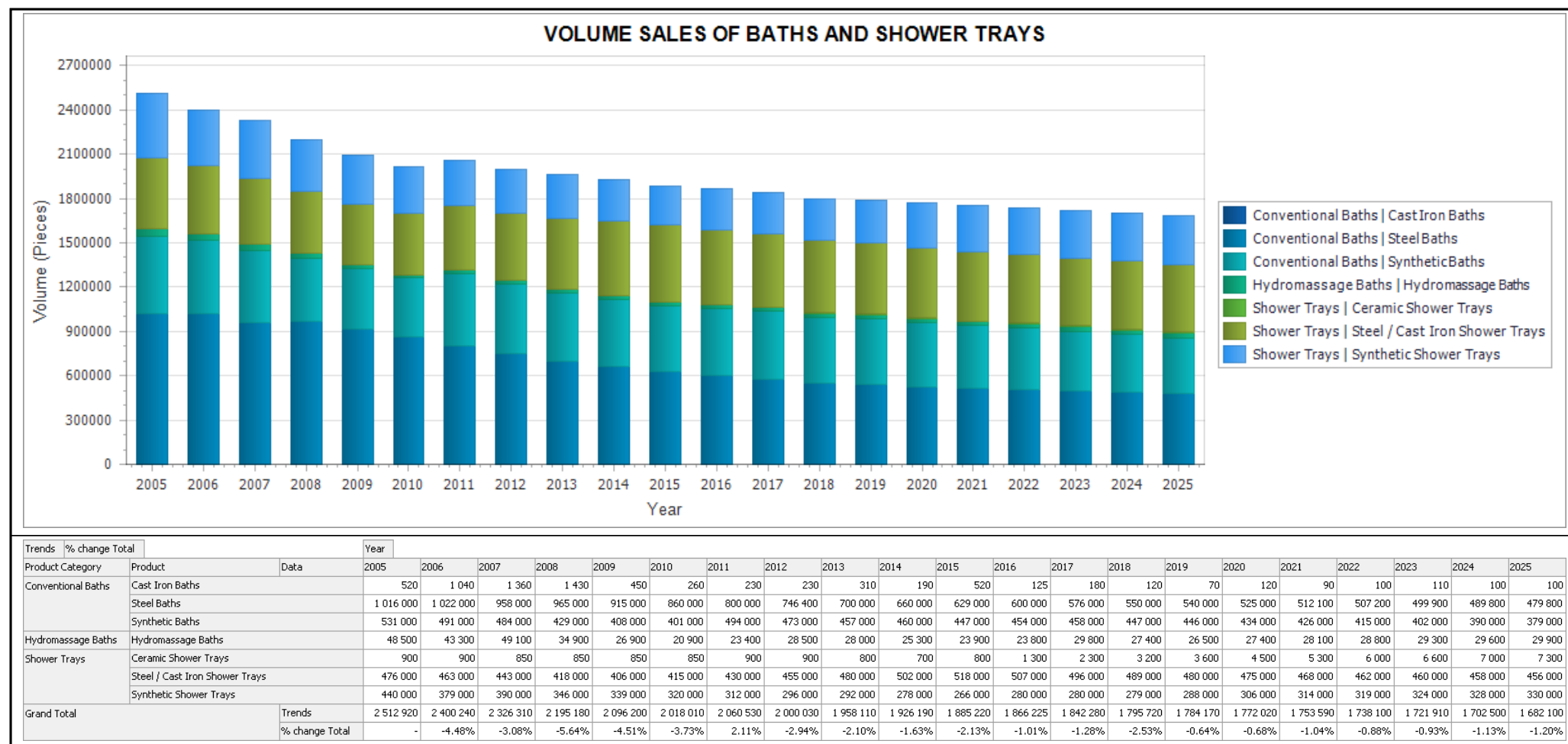
It is expected that new housing will decline significantly towards the end of the forecast period. This development is the result of the ongoing pandemic and the financial implications of the lockdown measures. Apart from that, both the product shift from baths towards shower trays and the trend for higher priced products and materials such as mineral cast and solid surface are likely to continue further, as well as the rising popularity of floor level shower trays, which already are a standard in the new build segment.

According to the 7th fundamental study of bathrooms in Germany by VDS from 2017, 17% of all existing bathrooms were equipped with floor level showers in 2017 as opposed to only 7% in 2011. In line with the evolution of floor level shower trays the share of walk-in showers is expected to stabilise at a small growth rate only. Manufacturers launch improved products that are easy to clean and visually pleasing. Bath sales volumes are expected to remain important in the German market. It is expected that baths and shower trays are likely to shift towards either

the economy or the upper/luxury price segments. A similar shift is also predicted for the hydromassage baths. The lifestyle of the German people is changing, and bathrooms are getting bigger, that is why the sales number of high-quality baths, which often are free-standing, is expected to increase during the forecast period.

1.2 Historical Trends and Forecasts

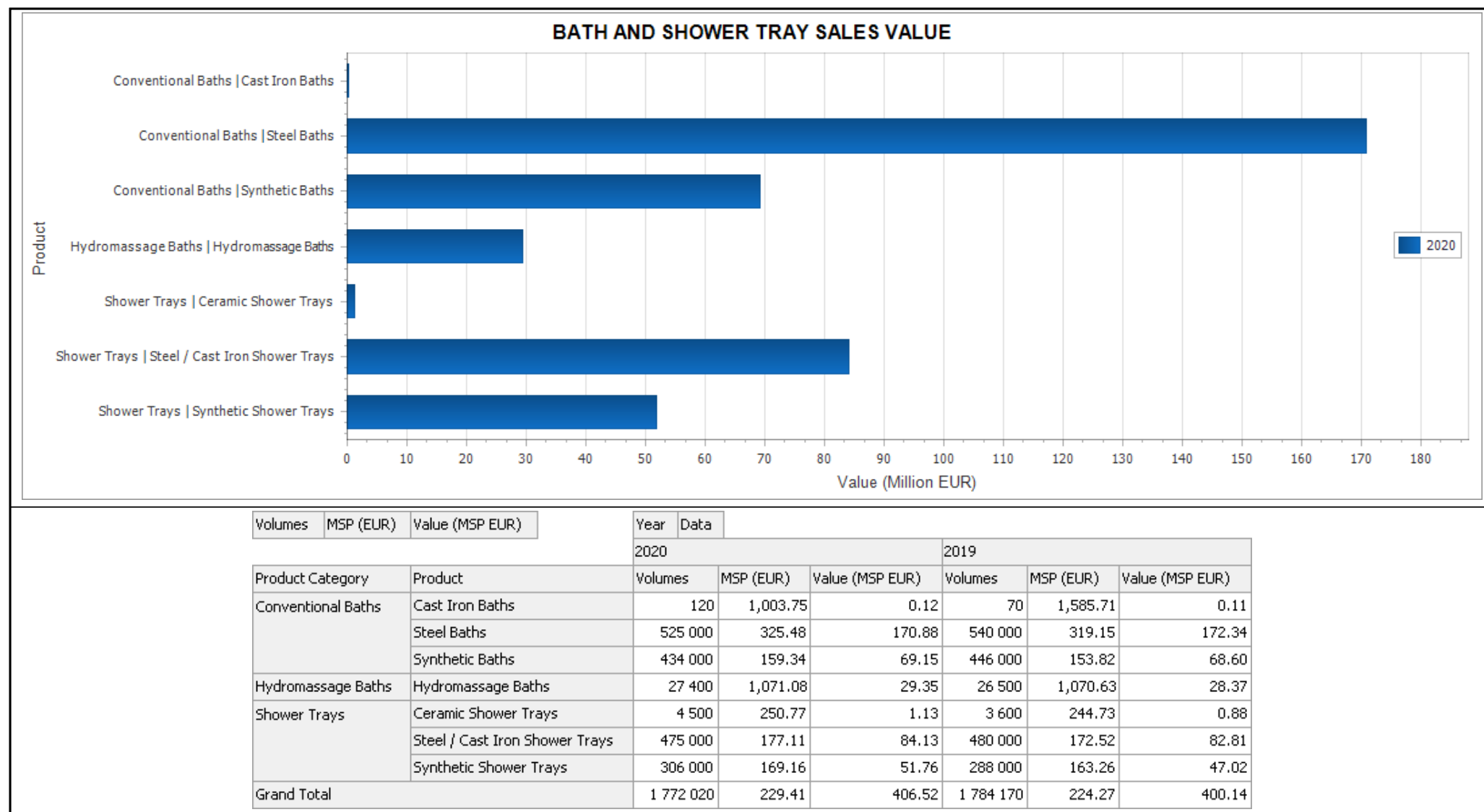
DE - SUMMARY OF TRENDS IN THE BATH AND SHOWER TRAY MARKET : HISTORICAL AND FORECASTS



Source: BRG based on industry interviews.

1.3.1 Prices and Market Values : Baths and Shower Trays

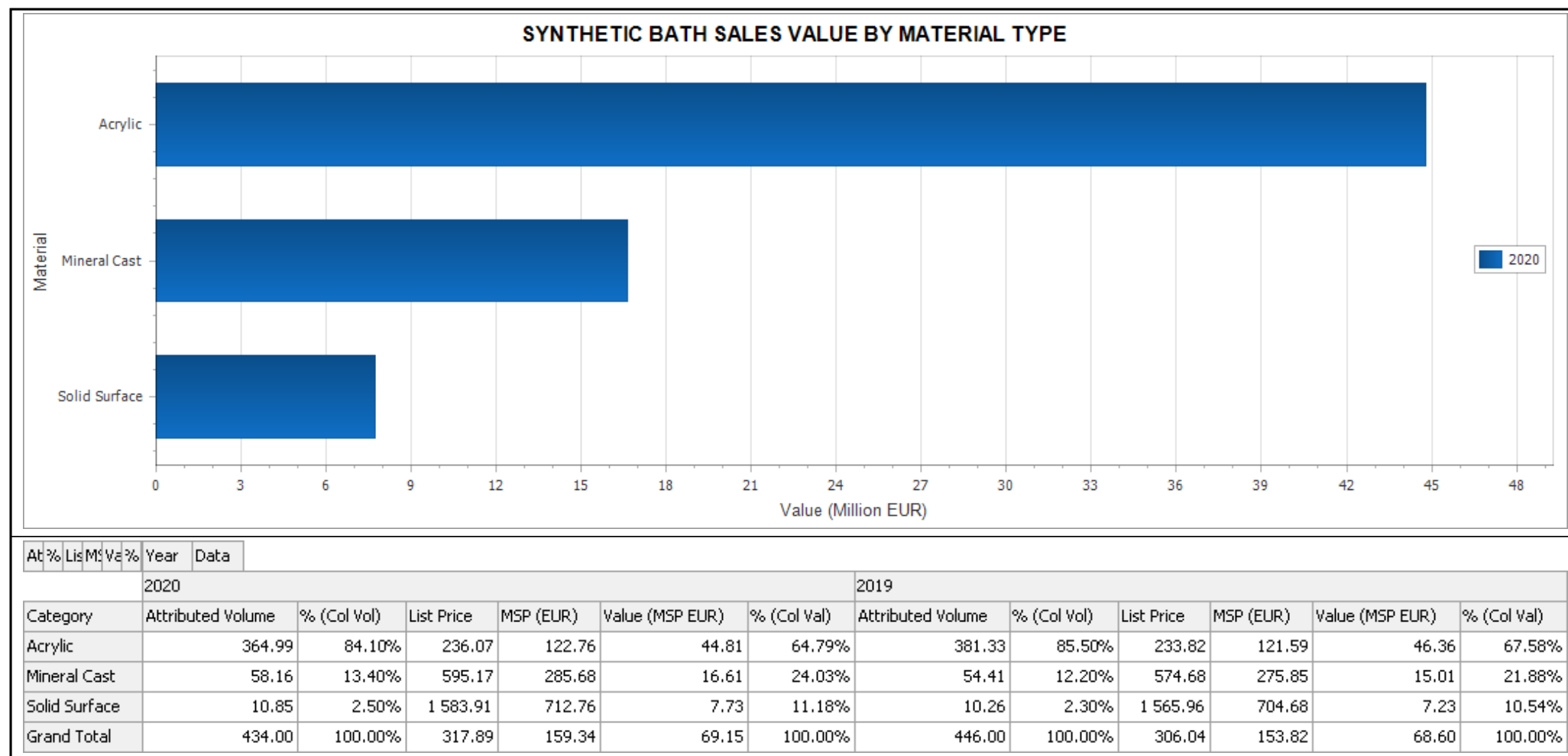
DE - ESTIMATED VALUES OF THE BATH AND SHOWER TRAY MARKET



Source: BRG based on industry interviews. Values Shown in Millions (at MSP = Manufacturers' Selling Price)

1.3.2 Prices and Market Values : Synthetic Baths

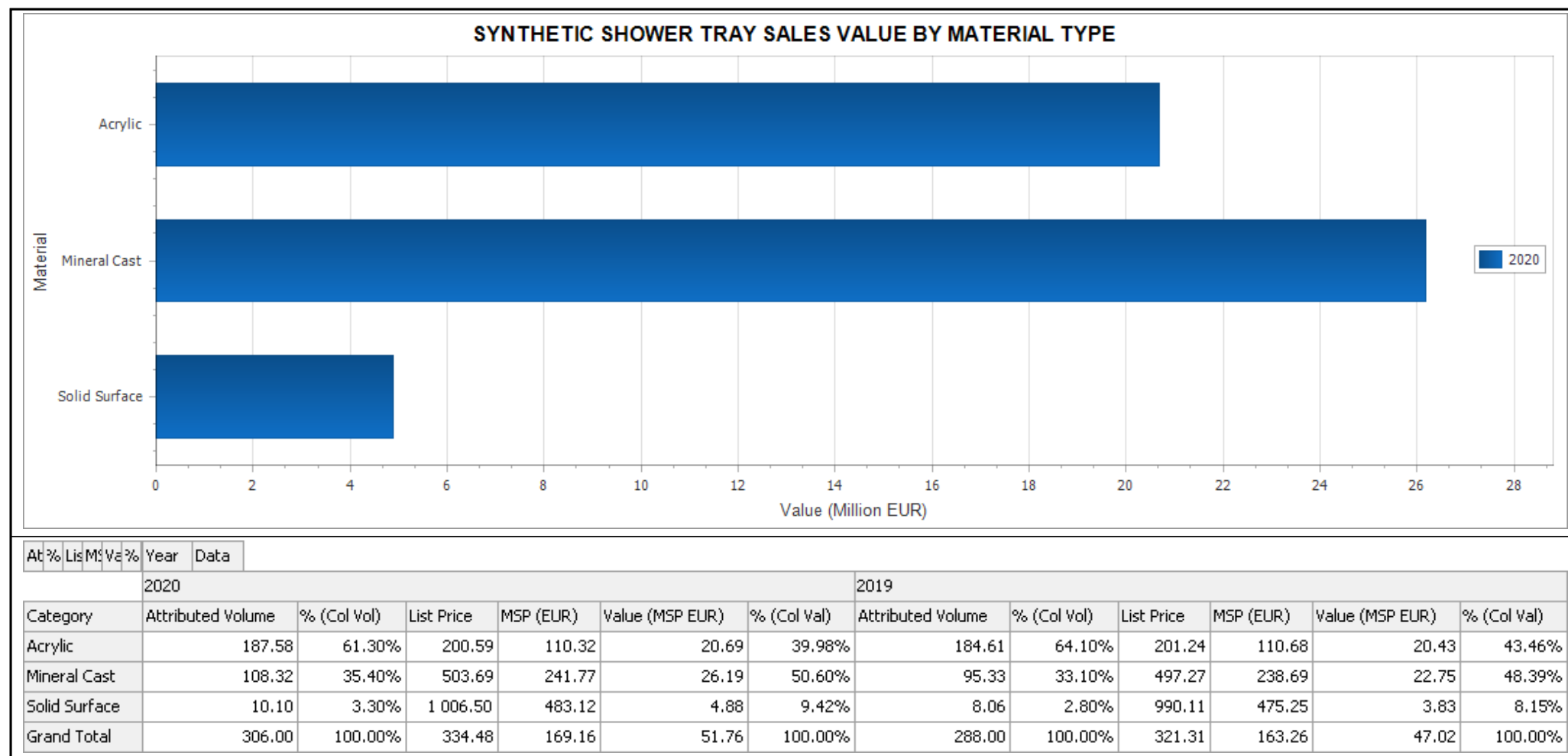
DE - ESTIMATED VALUES OF THE SYNTHETIC BATH MARKET BY TYPE OF MATERIAL



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Values Shown in Millions (at MSP = Manufacturers' Selling Price)

1.3.3 Prices and Market Values : Synthetic Shower Trays

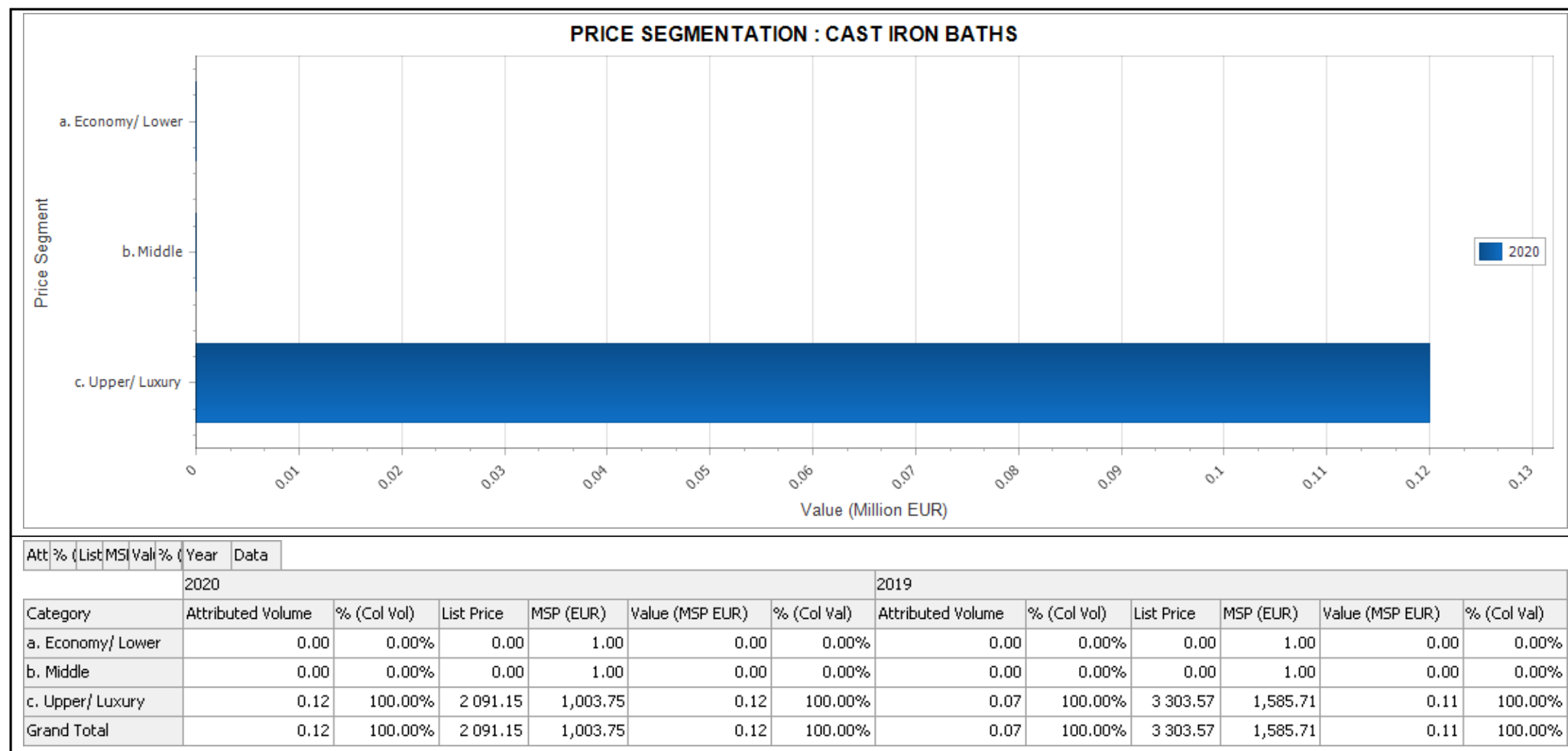
DE - ESTIMATED VALUES OF THE SYNTHETIC SHOWER TRAYS MARKET BY TYPE OF MATERIAL



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Values Shown in Millions (at MSP = Manufacturers' Selling Price)

1.3.4 Price Segmentation : Cast Iron Baths

DE - ESTIMATED PRICE SEGMENTATION FOR CAST IRON BATHS

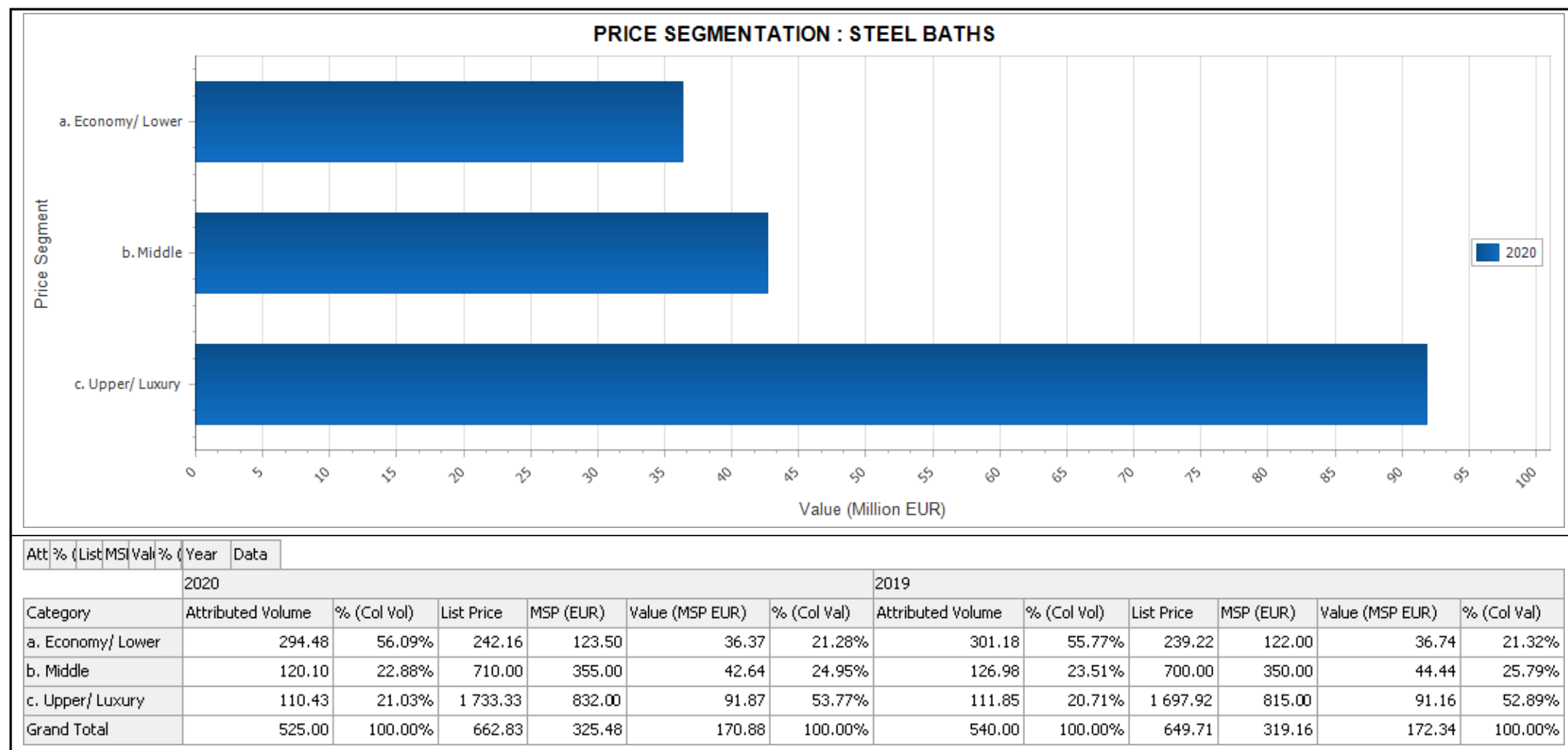


Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€400, Middle €400-€800, Upper/ Luxury >€800

1.3.5 Price Segmentation : Steel Baths

DE - ESTIMATED PRICE SEGMENTATION FOR STEEL BATHS

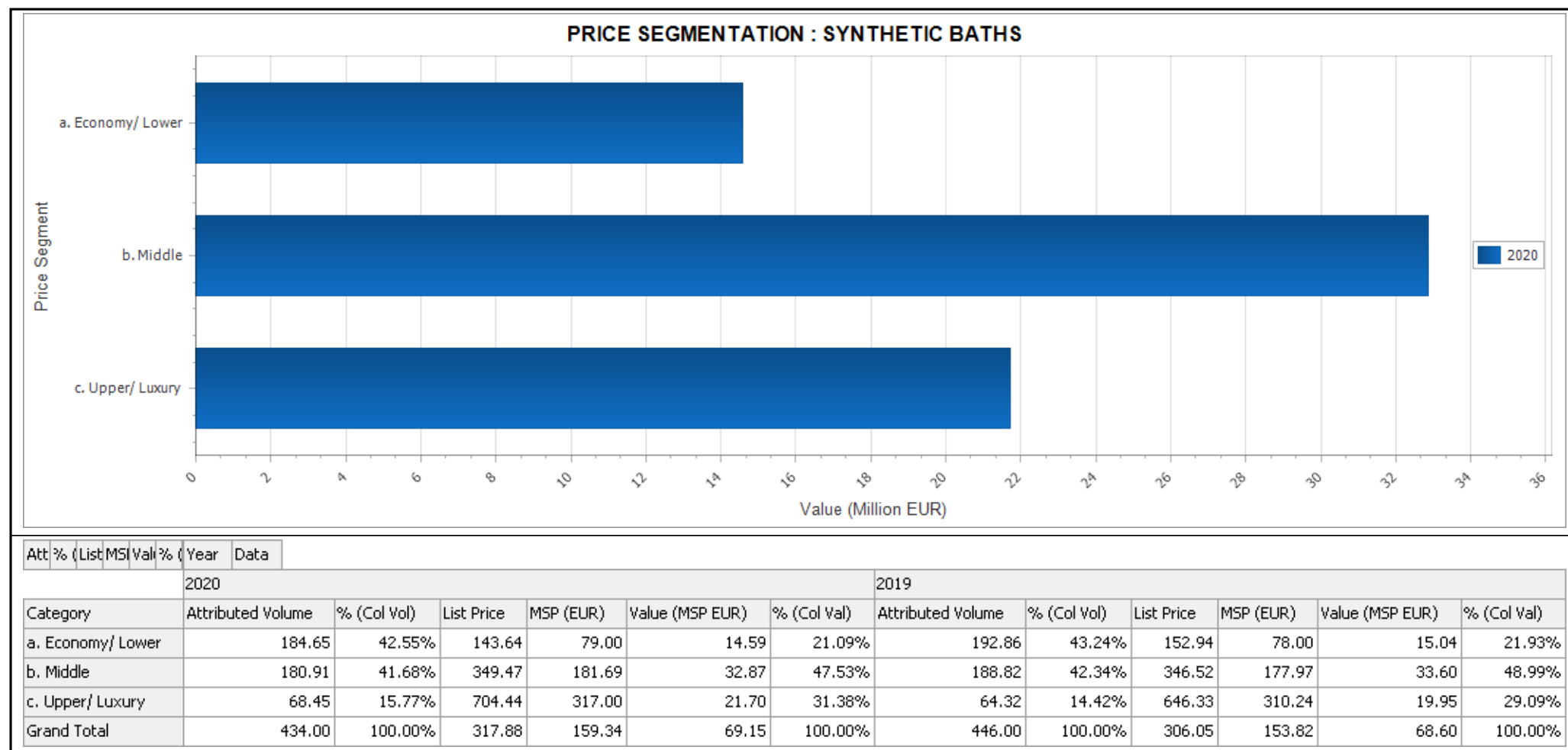


Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€250, Middle €250-€500, Upper/ Luxury >€500

1.3.6 Price Segmentation : Synthetic Baths

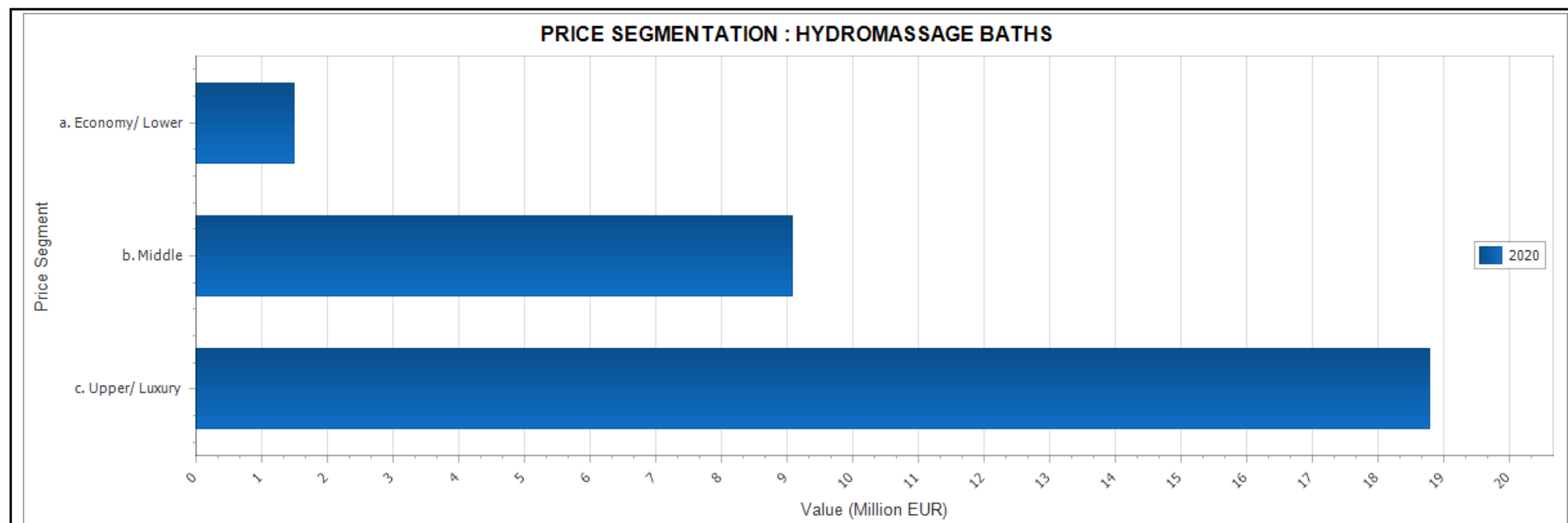
DE - ESTIMATED PRICE SEGMENTATION FOR SYNTHETIC BATHS



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€125, Middle €125-€250, Upper/ Luxury >€125

DE - ESTIMATED PRICE SEGMENTATION FOR HYDROMASSAGE BATHS



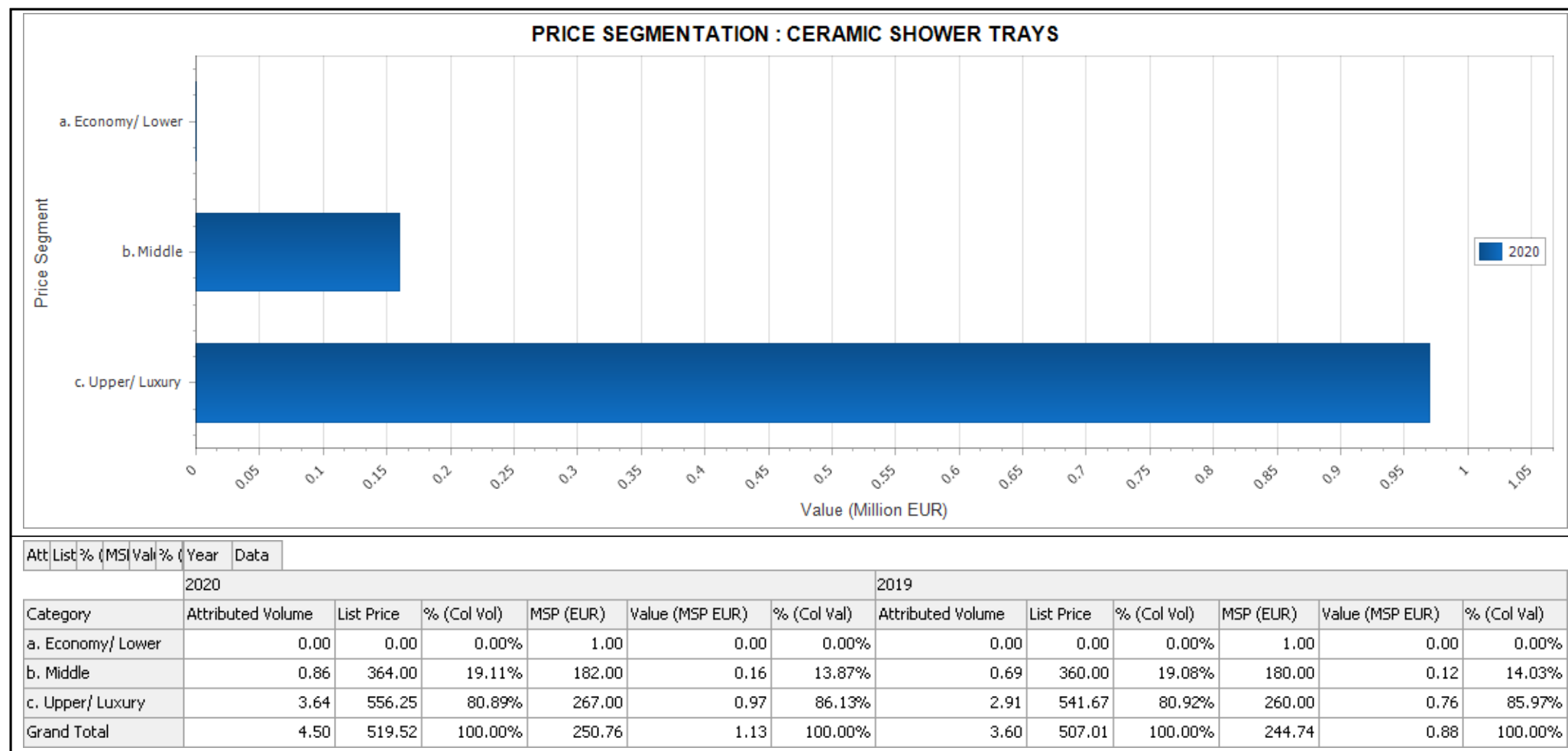
Att	%	List	MS	Val	%	(Year	Data					
2020							2019					
Category	Attributed Volume	% (Col Vol)	List Price	MSP (EUR)	Value (MSP EUR)	% (Col Val)	Attributed Volume	% (Col Vol)	List Price	MSP (EUR)	Value (MSP EUR)	% (Col Val)
a. Economy/ Lower	5.99	21.87%	498.40	249.20	1.49	5.09%	5.52	20.84%	496.00	248.00	1.37	4.83%
b. Middle	10.55	38.51%	1 755.10	860.00	9.08	30.92%	10.45	39.42%	1 746.94	856.00	8.94	31.52%
c. Upper/ Luxury	10.86	39.62%	3 460.00	1,730.00	18.78	63.99%	10.53	39.74%	3 430.00	1,715.00	18.06	63.66%
Grand Total	27.40	100.00%	2 155.69	1,071.09	29.35	100.00%	26.50	100.00%	2 155.00	1,070.62	28.37	100.00%

Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€500, Middle €500-€1,000, Upper/ Luxury >€1,000

1.3.8 Price Segmentation : Ceramic Shower Trays

DE - ESTIMATED PRICE SEGMENTATION FOR CERAMIC SHOWER TRAYS

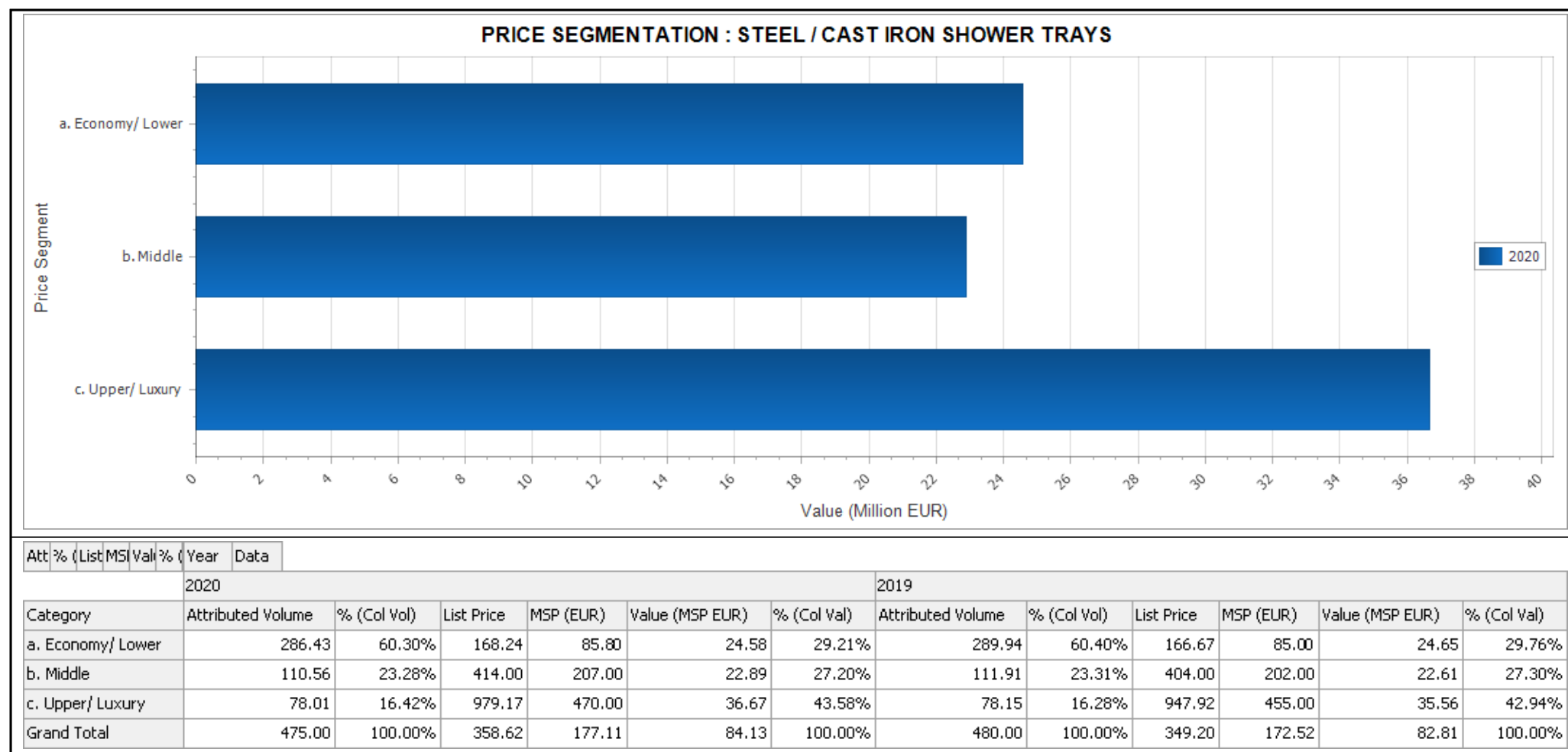


Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€120, Middle €120-€240, Upper/ Luxury >€240

1.3.9 Price Segmentation : Steel/ Cast Iron Shower Trays

DE - ESTIMATED PRICE SEGMENTATION FOR STEEL / CAST IRON SHOWER TRAYS

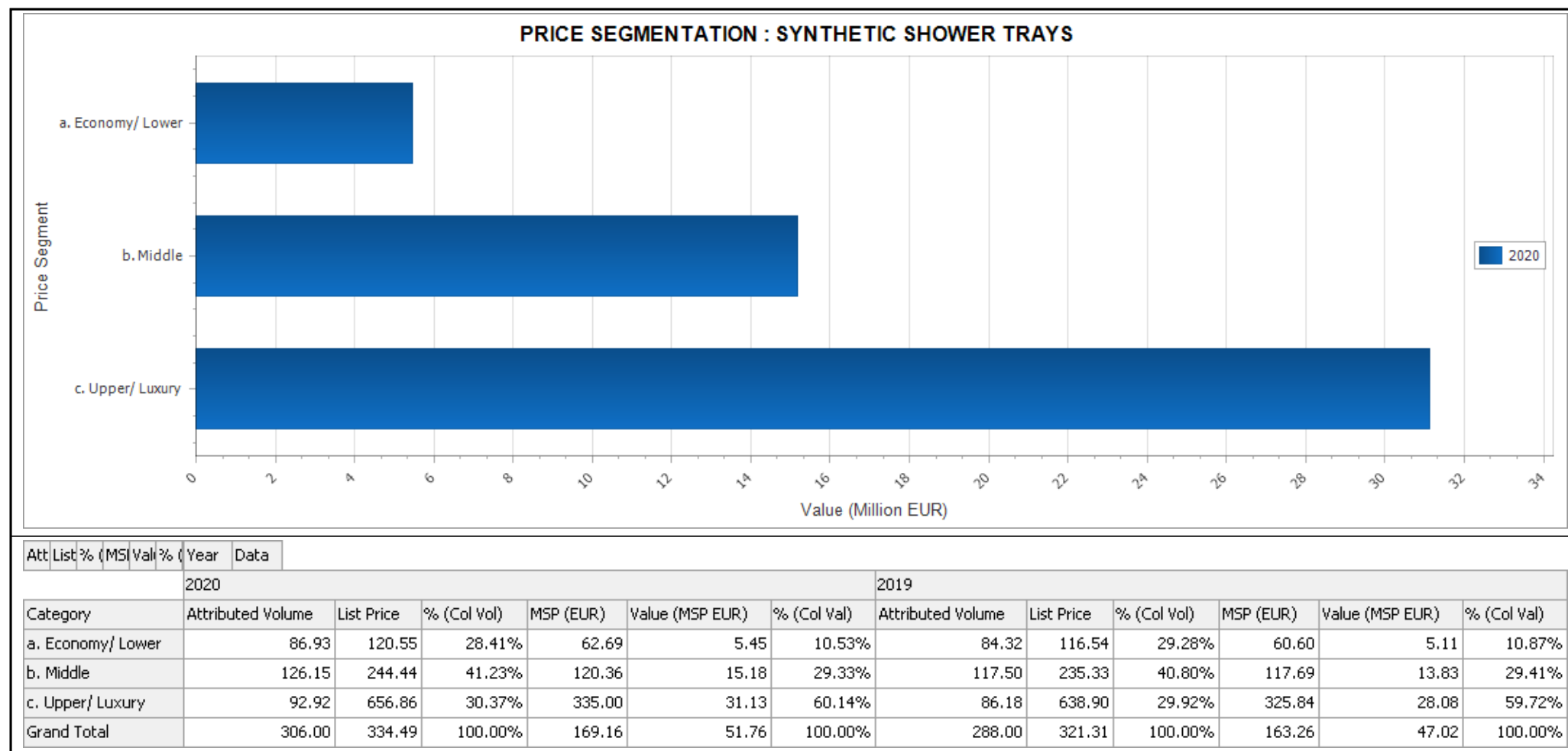


Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€150, Middle €150-€300, Upper/ Luxury >€300

1.3.10 Price Segmentation : Synthetic Shower Trays

DE - ESTIMATED PRICE SEGMENTATION FOR SYNTHETIC SHOWER TRAYS

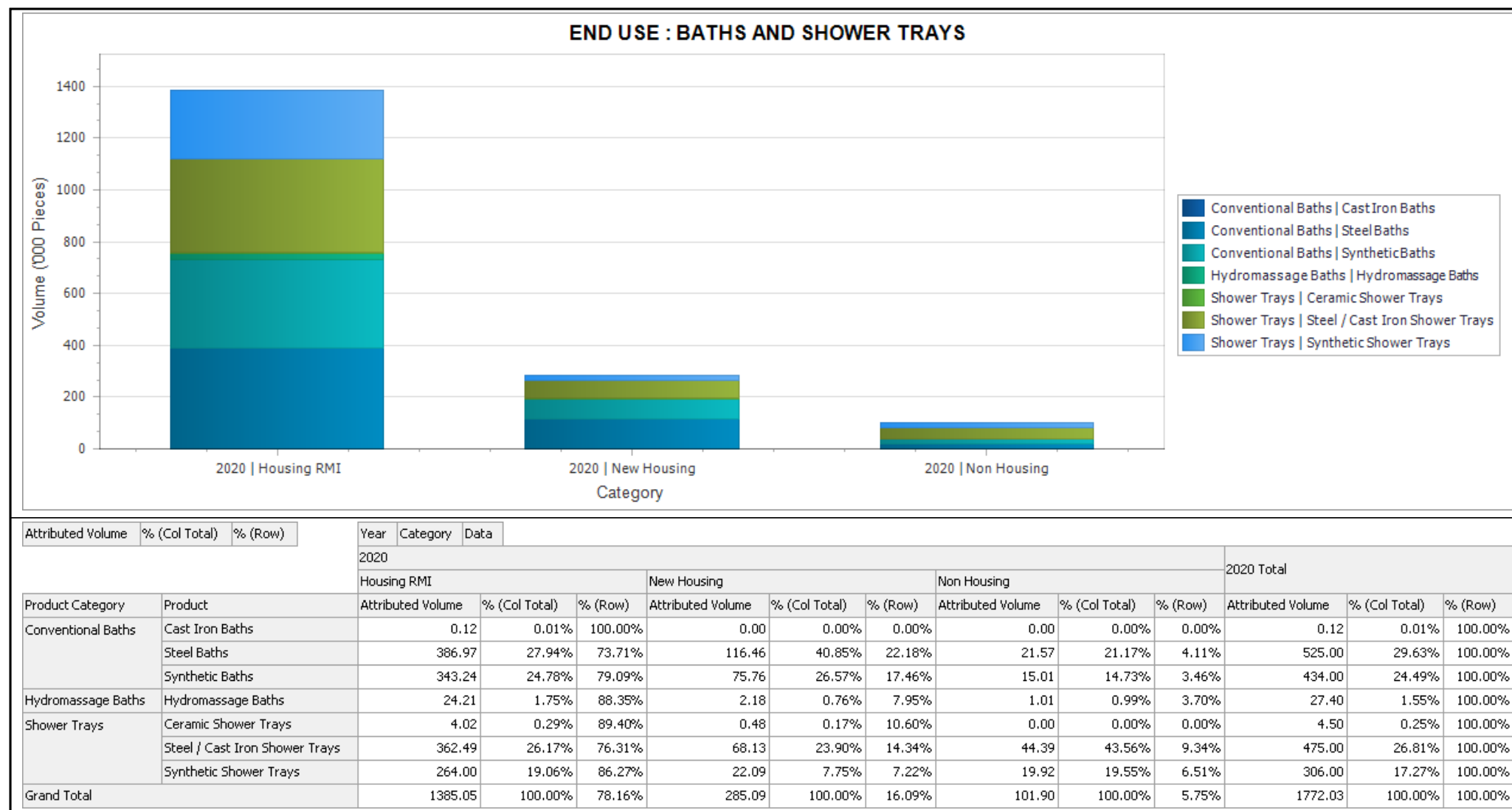


Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€75, Middle €75-€150, Upper/ Luxury >€150

1.4.1 End Use : Base Year

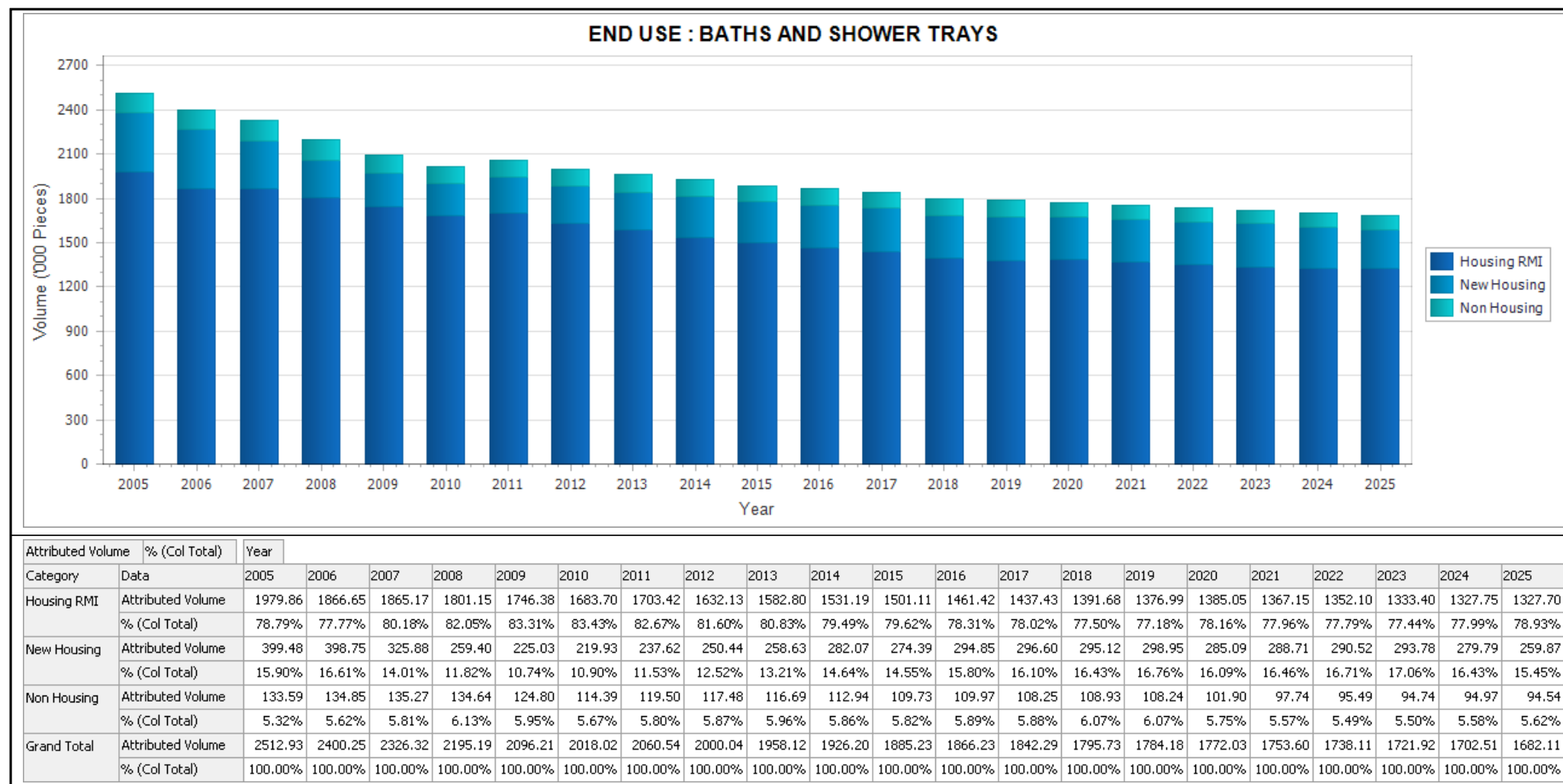
DE - SALES OF BATHS AND SHOWER TRAYS BY TYPE AND END USE SEGMENT ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

1.4.2 End Use : Historical and Forecasts

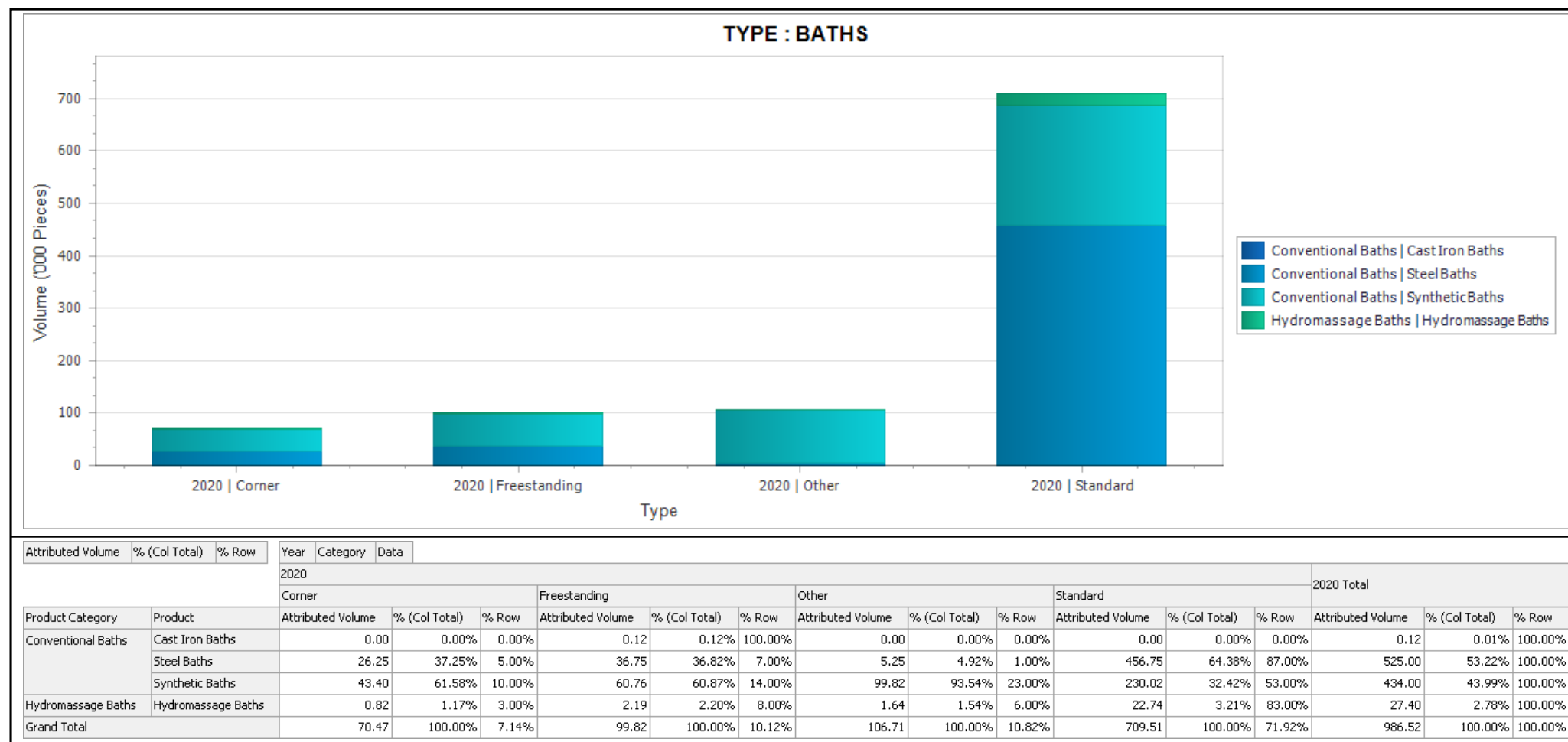
DE - SALES OF BATHS AND SHOWERS TRAYS BY END USE SEGMENT ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

1.4.3 Sales by Type : Baths

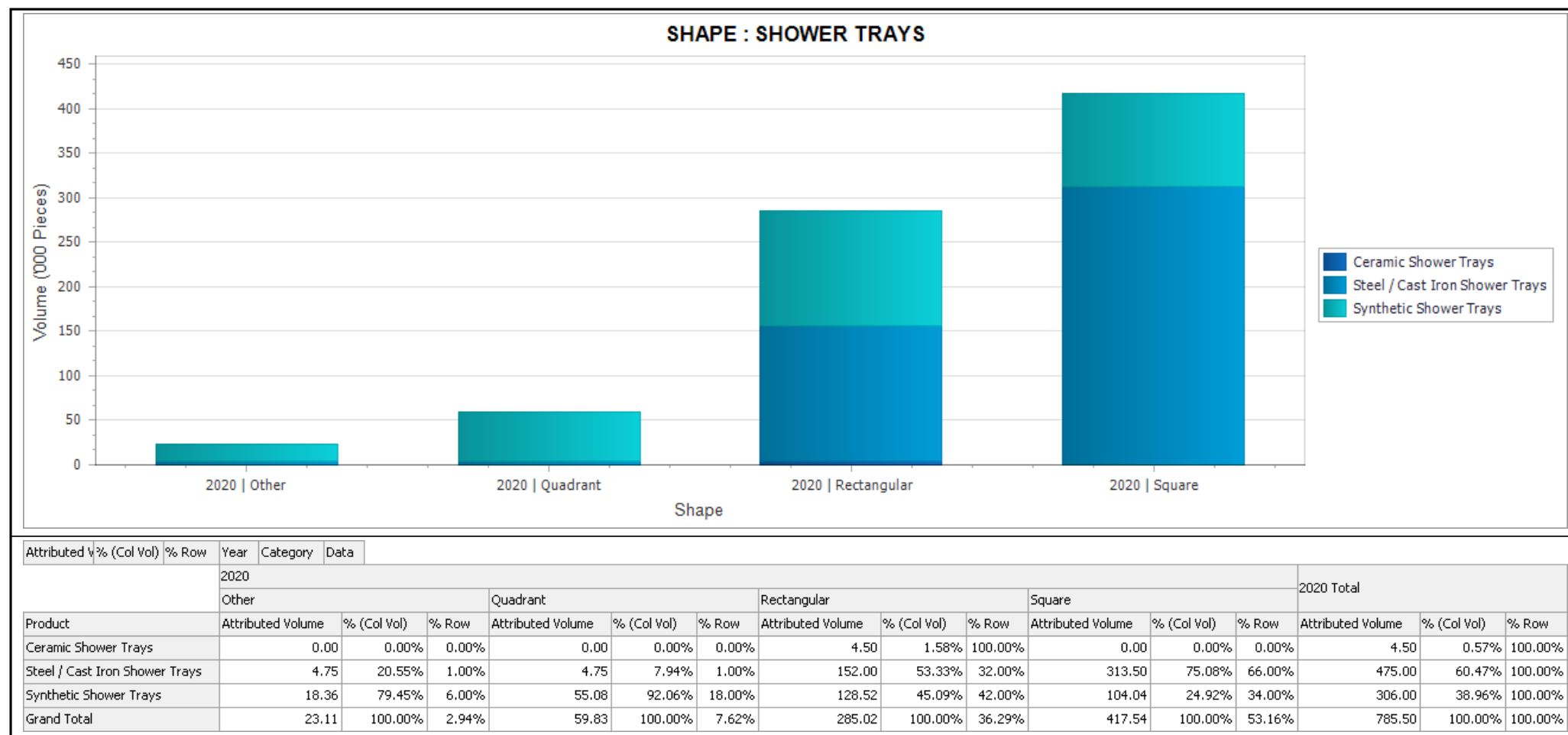
DE - BATH SALES BY TYPE ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

1.4.4 Sales by Shape : Shower Trays

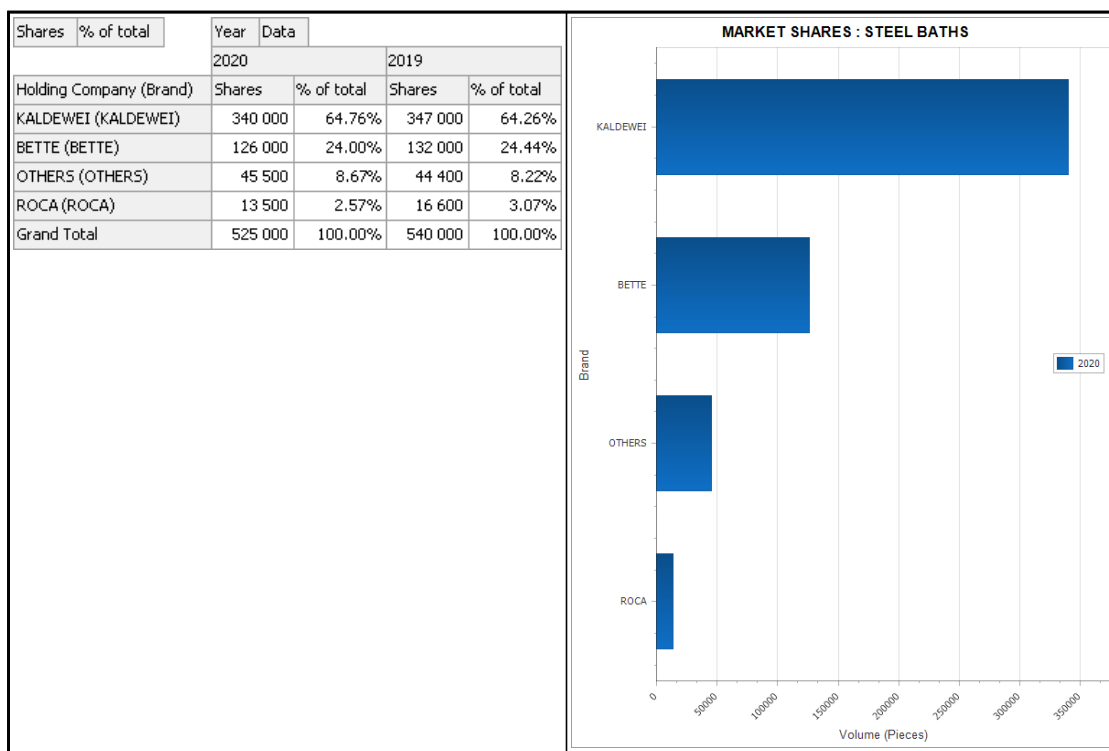
DE - SHOWER TRAY SALES BY SHAPE ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

1.5.1 Market Shares : Steel Baths

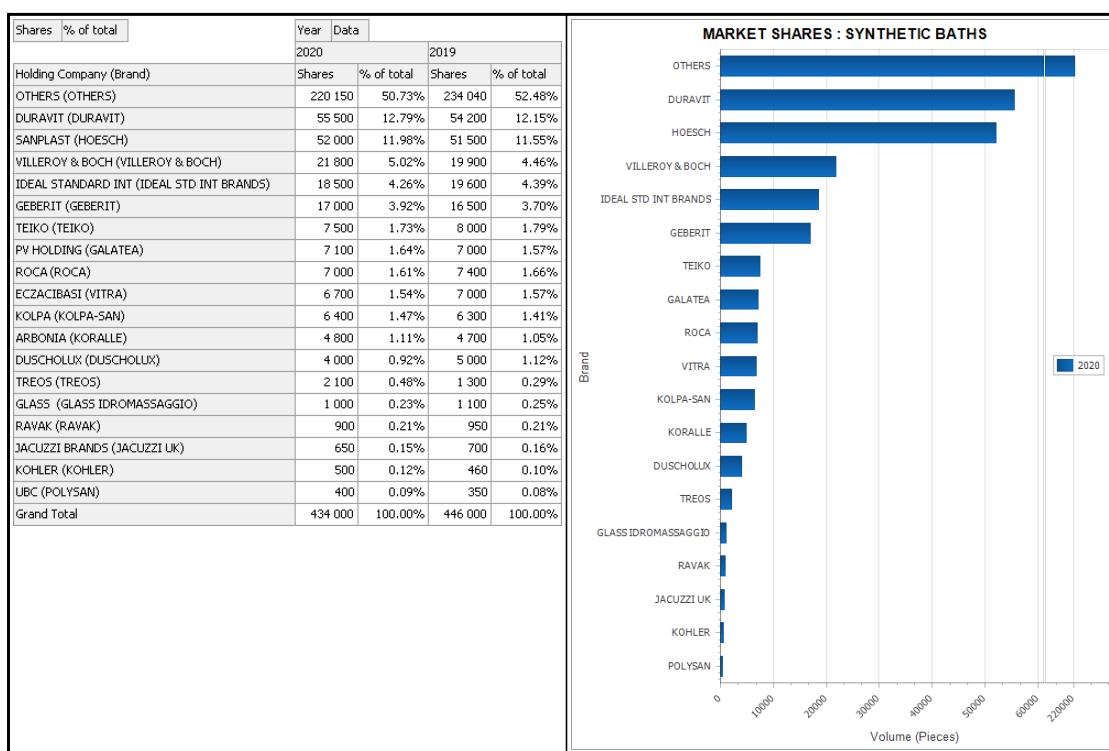
DE - ESTIMATED MARKET SHARES FOR STEEL BATHS



Source: BRG based on industry interviews.

1.5.2 Market Shares : Synthetic Baths

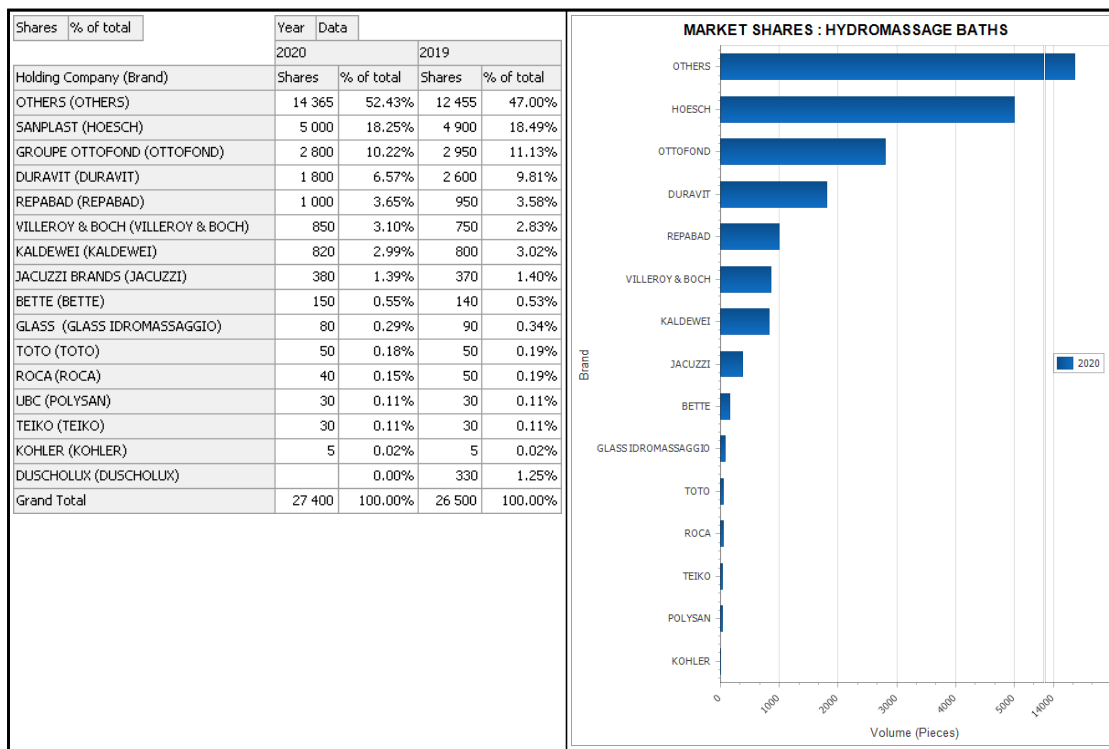
DE - ESTIMATED MARKET SHARES FOR SYNTHETIC BATHS



Source: BRG based on industry interviews.

1.5.3 Market Shares : Hydromassage Baths

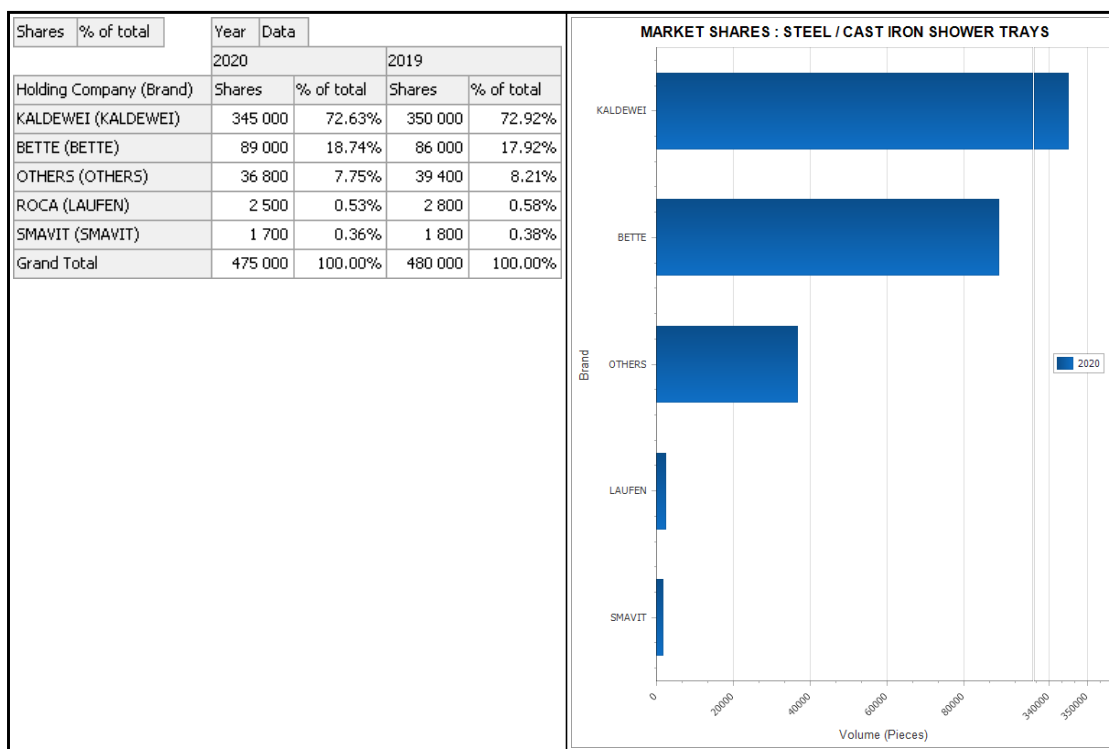
DE - ESTIMATED MARKET SHARES FOR HYDROMASSAGE BATHS



Source: BRG based on industry interviews.

1.5.4 Market Shares : Steel / Cast Iron Shower Trays

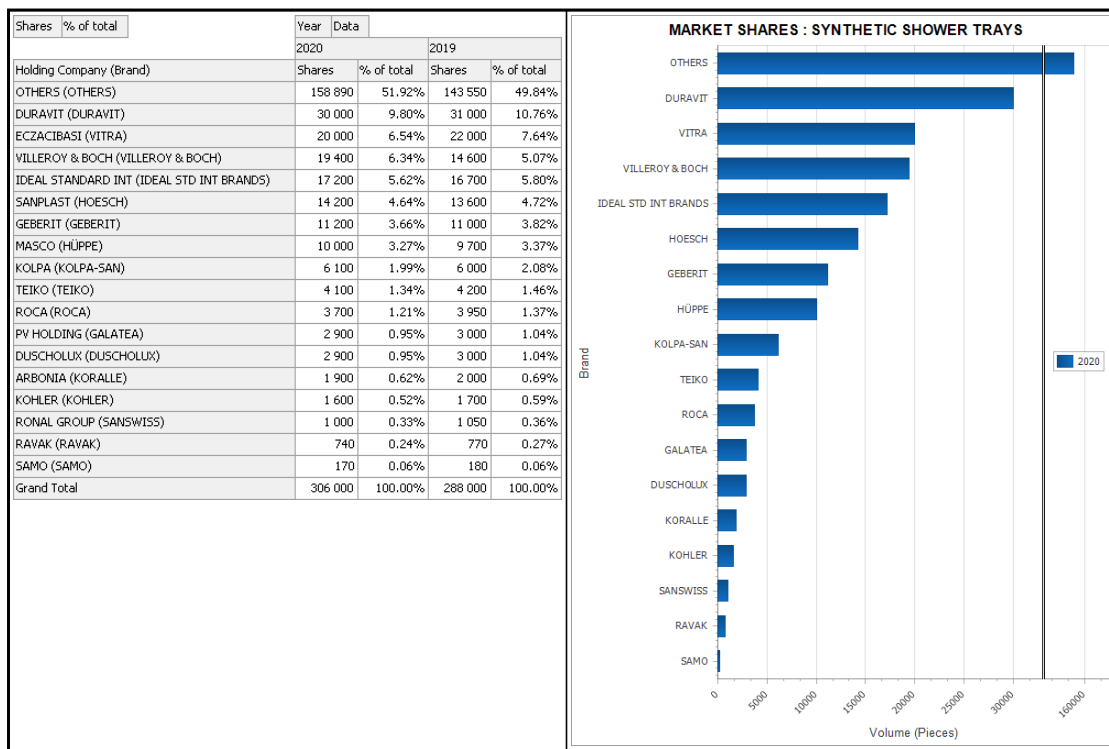
DE - ESTIMATED MARKET SHARES FOR STEEL / CAST IRON SHOWER TRAYS



Source: BRG based on industry interviews.

1.5.5 Market Shares : Synthetic Shower Trays

DE - ESTIMATED MARKET SHARES FOR SYNTHETIC SHOWER TRAYS



Source: BRG based on industry interviews.

1.6.1 DE - IMPORTS OF BATHS AND SHOWER TRAYS - CAST IRON 2020

Source Country	2020				2019			
	Volume		Value 000		Volume		Value 000	
	Tonnes	%	Euros	%	Tonnes	%	Euros	%
Poland	18	52.9%	35	44.3%	127	86.4%	207	76.4%
China	10	29.4%	29	36.7%	12	8.2%	42	15.5%
Portugal	4	11.8%	2	2.5%	7	4.8%	12	4.4%
Italy	1	2.9%	11	13.9%	1	0.7%	10	3.7%
Spain	1	2.9%	2	2.5%	0	0.0%	0	0.0%
Total	34	100.0%	79	100.0%	147	100.0%	271	100.0%

Source : Trademap

1.6.2 DE - EXPORTS OF BATHS AND SHOWER TRAYS - CAST IRON 2020

Destination	2020				2019			
	Volume		Value 000		Volume		Value 000	
	Tonnes	%	Euros	%	Tonnes	%	Euros	%
Switzerland	5	100.0%	86	79.6%	5	18.5%	90	60.4%
OTHERS	0	0.0%	22	20.4%	22	81.5%	59	39.6%
Total	5	100.0%	108	100.0%	27	100.0%	149	100.0%

Source : Trademap

1.6.3 DE - IMPORTS OF BATHS AND SHOWER TRAYS - STEEL 2020

Source Country	2020				2019			
	Volume		Value 000		Volume		Value 000	
	Tonnes	%	Euros	%	Tonnes	%	Euros	%
Spain	1 605	83.9%	2 647	80.7%	1 514	84.6%	2 427	79.8%
Poland	105	5.5%	106	3.2%	55	3.1%	101	3.3%
Czech Republic	35	1.8%	54	1.6%	16	0.9%	33	1.1%
Austria	32	1.7%	115	3.5%	66	3.7%	219	7.2%
China	25	1.3%	91	2.8%	10	0.6%	34	1.1%
OTHERS	111	5.8%	266	8.1%	129	7.2%	229	7.5%
Total	1 913	100.0%	3 279	100.0%	1 790	100.0%	3 043	100.0%

Source : Trademap

1.6.4 DE - EXPORTS OF BATHS AND SHOWER TRAYS - STEEL 2020

Destination	2020				2019			
	Volume		Value 000		Volume		Value 000	
	Tonnes	%	Euros	%	Tonnes	%	Euros	%
United Kingdom	3 664	21.5%	10 067	16.1%	4 500	23.1%	12 287	17.7%
Switzerland	2 859	16.8%	8 353	13.4%	3 070	15.8%	8 968	12.9%
Russian Federation	1 793	10.5%	5 168	8.3%	1 471	7.6%	3 747	5.4%
Austria	1 450	8.5%	7 501	12.0%	1 595	8.2%	7 249	10.4%
China	1 278	7.5%	6 065	9.7%	1 568	8.1%	7 466	10.8%
France	860	5.0%	3 245	5.2%	1 084	5.6%	3 932	5.7%
Ukraine	755	4.4%	1 408	2.3%	824	4.2%	1 456	2.1%
Czech Republic	700	4.1%	2 298	3.7%	680	3.5%	2 240	3.2%
Hong Kong, China	381	2.2%	1 373	2.2%	485	2.5%	1 834	2.6%
United States of Amer	324	1.9%	1 587	2.5%	519	2.7%	2 352	3.4%
Poland	321	1.9%	1 261	2.0%	261	1.3%	1 172	1.7%
OTHERS	2 665	15.6%	14 171	22.7%	3 408	17.5%	16 705	24.1%
Total	17 050	100.0%	62 497	100.0%	19 465	100.0%	69 408	100.0%

Source : Trademap

1.6.5 DE - IMPORTS OF BATHS AND SHOWER TRAYS - SYNTHETIC 2020

Source Country	2020				2019			
	Volume		Value 000		Volume		Value 000	
	Tonnes	%	Euros	%	Tonnes	%	Euros	%
Poland	4 694	16.1%	16 775	10.5%	4 282	15.3%	16 015	10.9%
China	4 406	15.1%	15 110	9.5%	4 001	14.3%	11 987	8.1%
Egypt	4 092	14.0%	14 821	9.3%	5 934	21.2%	19 048	12.9%
Slovakia	2 589	8.9%	9 947	6.2%	1 771	6.3%	5 809	3.9%
Turkey	2 292	7.8%	7 908	5.0%	750	2.7%	2 433	1.7%
Belgium	1 754	6.0%	11 695	7.3%	1 738	6.2%	10 906	7.4%
Croatia	1 500	5.1%	11 104	7.0%	1 183	4.2%	8 790	6.0%
Italy	1 236	4.2%	13 843	8.7%	1 407	5.0%	13 233	9.0%
Netherlands	1 192	4.1%	8 753	5.5%	1 181	4.2%	8 678	5.9%
Austria	1 003	3.4%	18 748	11.7%	924	3.3%	17 032	11.6%
Czech Republic	843	2.9%	5 288	3.3%	658	2.4%	4 616	3.1%
OTHERS	3 615	12.4%	25 604	16.0%	4 166	14.9%	28 651	19.5%
Total	29 216	100.0%	159 596	100.0%	27 995	100.0%	147 198	100.0%

Source : Trademap

1.6.6 DE - EXPORTS OF BATHS AND SHOWER TRAYS - SYNTHETIC 2020

Destination	2020				2019			
	Volume		Value 000		Volume		Value 000	
	Tonnes	%	Euros	%	Tonnes	%	Euros	%
Austria	1 276	16.6%	14 217	17.4%	1 364	15.9%	14 540	16.6%
Netherlands	1 023	13.3%	8 497	10.4%	827	9.6%	7 603	8.7%
Switzerland	778	10.1%	11 705	14.3%	764	8.9%	11 202	12.8%
France	755	9.8%	8 520	10.4%	697	8.1%	8 040	9.2%
Belgium	727	9.5%	6 676	8.2%	902	10.5%	6 966	8.0%
Poland	415	5.4%	2 829	3.5%	890	10.4%	4 161	4.8%
China	368	4.8%	3 375	4.1%	227	2.6%	2 304	2.6%
Italy	301	3.9%	4 326	5.3%	348	4.1%	5 155	5.9%
United Kingdom	202	2.6%	3 013	3.7%	275	3.2%	4 422	5.1%
Czech Republic	180	2.3%	1 535	1.9%	223	2.6%	1 713	2.0%
Spain	165	2.1%	1 596	2.0%	189	2.2%	1 986	2.3%
OTHERS	1 487	19.4%	15 379	18.8%	1 877	21.9%	19 289	22.1%
Total	7 677	100.0%	81 668	100.0%	8 583	100.0%	87 381	100.0%

Source : Trademap

1.7.1

Route to Market: Baths
DE - ESTIMATED DISTRIBUTION FOR BATHS: 2020

Manufacturer / Importer	Manufacturers / Importers
	100.0%

Wholesale Level	78.5%
	Wholesalers / Merchants
	78.5%

Retail Level	Manufacturer 2.2%	Wholesaler 3.8%	Manufacturer 12.5%	Wholesaler 5.0%	Manufacturer 1.2%	Wholesaler 1.3%
	Internet Retailers		DIY Retailers		Specialist Retailers	
	6.0%		17.5%		2.5%	

Manufacturer	Wholesaler	Internet	DIY	Specialist
2.3%	68.0%	3.0%	1.5%	0.1%
Installers				
74.8%				

End Use Level	Manufacturer 1.2%	Wholesaler 0.3%	Internet 3.0%	DIY 16.0%	Specialist 2.4%	Installer 73.2%
	Residential					
	96.2%					
	New Build		Residential RMI			
	19.7%		76.5%			
	Project	Non-Project	Project	Non-Project		
	7.0%	12.7%	2.0%	74.5%		
	Non-Residential					
	3.8%					
	New Build		Non-Residential RMI			
	1.2%		2.6%			
	Project	Non-Project	Project	Non-Project		
	1.0%	0.2%	1.4%	1.2%		

1.7.2

Route to Market: Baths
DE - ESTIMATED DISTRIBUTION FOR BATHS: 2025

Manufacturer / Importer	Manufacturers / Importers
	100.0%

Wholesale Level	78.0%
	Wholesalers / Merchants
	78.0%

Retail Level	Manufacturer 2.1%	Wholesaler 5.9%	Manufacturer 12.0%	Wholesaler 6.2%	Manufacturer 1.2%	Wholesaler 0.3%
	Internet Retailers		DIY Retailers		Specialist Retailers	
	8.0%		18.2%		1.5%	

Manufacturer	Wholesaler	Internet	DIY	Specialist
3.0%	65.2%	4.0%	2.2%	0.1%
Installers				
74.5%				

End Use Level	Manufacturer 1.7%	Wholesaler 0.3%	Internet 4.0%	DIY 16.0%	Specialist 1.4%	Installer 72.7%
	Residential					
	96.1%					
	New Build		Residential RMI			
	19.7%		76.4%			
	Project	Non-Project	Project	Non-Project		
	8.0%	11.7%	3.0%	73.4%		
	Non-Residential					
	3.9%					
	New Build		Non-Residential RMI			
	1.4%		2.5%			
	Project	Non-Project	Project	Non-Project		
	1.0%	0.4%	1.4%	1.1%		

Source: BRG Based on Industry Interviews. - Totals may vary due to roundings

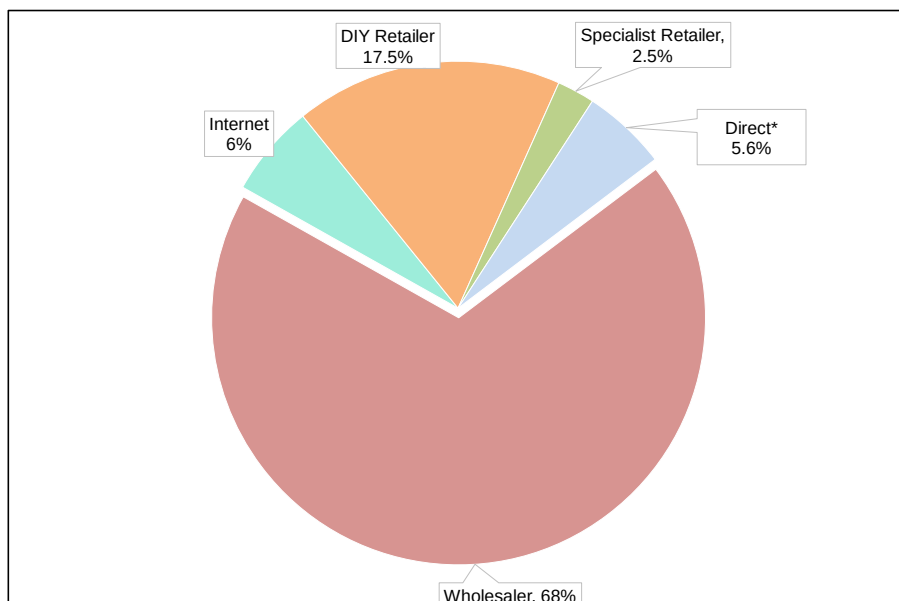
1.7.3 Route to Market - Consolidated View : Baths
DE - ESTIMATED DISTRIBUTION FOR BATHS : 2020

Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
68.4%	6.0%	17.5%	2.5%	5.6%	9.0%	2.4%	87.2%	1.4%
100%					11.4%		88.6%	

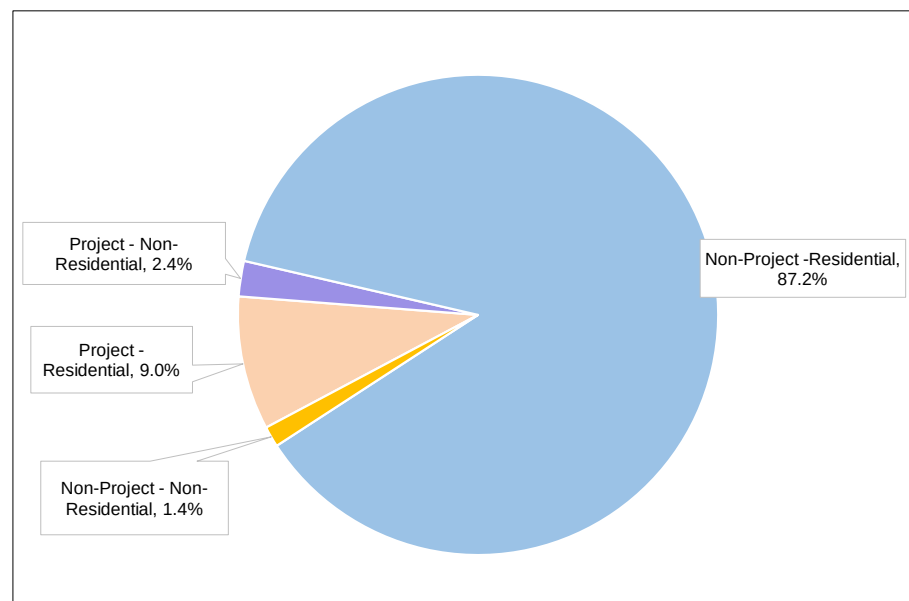
1.7.4 Route to Market - Consolidated View : Baths
DE - ESTIMATED DISTRIBUTION FOR BATHS : 2025

Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
65.6%	8.0%	18.2%	1.5%	6.7%	11.0%	2.4%	85.1%	1.5%
100%					13.4%		86.6%	

DE - SALES OF BATHS BY DISTRIBUTION CHANNEL: 2020



DE - SALES OF BATHS BY POINT OF DECISION AND END-USE SECTOR: 2020



Source: BRG Based on Industry Interviews.

*Direct includes sales going directly from manufacturers to installers, contractors and private customers.

1.7.5

Route to Market: Shower Trays
DE - ESTIMATED DISTRIBUTION FOR SHOWER TRAYS: 2020

Manufacturer / Importer	Manufacturers / Importers
	100.0%

Wholesale Level	81.5%
	Wholesalers / Merchants
	81.5%

Retail Level	Manufacturer	Wholesaler	Manufacturer	Wholesaler	Manufacturer	Wholesaler
	3.4%	6.0%	9.7%	7.6%	0.5%	1.5%
	Internet Retailers		DIY Retailers		Specialist Retailers	
	9.4%		17.3%		2.0%	

Manufacturer	Wholesaler	Internet	DIY	Specialist
2.6%	66.1%	3.0%	1.5%	0.1%
Installers				
73.3%				

End Use Level	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer
	0.6%	0.2%	5.2%	14.3%	1.9%	69.6%
	Residential					
	91.8%					

New Build			Residential RMI		
11.5%			80.3%		
Project	Non-Project		Project	Non-Project	
5.0%	6.5%		2.0%	78.3%	

Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer
1.7%	0.1%	1.2%	1.5%	0.0%	3.7%
Non-Residential					
8.2%					

New Build			Non-Residential RMI		
3.0%			5.2%		
Project	Non-Project		Project	Non-Project	
2.0%	1.0%		3.0%	2.2%	

1.7.6

Route to Market: Shower Trays
DE - ESTIMATED DISTRIBUTION FOR SHOWER TRAYS: 2025

Manufacturer / Importer	Manufacturers / Importers
	100.0%

Wholesale Level	81.0%
	Wholesalers / Merchants
	81.0%

Retail Level	Manufacturer	Wholesaler	Manufacturer	Wholesaler	Manufacturer	Wholesaler
	4.4%	8.0%	9.0%	9.0%	0.5%	0.0%
	Internet Retailers		DIY Retailers		Specialist Retailers	
	12.4%		18.0%		0.5%	

Manufacturer	Wholesaler	Internet	DIY	Specialist
2.5%	63.7%	4.2%	2.0%	0.1%
Installers				
72.5%				

End Use Level	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer
	1.0%	0.2%	6.7%	14.6%	0.4%	69.5%
	Residential					
	92.4%					

New Build		Residential RMI	
10.7%		81.7%	
Project	Non-Project	Project	Non-Project
5.4%	5.3%	2.0%	79.7%

Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer
1.6%	0.1%	1.5%	1.4%	0.0%	3.0%
Non-Residential					
7.6%					

New Build		Non-Residential RMI	
3.5%		4.1%	
Project	Non-Project	Project	Non-Project
2.8%	0.7%	2.7%	1.4%

Source: BRG Based on Industry Interviews. - Totals may vary due to roundings

1.7.7 Route to Market - Consolidated View : Shower Trays

DE - ESTIMATED DISTRIBUTION FOR SHOWER TRAYS : 2020

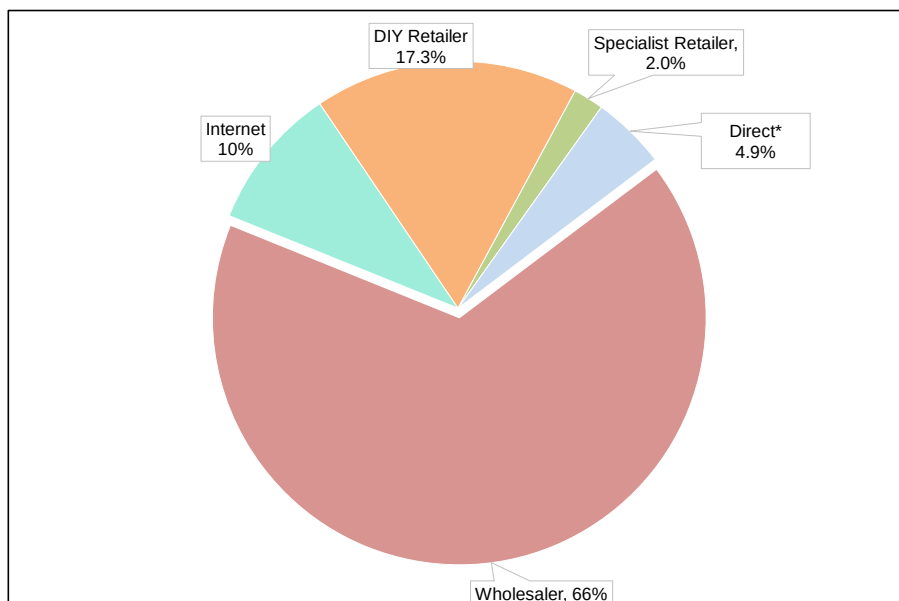
Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
66.4%	9.4%	17.3%	2.0%	4.9%	7.0%	5.0%	84.8%	3.2%
100%					12.0%		88.0%	

1.7.8 Route to Market - Consolidated View : Shower Trays

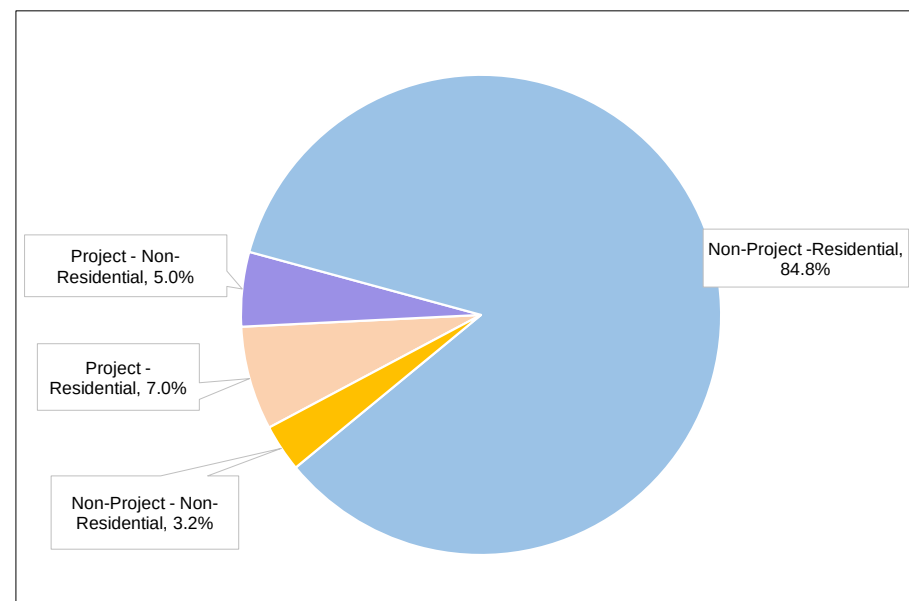
DE - ESTIMATED DISTRIBUTION FOR SHOWER TRAYS : 2025

Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
64.0%	12.4%	18.0%	0.5%	5.1%	7.4%	5.5%	85.0%	2.1%
100%					12.9%		87.1%	

DE - SALES OF SHOWER TRAYS BY DISTRIBUTION CHANNEL: 2020



DE - SALES OF SHOWER TRAYS BY POINT OF DECISION AND END-USE SECTOR: 2020



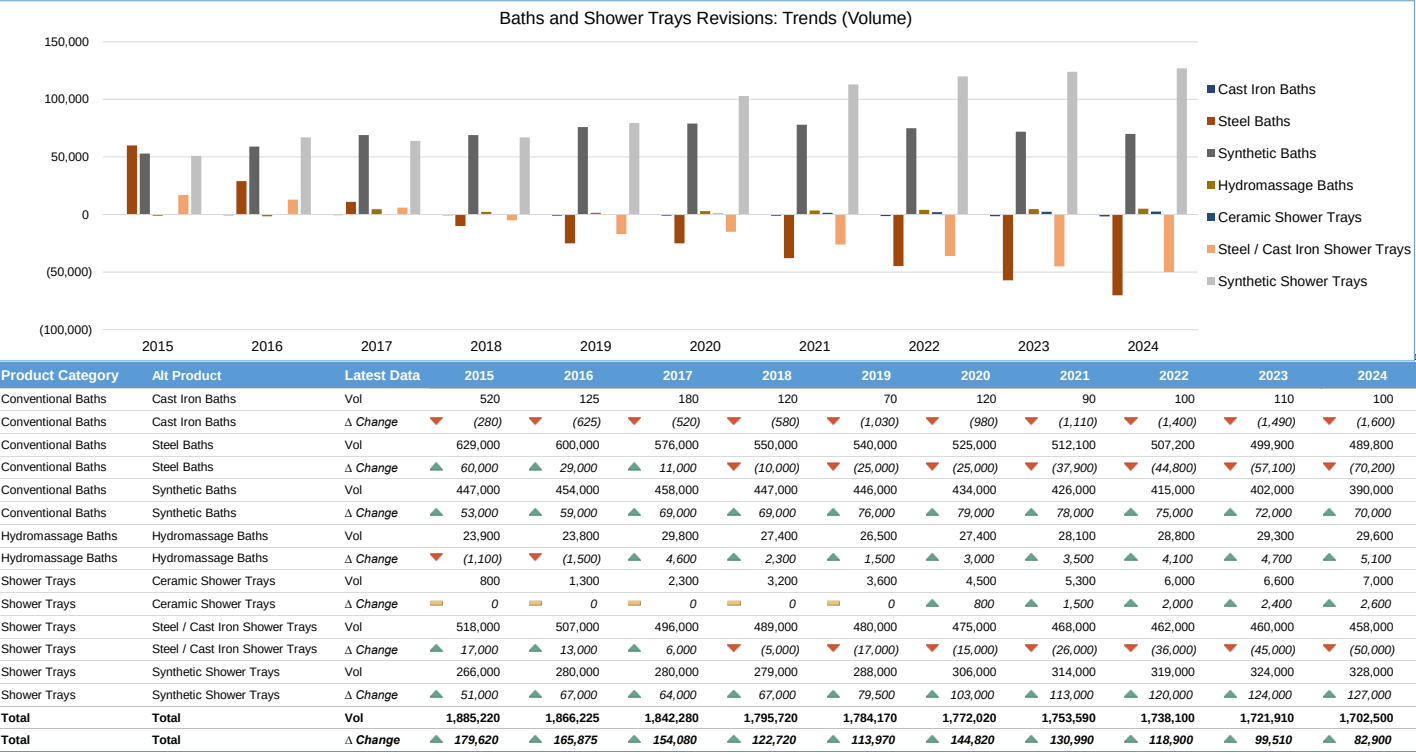
Source: BRG Based on Industry Interviews.

*Direct includes sales going directly from manufacturers to installers, contractors and private customers.

1.8 Revisions in the German Baths and Shower Trays Market (April 2021)

DE - REVISIONS IN THE GERMAN BATHS AND SHOWER TRAYS MARKET

Data Comparison: October 2020



The market volume for cast iron baths was adjusted for all years based on production statistics for the main supplying countries such as Italy, Portugal, Spain, France and the UK. Previously the volume was over-estimated, and it is believed that demand has suffered in recent years due to the availability of high-quality free-standing baths that are made of mineral cast and solid surface.

The market for steel baths was revised up historically according to official production statistics combined with input from manufacturers. The result of the change summarises the past development of the market in a better way, since demand for baths has been strongly declining in recent years. According to industry feedback, the decline has been stronger than previously anticipated and the total market volume had to be slightly reduced for the years 2018, 2019 and 2020. It is believed that steel baths sales are likely to remain in decline during the forecast years. In 2021 this decline is caused by the strong price increase of steel products, the growing competition for mineral cast and solid surface products along with the overall trend towards larger showers in renovated bathrooms.

The market volumes for synthetic baths, including hydromassage baths, was revised up for all years based on official production statistics combined with a detailed analysis of import statistics. The share of 'Others' increased, which reflects the important of manufacturers active in DIY stores and e-commerce, e.g. OTTOFOND, SANOTECHNIK, MAUERSBERGER and REPABAD.

Similar to the synthetic baths market, the hydromassage baths market was revised up for all years. Manufacturer's feedback suggests that the market volume is different and that demand has started to pick up again. The influence of the pandemic on consumer behaviour and the wish for more wellness at home has started to generate more demand for hydromassage baths in 2020. This trend is expected to continue during the forecast period for as long as the pandemic lasts. Moreover, the wish for more wellness at home is likely to drive demand in the long run too.

The MSP for cast iron baths was revised up for all years, largely due to the influence of imports from Portugal, Spain and Italy on average prices. Cast iron baths are no longer standard baths but are becoming more luxurious, affecting average price developments.

The MSP for steel baths was revised up for 2019 following input from manufacturers; steel baths are continuing to become more expensive, affected by changes in the supply side, with declining numbers sold in the lower price segments.

Some minor changes were made to the MSP for synthetic and hydromassage baths, mostly because of changes made to the sales per price segment, combined with a more detailed analysis of sales by material type (e.g., acrylic vs. mineral cast), affecting the overall MSP.

GERMANY

SANITARY WARE

THE EUROPEAN BATHROOM & KITCHEN PRODUCT MARKETS



Market Characteristics and Technological Trends

Ceramic sanitary ware (CSW) is one of the oldest segments on the overall German bathroom market. The material was already recognised in the 19th century as easy to clean and ideal for tiling in hospitals and publicly accessible baths. Health legislation and the increasing penetration of canalisation slowly led to an increase in public water closets (WCs). It was, however, only in the 1950s that almost every household started to have its own bathroom inside the dwelling. The toilet would often be in a separate room. In Germany, the tradition of having a separate toilet room has perpetuated until today, which is why most one- and two-family houses and many larger apartments have both a bathroom with a WC and a separate guest toilet.

Unlike in many other European countries, the German WC market shows the peculiarity of a high share of 'washout' models (having a flat platform). Although the park of these types of toilets remains large, the trend is negative and, nowadays, they are replaced with 'washdown' models which are often wall hung. Typically, these are installed in dry (pre-wall) installation modules, in combination with a concealed plastic cistern. The penetration of ceramic cisterns is marginal, and nearly all these ceramic cisterns are sold with close coupled WCs. The German bidet and urinal markets are small; urinals are usually installed in public places such as airports and the hospitality sector, and rarely installed in private households.

A more recent product innovation is the so-called 'rimless' WC, which eliminates the need for a hidden rim or channel. These models were introduced to the German market in early 2012 by the domestic KERAMAG (now GEBERIT) and were a success, despite relatively higher prices when compared to standard wall hung models. The German consumer is quality oriented and willing to pay a premium for a clear distinguishable product advantage such as hygiene. The market for rimless WCs developed so well that almost all market players, including VILLEROY & BOCH, DURAVIT and LAUFEN, entered the market soon after. Nowadays, it is estimated that the share of rimless WCs on the German market has reached almost half of the total WC market.

Washbasins, after WCs, account for the second largest share of total sanitary ware sales in Germany. There have been some displacement sales from ceramics to other materials such as synthetic and enamelled steel washbasins. Built-in washbasins and universal washbasins, which can either be wall hung or built into furniture, are gaining popularity, benefiting from an increasing focus on bathroom furniture. Synthetic products seem to have benefited more than ceramic products. Pedestal sales depend on the development of the washbasin market. With the increasing popularity of bathroom furniture, full pedestals have almost disappeared from the market. The main colour used in German bathrooms continues to be white, especially in the economy and lower-priced segments. Consumers tend to choose white pieces of ceramic and add colour touches to their bathroom through

coloured ceramic tiles, furniture and accessories. However, manufacturers are starting to introduce colours into their assortments and black seems to be having the greatest success. End users tend to customise their bathroom according to their needs and taste. This new trend towards 'bathroom personalisation' was noticeable at the ISH 2019 in Frankfurt, where the bathroom was transformed into a lifestyle space with customised products in shapes and colours.

Recent product innovations have been in the field of ceramic surfaces (non-stick and easy to clean) and accessible ceramic sanitary ware ('*barrierefrei*') which is adapted to the needs of the elderly (e.g. higher and broader WCs, roll under washbasins and shower trays with extra grip). The topic of health and hygiene has gained in significance in the light of the COVID-19 pandemic. End users are more aware of the importance of hygiene and are likely to spend more for products that can help them feel safer against infections.

Typical for a Western European country, the trend in Germany is towards concealed plastic cisterns at the expense of the exposed cisterns, both ceramic and plastic. Concealed plastic cisterns are usually found in commercial and public buildings. However, these are challenged by (self-closing or manual) flushing valves, which can be used as an alternative. Exposed cisterns are used during bathroom refurbishments, either for price reasons or if a concealed alternative is technically not possible and, in the case of very cost sensitive builders (e.g. construction companies), as they are the most affordable replacement products. Other than that, concealed plastic cisterns are the standard solution in every modern bathroom.

Recent Market Trends

The COVID-19 pandemic shaped the developments in the year 2020 and 'business as usual' was replaced by government measures. In Germany the official lockdown, which is better understood as a 'prohibition of contact', started at the end of March 2020. Bavaria was the first state to implement measures to contain the spread of the virus, which was followed by the remaining states towards the end of March. For a few weeks the German population was advised to work from home when possible, with most non-essential stores closing for a limited time only. Measures implemented varied by state (*Bundesländer*). Most stores, including DIY stores, were allowed to open again from the beginning of April 2020 onwards, with differences from state to state in the level of restrictions and measures. In certain regions, shopping in DIY stores was limited to professionals only.

During 2020, new housing construction continued without much interruption, with only large-scale project sites halted to implement protective measures. The overall production per employee is believed to have shrunk significantly, and remains to be the case, since many employers introduced measures to protect the construction workers. The bathroom industry was declared part of the crisis infrastructure, but installers could continue working without interruptions. Overall, there has been relatively little impact of the COVID-19 pandemic on the building related industry, including the bathroom market, in Germany. In some cases, end users may have

postponed, or cancelled, bathroom related projects due to the uncertainty on the market, but overall demand remained relatively strong, as there was a large backlog of works from the past, benefiting sales of bathroom and kitchen equipment. At the same time, in certain instances, the COVID-19 pandemic may have contributed to additional demand. Since consumers were spending more time indoors, they had the opportunity to dedicate themselves to repair or upgrade works, whereas others might have decided to invest into their property by installing a new bathroom or kitchen due to (international) holidays being difficult because of the current pandemic.

In 2020 the German sanitary ware sales by manufacturers strongly declined when compared to 2019. The drop was caused by fewer sales of all ceramic sanitary ware products except for non-ceramic washbasins and concealed plastic cisterns. The impact of the COVID-19 pandemic reduced consumers' interest in having a bathroom renovated. During lockdown end users preferred to keep social distance and avoid unnecessary workmen entering their private space. At the same time, there was a stronger competition for the installation of heating products. Significantly more installers than in the previous years decided to carry out heating jobs instead of sanitary ware works.

To a certain extent, other distribution channels such as DIY stores and the Internet could benefit from more sales, especially of products that a skilled private end user can install, e.g. WCs, washbasins, furniture and bathroom accessories. The German sanitary ware market generates most of its sales in housing RMI. With slightly fewer new housing completions in 2020, there was also lower demand from the new housing segment. According to industry feedback, it was mainly the upper/luxury price segment, which contributed to growth in terms of value in 2020. Instead, the overall volume declined as renovation from more price-conscious homeowners as well as commercial customers might have been postponed due to economic uncertainty.

One important trend regarding recent product developments introduced onto the market are shower toilets and bidet seats, which are slowly gaining popularity among German consumers. Almost 60,000 shower toilets and bidet seats are believed to have been sold in Germany in 2020. This number includes bidet seats, which consumers can buy separately to equip existing WCs and shower toilets, which are mainly purchased during a full bathroom renovation or for a new dwelling. The purchase of a shower toilet is likely to replace an existing bidet or prevent end users from installing a new one.

The category 'Other CSW' has been decreasing for years. It consists of products that are declining in sales, e.g. bidets, close-coupled cisterns as well as full and half pedestals. With the trend towards bathroom furniture, full pedestal sales have almost entirely vanished. Half pedestals can still be found on the market, e.g. in secondary bathrooms or hotels (both being places that usually do not require bathroom furniture). Bidets are regarded to be an out-dated product, which is gradually being replaced by new shower toilets. Urinals are rarely used in residential houses and their demand primarily depends on the development of non-

housing constructions. Rimless WCs are the standard type in new housing and are expected to force out the traditional WC in the long turn.

Prices and Market Values

As for most other segments on the German bathroom market, prices for sanitary ware products continued to increase in 2020. The overall price rise of sanitary ware products, such as WCs and washbasins, is a natural consequence of inflation adjustments and increasing labour costs. Prices for exposed and concealed cisterns similarly increased when compared to the previous year. In addition to a regular price increase, there was a notable value increase because of product mix changes. The rimless WCs have been rapidly gaining popularity among German end users, who value products with superior hygienic features and tend to pay a premium price for it. This increase in demand for rimless WCs continues to positively influence the average prices. Rimless WCs were first intended for the non-residential market, such as hotels and hospitals, but quickly gained ground in the upper residential markets in which clients are willing to invest in high quality products. Shower toilets are not part of the report and the MSP increase for WCs does not reflect their growing share.

Higher-priced products contributed to more sales in the upper and luxury price segments in 2020, which resulted in a higher MSP for most sanitary ware products. The middle-priced segment continued to be the biggest price segment in terms of volume for all sanitary ware products in Germany. The only exceptions are exposed plastic cisterns, which have the strongest share in the economy/lower price segment.

Concealed plastic WC cisterns are the industry standard due to an easy and fast installation, as well as being aesthetically pleasing. Prices of such models are consistently higher than those of exposed cisterns. This is mostly due to the fact that their average MSP also includes installation modules, which are the expensive part. The price of a concealed plastic cistern depends, to a large extent, on the pre-wall installation system sold with the cistern, whereas the separate cisterns themselves would be around the same price as a cheap exposed plastic model. The latter can be bought for as little as 20 euros in DIY stores.

New Housing

The German new build market developed negatively in 2020 and demand for all sanitary ware decreased when compared to the previous year. There is a trend in new housing, that two bathrooms are installed, rather than one main bathroom and a guest toilet. So even a 100 m² apartment could have two bathrooms with two showers. Moreover, it is common to have two washbasins installed in one bathroom. Such trends somewhat compensate for a decline in new completions. Nowadays, most new bathrooms are equipped with bathroom furniture and pedestals are not used anymore, except for classical bathroom designs only. Half pedestals are sometimes installed in hotels and places where furniture is not needed. Urinals are rarely installed in private houses and sales are mainly

dedicated for non-housing. Overall, demand for other ceramic sanitary ware declined stronger than for washbasins and WCs. Concealed plastic cisterns developed in relation to WCs and demand was similarly negative. Demand for non-ceramic washbasins recorded a smaller decline as they are gaining share from the ceramic sanitary ware market.

Housing RMI

Replacement demand for sanitary ware products has the highest volumes for WCs and ceramic washbasins. Wall hung WCs often replace floor standing models, which are old fashioned. However, not every German consumer is willing to invest more into a concealed installation module, which requires a complete bathroom refurbishment. The popularity of rimless models continues to grow, and end users appreciate their hygiene benefits. Product innovations, such as rimless and washlet WCs, shorten the lifespan of traditional WCs as end users feel the need to replace them with a more advanced WC. Another trend is the adaptation of bathrooms, often sustained by housing associations, to cater for the needs of elderly people. In line with government incentives to fuel demand for bathroom refurbishments, sanitary ware products such as comfortable WCs, lower washbasins and safety handles are being installed. If the latter are made from ceramics, they are counted as 'Others' in other ceramic sanitary ware. Sales of other ceramic sanitary ware products to housing RMI declined in 2020 being negatively driven by their outdated design and functionality.

Over the past years economic developments have made it more interesting to invest into properties instead of leaving savings in a bank account with low interest rates. This trend had a positive influence on private and public investments into bathroom renovations. However, the shortage of professional installers limits sales to the RMI segment as consumers must take long waiting lists into consideration. This leads to delays and postponed investments for products sourced from the professional distribution channel. DIY stores such as HORNBACH and OBI have recognised this potential by launching marketing campaigns to position themselves as a one-stop solution for basic bathroom renovations. In addition to offering a complete range of sanitary products, they offer supplementary services such as planning, 3d visualisation of the design and installation services. According to industry feedback, the DIY channel is growing; however, the growth rate is not dramatic. Despite the strong growth rates of DIY stores during the pandemic, it is believed that ceramic sanitary products only marginally profited from that boom. It was mainly products such as paints, wood, garden furniture and construction materials which recorded the strongest growth rates.

In 2020, demand for total ceramic sanitary ware dropped significantly. Ceramic washbasins and WCs sold by manufacturers decreased in volume and the decline strongly affected overall sales. Sales of concealed plastic cisterns increased in 2020 into the RMI segment, which underlines the growing share of wall hung toilets when renovating a bathroom. Demand for exposed plastic cisterns declined similarly to the previous year and sales of non-ceramic washbasins increased with a stronger growth rate than in 2019. This can be explained by stronger demand

from DIY and online stores, which have a greater choice for non-ceramic materials.

Non-Housing

The non-housing sector generated significantly less demand for sanitary ware products in 2020 when compared to 2019. Most investments into buildings such as hotels, wellness facilities and offices, which are main drivers for sanitary ware purchases, receded in 2020 due to the COVID-19 pandemic. There was a slightly positive investment into schools and healthcare buildings, which was not enough to compensate for the overall declining trend. Regarding health buildings, it is necessary to distinguish between public owned hospitals, which are under a more restricted budget, and the privately owned ones. The latter are meant to be the main drivers for sanitary ware products. These types of buildings have massively relied on the hygienic product solutions, for example rimless WCs and hygienic surfaces, which have reached an increasing popularity over the last couple of years. New product innovations cater for the needs of an aging population with limited motor skills and increasing hygiene standards. The commercial sector represents the biggest purchasing pool of low-priced CSW models. Both commercial and public buildings are normally equipped with concealed plastic cisterns and the share of exposed cisterns is low within non-housing buildings in Germany. The slowly increasing popularity of self-closing flushing valves could represent a threat for the cistern markets.

Supply Structure

The supply structure in the sanitary ware market is shifting towards online sales, especially of products that can be easily transported and installed. Wholesalers complained about stagnating sales due to a bottleneck of installers, which in 2020 favoured the installation of heating products due to lucrative government funding. Moreover, end users tried to avoid having unnecessary renovations done in their private spaces to maintain social distancing. This again benefited the trend towards heating systems, which can be replaced by max. two people working in the basement.

In addition to the traditional installers, consumers are seeking alternative distribution channels such as DIY and internet stores. Online websites such as REUTER and MEGABAD lure end users with large discounts on products, that would usually be only sold by professional installers. It is believed that smaller installers might also agree to carry out works for end users, who purchased their own products online. The Internet not only offers cheaper prices, it also has a wide range of brands and models at hand.

The market for ceramic sanitary ware is fragmented, with many domestic and foreign suppliers competing with cheap imported products. German manufacturers, including KERAMAG, which from 2019 onwards was being sold under the brand name GEBERIT, VILLEROY & BOCH and DURAVIT, still account for most sales. Imports come from countries such as Egypt, Poland, Romania. Hungary, China,

Italy, Austria, the Czech Republic, Bulgaria, France and Turkey. VILLEROY & BOCH produces in 17 different countries and the main production locations for the German market are in Germany and France, as well as Hungary and Romania. GEBERIT has a production site in Poland, which is among the most important countries in terms of imported volume to Germany after Egypt. The share of Chinese imports is usually sold in DIY stores.

The cistern market is dominated by the Swiss manufacturer GEBERIT, both in the concealed and exposed market segments. Most imports come from other European countries such as Poland, Portugal and Switzerland. GEBERIT has a production site in Austria. Imports from the Netherlands include WISA. GROHE introduced ceramic sanitary ware products in autumn 2017. The market leader for taps and mixers sees the advantage in the brand name itself. A brand such as GROHE conveys trust and if you renovate your bathroom every 19 years on average having a strong brand recognition is a great advantage. GROHE's ceramic sanitary ware is sold through its established distribution channels, the same as taps and mixers. According to a change in GEBERIT's brand portfolio, the group has rebranded all bathroom products from KERAMAG into GEBERIT from 2019. The same process is going to be applied to established foreign brands such as SPHINX, ALLIA and POZZI GINORI.

Forecasts

The outlook for the German sanitary ware market for 2021 is positive. Sales to the RMI and new housing sectors are expected to develop positively. Non-housing sales will be negatively affected by the continuation of the lockdown. The COVID-19 crisis is likely to have a strong negative impact on the non-housing segment in 2021, too. In contrast, demand from the new and RMI end-use segments is believed to turn positive during most years. From 2021 until 2023, new build construction of apartment buildings is believed to have a positive effect on sanitary ware sales. The number of one- and two-family home constructions is expected to decline during most forecast years.

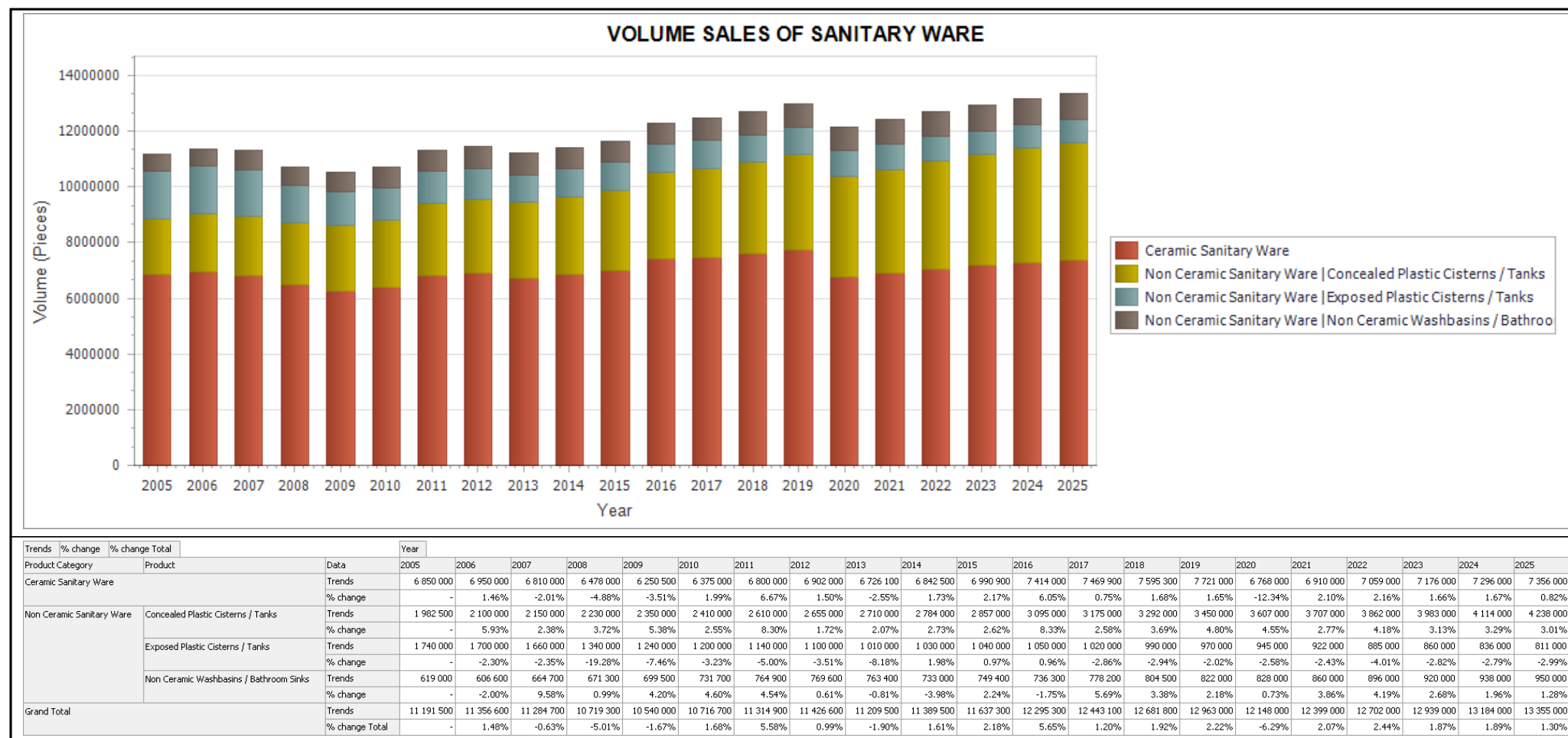
Demand for ceramic and non-ceramic washbasins is expected to remain positive. Other CSW sales are expected to rebound in 2021, before starting to decline again from 2023 onwards. Exposed plastic cisterns sales are expected to decline continuously throughout the forecast period. The growth rate for sanitary ware products not only depends on the acceptance of product innovations, but also on the availability of professionals to install them. Manufacturers are launching new products that combat the lack of installers by easy and quick to install products. Overall, the economic situation is believed to slightly recover from the COVID-19 pandemic in 2020. At the same time, a further shift towards high quality products and alternative distribution levels is anticipated. Demand from the middle price segment is likely to weaken as the average German consumer is more likely to postpone a bathroom renovation in financially uncertain times.

Despite the COVID-19 pandemic, full bathroom renovations, which generate the strongest demand of sanitary ware in Germany, are expected to continue to grow.

The RMI segment is believed to be boosted by the launch of new products and the need for living space in popular regions and cities. In terms of product mix, concealed cisterns are likely to maintain their positive trend, at the expense of the already declining exposed cistern market. Sales are expected to shift towards higher-priced sanitary ware products, which offer additional health benefits through antibacterial coatings for example. The lack of installers is likely to get more tense during the forecast period as the competition from the heating sector increases. Government funding is very attractive at the moment and the ambition to increase the energy efficiency of existing buildings is likely to drive demand on the heating side. This will further limit the availability of installers, who are willing to carry out bathroom projects.

2.2.1 Historical Trends and Forecasts : Sanitary Ware

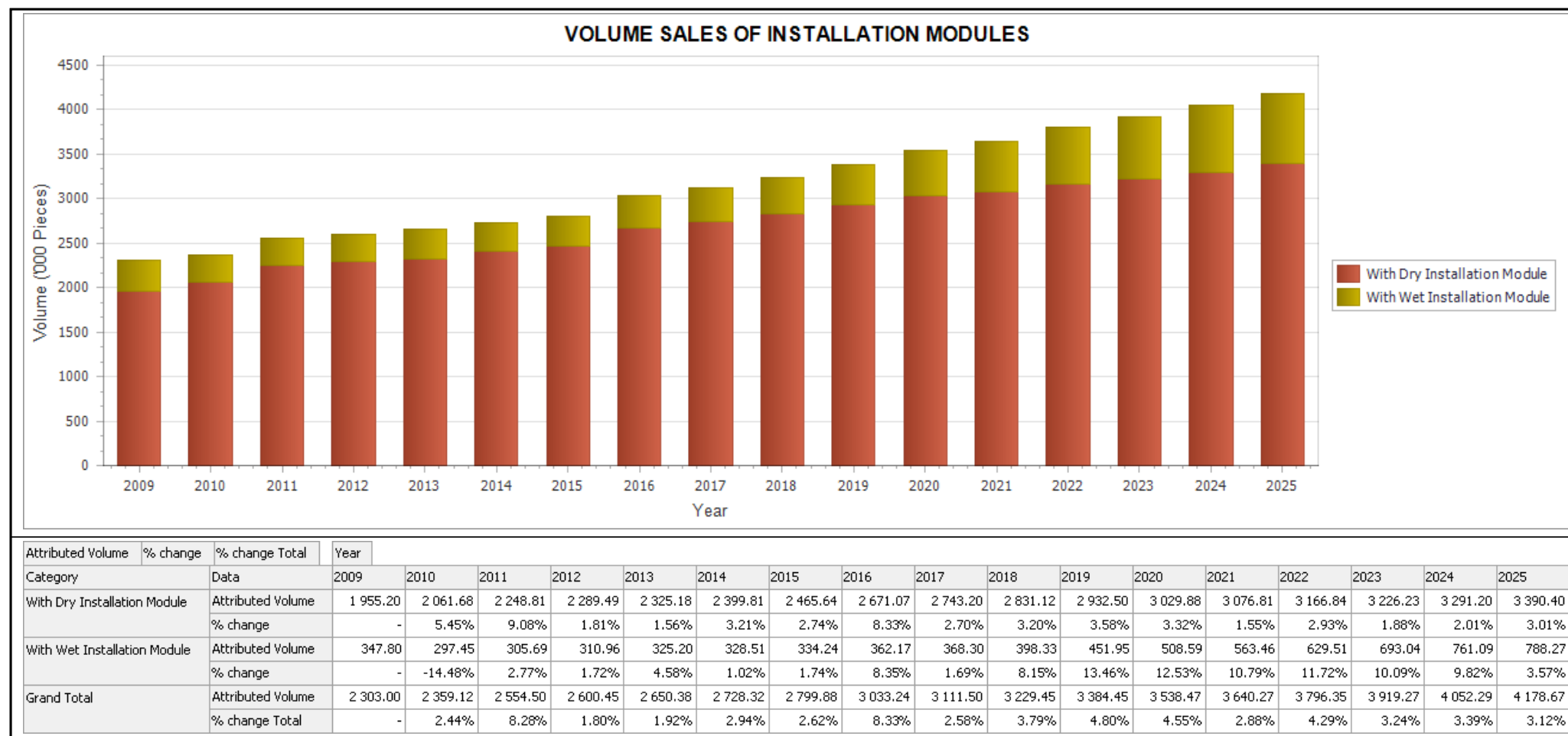
DE - SUMMARY OF TRENDS IN THE SANITARY WARE MARKET : HISTORICAL AND FORECASTS



Source: BRG based on industry interviews. NB: Ceramic Sanitary Ware Includes Ceramic Shower Trays.

2.2.2 Historical Trend and Forecasts: Installation Mod.

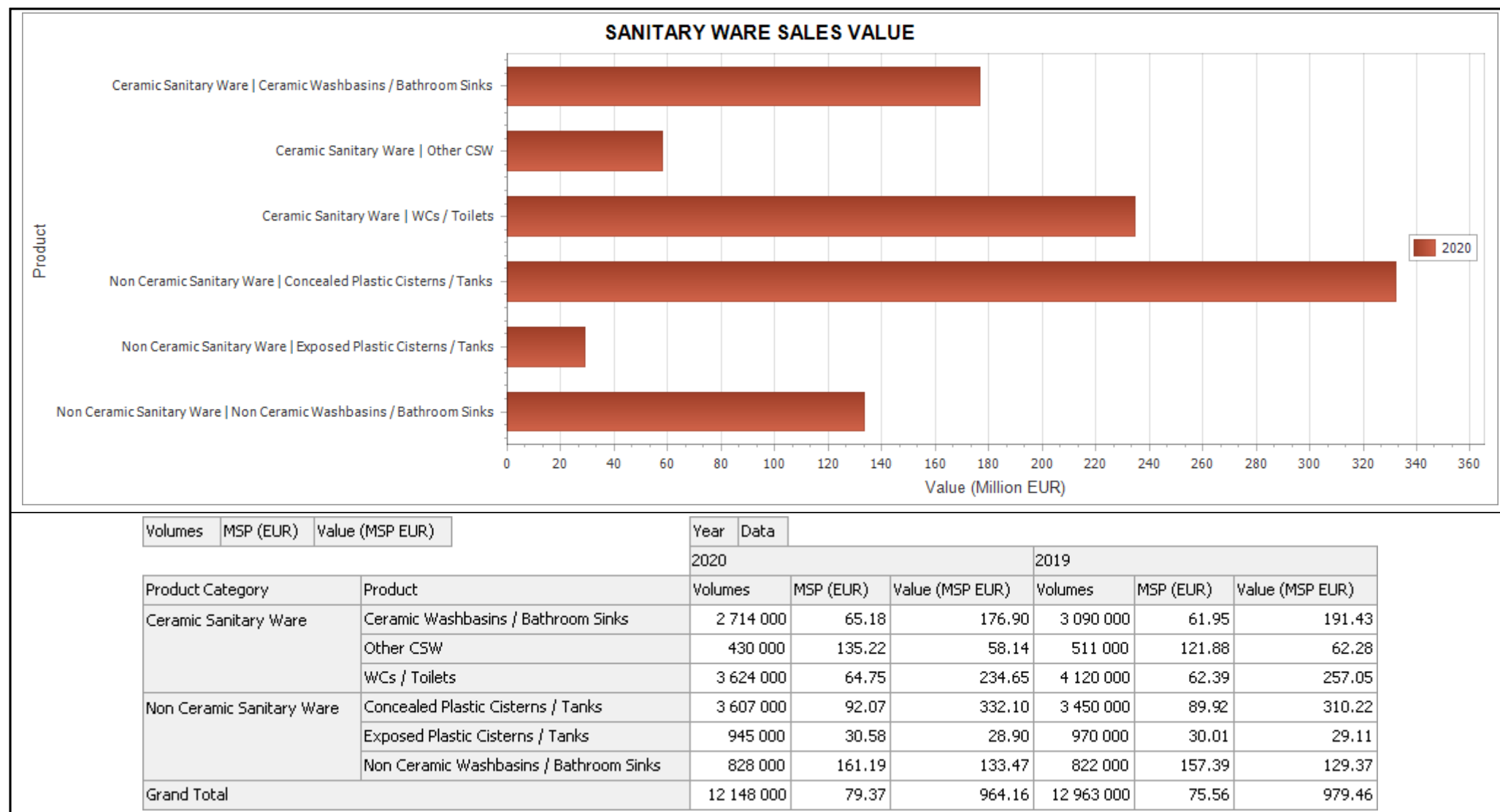
DE - SUMMARY OF TRENDS IN THE INSTALLATION MODULE MARKET : HISTORICAL AND FORECASTS ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

2.3.1 Prices and Market Values : Sanitary Ware

DE - ESTIMATED VALUES OF THE SANITARY WARE MARKET



Source: BRG based on industry interviews. Values Shown in Millions (at MSP = Manufacturers' Selling Price)

2.3.2 Prices and Market Values by Type : CSW

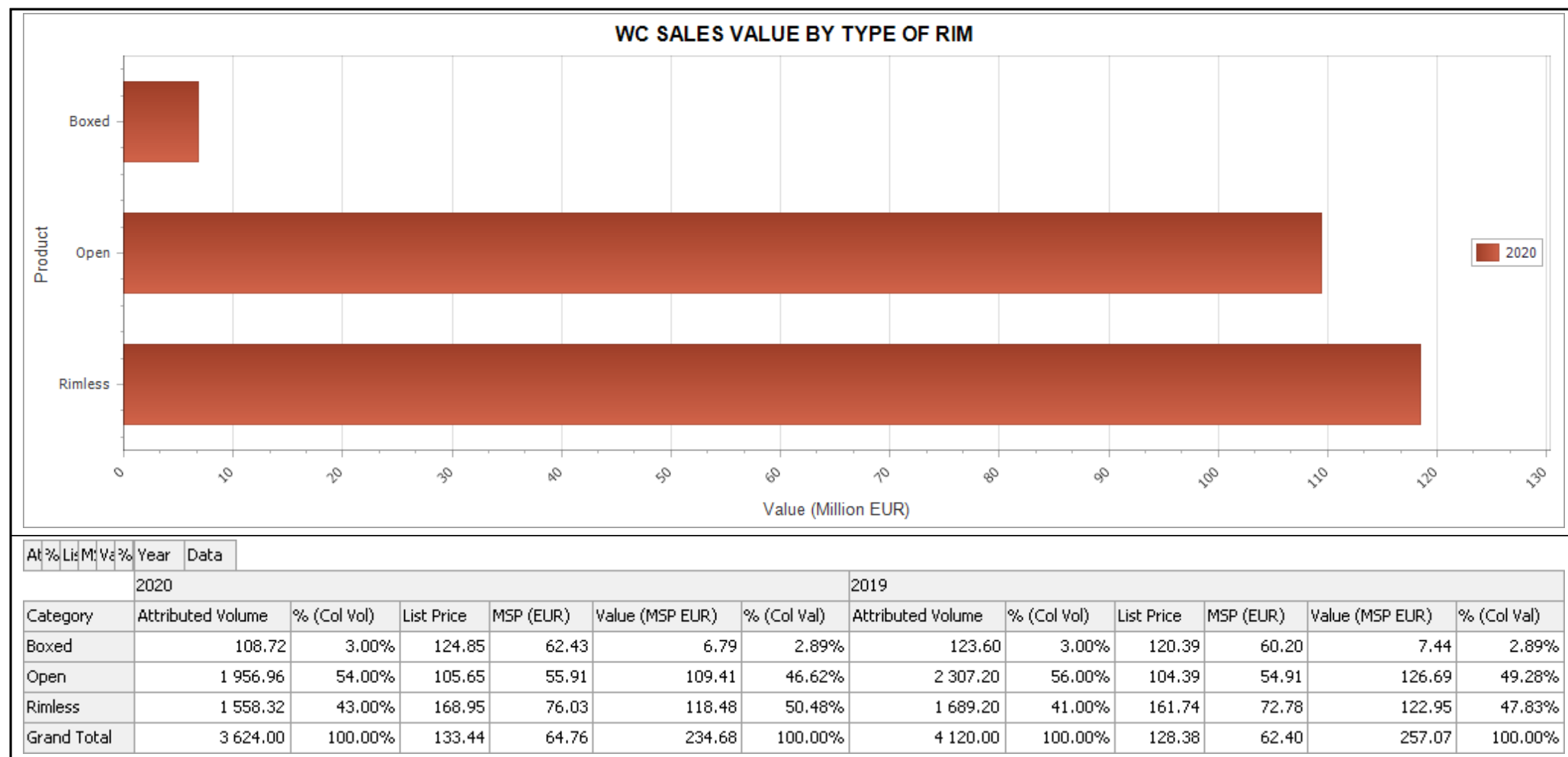
DE - ESTIMATED VALUES OF THE CERAMIC SANITARY WARE MARKET BY TYPE OF PRODUCT

Attributed %	% (Col Vol)	List Price	MSP (EUR)	Value (MSP	% (Col Val)	Year	Data						
2020								2019					
Alt Product	Category	Attributed Volume	% (Col Vol)	List Price	MSP (EUR)	Value (MSP EUR)	% (Col Val)	Attributed Volume	% (Col Vol)	List Price	MSP (EUR)	Value (MSP EUR)	% (Col Val)
Ceramic Washbasins / Bathroom Sinks	Countertop Washbasins	658.96	24.28%	236.00	106.20	69.98	39.56%	736.54	23.84%	230.67	103.80	76.45	39.94%
	Hand Washbasins	672.53	24.78%	56.70	29.48	19.83	11.21%	763.23	24.70%	54.62	28.40	21.68	11.32%
	Inset Washbasins	81.42	3.00%	140.74	76.00	6.19	3.50%	92.70	3.00%	139.07	75.10	6.96	3.64%
	Pedestal Washbasins	60.79	2.24%	111.54	58.00	3.53	1.99%	71.50	2.31%	110.00	57.20	4.09	2.14%
	Semi-Recessed Washbasins	8.14	0.30%	175.56	79.00	0.64	0.36%	9.27	0.30%	174.23	78.40	0.73	0.38%
	Undermount Washbasins	54.28	2.00%	144.89	65.20	3.54	2.00%	61.80	2.00%	144.00	64.80	4.00	2.09%
	Wall Hung Washbasins	1 177.88	43.40%	119.51	62.15	73.20	41.38%	1 354.97	43.85%	110.00	57.20	77.50	40.49%
Other CSW	Bidets	30.53	7.10%	121.95	50.00	1.53	2.63%	36.28	7.10%	120.58	49.44	1.79	2.88%
	Close-Coupled Cisterns	22.34	5.20%	168.29	69.00	1.54	2.65%	49.05	9.60%	167.05	68.49	3.36	5.39%
	Kitchen Sinks	99.00	23.02%	699.32	320.57	31.74	54.58%	95.80	18.75%	682.99	313.22	30.01	48.18%
	Others	21.03	4.89%	57.07	23.40	0.49	0.85%	25.55	5.00%	56.14	23.02	0.59	0.94%
	Pedestals & Half Pedestals	61.25	14.25%	102.44	42.00	2.57	4.42%	73.33	14.35%	101.61	41.66	3.05	4.90%
	Shower Trays	4.50	1.05%	519.55	250.79	1.13	1.94%	3.60	0.70%	507.00	244.73	0.88	1.41%
	Urinals	184.90	43.00%	239.02	98.00	18.12	31.16%	219.73	43.00%	237.44	97.35	21.39	34.35%
WCs / Toilets	Utility / Laundry Sinks	6.45	1.50%	390.24	160.00	1.03	1.77%	7.67	1.50%	384.15	157.50	1.21	1.94%
	Close-Coupled WC Bowls	21.22	0.59%	145.77	75.80	1.61	0.69%	46.59	1.13%	144.19	74.98	3.49	1.36%
	Floor Standing WC Bowls	579.84	16.00%	104.23	54.20	31.43	13.39%	700.40	17.00%	103.79	53.97	37.80	14.70%
Wall Hung WC Bowls		3 022.94	83.41%	138.96	66.70	201.63	85.92%	3 373.01	81.87%	133.27	63.97	215.77	83.94%
Grand Total		6 768.01	100.00%	144.62	69.40	469.72	100.00%	7 721.00	100.00%	137.89	66.15	510.76	100.00%

Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price).

2.3.3 Prices and Market Values by Type of Rim : WCs

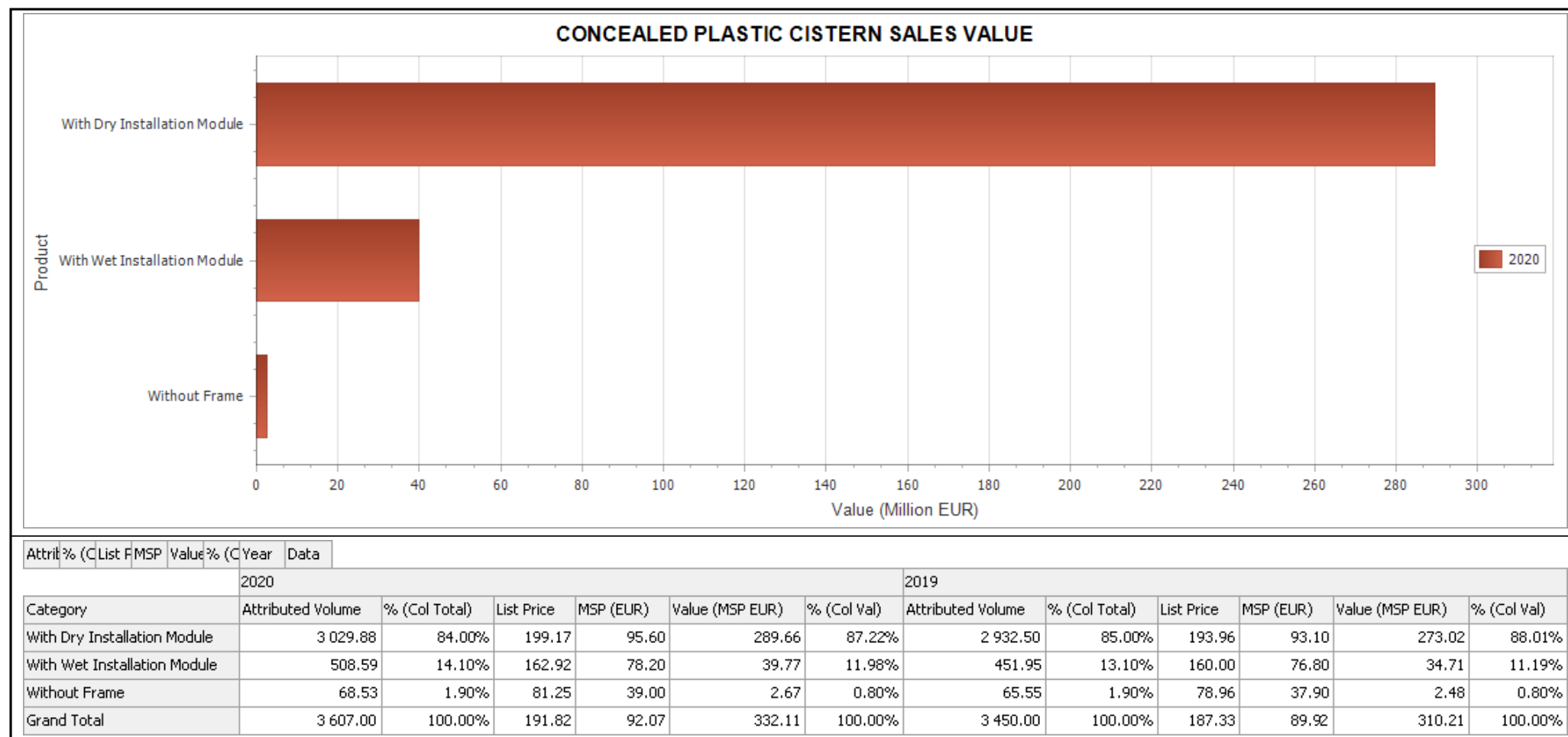
DE - ESTIMATED VALUES OF THE WC MARKET BY TYPE OF RIM



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price).

2.3.4 Prices and Market Values by Type : Conc. Cisterns

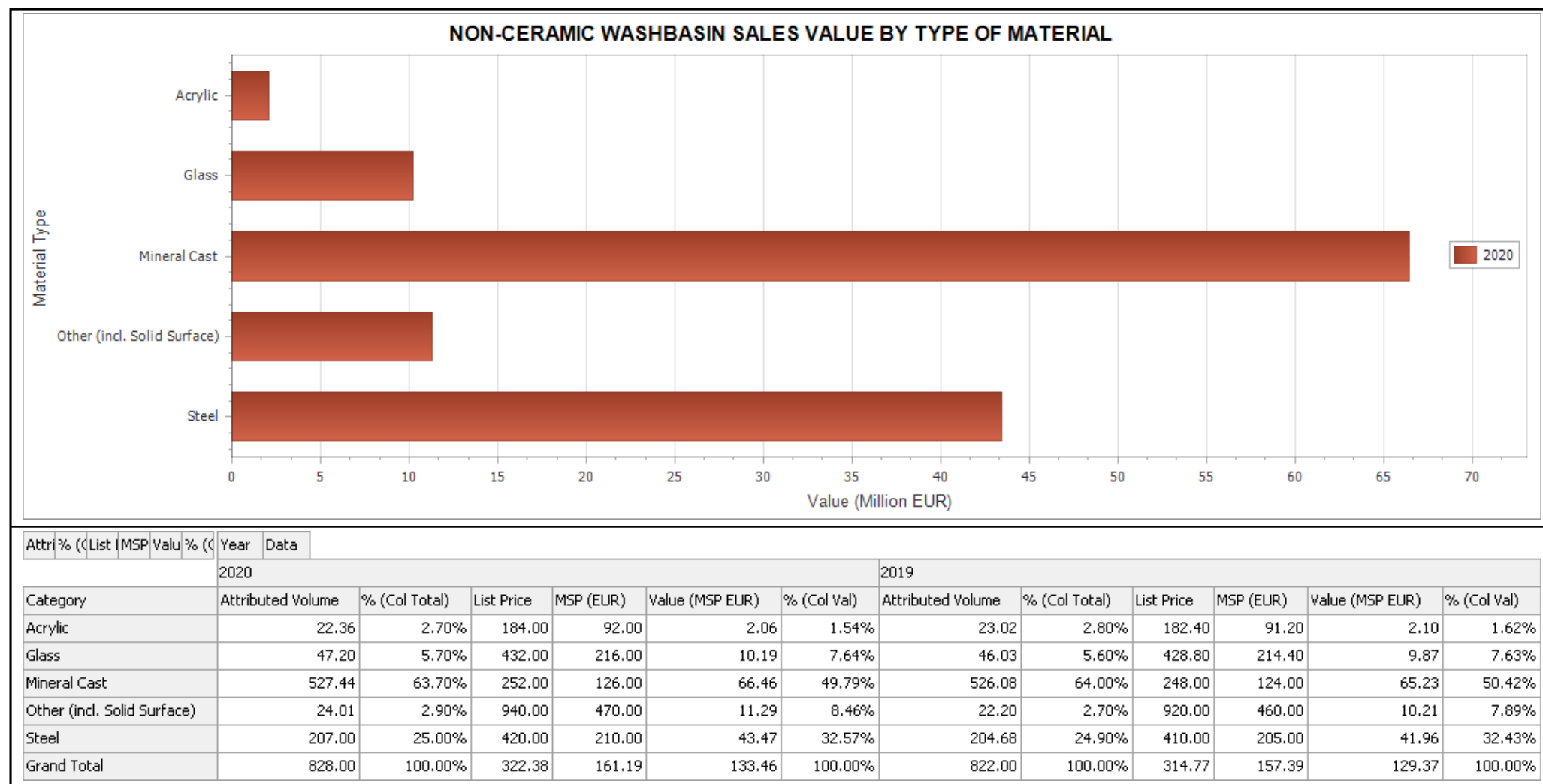
DE - ESTIMATED VALUE OF THE CONCEALED PLASTIC CISTERNS MARKET BY TYPE OF INSTALLATION



Source: BRG based on industry interviews. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price).

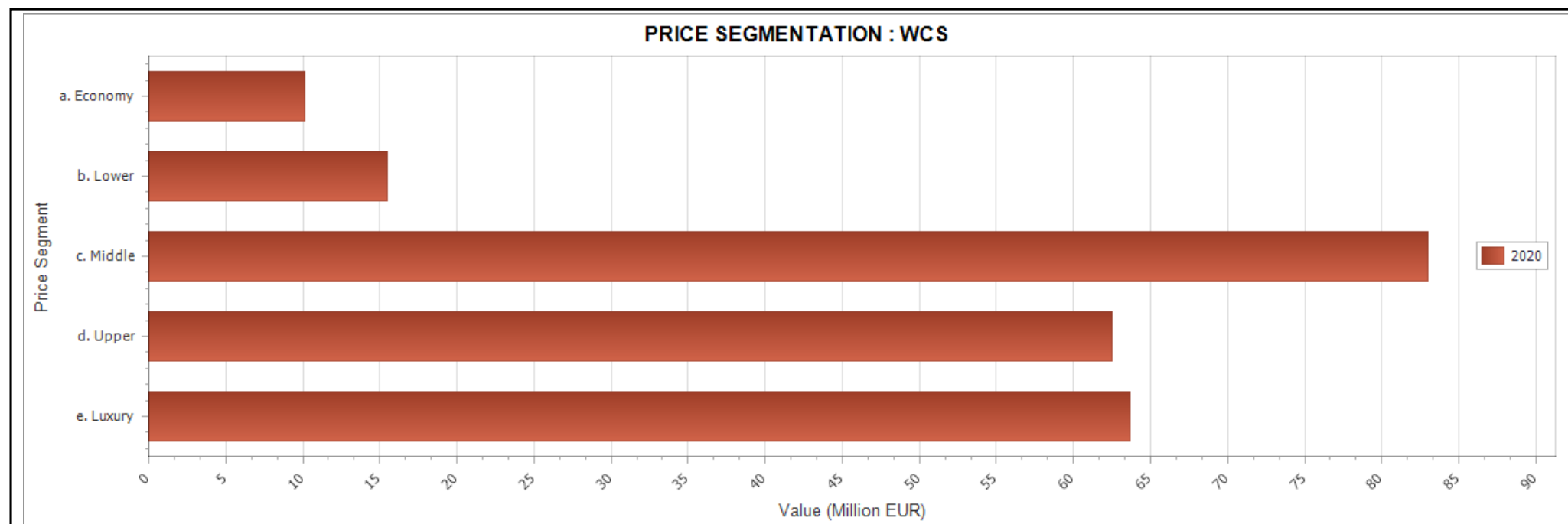
2.3.5 Prices and Market Values : Non-Ceramic Washbasins

DE - ESTIMATED VALUE OF THE NON-CERAMIC WASHBASIN MARKET BY MATERIAL TYPE



Source: BRG based on industry interviews. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price).

DE - ESTIMATED PRICE SEGMENTATION FOR WC

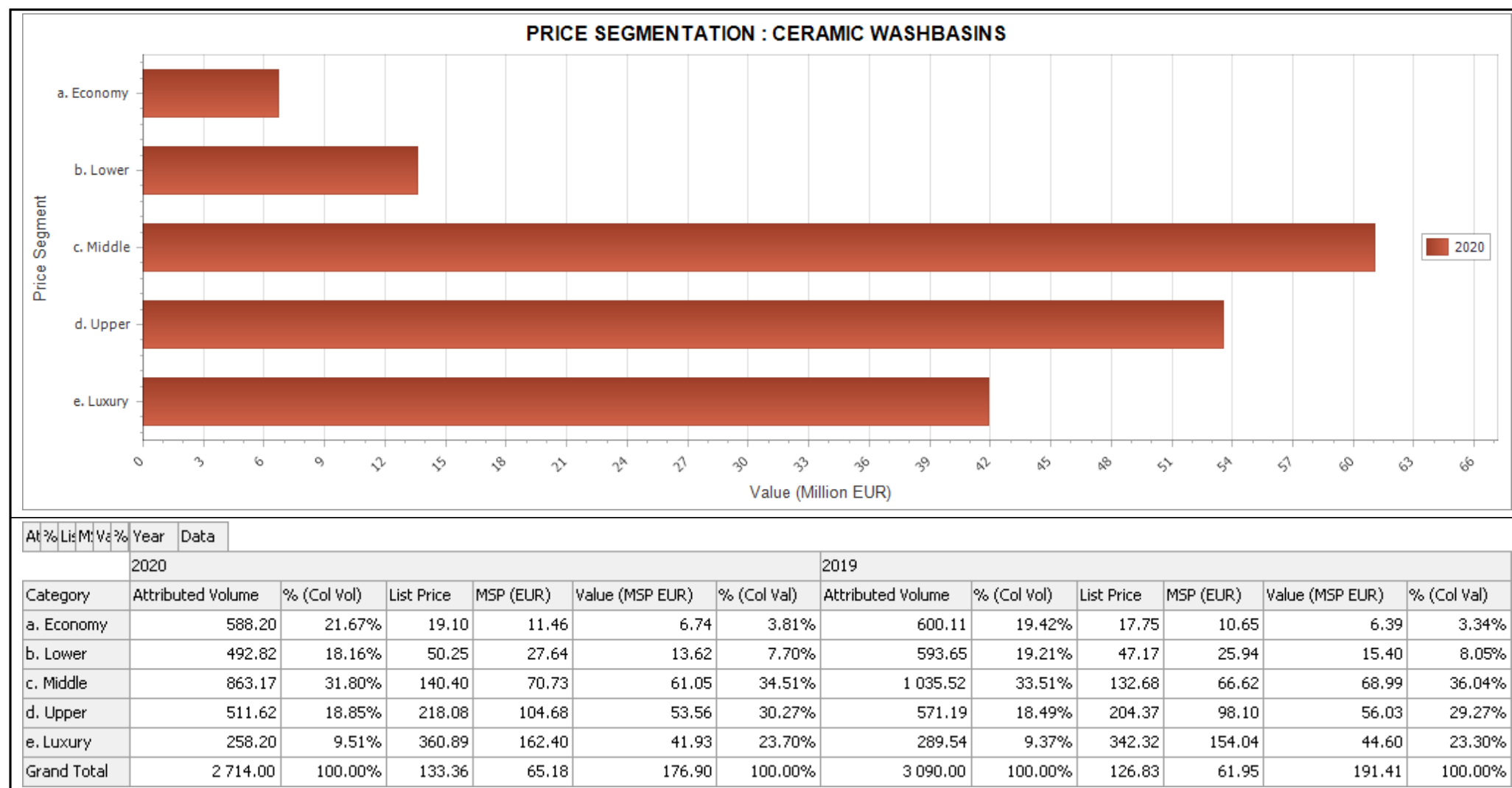


At %	List M %	Vol %	Year	Data								
2020							2019					
Category	Attributed Volume	% (Col Vol)	List Price	MSP (EUR)	Value (MSP EUR)	% (Col Val)	Attributed Volume	% (Col Vol)	List Price	MSP (EUR)	Value (MSP EUR)	% (Col Val)
a. Economy	592.74	16.36%	28.37	17.02	10.09	4.30%	651.84	15.82%	27.00	16.20	10.56	4.11%
b. Lower	532.90	14.70%	52.73	29.00	15.45	6.59%	662.69	16.08%	50.91	28.00	18.56	7.22%
c. Middle	1 497.77	41.33%	111.50	55.39	82.97	35.35%	1 693.70	41.11%	107.06	53.35	90.36	35.15%
d. Upper	628.02	17.33%	207.19	99.45	62.46	26.61%	696.76	16.91%	204.17	98.00	68.28	26.56%
e. Luxury	372.57	10.28%	379.96	170.98	63.70	27.15%	415.01	10.07%	371.11	167.00	69.31	26.96%
Grand Total	3 624.00	100.00%	133.44	64.75	234.67	100.00%	4 120.00	100.00%	128.38	62.39	257.06	100.00%

Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Includes all WC Types. Volumes Shown in Thousands and Values Shown in Millions. Economy <€20, Lower €20-€40, Middle €40-€80, Upper >€80-€120, Luxury >€120

2.3.7 Price Segmentation : Ceramic Washbasins

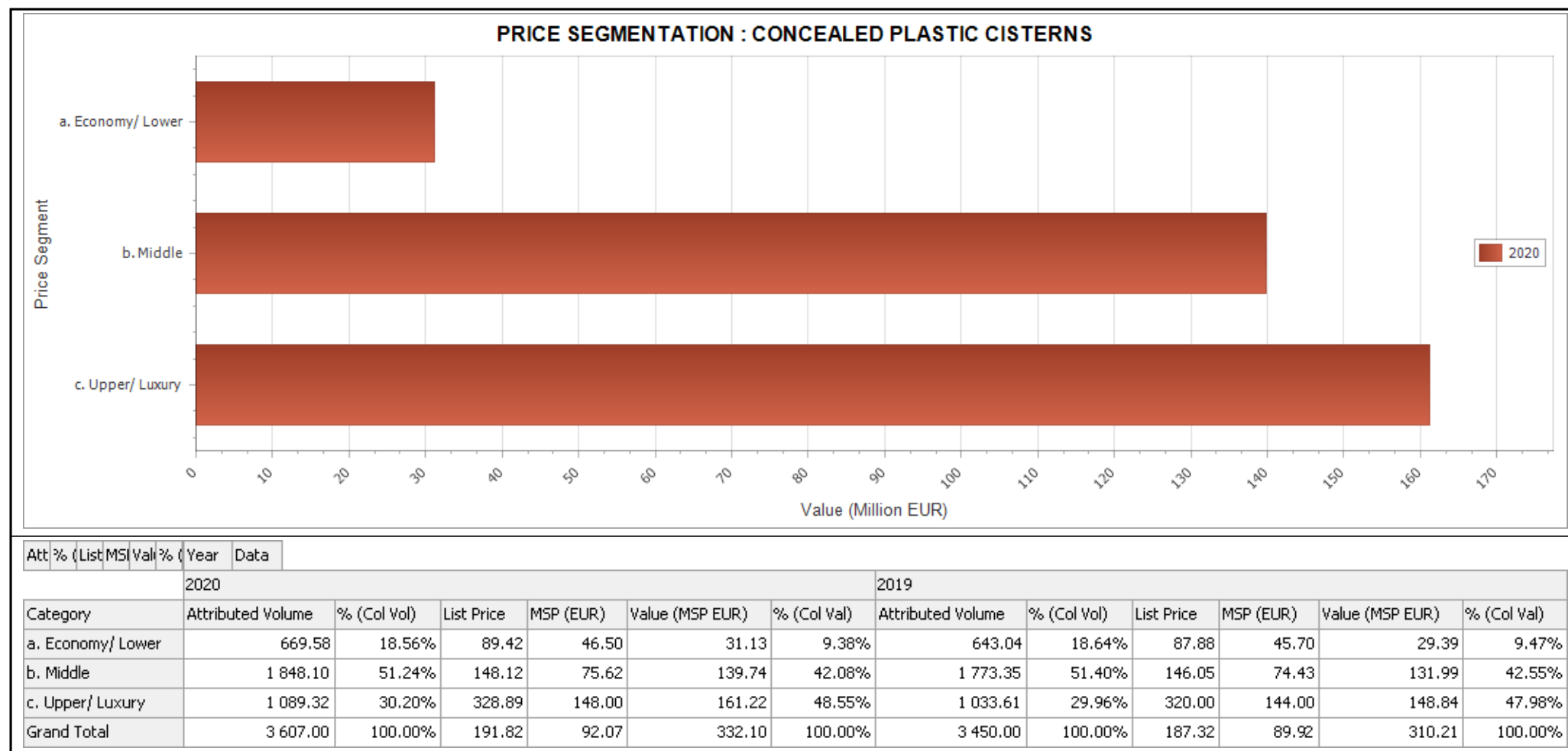
DE - ESTIMATED PRICE SEGMENTATION FOR CERAMIC WASHBASINS



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Includes All Washbasin Types. Volumes Shown in Thousands and Values Shown in Millions. Economy <€20, Lower €20-€40, Middle €40-€80, Upper >€80-€120, Luxury >€120

2.3.8 Price Segmentation : Concealed Plastic Cisterns

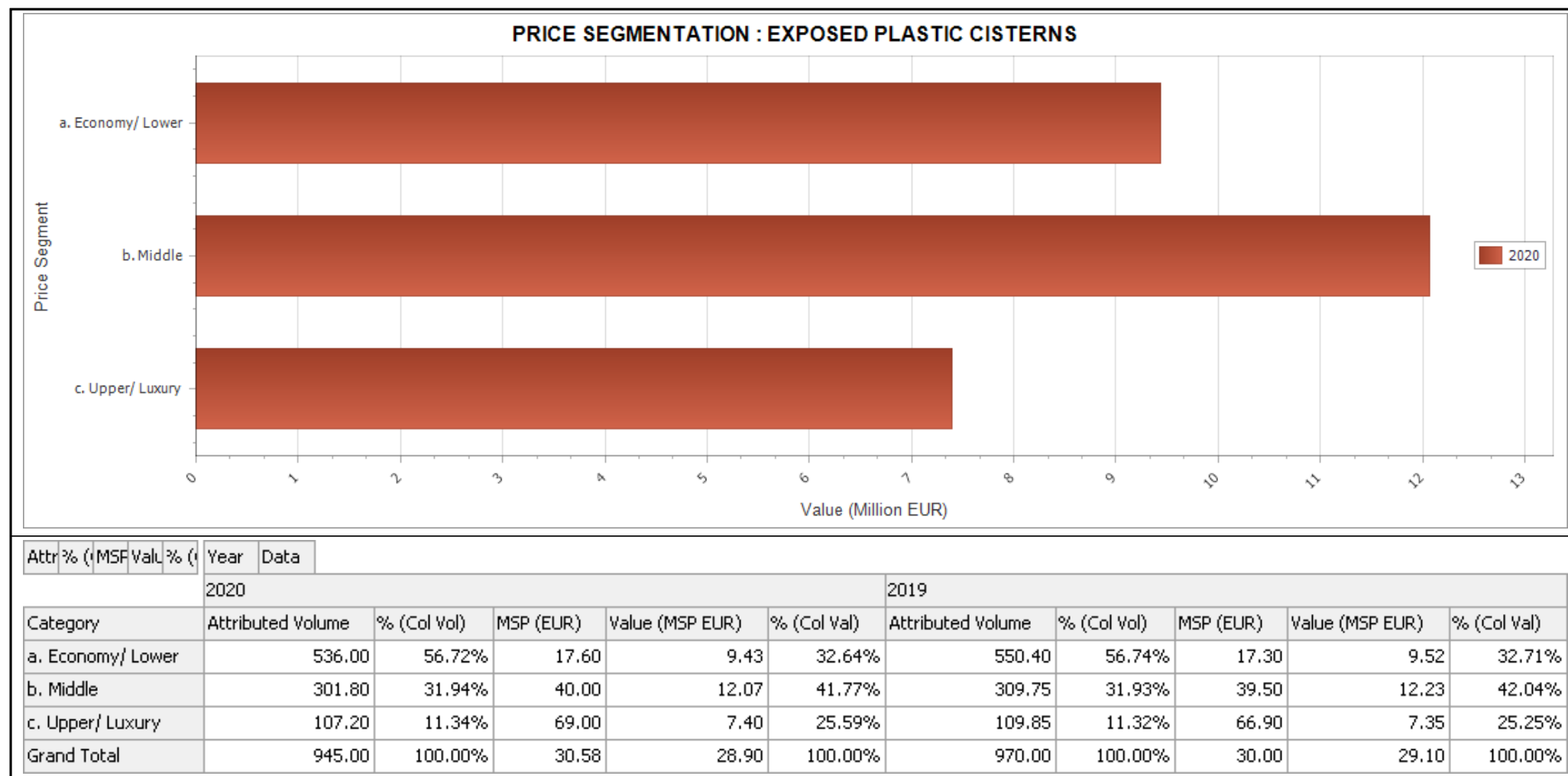
DE - ESTIMATED PRICE SEGMENTATION FOR CONCEALED PLASTIC CISTERNS



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Volumes Shown in Thousands and Values Shown in Millions. Economy/ Lower <€60, Middle €60-€120, Upper/ Luxury >€120

2.3.9 Price Segmentation : Exposed Plastic Cisterns

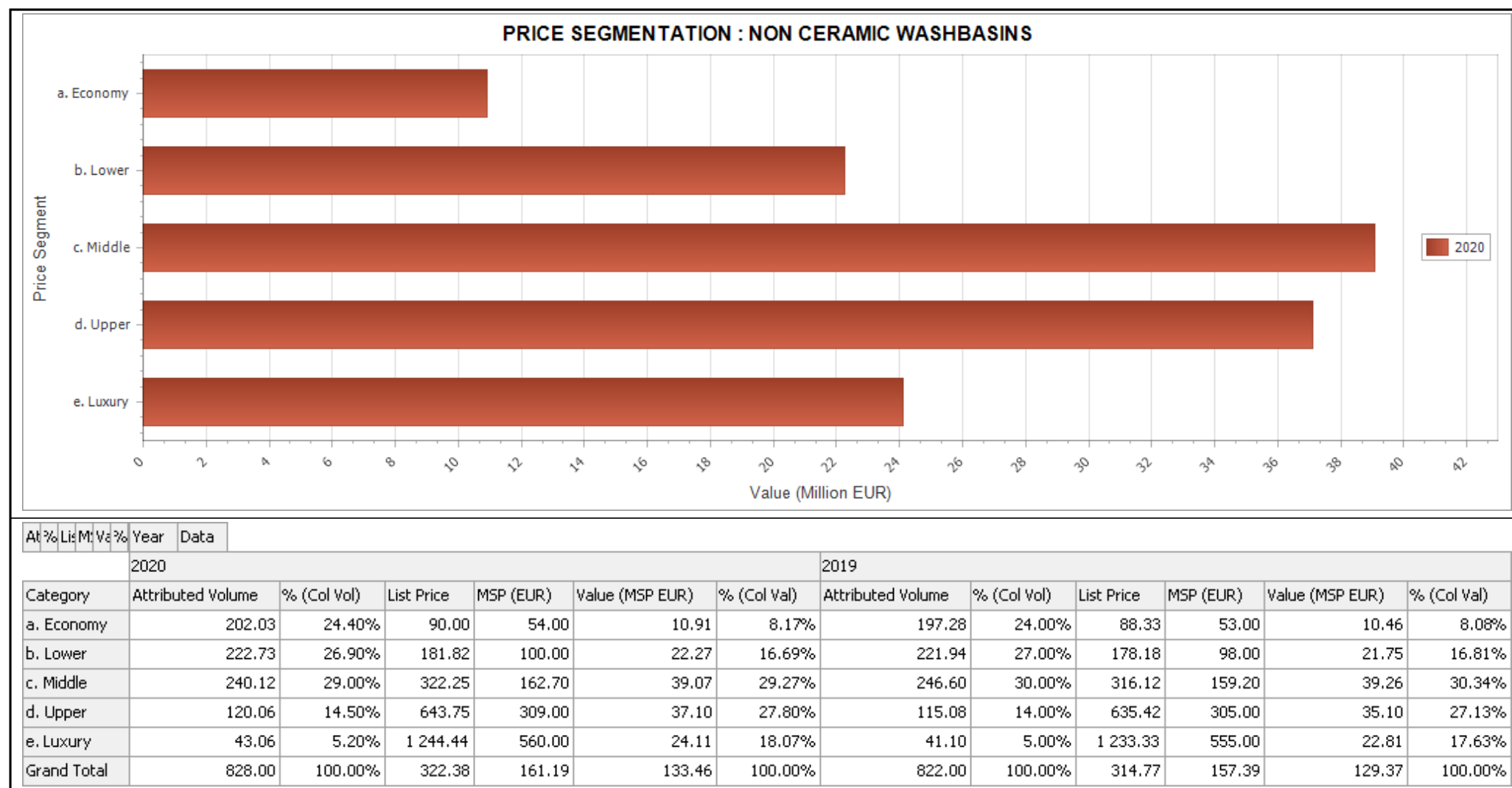
DE - ESTIMATED PRICE SEGMENTATION FOR EXPOSED PLASTIC CISTERNS



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Volumes Shown in Thousands and Values Shown in Millions. Economy/ Lower <€25, Middle €25-€50, Upper/ Luxury >€50

2.3.10 Price Segmentation : Non Ceramic Washbasins

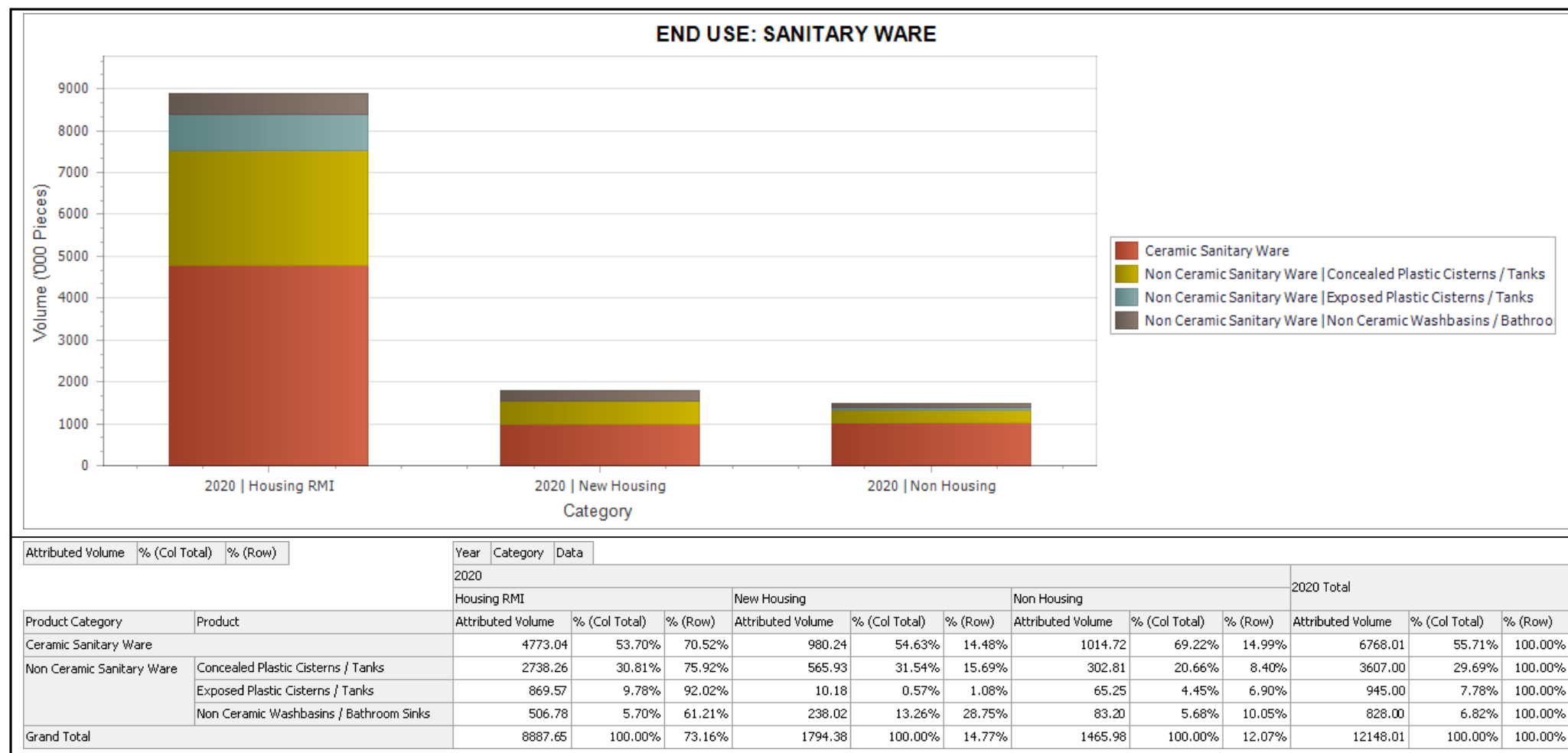
DE - ESTIMATED PRICE SEGMENTATION FOR NON CERAMIC WASHBASINS



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Includes all Washbasin Types. Volumes Shown in Thousands and Values Shown in Millions. Economy <€60, Lower €60-€120, Middle €120-€200, Upper €200-€400, Luxury >€400

2.4.1 End Use : Base Year

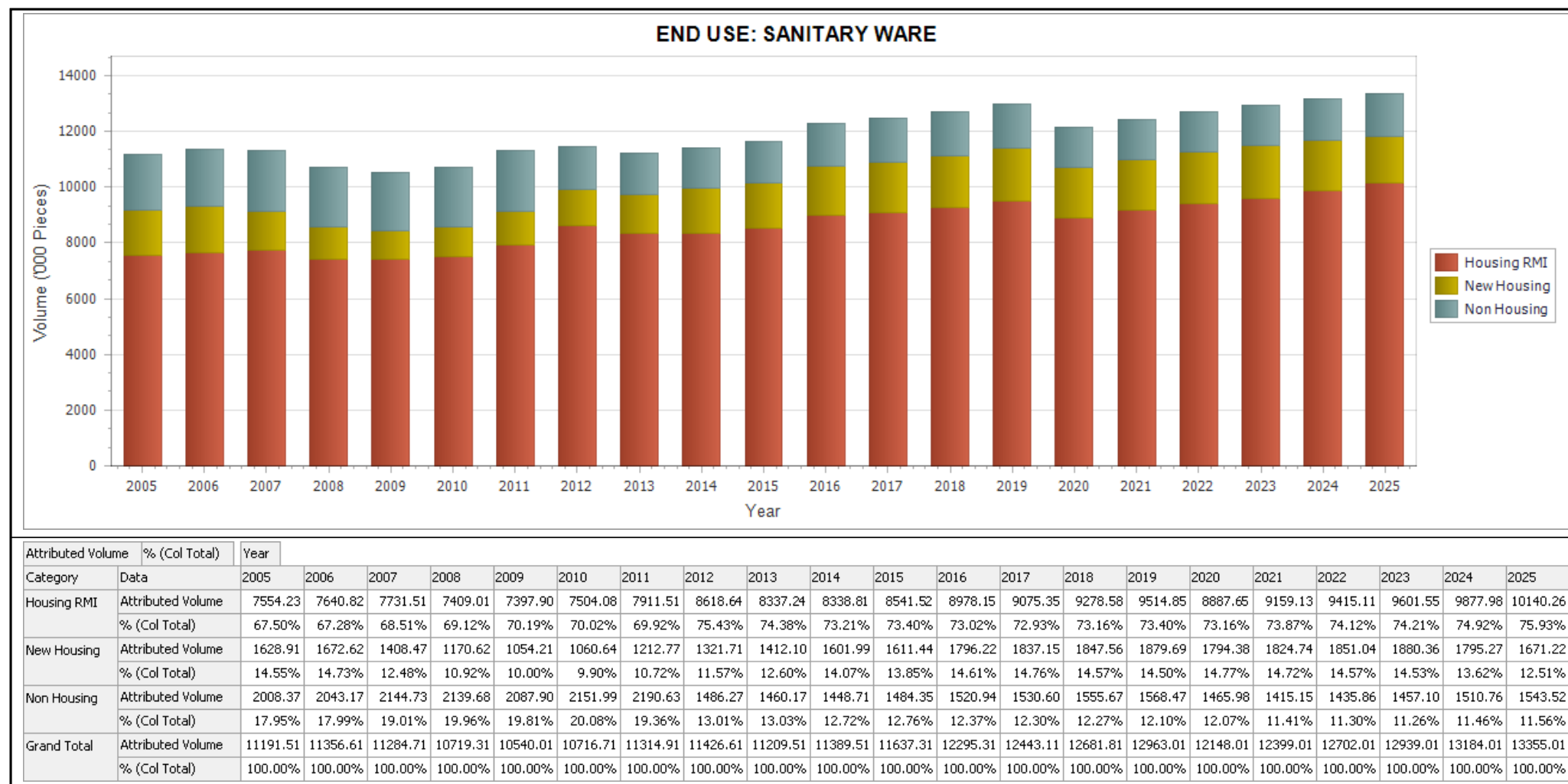
DE - SALES OF SANITARY WARE BY TYPE AND END USE SEGMENT ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

2.4.2 End Use : Historical and Forecasts

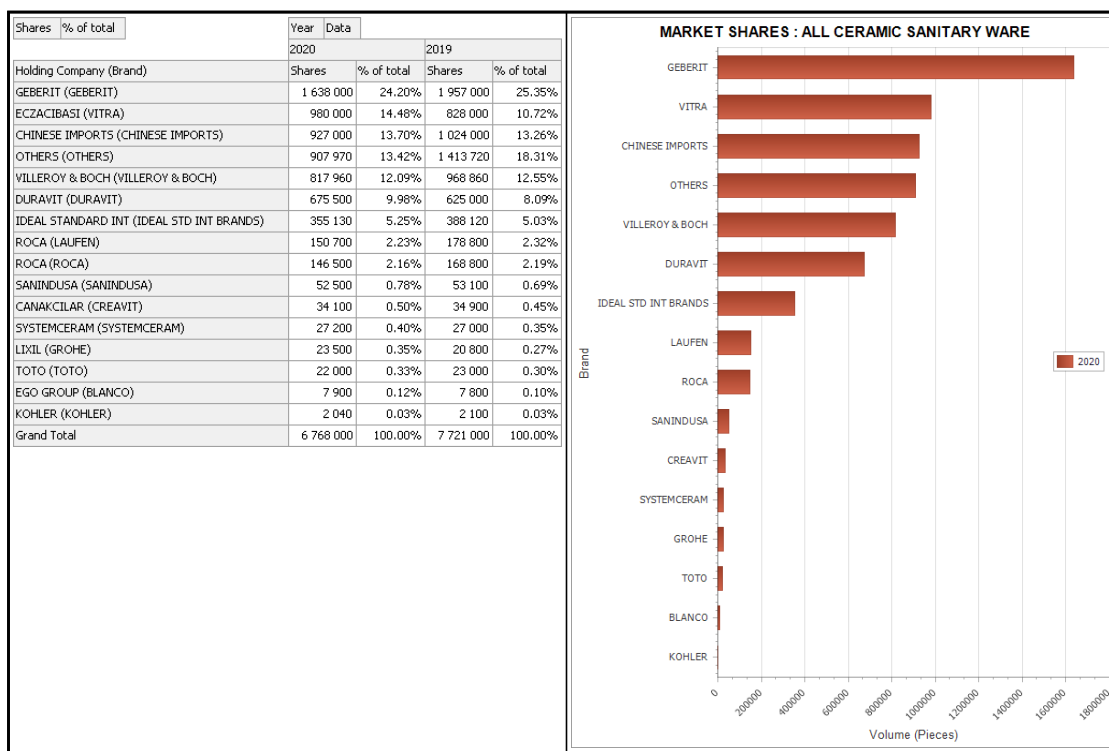
DE - SALES OF SANITARY WARE BY END USE SEGMENT ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

2.5.1 Market Shares : All Ceramic Sanitary Ware

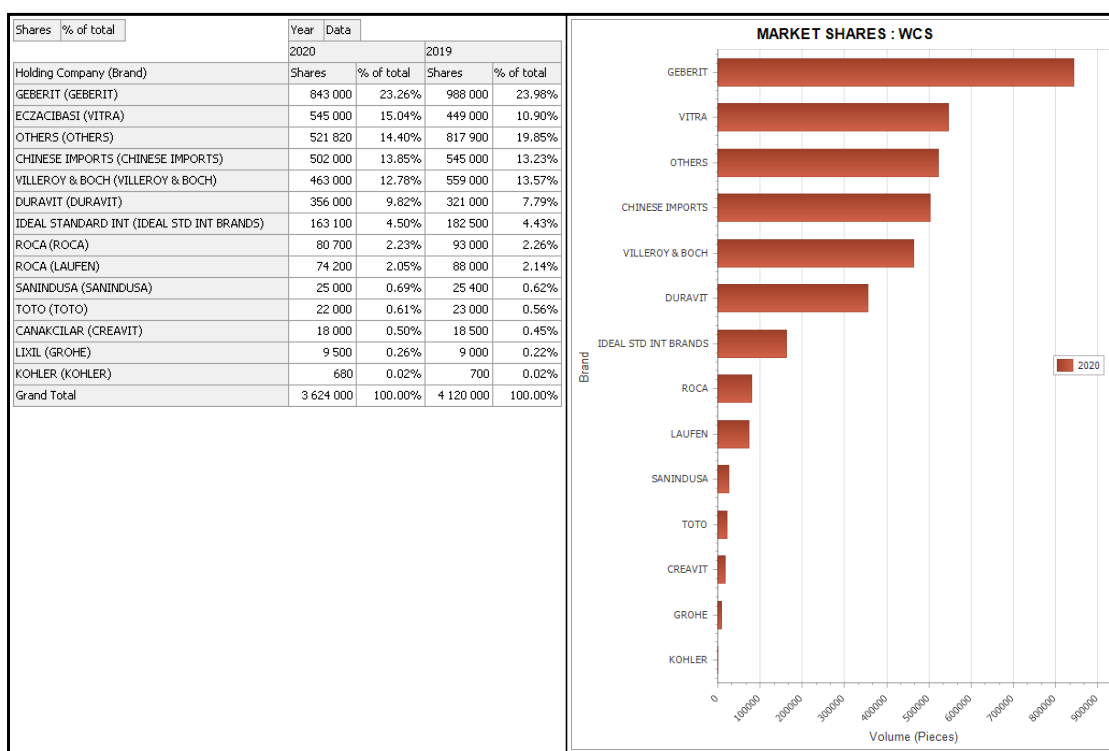
DE - ESTIMATED MARKET SHARES FOR ALL CERAMIC SANITARY WARE



Source: BRG based on industry interviews

2.5.2 Market Shares : WCs

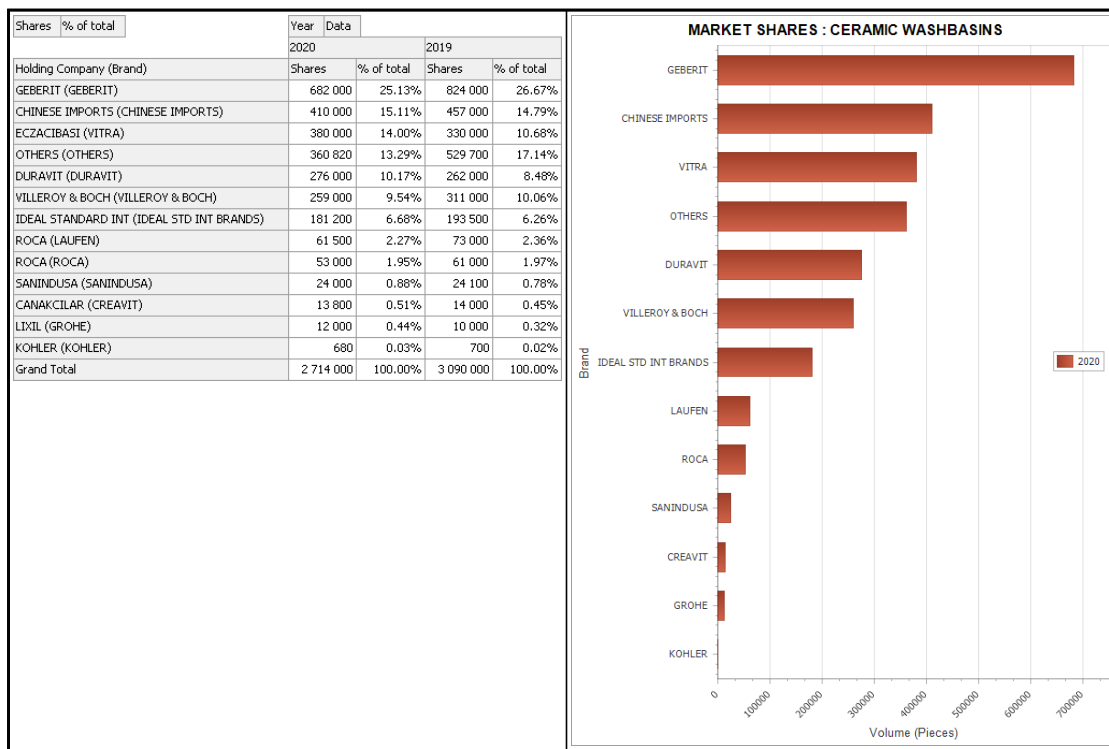
DE - ESTIMATED MARKET SHARES FOR WCS



Source: BRG based on industry interviews.

2.5.3 Market Shares : Ceramic Washbasins

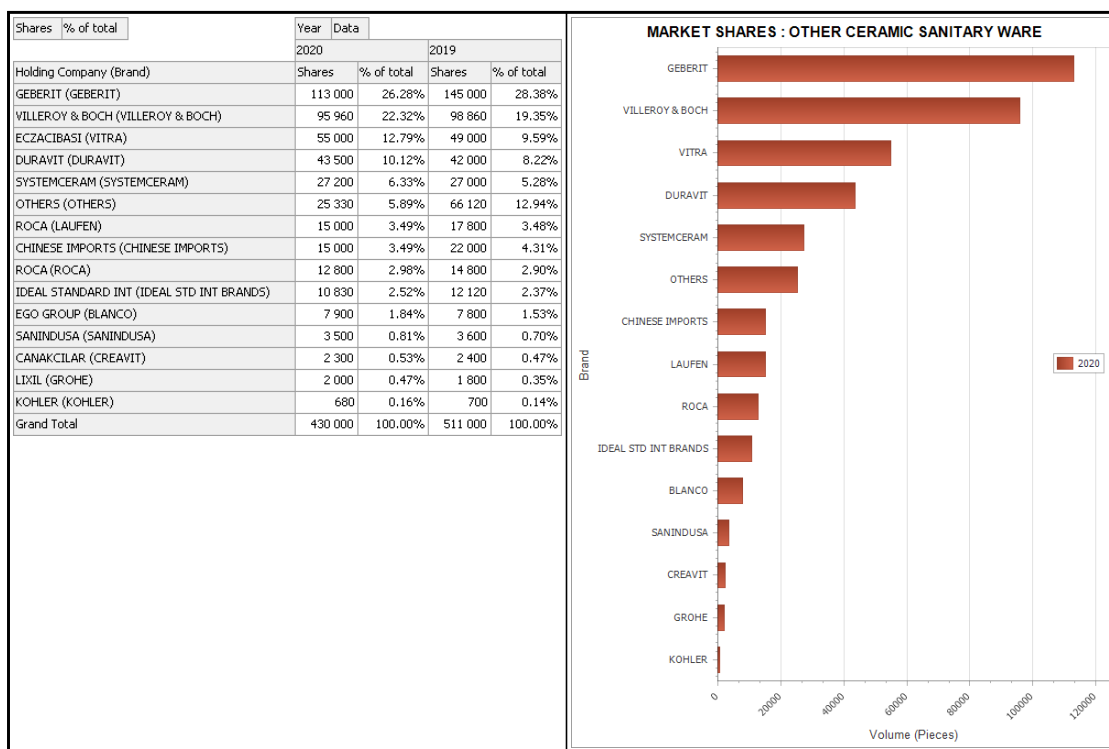
DE - ESTIMATED MARKET SHARES FOR CERAMIC WASHBASINS



Source: BRG based on industry interviews

2.5.4 Market Shares : Other Ceramic Sanitary Ware

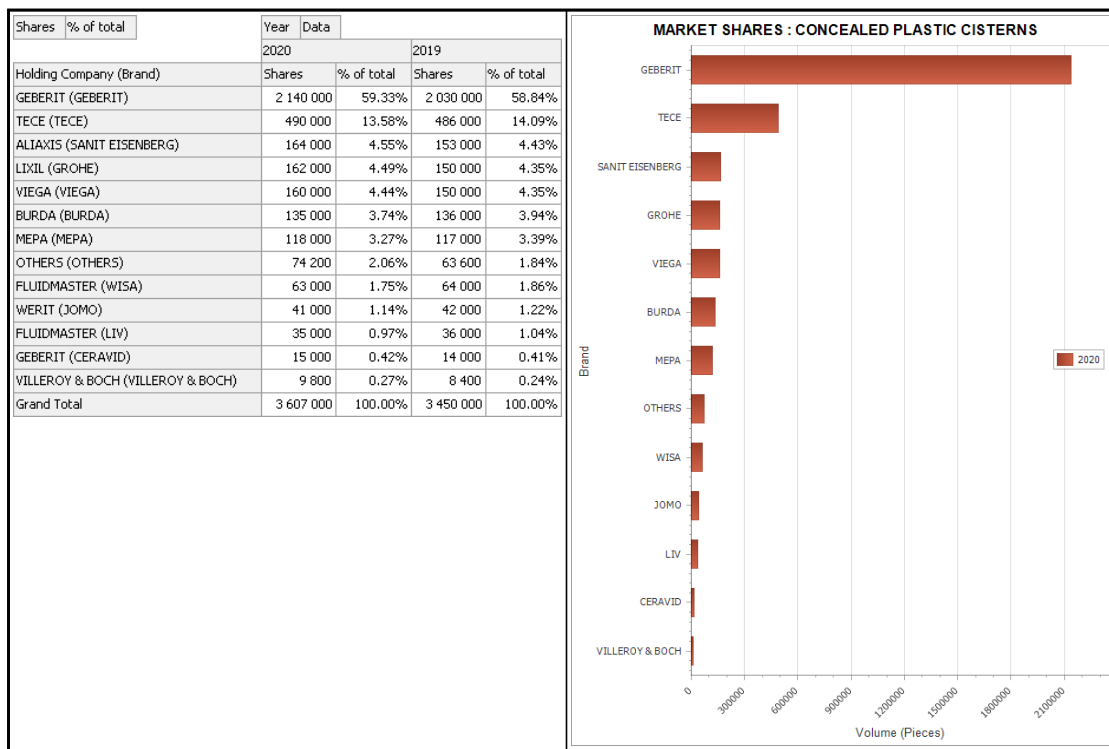
DE - ESTIMATED MARKET SHARES FOR OTHER CERAMIC SANITARY WARE



Source: BRG based on industry interviews

2.5.5 Market Shares : Concealed Plastic Cisterns

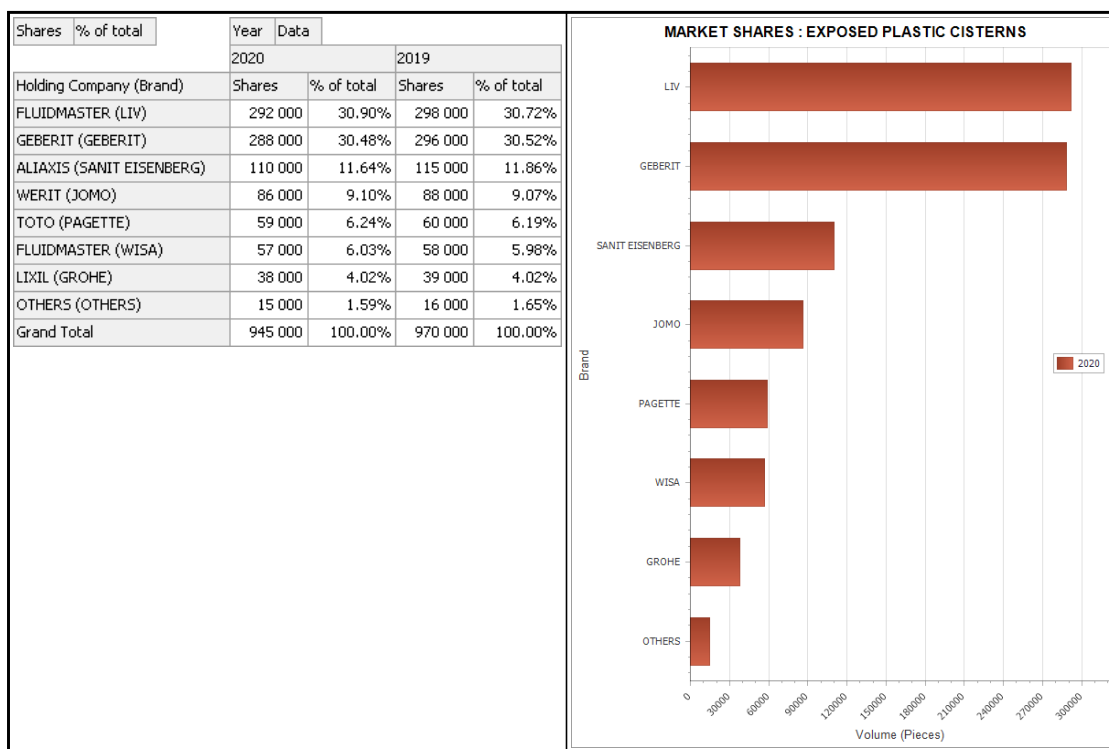
DE - ESTIMATED MARKET SHARES FOR CONCEALED PLASTIC CISTERNS



Source: BRG based on industry interviews

2.5.6 Market Shares : Exposed Plastic Cisterns

DE - ESTIMATED MARKET SHARES FOR EXPOSED PLASTIC CISTERNS



Source: BRG based on industry interviews.

2.6.1 DE - IMPORTS OF ALL CERAMIC SANITARY WARE 2020

Source Country	2020				2019			
	Volume		Value 000		Volume		Value 000	
	Tonnes	%	Euros	%	Tonnes	%	Euros	%
Egypt	24 109	16.3%	31 750	8.2%	24 648	15.7%	29 743	7.8%
Poland	22 432	15.2%	42 912	11.1%	20 984	13.4%	33 386	8.7%
Turkey	22 224	15.0%	40 862	10.6%	21 027	13.4%	41 061	10.7%
China	12 287	8.3%	39 062	10.1%	12 017	7.6%	35 106	9.2%
Romania	9 783	6.6%	25 186	6.5%	10 096	6.4%	25 434	6.6%
Hungary	8 070	5.5%	20 348	5.3%	8 747	5.6%	23 036	6.0%
Italy	5 553	3.8%	24 158	6.3%	7 181	4.6%	29 170	7.6%
France	5 181	3.5%	27 416	7.1%	6 420	4.1%	32 222	8.4%
Austria	5 175	3.5%	28 958	7.5%	5 770	3.7%	29 716	7.8%
Czech Republic	3 912	2.6%	10 186	2.6%	5 030	3.2%	12 498	3.3%
Portugal	3 152	2.1%	6 750	1.7%	3 632	2.3%	6 407	1.7%
OTHERS	25 844	17.5%	88 375	22.9%	31 599	20.1%	85 145	22.2%
Total	147 722	100.0%	385 963	100.0%	157 151	100.0%	382 924	100.0%

Source : Trademap

2.6.2 DE - EXPORTS OF ALL CERAMIC SANITARY WARE 2020

Destination	2020				2019			
	Volume		Value 000		Volume		Value 000	
	Tonnes	%	Euros	%	Tonnes	%	Euros	%
Netherlands	10 971	14.5%	56 781	14.1%	11 951	14.9%	61 139	15.0%
France	8 796	11.6%	33 324	8.3%	6 824	8.5%	24 642	6.0%
Austria	6 210	8.2%	39 139	9.8%	7 120	8.8%	39 037	9.6%
Belgium	5 775	7.6%	25 101	6.3%	6 862	8.5%	28 284	6.9%
United Kingdom	5 290	7.0%	27 746	6.9%	4 999	6.2%	28 737	7.1%
Poland	4 031	5.3%	20 260	5.0%	6 450	8.0%	27 051	6.6%
Switzerland	3 749	5.0%	48 845	12.2%	3 437	4.3%	42 024	10.3%
Russian Federation	3 285	4.3%	11 449	2.9%	2 608	3.2%	10 360	2.5%
Italy	2 510	3.3%	12 964	3.2%	2 593	3.2%	12 845	3.2%
China	1 873	2.5%	12 335	3.1%	2 004	2.5%	13 164	3.2%
Luxembourg	1 870	2.5%	11 225	2.8%	1 822	2.3%	10 999	2.7%
OTHERS	21 369	28.2%	102 212	25.5%	23 782	29.6%	109 249	26.8%
Total	75 729	100.0%	401 381	100.0%	80 452	100.0%	407 531	100.0%

Source : Trademap

2.6.3 DE - IMPORTS OF PLASTIC CISTERNS 2020

Source Country	2020				2019			
	Volume		Value 000		Volume		Value 000	
	Tonnes	%	Euros	%	Tonnes	%	Euros	%
Poland	3 223	18.5%	16 198	10.5%	5 543	27.9%	23 106	14.7%
Switzerland	2 404	13.8%	36 074	23.3%	2 398	12.1%	34 610	22.0%
Portugal	1 888	10.8%	5 832	3.8%	2 214	11.1%	4 844	3.1%
China	1 662	9.5%	10 989	7.1%	1 716	8.6%	11 470	7.3%
Austria	1 658	9.5%	25 918	16.7%	1 849	9.3%	25 998	16.6%
Netherlands	1 648	9.5%	20 325	13.1%	1 299	6.5%	15 817	10.1%
Slovenia	979	5.6%	4 759	3.1%	946	4.8%	4 701	3.0%
Italy	963	5.5%	9 037	5.8%	1 009	5.1%	9 796	6.2%
Czech Republic	827	4.7%	5 956	3.8%	757	3.8%	6 051	3.9%
Turkey	684	3.9%	4 431	2.9%	685	3.4%	4 050	2.6%
France	258	1.5%	2 852	1.8%	290	1.5%	4 195	2.7%
OTHERS	1 222	7.0%	12 384	8.0%	1 187	6.0%	12 419	7.9%
Total	17 416	100.0%	154 755	100.0%	19 893	100.0%	157 057	100.0%

Source : Trademap

2.6.4 DE - EXPORTS OF PLASTIC CISTERNS 2020

Destination	2020				2019			
	Volume		Value 000		Volume		Value 000	
	Tonnes	%	Euros	%	Tonnes	%	Euros	%
France	9 077	12.8%	49 283	9.9%	10 059	14.1%	55 457	10.9%
Netherlands	5 930	8.4%	44 839	9.0%	5 461	7.7%	40 462	8.0%
Poland	5 571	7.9%	31 644	6.3%	5 067	7.1%	31 501	6.2%
Russian Federation	4 757	6.7%	26 427	5.3%	3 994	5.6%	24 652	4.9%
Italy	4 446	6.3%	56 025	11.2%	4 886	6.9%	61 176	12.1%
Belgium	4 053	5.7%	30 078	6.0%	3 854	5.4%	28 934	5.7%
Switzerland	3 750	5.3%	51 448	10.3%	3 735	5.3%	43 809	8.6%
Saudi Arabia	3 266	4.6%	15 399	3.1%	1 833	2.6%	9 196	1.8%
Czech Republic	2 275	3.2%	17 755	3.6%	2 165	3.0%	16 529	3.3%
Austria	2 052	2.9%	22 566	4.5%	2 255	3.2%	22 491	4.4%
United Kingdom	1 878	2.6%	14 020	2.8%	2 197	3.1%	16 169	3.2%
OTHERS	23 826	33.6%	140 168	28.1%	25 617	36.0%	156 795	30.9%
Total	70 881	100.0%	499 652	100.0%	71 123	100.0%	507 171	100.0%

Source : Trademap

2.7.1

Route to Market: Ceramic Sanitary Ware
DE - ESTIMATED DISTRIBUTION FOR CERAMIC SANITARY WARE: 2020

Manufacturer / Importer	Manufacturers / Importers
	100.0%

Wholesale Level	77.9%
	Wholesalers / Merchants
	77.9%

Retail Level	Manufacturer	Wholesaler	Manufacturer	Wholesaler	Manufacturer	Wholesaler
	4.9%	4.0%	12.0%	10.0%	0.9%	0.2%
	Internet Retailers		DIY Retailers		Specialist Retailers	
	8.9%		22.0%		1.1%	

Manufacturer	Wholesaler	Internet	DIY	Specialist
2.4%	63.5%	4.0%	1.0%	0.3%
Installers				
71.2%				

End Use Level	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer
	0.5%	0.1%	4.3%	21.0%	0.8%	58.4%	1.5%	0.1%	0.6%	0.0%	0.0%	12.8%
	Residential						Non-Residential					
	85.0%						15.0%					
New Build			Residential RMI			New Build			Non-Residential RMI			
14.5%			70.5%			5.9%			9.1%			
Project	Non-Project		Project	Non-Project		Project	Non-Project		Project	Non-Project		
5.7%	8.8%		2.0%	68.5%		3.7%	2.2%		3.0%	6.1%		

2.7.2

Route to Market: Ceramic Sanitary Ware
DE - ESTIMATED DISTRIBUTION FOR CERAMIC SANITARY WARE: 2025

Manufacturer / Importer	Manufacturers / Importers
	100.0%

Wholesale Level	76.8%
	Wholesalers / Merchants
	76.8%

Retail Level	Manufacturer	Wholesaler	Manufacturer	Wholesaler	Manufacturer	Wholesaler
	5.0%	6.0%	10.0%	9.0%	0.7%	0.0%
	Internet Retailers		DIY Retailers		Specialist Retailers	
	11.0%		19.0%		0.7%	

Manufacturer	Wholesaler	Internet	DIY	Specialist
4.3%	61.6%	5.0%	1.5%	0.1%
Installers				
72.5%				

End Use Level	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer
	0.8%	0.1%	6.0%	17.5%	0.6%	60.6%	2.4%	0.1%	0.0%	0.0%	0.0%	11.9%
	Residential						Non-Residential					
	85.6%						14.4%					
	New Build			Residential RMI			New Build			Non-Residential RMI		
	12.2%			73.4%			4.9%			9.5%		
	Project		Non-Project	Project		Non-Project	Project		Non-Project	Project		Non-Project
	5.5%		6.7%	2.0%		71.4%	2.4%		2.5%	3.2%		6.3%

Source: BRG Based on Industry Interviews. - Totals may vary due to roundings

2.7.3 Route to Market - Consolidated View : Ceramic Sanitary Ware

DE - ESTIMATED DISTRIBUTION FOR CERAMIC SANITARY WARE : 2020

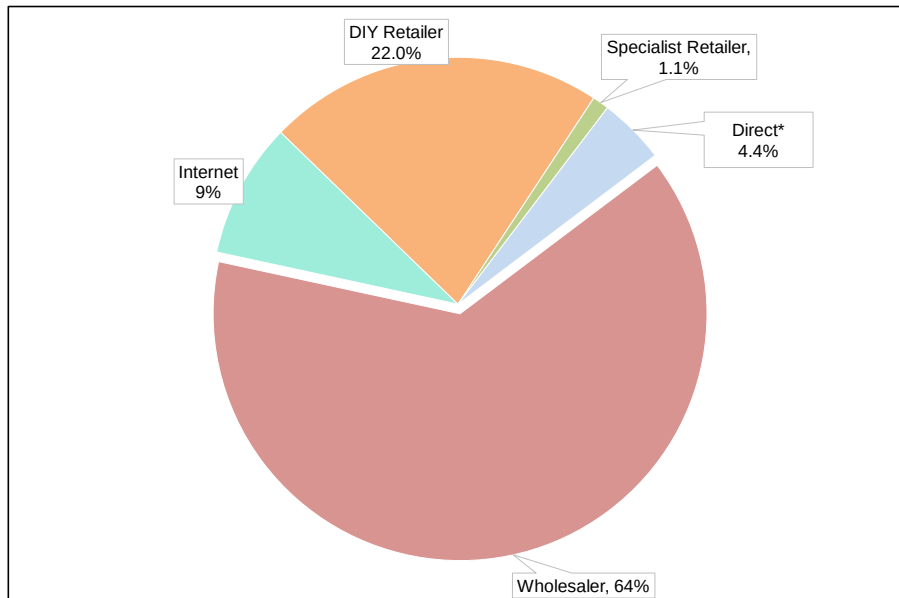
Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
63.7%	8.9%	22.0%	1.1%	4.4%	7.7%	6.7%	77.3%	8.3%
100%					14.4%		85.6%	

2.7.4 Route to Market - Consolidated View : Ceramic Sanitary Ware

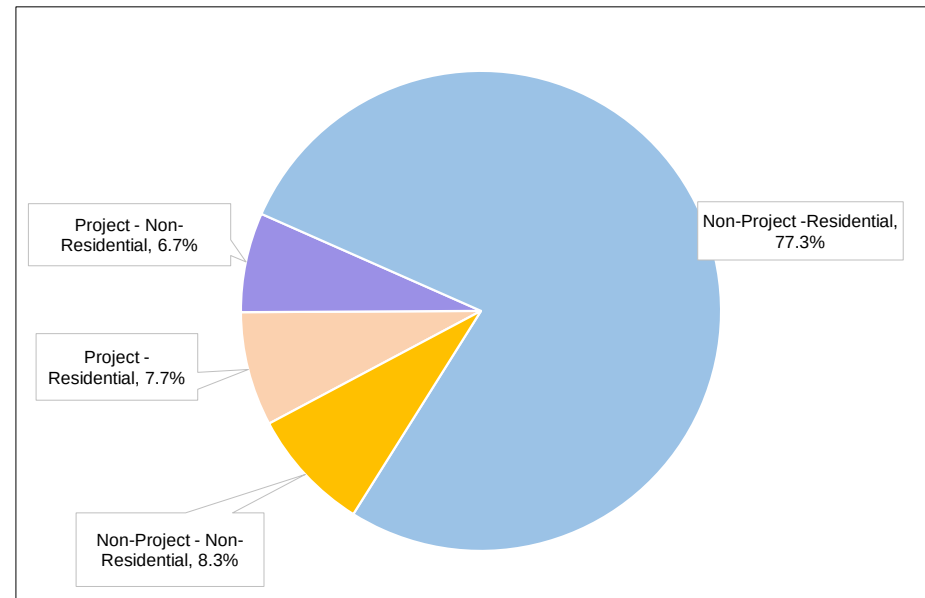
DE - ESTIMATED DISTRIBUTION FOR CERAMIC SANITARY WARE : 2025

Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
61.8%	11.0%	19.0%	0.7%	7.5%	7.5%	5.6%	78.1%	8.8%
100%					13.1%		86.9%	

DE - SALES OF CERAMIC SANITARY WARE BY DISTRIBUTION CHANNEL: 2020



DE - SALES OF CERAMIC SANITARY WARE BY POINT OF DECISION AND END-USE SECTOR: 2020



Source: BRG Based on Industry Interviews.

*Direct includes sales going directly from manufacturers to installers, contractors and private customers.

2.7.5 Route to Market: Plastic Cisterns
DE - ESTIMATED DISTRIBUTION FOR PLASTIC CISTERNS: 2020

Manufacturer / Importer	Manufacturers / Importers
	100.0%

Wholesale Level	84.5%
	Wholesalers / Merchants
	84.5%

Retail Level	Manufacturer 3.6%	Wholesaler 3.2%	Manufacturer 6.9%	Wholesaler 6.2%	Manufacturer 0.1%	Wholesaler 0.0%
	Internet Retailers		DIY Retailers		Specialist Retailers	
	6.8%		13.1%		0.1%	

Manufacturer	Wholesaler	Internet	DIY	Specialist
4.9%	74.8%	3.0%	1.0%	0.0%
Installers				
83.7%				

End Use Level	Manufacturer 0.0%	Wholesaler 0.2%	Internet 3.0%	DIY 11.8%	Specialist 0.1%	Installer 76.8%
	Residential					
	91.9%					
	New Build		Residential RMI			
	12.7%		79.3%			
	Project	Non-Project	Project	Non-Project		
	5.0%	7.7%	2.0%	77.3%		
	Non-Residential					
	8.1%					
	New Build		Non-Residential RMI			
	3.2%		4.9%			
	Project	Non-Project	Project	Non-Project		
	2.1%	1.1%	1.5%	3.4%		

2.7.6 Route to Market: Plastic Cisterns
DE - ESTIMATED DISTRIBUTION FOR PLASTIC CISTERNS: 2025

Manufacturer / Importer	Manufacturers / Importers
	100.0%

Wholesale Level	84.0%
	Wholesalers / Merchants
	84.0%

Retail Level	Manufacturer 3.7%	Wholesaler 4.5%	Manufacturer 6.7%	Wholesaler 6.0%	Manufacturer 0.1%	Wholesaler 0.0%
	Internet Retailers		DIY Retailers		Specialist Retailers	
	8.2%		12.7%		0.1%	

Manufacturer	Wholesaler	Internet	DIY	Specialist
5.5%	73.0%	4.0%	1.0%	0.0%
Installers				
83.5%				

End Use Level	Manufacturer 0.0%	Wholesaler 0.3%	Internet 3.2%	DIY 11.4%	Specialist 0.1%	Installer 77.3%
	Residential					
	92.3%					
	New Build		Residential RMI			
	10.7%		81.6%			
	Project	Non-Project	Project	Non-Project		
	5.0%	5.7%	2.0%	79.6%		
	Non-Residential					
	7.7%					
	New Build		Non-Residential RMI			
	3.0%		4.7%			
	Project	Non-Project	Project	Non-Project		
	2.0%	1.0%	3.5%	1.2%		

Source: BRG Based on Industry Interviews. N.B. Include both Exposed and Concealed Plastic Cisterns - Totals may vary due to roundings

2.7.7 Route to Market - Consolidated View : Plastic Cisterns

DE - ESTIMATED DISTRIBUTION FOR PLASTIC CISTERNS : 2020

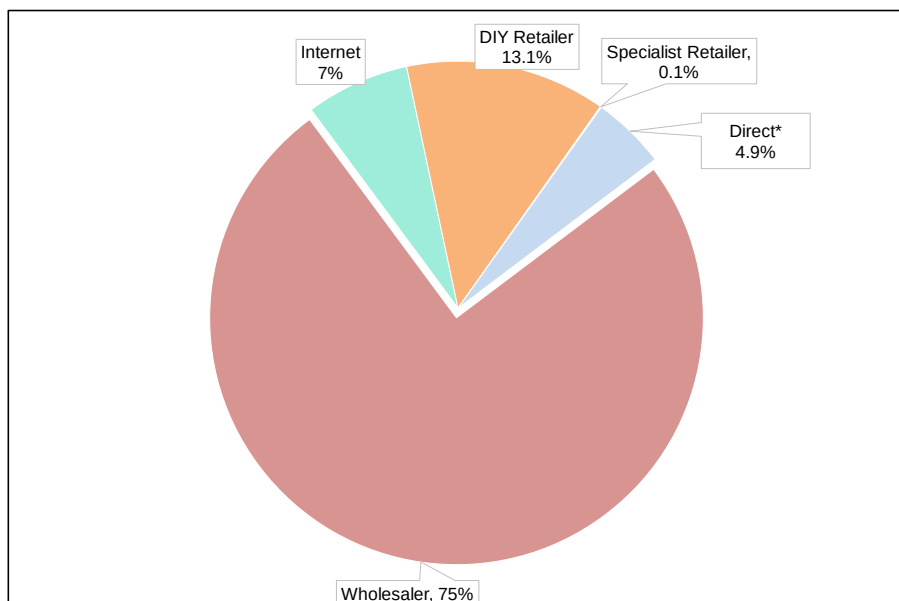
Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
75.1%	6.8%	13.1%	0.1%	4.9%	7.0%	3.6%	84.9%	4.5%
100%					10.6%		89.4%	

2.7.8 Route to Market - Consolidated View : Plastic Cisterns

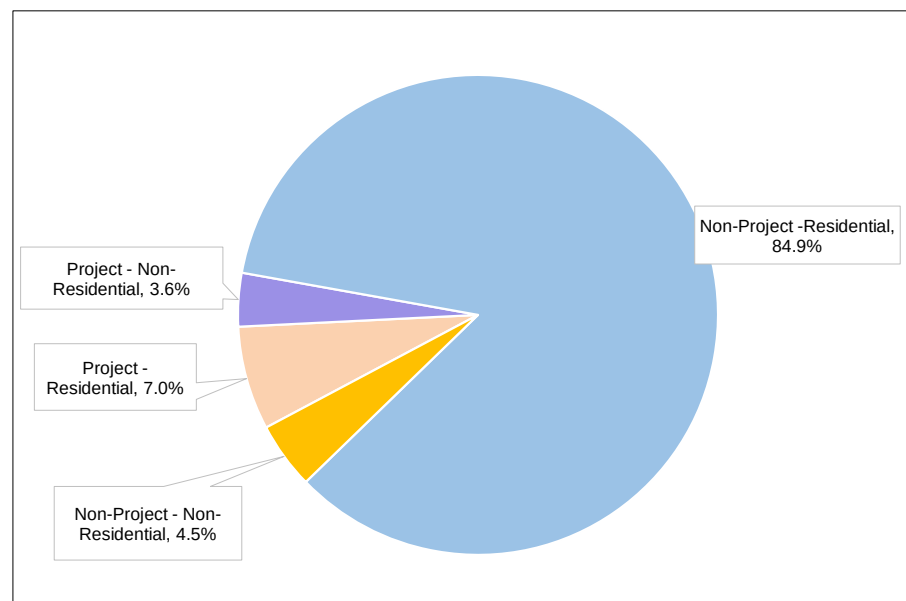
DE - ESTIMATED DISTRIBUTION FOR PLASTIC CISTERNS : 2025

Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
73.5%	8.2%	12.7%	0.1%	5.5%	7.0%	5.5%	85.3%	2.2%
100%					12.5%		87.5%	

DE - SALES OF PLASTIC CISTERNS BY DISTRIBUTION CHANNEL: 2020



DE - SALES OF PLASTIC CISTERNS BY POINT OF DECISION AND END-USE SECTOR: 2020



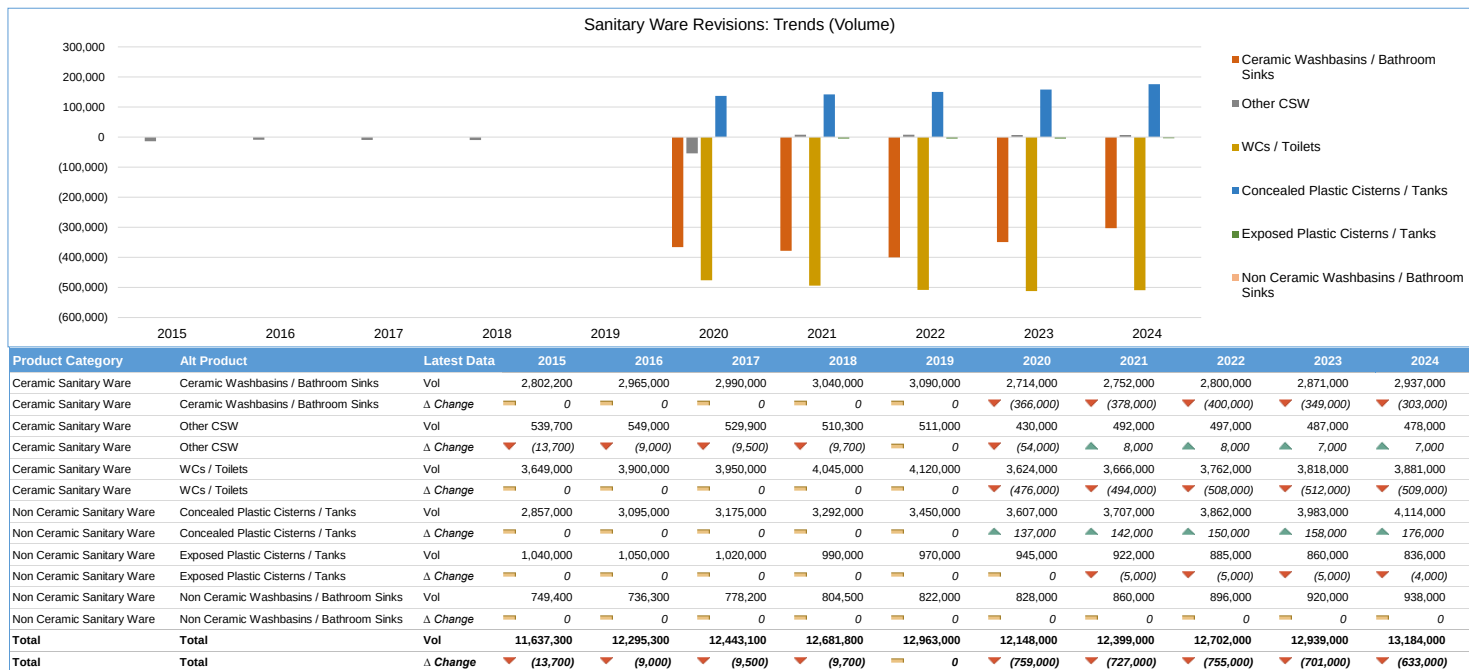
Source: BRG Based on Industry Interviews.

*Direct includes sales going directly from manufacturers to installers, contractors and private customers.

2.8 Revisions in the German Sanitary Ware Market (April 2021)

DE - REVISIONS IN THE GERMAN SANITARY WARE MARKET

Data Comparison: October 2020



Following industry feedback, a few minor changes were made to the historical market sizes for Other CSW.

No changes have been made to historical values for this market.

GERMANY

TAPS AND MIXERS

THE EUROPEAN BATHROOM & KITCHEN PRODUCT MARKETS



Market Characteristics and Technological Trends

The overall German taps and mixers market is a mature market, like that for ceramic sanitary ware, with sales' developments correlated to the overall economic climate, RMI investments and construction expenditure. Sales grew particularly strongly during the end of the 1980s, due to the positive effects of the reunification and revived consumer expenditure. The two long-term trends in the market are from two head to one head mixers and, more recently, from one and two head mixers to thermostatic mixers. Concealed models, mostly thermostatic, have also become more popular in Germany, in particular in the southern regions. The share of concealed models differs strongly per manufacturer. At the ISH 2019 in Frankfurt one of the main topics were the colour trends. Most manufacturers presented a colourful assortment of taps and mixers at the exhibition. According to industry feedback, there has been an increasing interest in colours such as black and gold to some extent. Sales of coloured mixers represented only a marginal share of the total market in recent years. However, in 2020 manufacturers reported that the product has established itself in the market and the share accounts for more than 10%.

For the kitchen sinks, the most popular product is the one head mixer. Although two head mixers were historically a popular choice for installation in kitchen sinks, less so in bathroom applications, sales have gradually been declining. DIY stores had originally encouraged the development from two head to one head models, both through their product range and low retail prices. Thermostatic mixers were introduced to the market for safety and comfort reasons, being mostly used in shower applications, these are now slowly gaining popularity also in the bath segment. The share of thermostatic mixers in showers continues to increase, in particular in combination with shower pipes. Since the concealed models have arrived in the DIY stores, sales numbers are slowly growing, and they are taking further share from exposed ones. Also, electronic self-closing taps and mixers are gaining market shares from manual taps and mixers, especially in the case of non-residential applications. The COVID-19 pandemic has significantly accelerated this trend and a small number of electronic self-closing taps and mixers is also installed in residential dwellings.

Flushing valves are a substitute product for WC cisterns. In the past, they were used both in residential and non-residential buildings. As they are directly connected to water pipes, flushing valves require a high pressure in the pipe work, which cannot be achieved in all residential dwellings. The longer and smaller the pipes, the more difficult it is to establish the necessary pressure. In older bathrooms, there is often a separate pipe for the WC installed to prevent interference with the water supply for the shower. Therefore, over the years, cisterns replaced most flushing valves in residential buildings. Moreover, flushing valves are much louder than cisterns and less appreciated in homes. In commercial and public buildings, however, there is still demand for this type of system, as it is generally said to be less likely to be vandalised and to use less water. Flushing

valves also require no waiting time before the next flush. Many public urinals are equipped with flushing valves. Electric systems introduced in the 1990s are very hygienic and popular for public buildings with many visitors. Self-closing and electronic taps are designed to stop the water flow after a predetermined time, and be operated either manually or automatically, or without physical contact, and are state of the art when it comes to water saving and ease of operation in public, semi-public and commercial sanitary and kitchen areas.

Self-closing taps have historically been installed in public spaces, but usually with manual mechanisms. Different articles within German hygiene law demand the use of contactless systems in certain environments such as restaurants, food processing workplaces, hospitals and public swimming pools. New products include models that are a combination of one head and electronic. The overall demand for both self-closing and electric taps is dependent upon the development from the non-residential market, as residential installations are still limited. Digital mixers, which are fitted with a digital and wireless mixing device, have also been introduced to the German market, such as GROHE's Allure F-Digital model, but the uptake of these kind of products has been relatively slow. These models are available for installation in both kitchens and bathrooms. The water flow is operated by pressing a button that is mounted on the wall or edge of the application, and the water flow can easily be adjusted.

Recent Market Trends

The COVID-19 pandemic shaped the developments in the year 2020 and 'business as usual' was replaced by government measures. In Germany the official lockdown, which is better understood as a 'prohibition of contact', started at the end of March 2020. Bavaria was the first state to implement measures to contain the spread of the virus, which was followed by the remaining states towards the end of March. For a few weeks the German population was advised to work from home when possible, with most non-essential stores closing for a limited time only. Measures implemented varied by state (*Bundesländer*). Most stores, including DIY stores, were allowed to open again from the beginning of April 2020 onwards, with differences from state to state in the level of restrictions and measures. In certain regions, shopping in DIY stores was limited to professionals only.

During 2020, new housing construction continued without much interruption, with only large scale project sites halted to implement protective measures. That being said, the overall production per employee is believed to have shrunk significantly, and remains to be the case, since many employers introduced measures to protect the construction workers. The bathroom industry was declared part of the crisis infrastructure, but installers could still continue to work without many interruptions. Overall, there has been relatively little impact of the COVID-19 pandemic on the building related industry, including the bathroom market, in Germany. In some cases, end users may have postponed, or cancelled, bathroom related projects due to the uncertainty on the market, but overall demand remained relatively strong, as there was a large backlog of works from the past, benefiting sales of bathroom and kitchen equipment. At the same time, in certain instances, the COVID-19 pandemic

may have contributed to additional demand. Since consumers were spending more time indoors, they had the opportunity to dedicate themselves to repair or upgrade works, whereas others might have decided to invest into their property by installing a new bathroom or kitchen due to (international) holidays being difficult because of the current pandemic.

In 2020, the German taps and mixers market increased, however with a weaker growth rate when compared to the previous year. There was a weaker demand from the new and non-housing segments. In contrast, sales in the RMI segment increased with a stronger growth rate, which compensated for the loss in the other two end-use segments. It is believed that potential business was procrastinated due to social distancing and lockdown measures, high demand of installers and the growing competition from the heating industry. The intention of installers is not to do many bathrooms, but rather to focus on projects that are most profitable. With many people working shorter hours and spending more time at home, sales through DIY and online stores gained share. Overall, sales of sanitary taps and mixers slightly increased in volume when compared to the previous year. The kitchen taps and mixers market performed much stronger than the sanitary market in terms of volume growth. This development underlines the recent trend of Germans investing into kitchens, especially when having to wait for up to half a year and longer for a bathroom renovation.

In 2020 it was mainly the upper and luxury price segments that were growing. The economy and middle market segments remained strongest across all taps and mixers applications. Demand increased from the premium price segment because consumers had more time available to carry out bathroom installations. At the same time installers prefer to purchase and install upper price models which allow them to receive higher profit margins. Since there is more work than availability, installers are in the position to choose jobs that bring more profit. For minor jobs such as replacing broken taps and mixers, a German consumer would go to DIY stores or order online. Since new housing completions are still on a high level, there is significant demand from new bathrooms. At the same time there is still a large backlog of accepted construction permits that are being delayed because of limited capacity and the impact of the COVID-19 crisis on the economy.

Electronic self-closing taps and mixers are believed to be the best performing segment after thermostatic mixers. Thermostatic mixers are becoming the preferred type installed in new or fully renovated bathrooms, especially in shower applications. They mostly gain share from one head mixers which, consequently, hampered growth for one head mixers. Since thermostatic mixers have been on the market for many years, their volumes nowadays only increase gradually with a growth rate that is slightly above the market average. Demand depends on a region's water quality. If sanitary water is rich in limescale, then end users are less likely to spend more money for a thermostatic mixer.

Demand for two head mixers had the strongest negative rate after self-closing taps and mixers when compared to 2019. Two head mixers are an outdated product and demand is mainly limited to classical designs from the '20s and '30s bathroom

styles. The share of electronic taps and mixers increased with the strongest growth rate at the expense of self-closing taps due to demand from non-residential buildings such as hospitals, offices and recreational buildings. In response to stricter hygiene requirements related to the COVID-19 virus, electronic taps and mixers are becoming the preferred option in public places. At the same time, there is a growing awareness for this product in residential bathrooms, too. There is a trend to install taps in a complete system, together with furniture and washbasin. While before a single washbasin was replaced together with a tap, complete systems including furniture are an increasing choice by German end users. Such complete systems are offered by DIY and furniture stores and can easily be ordered online.

Demand for kitchen and shower taps and mixers increased in 2020. Demand from kitchens generated the strongest growth rate, followed by showers. Demand for kitchen taps and mixers was particularly strong due to more demand from the RMI segment. With end users spending more time at home and having financial savings from not going on holidays, the kitchen segment boomed in 2020. Demand for showers can be explained by the ongoing trend towards showering. End users prefer to have a larger shower and almost all new and renovated bathrooms have a shower. However, the popularity of shower pipes, which often have a thermostatic mixer installed, are taking away market share from more traditional shower taps and mixers. Moreover, there is a trend in new housing to install two bathrooms with two showers, rather than one main bathroom and a guest toilet. The number of showers is increasing in new dwellings and from bathroom renovations, where an old bath is replaced by a large shower. Demand for bath and bidet taps and mixers was negative in 2020 as both products are outdated. Not only are they used less frequently, but during a renovation a bath is likely to be substituted with a larger shower and an existing bidet with a shower toilet. End users might even not be bothered to replace broken taps and mixers because they hardly make use of them anyway.

Prices and Market Values

BRG price segmentation and value data is structured in terms of applications such as baths, bidets, washbasins, showers and kitchens. Average MSPs for most applications increased in 2020, according to the normal price increase to compensate for inflation, higher production and material costs. In addition, the MSP increased slightly stronger than in 2019 due to a growing demand of coloured taps and mixers. Since the share is growing rapidly, the strong increase in volume also influenced the MSP. Coloured taps and mixers are mainly found in the upper and luxury price segments, whereas supply in the lower price segments is still low.

Low-priced products are mainly found in DIY and online stores. Food discount stores such as ALDI and LIDL might also offer taps and mixers in weekly special deals. Low-cost products are imported from abroad. Some established manufacturers might offer the same taps and mixers through different distribution channels. According to the distribution channel, the same product might be offered at different price ranges. For example, a sanitary tap bought in a DYI or online

store might be up to three times less expensive than the same product distributed by an installer or showroom. While this might be true for some product ranges, it is more common for large manufacturers to have a lower priced product range to sell through alternative distribution channels.

Nowadays, there is a large offer of affordable taps and mixers on the market. The DIY offers not just the price entry segment, but also products with attractive design and good quality for an affordable price. Often, such products are from well-known manufacturers. Luxury taps and mixers are mainly supplied by German manufacturers that serve exclusive private and commercial customers with expensive branded products. The middle-priced categories are expanding their product portfolio with design-oriented products to make them more attractive for consumers. According to industry feedback, there is a trend away from the middle price segment towards either the higher or lower price segments. If an end user can afford it, the choice would be towards more expensive taps and mixers that reflect one's individual taste. On the other side, if an end user is more price sensitive, the choice would be towards a cheaper product with a good quality.

New Housing

In 2020, new housing completions decreased slightly in comparison to the previous year. The lower number of new dwellings resulted in fewer taps and mixers sales for all applications. Taps and mixers in new housing have a relatively small share of total sales, which is slightly lower than non-housing. Despite the decline of new completions, the German taps and mixers market is sustained by a high volume of new buildings. Positive demand for all types of taps and mixers applications is expected again from 2021 until 2023. Stronger demand is expected to be triggered from more completions of multi-family houses. Concealed taps and mixers are mostly sold in the new housing sector, and their share has increased. The installation in the existing bathrooms is less common due to costly modifications that are required to be carried out when a consumer decides to do so.

Housing RMI

Replacement demand accounted for most sales in the taps and mixers market and the growth rate was much smaller when compared to 2019. The limited number of installers and the lockdown restrictions in Germany slowed down demand for full bathroom renovations. At the same time like-for-like replacements are believed to have happened from alternative distribution channels. According to industry feedback, sales through DIY and e-commerce increased even more than in previous years. The COVID-19 pandemic changed consumer behaviour in such a way that buying a tap and mixer online has become common. Sales through the traditional three step remained flat and the lockdown has somewhat negatively contributed to a weaker demand. Despite lower volumes, however, the sales in value are believed to have increased. End users spent more money on higher quality products, which better fulfilled their needs, e.g. coloured taps and mixers.

The opinions of whether the lockdown hampered sales through showrooms is biased. Some industry feedback suggests that the lockdown had no effect on the face-to-face consultations of showrooms. End users could schedule in appointments and salespeople would visit them at home. Others reported that the limited access to showrooms had a strong negative effect on taps and mixers sales. Overall, installers are keener on accepting projects that involve a complete bathroom installation, which offers higher margins than the ones gained by replacing a single mixer. It is worth mentioning that the gap of end users in need of a bathroom renovation and the number of available installers has been growing in the recent years. Since the number of professional installers is not increasing at the same pace, there is an expanding potential for businesses to fill that gap. It is not just DIY brands that are aware of this growing demand, alternative business models seem to have caught the interest of consulting companies and entrepreneurs. One of a few examples is BADIA, an online service that allows end users to create an individual bathroom solution online. Various steps of the distribution process are coordinated in such a way that a bathroom can be installed at a lower cost and in less time.

It is a matter of fact that most taps and mixers have a shorter lifespan than, for example, ceramic sanitary ware. Most taps and mixers are sold as individual replacement products. In terms of product mix, one head taps continue to be the most sold product. Like-for-like replacements are common in the RMI sector. However, a slight shift towards the use of thermostatic mixers is attributed to the ongoing trend of showering, with a preference for extra-large showers. The popularity of shower pipes, which often have a thermostatic mixer included, is negatively influencing sales of traditional thermostatic and one head taps and mixers. Demand for pillar, one head, thermostatic, electronic and two head taps and mixers increased, whilst sales of self-closing declined in 2020. When looking at the type of applications, demand for kitchen taps and mixers increased the strongest, followed by showers and washbasins. The number of taps and mixers installed in baths only marginally increased and the replacement demand for bidets was slightly negative.

Non-Housing

Total non-residential construction output decreased in 2020, which was due to the negative impact of the COVID-19 pandemic. Project business such as hotels, cruise ships and restaurants were hit particularly hard by the crisis. Most public and commercial buildings were avoided and upcoming renovations and new constructions were put on hold. The only exception were buildings for education and health, which developed slightly positive in 2020. Overall, there was negative demand for all types of taps and mixer applications in 2020. Electronic mixers, which are mainly found in the non-housing segment were an exception. Demand for this product was positive in all non-housing application. Electronic mixers had the highest growth rate in the non-housing sector. The positive development can be explained from a technical point of view. Electronic taps are designed to stop the water flow after a predetermined time and are operated without physical contact. This makes them a safer product in terms of contamination in public spaces. Their

main advantages are saving water and preventing the spread of illness. The last point is particularly relevant in certain public buildings, such as hospitals and schools.

Thermostatic mixers are slowly gaining popularity in non-housing buildings such as hospitals and schools. They are installed mainly in showers and to a small extent in washbasin applications to comply with the German regulations for drinking water installations. The non-residential segments with the highest growth rates in 2020 was from education as the German government has allocated a huge budget for the modernisation of German schools, many of which are in a desperate condition. There is a great demand to renovate bathrooms in public schools. In contrast to previous years expenditure for healthcare buildings increased in 2020. Overall, the strong negative impact of the COVID-19 crisis on the non-housing segment resulted in fewer sales of taps and mixers, except for electronic ones.

Supply Structure

The supply structure in the German taps and mixers market is changing. The traditional three-step distribution channel is struggling to meet the growing demand from German consumers. Alternatives are slowly introduced to the German market. In 2018 REUTER opened their first showroom in Düsseldorf, where consumers can have a look at a selection of products before purchasing them online. Germany's main manufacturers are increasingly supplying REUTER as well as DIY stores directly with products that are on a comparable price range to unbranded products. Manufacturers tend to focus more on selling their own products, instead of only relying upon the wholesale. Some manufacturers believe that the role of wholesalers is likely to change in future. Most taps and mixers, especially in the upper-priced segments, are produced locally and the highest concentration of producers is located at the eastern end of the Ruhr, around Iserlohn and Herne.

All major European manufacturers also have manufacturing sites abroad, except for DORNBACHT which produces only in Germany. GROHE, for example, produces in Portugal, China and Thailand. HANS GROHE has production facilities in France and IDEAL STANDARD in Italy, Portugal, Bulgaria and the Netherlands. Domestic manufacturers continue to dominate the overall market for taps and mixers, although their share is slightly more for sanitary than it is for kitchens. The top three players in the traditional distribution channels, GROHE, HANS GROHE and HANSA, are all long established German manufacturers, although both GROHE and HANS GROHE are owned by foreign capital: Japanese LIXL and American MASCO, respectively. According to import statistics, most taps and mixers in Germany come from China, Portugal, Thailand, Italy, Bulgaria, Austria, the Netherlands and Poland. China, Portugal and Thailand account for almost three quarters of all imported taps and mixers.

Another important German manufacturer for taps and mixers is SEE (SCHÜTTE), which mainly sells through the DIY, e-commerce and discounter stores. SEE (SCHÜTTE) produces its own products outside of Europe, which are sold under the SCHÜTTE brand, other brands, and OEM products to furniture shops for example.

The sales in the market share table for SEE (SCHÜTTE) include all branded sales from SEE HOLDING and exclude OEM sales. In 2011 the manufacturer FRANZ JOSEPH SCHÜTTE was bought by KIRCHHOFF and since 2015 it is part of the SEE HOLDING, which employs more than 150 employees in Wallenhorst in Germany, Salzburg in Austria, and Hong Kong. SCHÜTTE produces all their products according to German quality standards in the Far East. SCHÜTTE is market leader and distributes most of its taps and mixers through DIY, followed by furniture specialist stores and occasional limited offers in food discount stores such as ALDI and LIDL.

Looking closer at the different application segments, either GROHE or HANSGROHE dominate all markets, except for the kitchen and taps market. The kitchen and taps market is believed to be dominated by the EGO GROUP (BLANCO) and SEE (SCHÜTTE). Looking closer at the different types of taps and mixers, it is believed that IDEAL STANDARD is the market leader for self-closing taps. GROHE and HANSGROHE are the strongest players for thermostatic mixers with an expected share of almost two thirds of the total market, with HANSA and KLUDI following quite far behind.

Another important player on the Germany DIY market is SANITOP WINGENROTH, which is a full-range supplier of DIY stores, project related sales and e-commerce websites. SANITOP WINGENROTH is believed to manufacture their own taps and mixers; however, the volume is believed to be marginal when compared to their total taps and mixers sales. The suppliers for the flushing valve market are LIXL (GROHE) and SCHELL, closely followed by GEBERIT and FRANKE (AQUAROTTER). There are also some smaller domestic companies, such as BENKISER and FRIEDRICH GAMPPER (the NIL brand, not to be mistaken for GAMPPER ARMATUREN).

Forecasts

Despite the ongoing COVID-19 crisis overall demand for taps and mixers is believed to turn positive during the forecast period. The main reason is a stronger growth from the RMI segment as well as a growing number of new completions. Demand from non-housing is believed to decline even stronger than in 2020 before turning positive again from 2022. For 2022 and 2023, a stronger total growth rate is anticipated as the total economy is expected to improve again. In 2024 and 2025 the total growth rate is expected to decline again in alignment with fewer new dwelling completions and a weakening RMI demand.

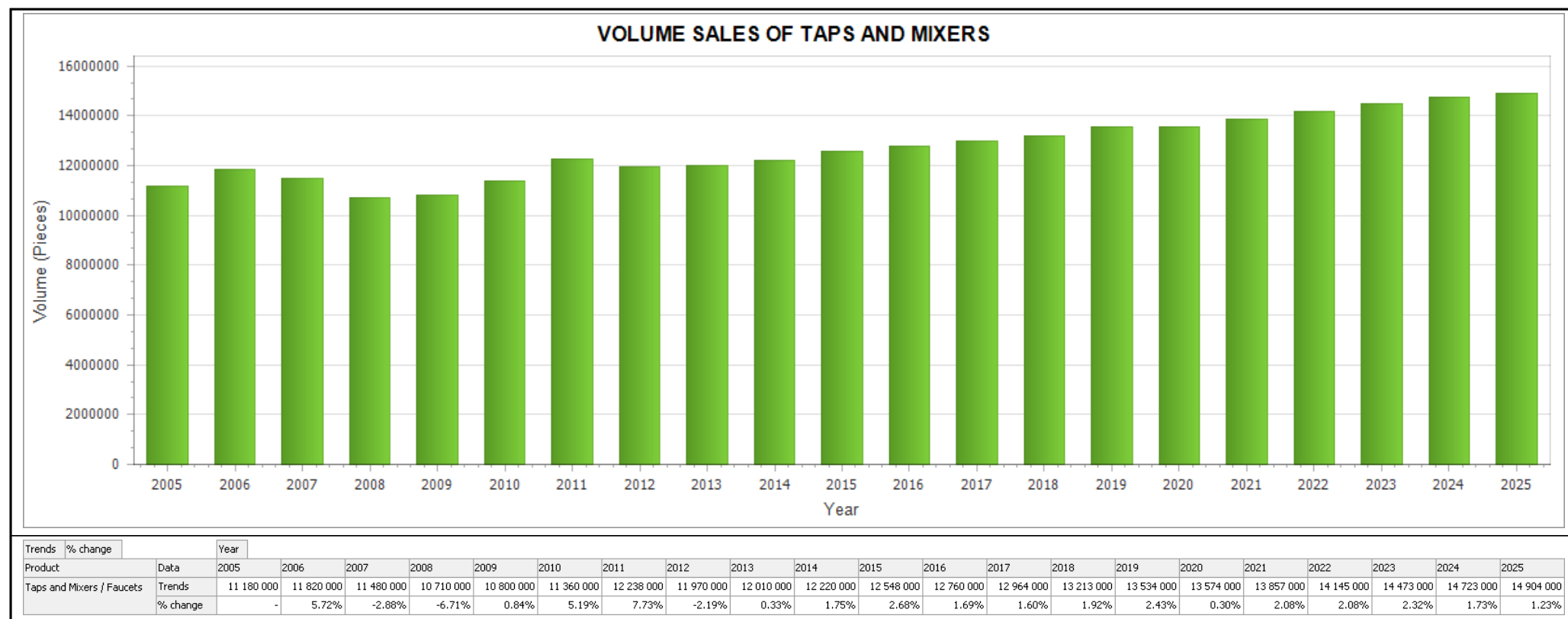
Government support during the COVID-19 crisis negatively affected the budget for future non-housing projects and renovations. At the same time, hotels and restaurants as well as many other commercial businesses are struggling and expansions are unlikely in near future. In the residential segment, the shift towards products in the upper and luxury price segments is expected to continue in the upcoming years, e.g. for wall concealed taps and mixers as well as coloured ones, which are usually in the upper price segment. The market for taps and mixers is expected to grow faster in terms of value than in volume. At the same time, there is

a trend towards cheaper products, which nowadays have a relatively good quality. A growing number of sales through DIY stores and e-commerce will likely put further price pressure on products sold by installers and showrooms.

The lack of professional installers is likely not going to change over the forecast period as the pool of qualified installers in Germany is further decreasing. At the same time there is a growing competition from the heating industry. Installers have shifted to the installation of new heating systems instead of bathrooms. This trend may prevent RMI sales from reaching full potential. Business alternatives are emerging to meet the demand without being dependent on installers and extreme long waiting periods. As regards complete refurbishments of bathrooms, it is expected that the number is likely to increase. Given the demographic changes, the number of bathroom renovations adapted to the needs of elderly people is likely to grow during the forecast period. The forecast for RMI activity is expected to remain positive towards the end of the forecast period.

3.2 Historical Trends and Forecasts : Taps and Mixers

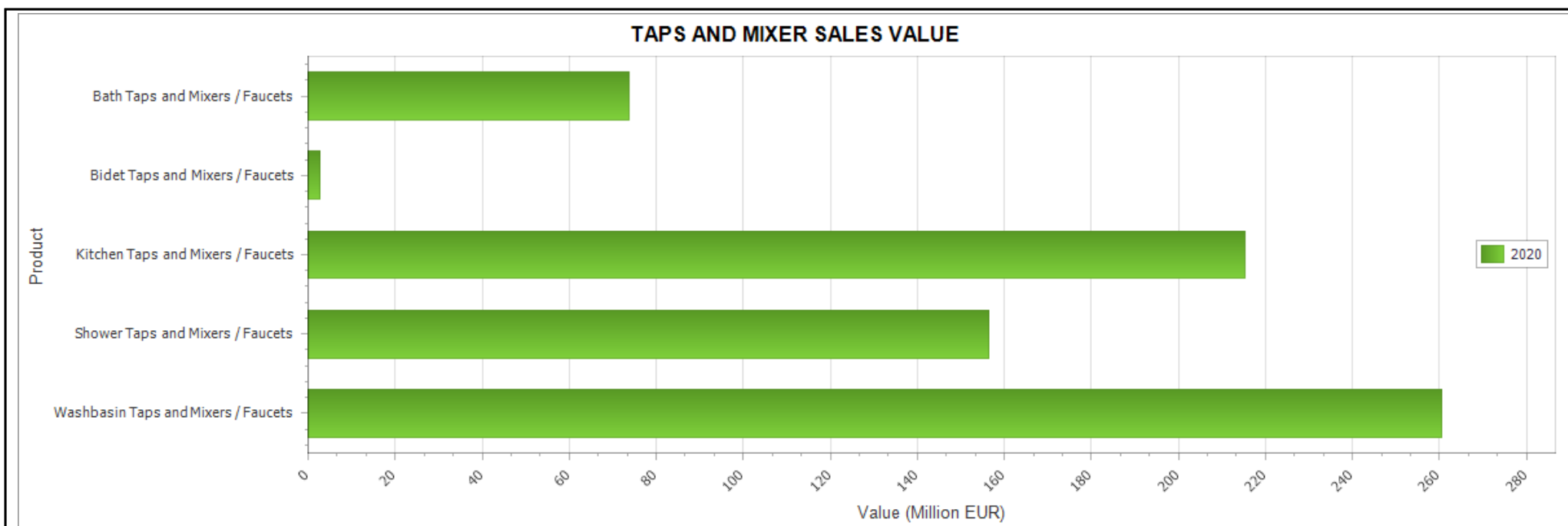
DE - SUMMARY OF TRENDS IN THE TAP AND MIXER MARKET : HISTORICAL AND FORECASTS



Source: BRG based on industry interviews. N.B. Includes Both Kitchen and Sanitary Taps & Mixers

3.3.1 Prices and Market Values : Taps and Mixers

DE - ESTIMATED VALUES OF THE TAP AND MIXER MARKET

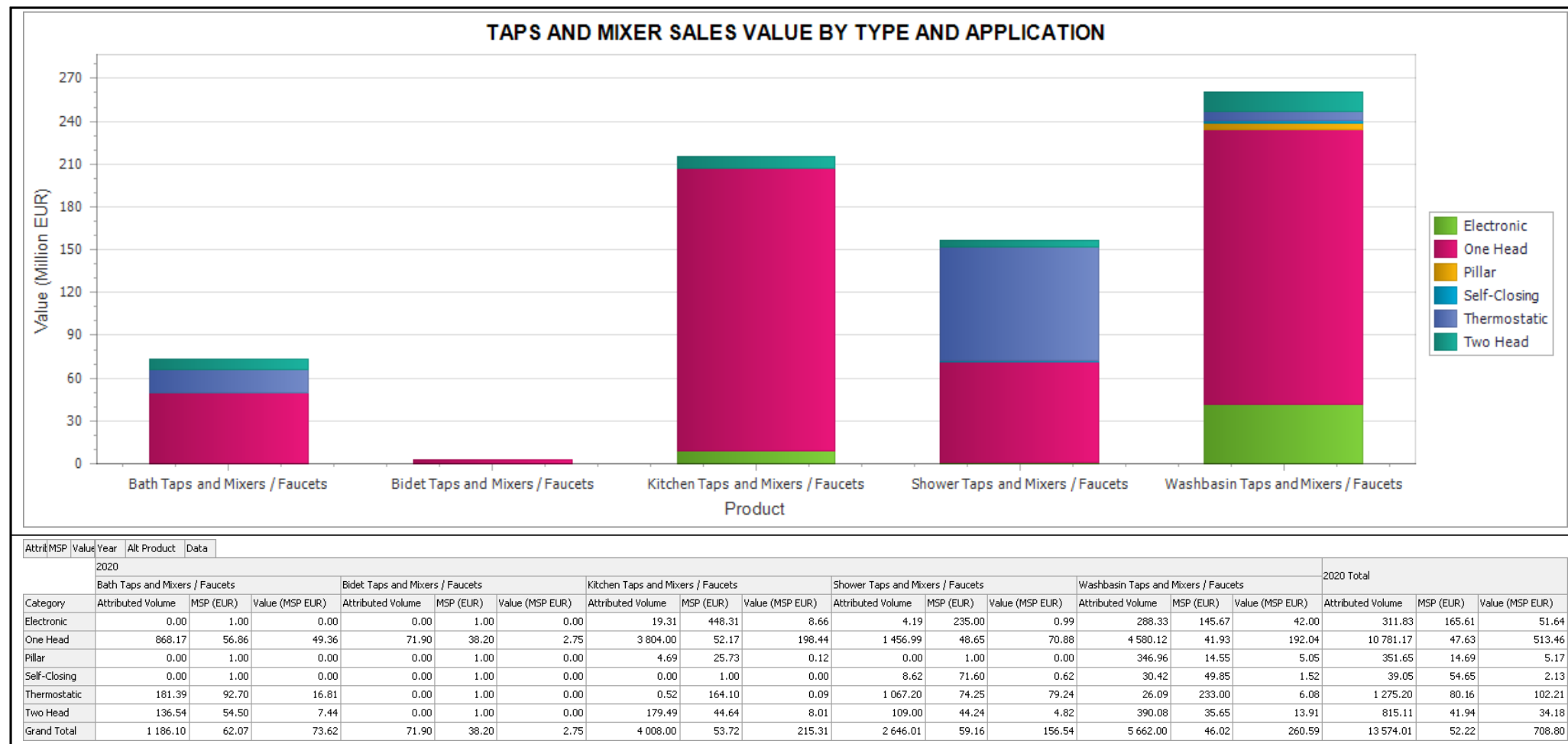


Volumes	MSP (EUR)	Value (MSP EUR)	Year	Data		
			2020	2019		
Alt Product	Volumes	MSP (EUR)	Value (MSP EUR)	Volumes	MSP (EUR)	Value (MSP EUR)
Bath Taps and Mixers / Faucets	1 186 100	62.07	73.62	1 229 700	61.40	75.50
Bidet Taps and Mixers / Faucets	71 900	38.20	2.75	74 300	38.58	2.87
Kitchen Taps and Mixers / Faucets	4 008 000	53.72	215.31	3 900 000	54.11	211.03
Shower Taps and Mixers / Faucets	2 646 000	59.16	156.54	2 640 000	58.27	153.83
Washbasin Taps and Mixers / Faucets	5 662 000	46.02	260.57	5 690 000	45.17	257.02
Grand Total	13 574 000	52.22	708.78	13 534 000	51.74	700.25

Source: BRG based on industry interviews. Values Shown in Millions (at MSP = Manufacturers' Selling Price).

3.3.2 Prices and Market Values by Type : Taps & Mixers

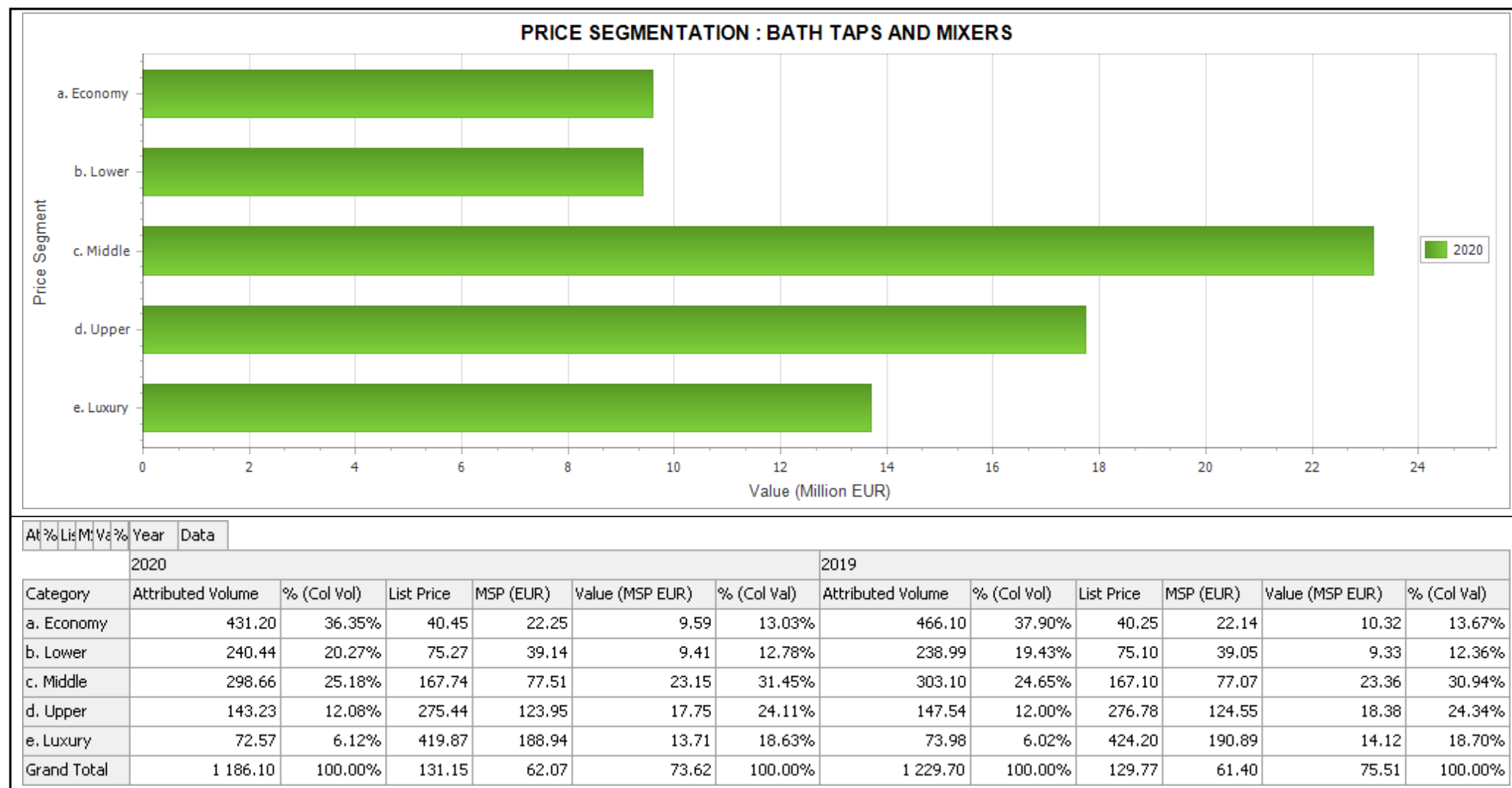
DE - ESTIMATED VALUES OF THE TAP AND MIXER MARKET BY TYPE AND APPLICATION



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price).

3.3.3 Price Segmentation : Bath Taps and Mixers

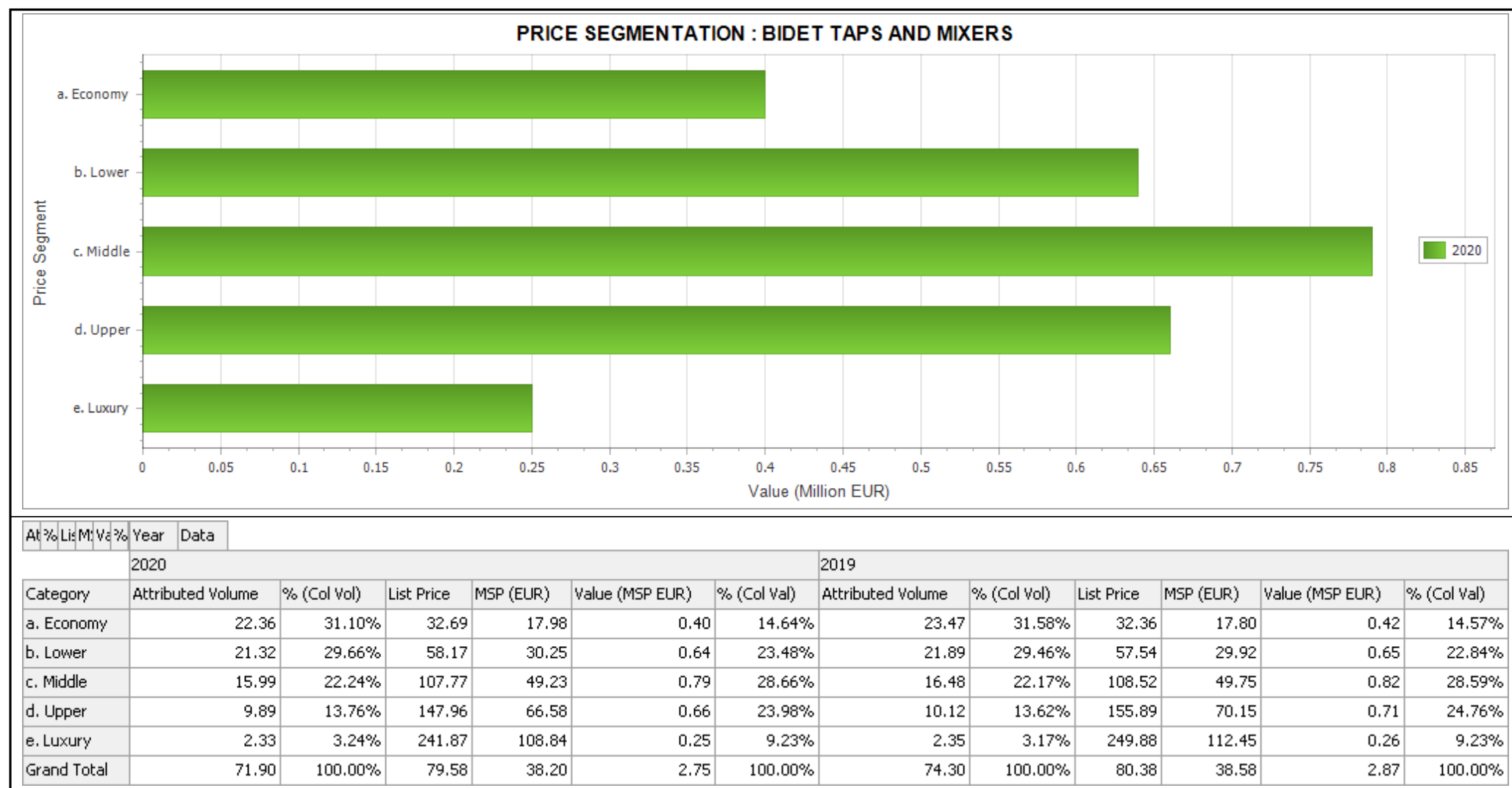
DE - ESTIMATED PRICE SEGMENTATION FOR BATH TAPS AND MIXERS



Source: BRG based on industry interviews. Total may show slight variance compared to market data due to rounding. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy <€30, Lower €30-€50, Middle €50-€100, Upper €100-€150 and Luxury >€150

3.3.4 Price Segmentation : Bidet Taps and Mixers

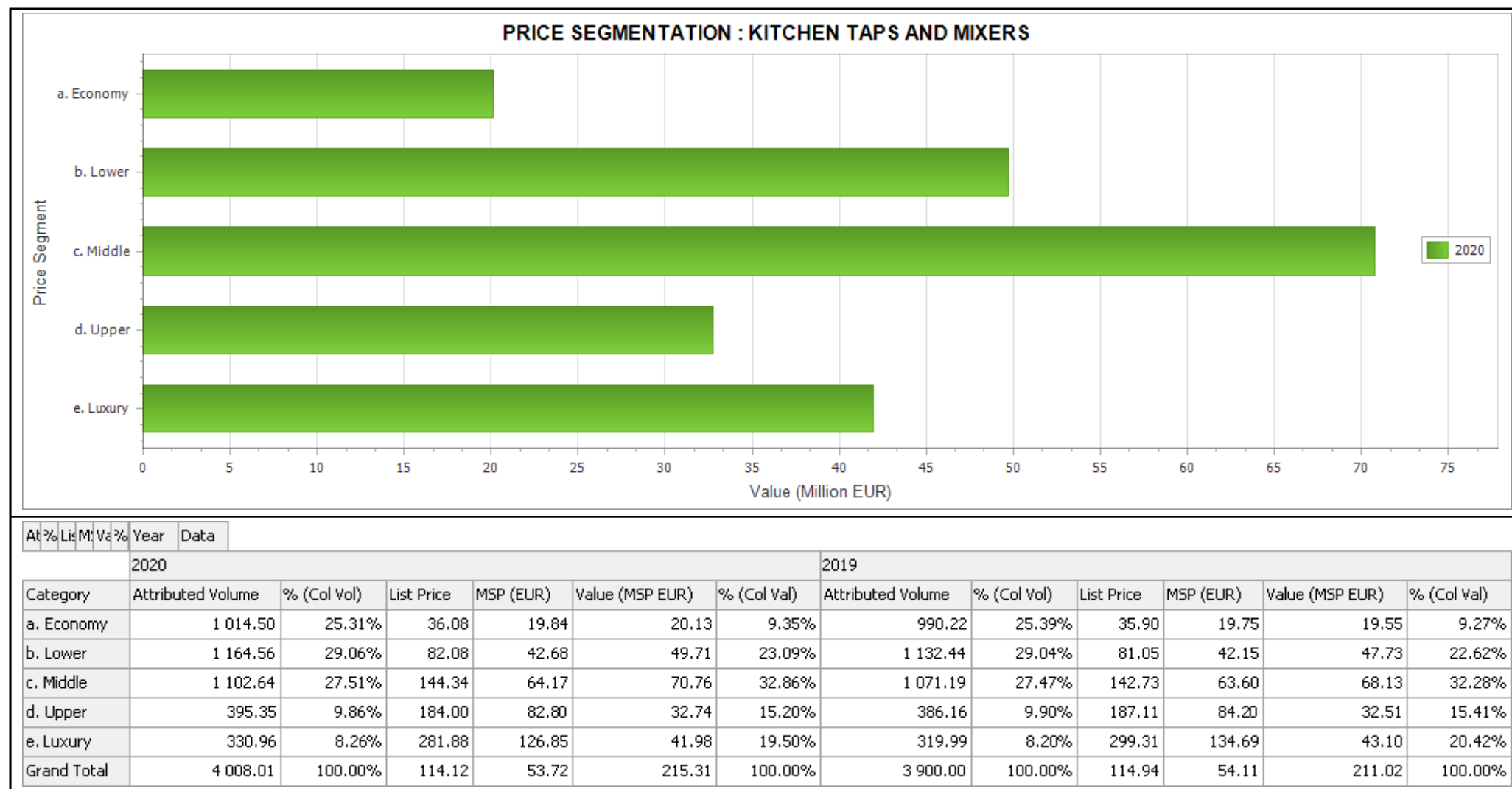
DE - ESTIMATED PRICE SEGMENTATION FOR BIDET TAPS AND MIXERS



Source: BRG based on industry interviews. Total may show slight variance compared to market data due to rounding. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy <€25, Lower €25-€40, Middle €40-€60, Upper €60-€100 and Luxury >€100

3.3.5 Price Segmentation : Kitchen Taps and Mixers

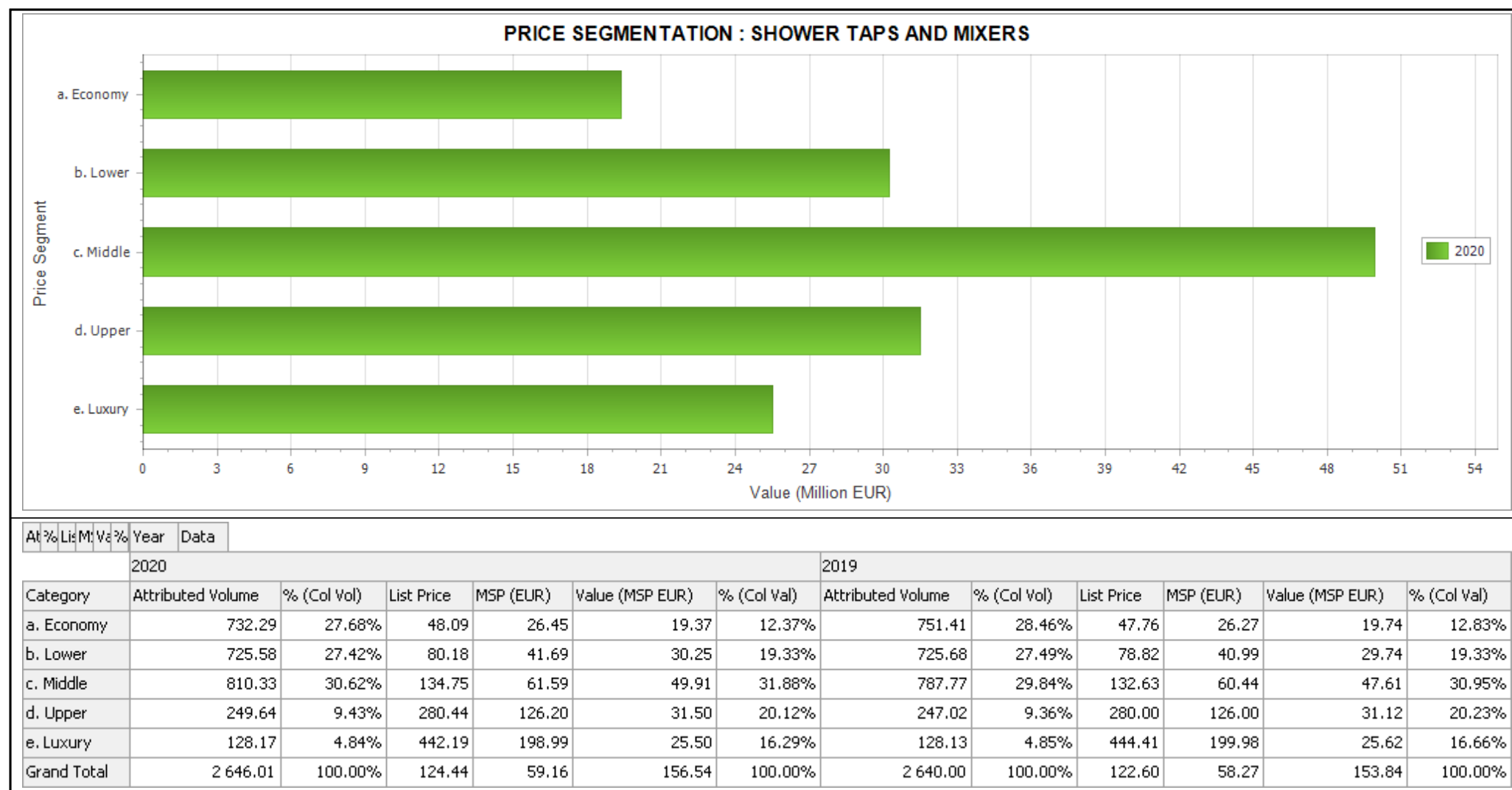
DE - ESTIMATED PRICE SEGMENTATION FOR KITCHEN TAPS AND MIXERS



Source: BRG based on industry interviews. Total may show slight variance compared to market data due to rounding. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy <€35, Lower €35-€50, Middle €50-€70, Upper €70-€110 and Luxury >€110

3.3.6 Price Segmentation : Shower Taps and Mixers

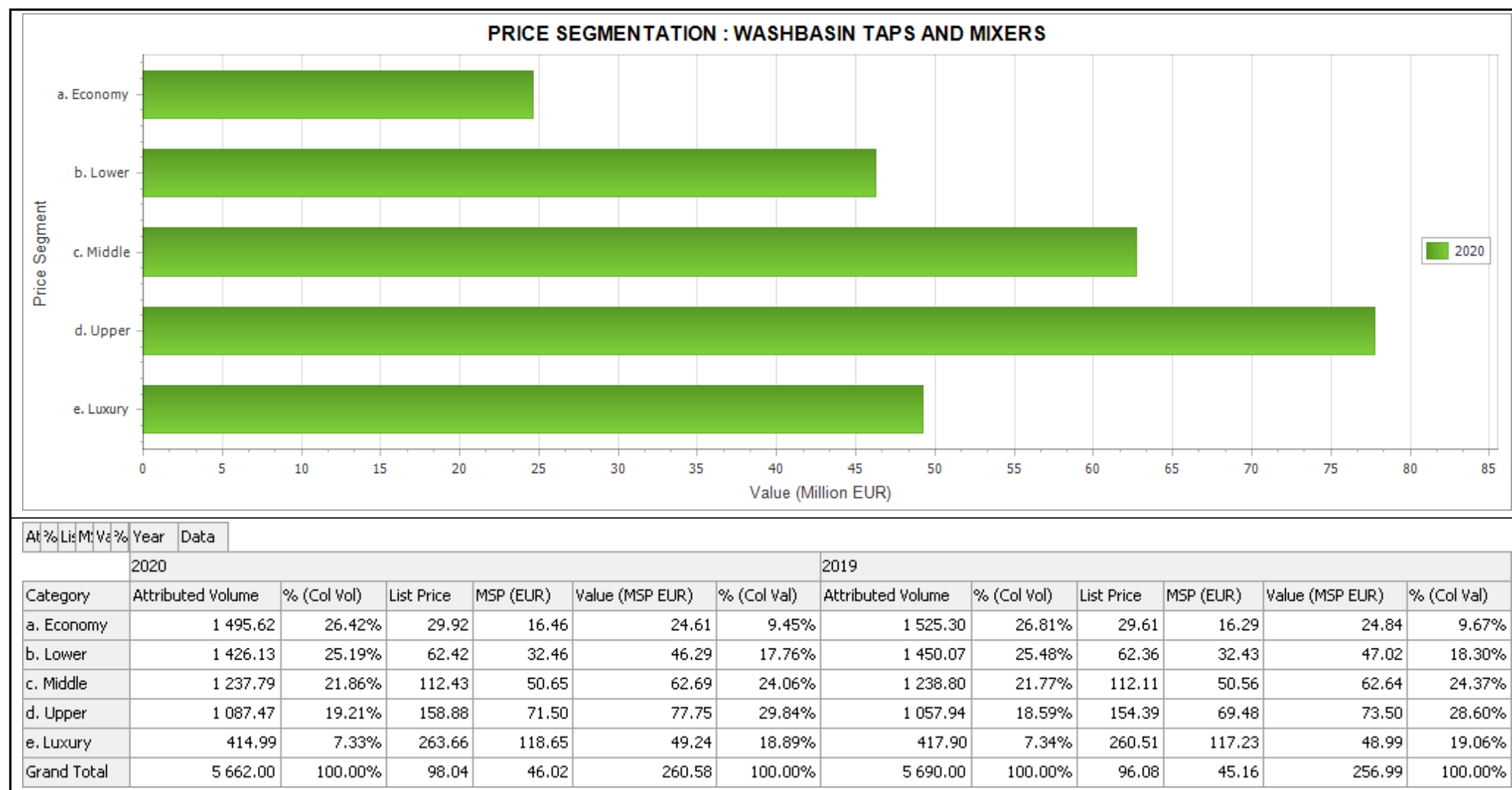
DE - ESTIMATED PRICE SEGMENTATION FOR SHOWER TAPS AND MIXERS



Source: BRG based on industry interviews. Total may show slight variance compared to market data due to rounding. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy <€30, Lower €30-€50, Middle €50-€100, Upper €100-€150 and Luxury >€150

3.3.7 Price Segmentation : Washbasin Taps and Mixers

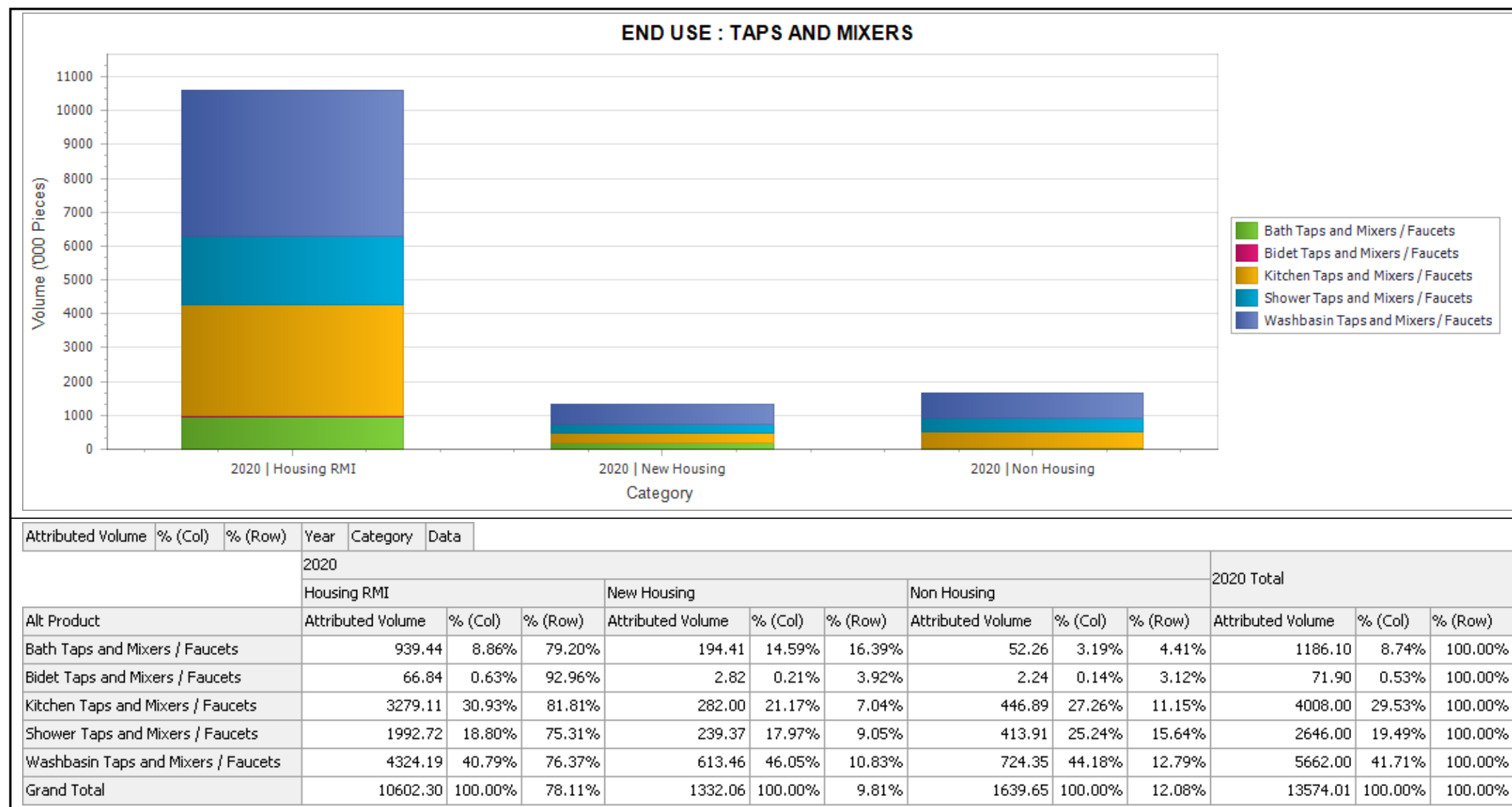
DE - ESTIMATED PRICE SEGMENTATION FOR WASHBASIN TAPS AND MIXERS



Source: BRG based on industry interviews. Total may show slight variance compared to market data due to rounding. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy <€20, Lower €20-€40, Middle €40-€60, Upper €60-€80 and Luxury >€80

3.4.1 End Use : Base Year

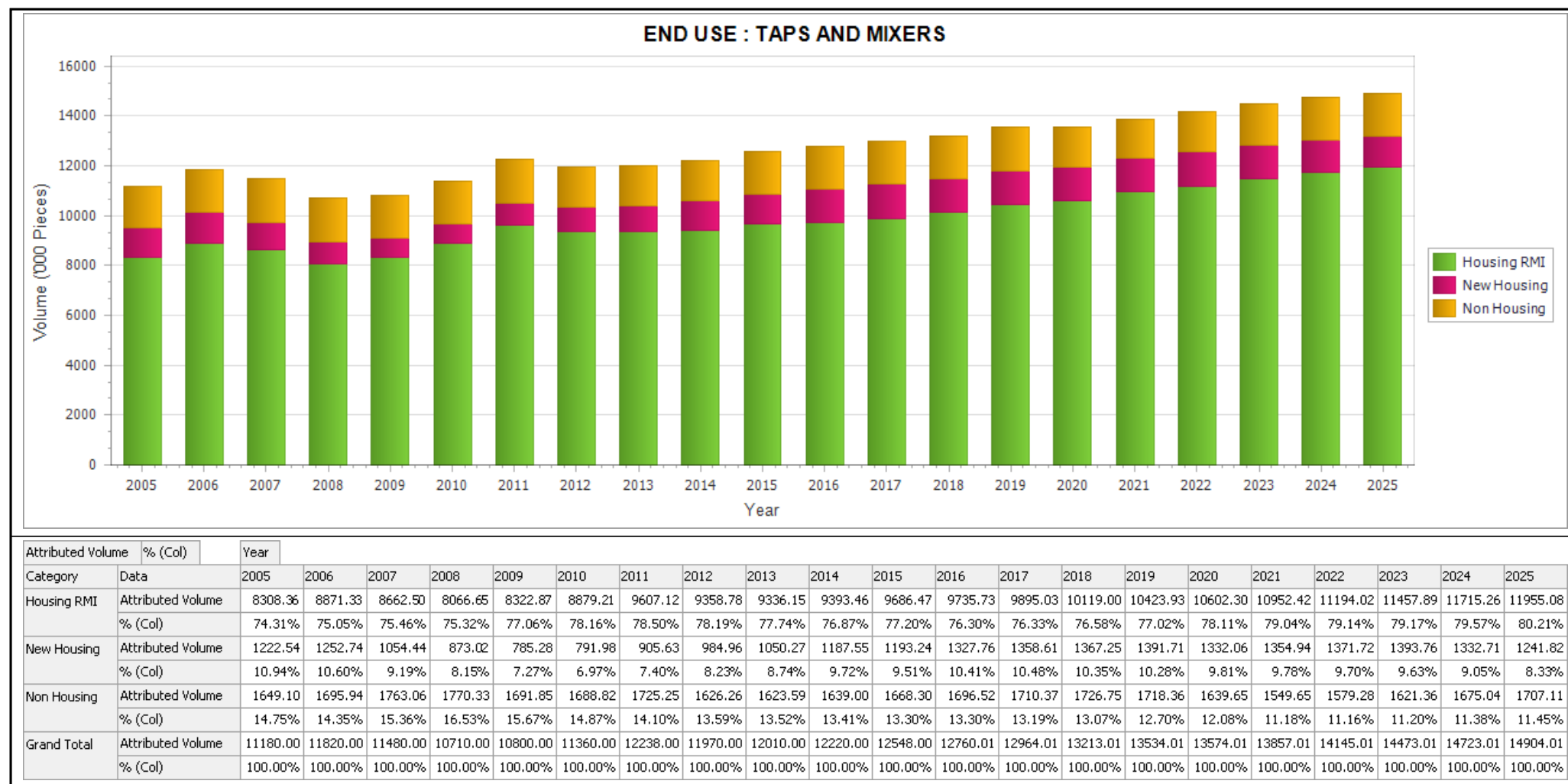
DE - SALES OF TAPS AND MIXERS BY TYPE AND END USE SEGMENT ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance due to rounding. N.B. Includes Kitchen and Sanitary Taps & Mixers

3.4.2 End Use : Historical and Forecasts

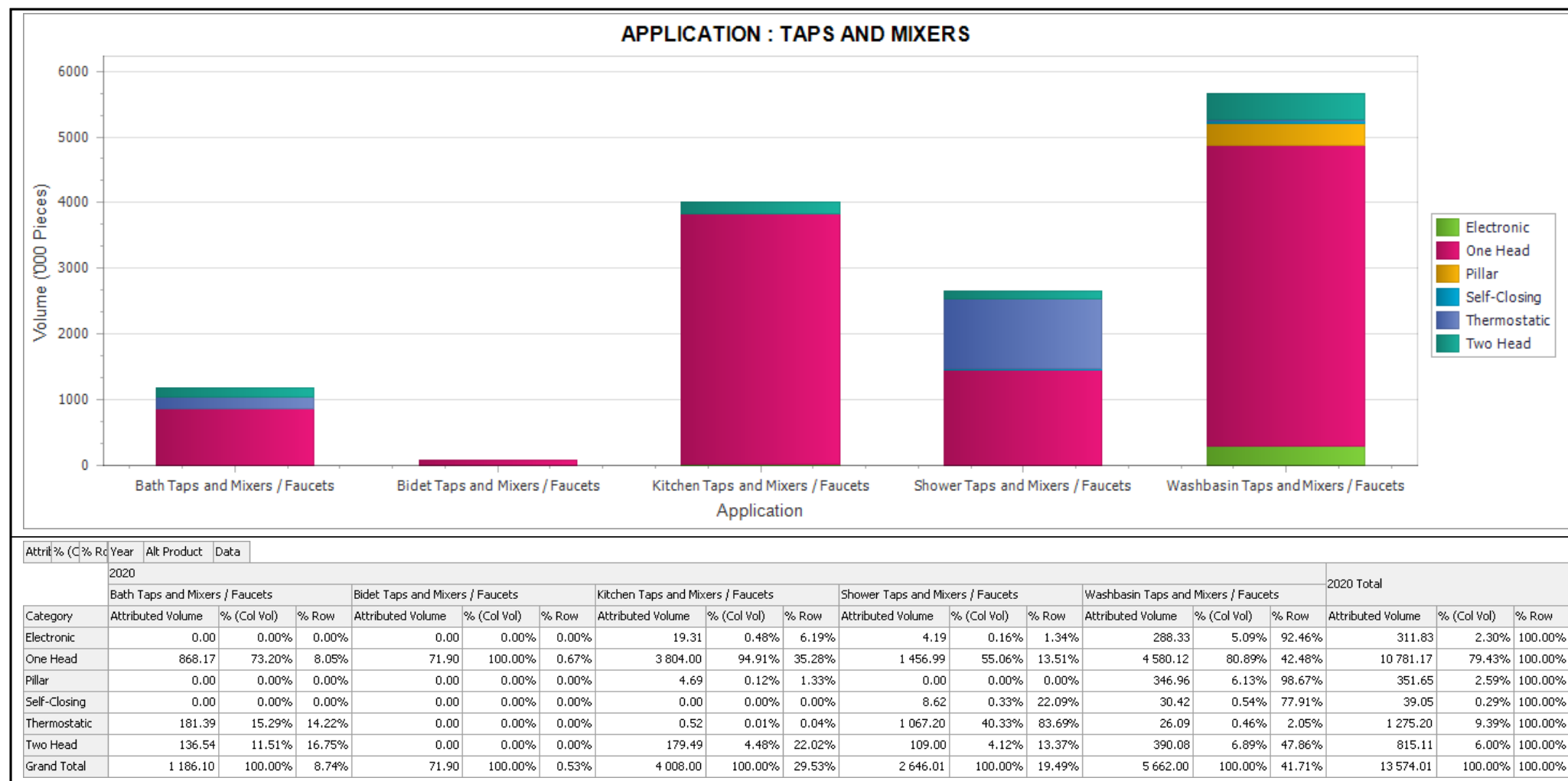
DE - SALES OF TAPS AND MIXERS BY END USE SEGMENT ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. N.B. Includes Both Kitchen and Sanitary Taps & Mixers

3.4.3 Sales by Type and Application : Taps and Mixers

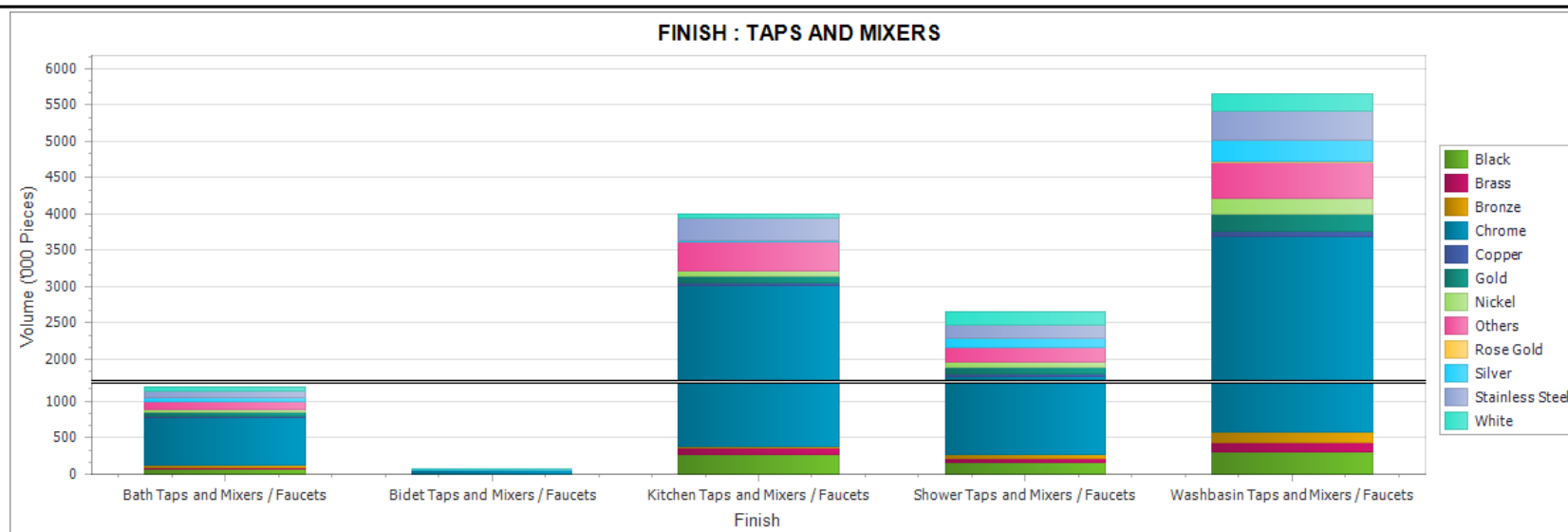
DE - TAP AND MIXER SALES BY TYPE AND APPLICATION ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

3.4.4 Sales by Type of Finish : Taps and Mixers

DE - TAP AND MIXER SALES BY TYPE OF FINISH AND APPLICATION ('000 PIECES)

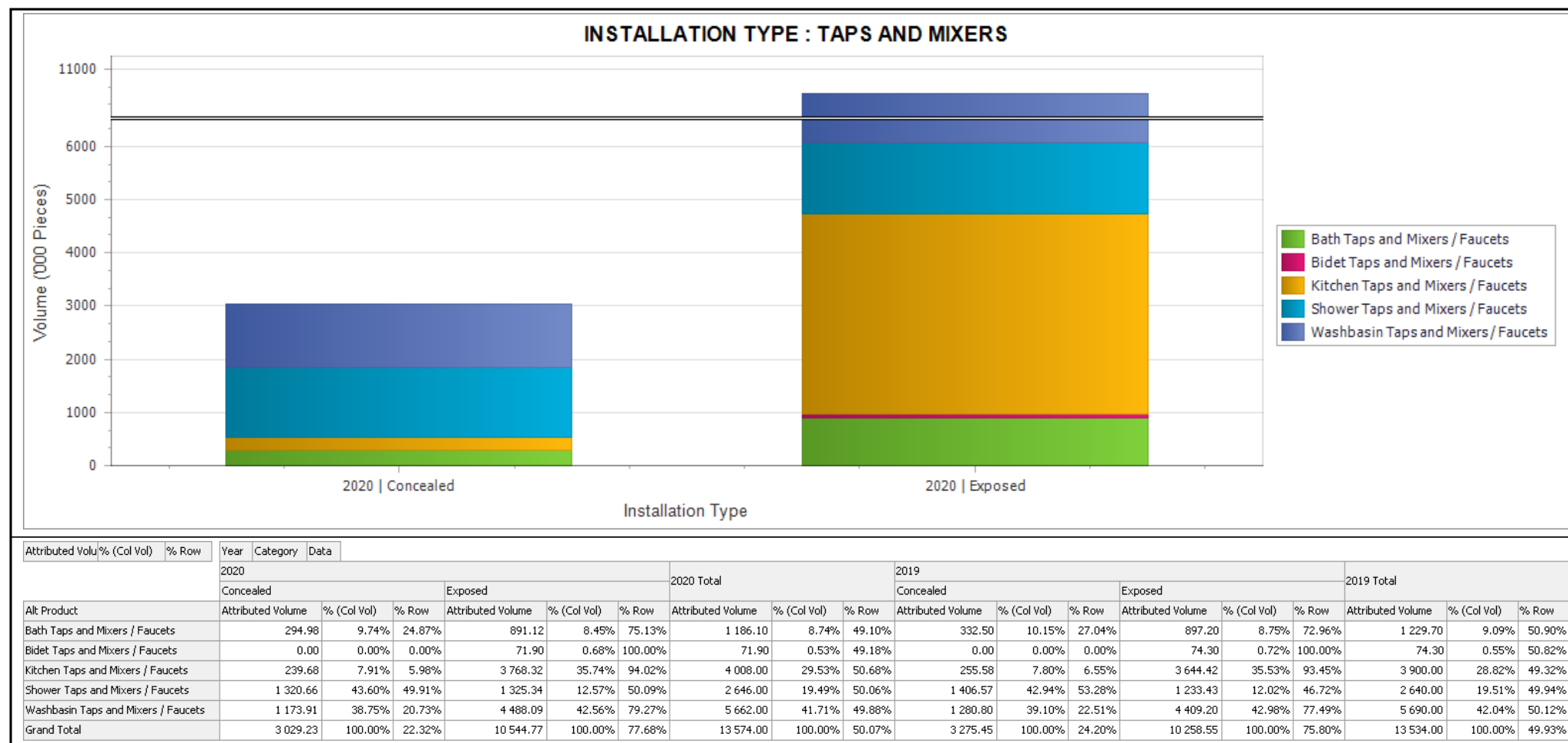


Attribute	% (Col Vol)	Year	Alt Product	Data								
2020											2020 Total	
Bath Taps and Mixers / Faucets		Bidet Taps and Mixers / Faucets		Kitchen Taps and Mixers / Faucets		Shower Taps and Mixers / Faucets		Washbasin Taps and Mixers / Faucets				
Category	Attributed Volume	% (Col Vol)	Attributed Volume	% (Col Vol)	Attributed Volume	% (Col Vol)	Attributed Volume	% (Col Vol)	Attributed Volume	% (Col Vol)	Attributed Volume	% (Col Vol)
Black	63.97	5.39%	4.65	6.47%	262.97	6.56%	155.77	5.89%	305.38	5.39%	792.73	5.84%
Brass	27.90	2.35%	1.83	2.54%	95.80	2.39%	52.14	1.97%	133.16	2.35%	310.82	2.29%
Bronze	29.38	2.48%	1.33	1.85%	24.44	0.61%	54.12	2.05%	140.27	2.48%	249.54	1.84%
Chrome	650.52	54.84%	41.51	57.73%	2 628.66	65.59%	1 504.18	56.85%	3 105.31	54.84%	7 930.18	58.42%
Copper	15.25	1.29%	0.83	1.16%	43.01	1.07%	29.04	1.10%	72.80	1.29%	160.93	1.19%
Gold	51.70	4.36%	5.98	8.31%	87.00	2.17%	93.72	3.54%	246.79	4.36%	485.19	3.57%
Nickel	44.63	3.76%	0.17	0.23%	77.23	1.93%	82.50	3.12%	213.06	3.76%	417.59	3.08%
Others	103.77	8.75%	4.82	6.70%	394.94	9.85%	196.02	7.41%	495.36	8.75%	1 194.91	8.80%
Rose Gold	1.86	0.16%	0.00	0.00%	2.93	0.07%	3.96	0.15%	8.88	0.16%	17.64	0.13%
Silver	63.23	5.33%	3.32	4.62%	11.73	0.29%	112.20	4.24%	301.83	5.33%	492.32	3.63%
Stainless Steel	84.06	7.09%	4.65	6.47%	315.75	7.88%	180.19	6.81%	401.26	7.09%	985.91	7.26%
White	49.84	4.20%	2.82	3.93%	63.54	1.59%	182.16	6.88%	237.91	4.20%	536.28	3.95%
Grand Total	1 186.10	100.00%	71.91	100.00%	4 008.01	100.00%	2 646.01	100.00%	5 662.00	100.00%	13 574.03	100.00%

Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

3.4.5 Sales by Type of Installation : Taps and Mixers

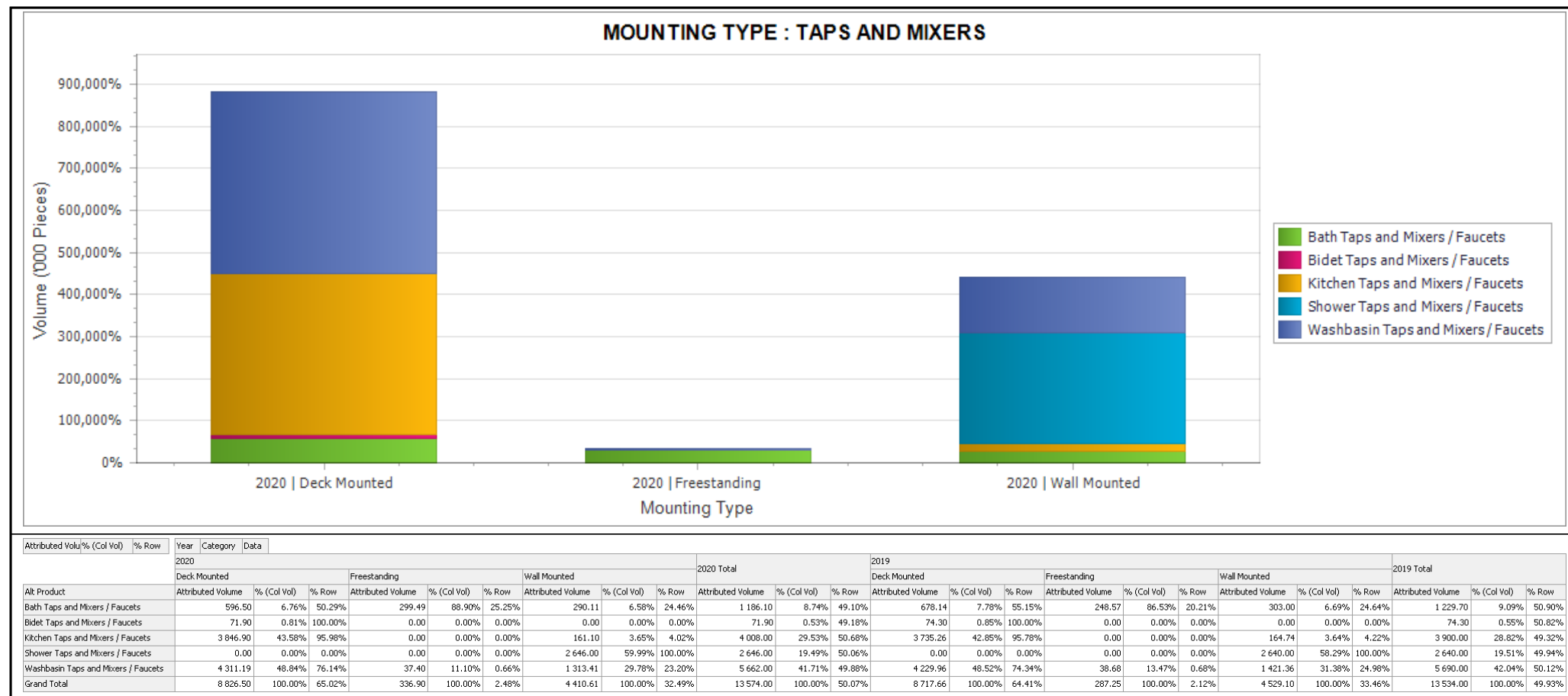
DE - TAP AND MIXER SALES BY INSTALLATION TYPE ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

3.4.6 Sales by Type of Mounting : Taps and Mixers

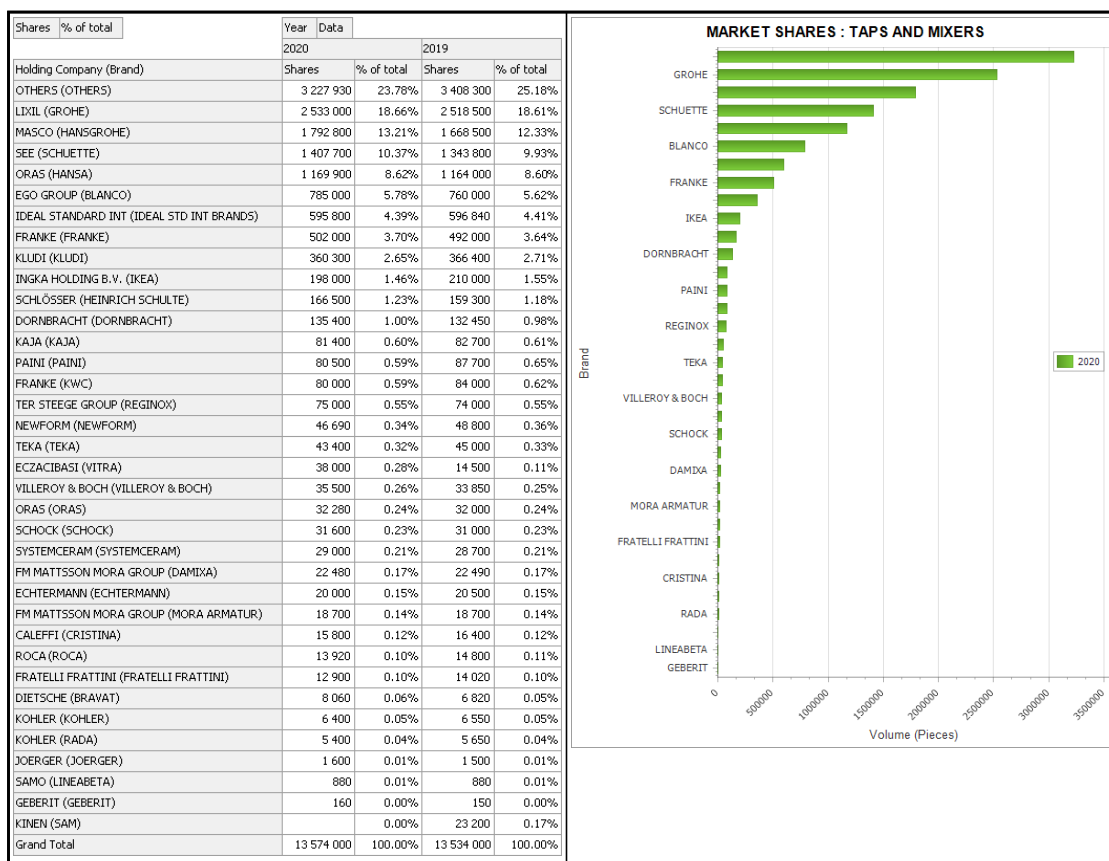
DE - TAP AND MIXER SALES BY MOUNTING TYPE ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

3.5 Market Shares : Taps and Mixers

DE - ESTIMATED MARKET SHARES FOR TAPS AND MIXERS



Source: BRG based on industry interviews. N.B. Includes Both Kitchen and Sanitary Taps & Mixers.

3.6.1 DE - IMPORTS OF TAPS & MIXERS 2020

Source Country	2020				2019			
	Volume		Value 000		Volume		Value 000	
	Tonnes	%	Euros	%	Tonnes	%	Euros	%
China	20 287	40.6%	248 638	31.8%	19 054	37.8%	245 908	31.2%
Portugal	8 996	18.0%	144 594	18.5%	9 514	18.9%	151 146	19.2%
Thailand	6 018	12.0%	72 413	9.3%	6 203	12.3%	83 542	10.6%
Italy	4 655	9.3%	99 619	12.7%	5 287	10.5%	103 809	13.2%
Bulgaria	2 354	4.7%	38 865	5.0%	2 458	4.9%	36 579	4.6%
Austria	1 606	3.2%	31 904	4.1%	1 618	3.2%	31 479	4.0%
Netherlands	1 337	2.7%	10 356	1.3%	1 061	2.1%	8 210	1.0%
Poland	959	1.9%	33 102	4.2%	558	1.1%	18 507	2.3%
France	669	1.3%	10 102	1.3%	1 044	2.1%	13 453	1.7%
Turkey	569	1.1%	7 725	1.0%	872	1.7%	8 269	1.0%
OTHERS	2 577	5.2%	85 439	10.9%	2 690	5.3%	87 769	11.1%
Total	50 027	100.0%	782 757	100.0%	50 359	100.0%	788 671	100.0%

Source : Trademap

3.6.2 DE - EXPORTS OF TAPS & MIXERS 2020

Destination	2020				2019			
	Volume		Value 000		Volume		Value 000	
	Tonnes	%	Euros	%	Tonnes	%	Euros	%
France	6 212	14.7%	162 549	13.3%	6 633	14.4%	168 799	13.0%
Poland	4 067	9.6%	91 201	7.5%	4 474	9.7%	95 563	7.3%
Netherlands	3 343	7.9%	97 466	8.0%	3 854	8.3%	105 161	8.1%
Austria	3 027	7.2%	83 542	6.8%	3 391	7.3%	88 036	6.8%
China	2 299	5.4%	73 214	6.0%	2 539	5.5%	75 028	5.8%
Belgium	2 078	4.9%	57 756	4.7%	2 213	4.8%	59 740	4.6%
Italy	1 973	4.7%	50 485	4.1%	2 385	5.2%	59 462	4.6%
United Kingdom	1 913	4.5%	62 198	5.1%	2 427	5.3%	78 862	6.1%
Spain	1 443	3.4%	42 251	3.5%	1 718	3.7%	48 650	3.7%
Russian Federation	1 375	3.3%	36 977	3.0%	1 269	2.7%	36 688	2.8%
Czech Republic	1 195	2.8%	39 832	3.3%	1 233	2.7%	37 601	2.9%
OTHERS	13 347	31.6%	422 382	34.6%	14 023	30.4%	448 906	34.5%
Total	42 272	100.0%	1 219 853	100.0%	46 159	100.0%	1 302 496	100.0%

Source : Trademap

3.7.1

Route to Market: Kitchen Taps & Mixers / Faucets
DE - ESTIMATED DISTRIBUTION FOR KITCHEN TAPS & MIXERS / FAUCETS: 2020

Manufacturer / Importer	Manufacturers / Importers
	100.0%

Wholesale Level	48.7%
	Wholesalers / Merchants
	48.7%

Retail Level	Manufacturer	Wholesaler	Manufacturer	Wholesaler	Manufacturer	Wholesaler
	3.9%	7.5%	17.1%	8.0%	24.5%	15.0%
	Internet Retailers		DIY Retailers		Specialist Retailers	
	11.4%		25.1%		39.5%	

Manufacturer	Wholesaler	Internet	DIY	Specialist
4.9%	18.0%	2.2%	1.0%	4.5%
Installers				
30.6%				

End Use Level	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer
	0.1%	0.1%	9.2%	24.1%	31.0%	24.3%	0.8%	0.1%	0.0%	0.0%	4.0%	6.3%
	Residential						Non-Residential					
	88.9%						11.1%					
New Build			Residential RMI			New Build			Non-Residential RMI			
7.0%			81.8%			4.0%			7.1%			
Project	Non-Project		Project	Non-Project		Project	Non-Project		Project	Non-Project		
2.8%	4.2%		2.0%	79.8%		1.9%	2.1%		2.0%	5.1%		

3.7.2

Route to Market: Kitchen Taps & Mixers / Faucets
DE - ESTIMATED DISTRIBUTION FOR KITCHEN TAPS & MIXERS / FAUCETS: 2025

Manufacturer / Importer	Manufacturers / Importers
	100.0%

Wholesale Level	47.0%
	Wholesalers / Merchants
	47.0%

Retail Level	Manufacturer	Wholesaler	Manufacturer	Wholesaler	Manufacturer	Wholesaler
	5.0%	11.0%	18.0%	7.0%	24.0%	16.0%
	Internet Retailers		DIY Retailers		Specialist Retailers	
	16.0%		25.0%		40.0%	

Manufacturer	Wholesaler	Internet	DIY	Specialist
4.8%	12.8%	5.0%	1.2%	5.2%
Installers				
29.0%				

End Use Level	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer
	0.2%	0.1%	11.0%	23.8%	31.0%	23.1%	1.0%	0.1%	0.0%	0.0%	3.8%	5.9%
	Residential						Non-Residential					
	89.2%						10.8%					
	New Build			Residential RMI			New Build			Non-Residential RMI		
	5.9%			83.3%			3.9%			6.9%		
	Project		Non-Project	Project		Non-Project	Project		Non-Project	Project		Non-Project
	2.8%		3.1%	2.0%		81.3%	1.8%		2.1%	2.1%		4.8%

Source: BRG Based on Industry Interviews. - Totals may vary due to roundings

3.7.3 Route to Market - Consolidated View : Kitchen Taps and Mixers / Faucets

DE - ESTIMATED DISTRIBUTION FOR KITCHEN TAPS AND MIXERS / FAUCETS : 2020

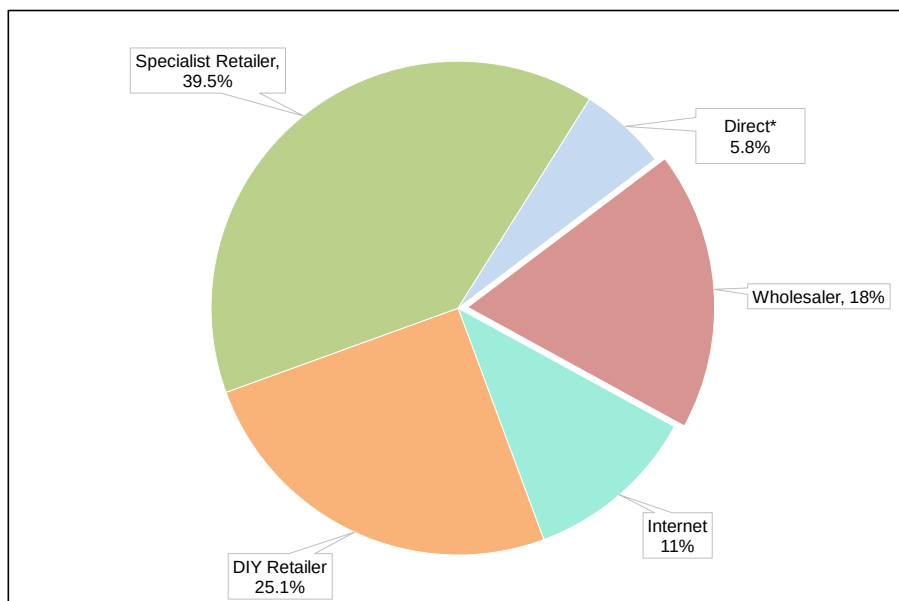
Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
18.2%	11.4%	25.1%	39.5%	5.8%	4.8%	3.9%	84.1%	7.2%
100%					8.7%		91.3%	

3.7.4 Route to Market - Consolidated View : Kitchen Taps and Mixers / Faucets

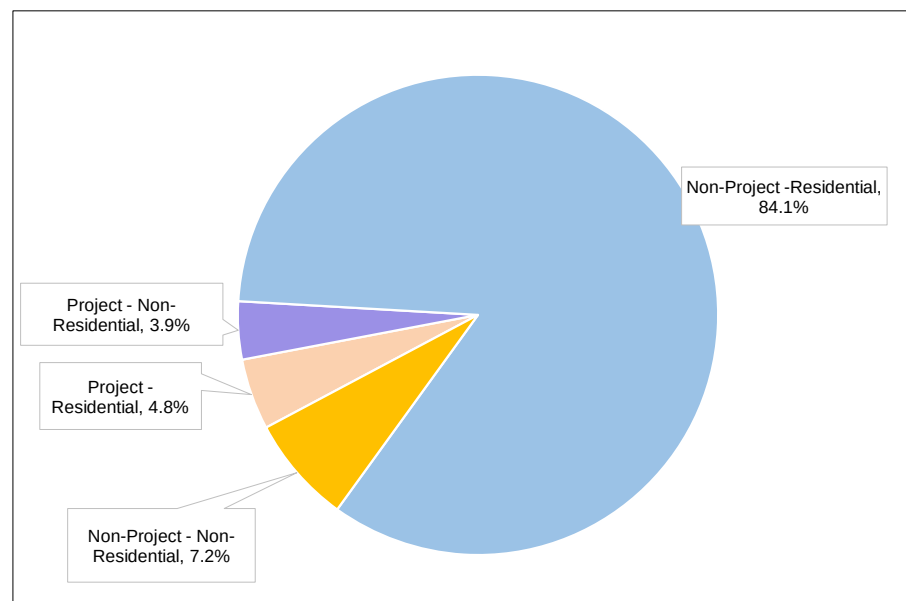
DE - ESTIMATED DISTRIBUTION FOR KITCHEN TAPS AND MIXERS / FAUCETS : 2025

Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
13.0%	16.0%	25.0%	40.0%	6.0%	4.8%	3.9%	84.4%	6.9%
100%					8.7%		91.3%	

DE - SALES OF KITCHEN TAPS AND MIXERS / FAUCETS BY DISTRIBUTION CHANNEL: 2020



DE - SALES OF KITCHEN TAPS AND MIXERS / FAUCETS BY POINT OF DECISION AND END-USE SECTOR: 2020



Source: BRG Based on Industry Interviews.

*Direct includes sales going directly from manufacturers to installers, contractors and private customers.

3.7.5

Route to Market: Sanitary Taps & Mixers / Faucets
DE - ESTIMATED DISTRIBUTION FOR SANITARY TAPS & MIXERS / FAUCETS: 2020

Manufacturer / Importer	Manufacturers / Importers
	100.0%

Wholesale Level	81.3%
	Wholesalers / Merchants
	81.3%

Retail Level	Manufacturer	Wholesaler	Manufacturer	Wholesaler	Manufacturer	Wholesaler
	5.7%	7.5%	8.8%	16.3%	3.1%	1.3%
	Internet Retailers		DIY Retailers		Specialist Retailers	
	13.2%		25.1%		4.4%	

Manufacturer	Wholesaler	Internet	DIY	Specialist
0.3%	56.0%	3.5%	2.1%	0.1%
Installers				
62.0%				

End Use Level	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer
	0.0%	0.1%	8.5%	23.0%	3.3%	52.6%	0.8%	0.1%	1.2%	0.0%	1.0%	9.4%
	Residential						Non-Residential					
	87.5%						12.5%					
New Build			Residential RMI			New Build			Non-Residential RMI			
11.0%			76.6%			2.5%			10.0%			
Project	Non-Project		Project	Non-Project		Project	Non-Project		Project	Non-Project		
4.5%	6.5%		2.0%	74.6%		1.2%	1.3%		3.0%	7.0%		

3.7.6

Route to Market: Sanitary Taps & Mixers / Faucets
DE - ESTIMATED DISTRIBUTION FOR SANITARY TAPS & MIXERS / FAUCETS: 2025

Manufacturer / Importer	Manufacturers / Importers
	100.0%

Wholesale Level	78.8%
	Wholesalers / Merchants
	78.8%

Retail Level	Manufacturer	Wholesaler	Manufacturer	Wholesaler	Manufacturer	Wholesaler
	6.0%	10.0%	9.0%	16.0%	3.2%	1.3%
	Internet Retailers		DIY Retailers		Specialist Retailers	
	16.0%		25.0%		4.5%	

Manufacturer	Wholesaler	Internet	DIY	Specialist
1.5%	51.2%	4.0%	2.2%	0.1%
Installers				
59.0%				

End Use Level	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer
	0.0%	0.2%	10.5%	22.8%	3.4%	51.3%	1.5%	0.1%	1.5%	0.0%	1.0%	7.7%
	Residential						Non-Residential					
	88.3%						11.7%					
	New Build			Residential RMI			New Build			Non-Residential RMI		
	9.4%			78.9%			2.4%			9.3%		
	Project	Non-Project		Project	Non-Project		Project	Non-Project		Project	Non-Project	
	4.0%	5.4%		2.0%	76.9%		1.2%	1.2%		2.4%	6.9%	

Source: BRG Based on Industry Interviews. - Totals may vary due to roundings

3.7.7 Route to Market - Consolidated View : Sanitary Taps and Mixers / Faucets

DE - ESTIMATED DISTRIBUTION FOR SANITARY TAPS AND MIXERS / FAUCETS : 2020

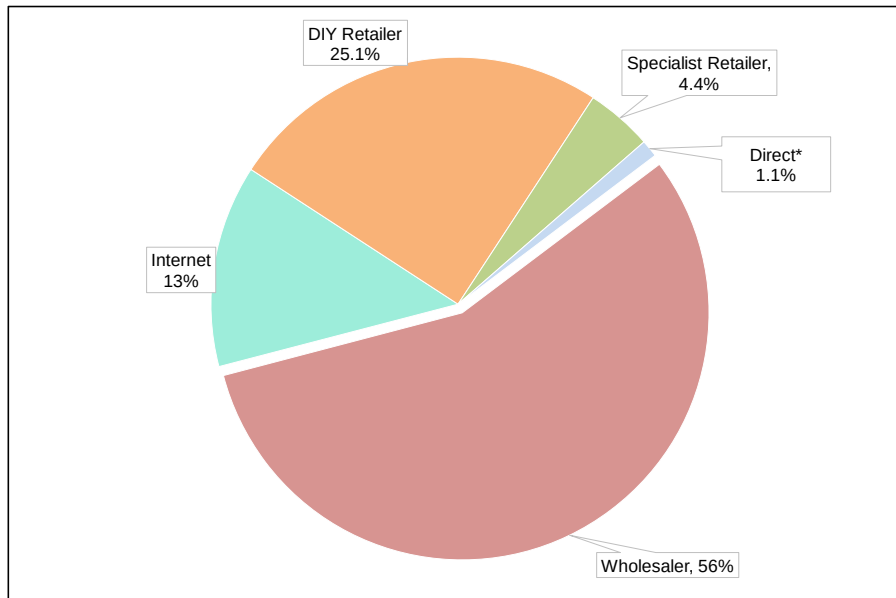
Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
56.2%	13.2%	25.1%	4.4%	1.1%	6.5%	4.2%	81.0%	8.3%
100%					10.7%		89.3%	

3.7.8 Route to Market - Consolidated View : Sanitary Taps and Mixers / Faucets

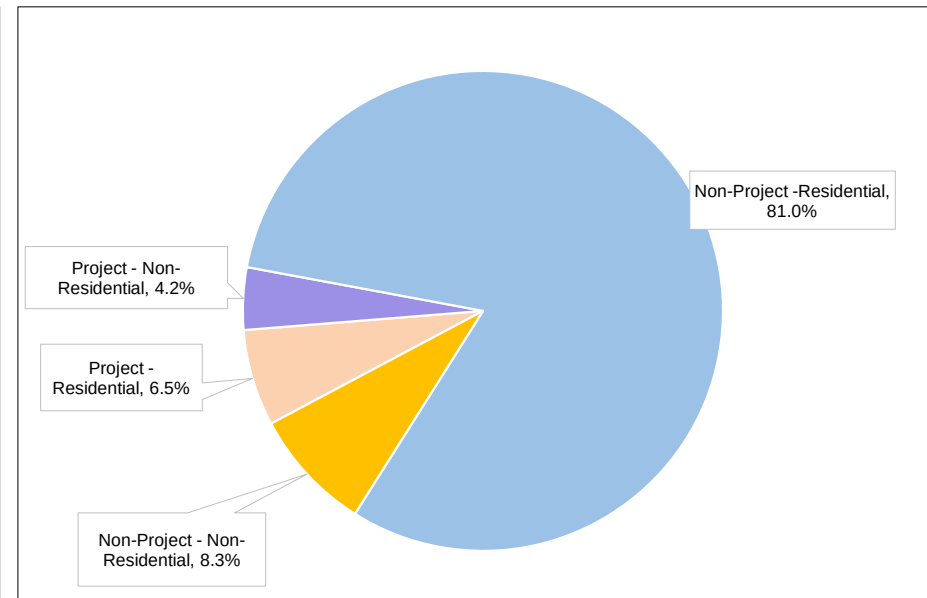
DE - ESTIMATED DISTRIBUTION FOR SANITARY TAPS AND MIXERS / FAUCETS : 2025

Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
51.5%	16.0%	25.0%	4.5%	3.0%	6.0%	3.6%	82.3%	8.1%
100%					9.6%		90.4%	

DE - SALES OF SANITARY TAPS AND MIXERS / FAUCETS BY DISTRIBUTION CHANNEL: 2020



DE - SALES OF SANITARY TAPS AND MIXERS / FAUCETS BY POINT OF DECISION AND END-USE SECTOR: 2020



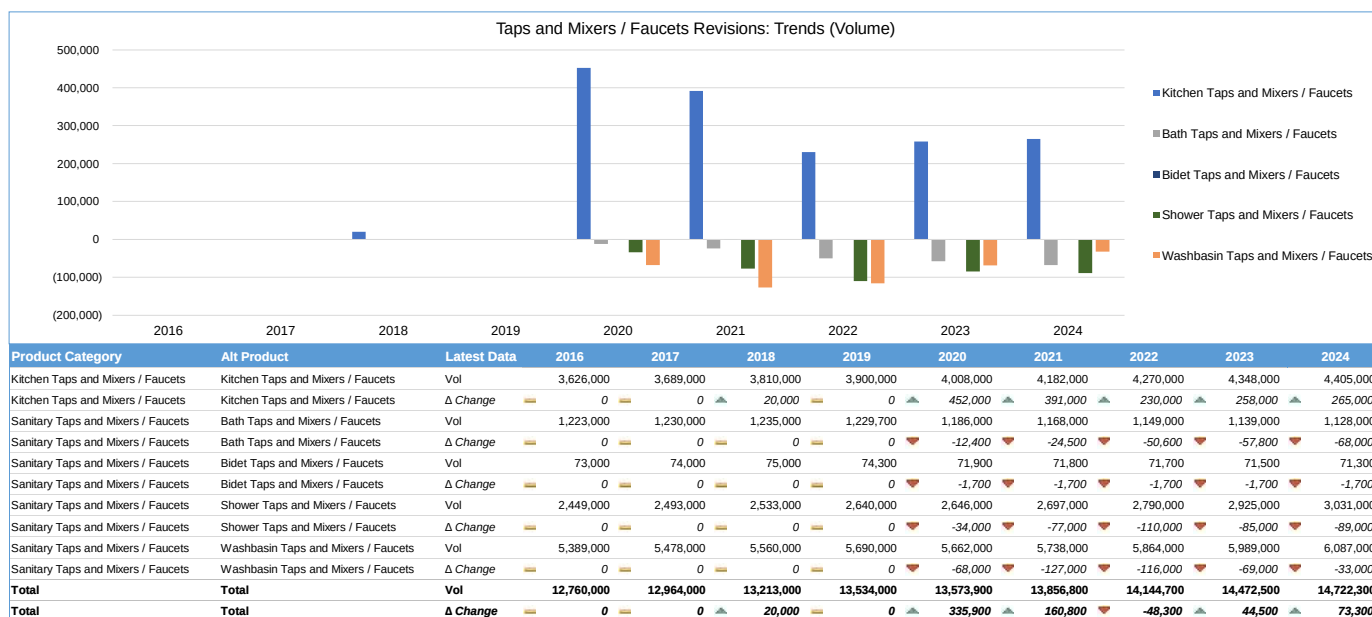
Source: BRG Based on Industry Interviews.

*Direct includes sales going directly from manufacturers to installers, contractors and private customers.

3.8 Revisions in the German Taps and Mixers / Faucets Market (April 2021)

DE - REVISIONS IN THE GERMAN TAPS AND MIXERS / FAUCETS MARKET

Data Comparison: October 2020

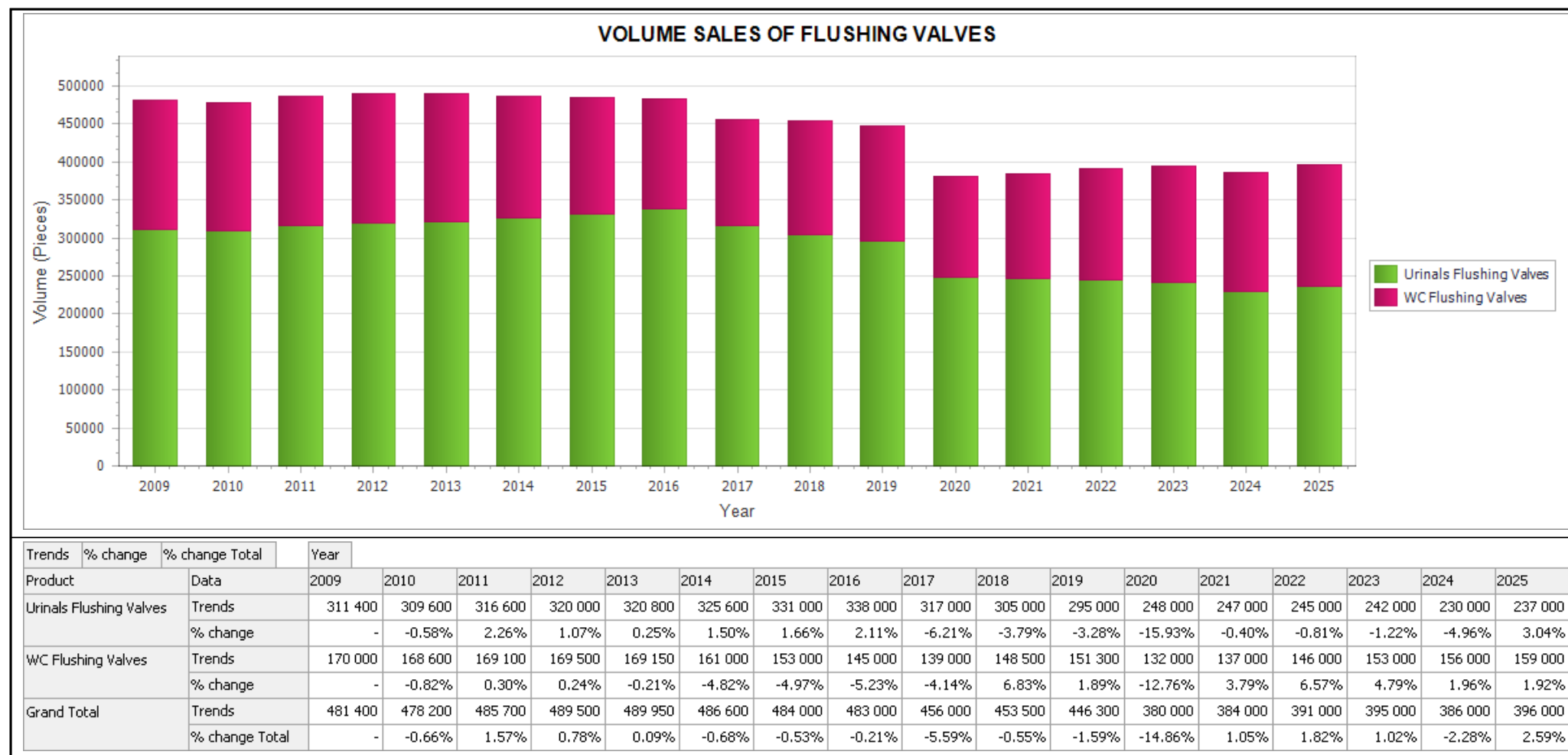


No or marginal changes have been made to historical volumes for this market.

Following a thorough detailed price analysis, by comparing domestically produced taps and mixers in contrast to imported taps and mixers, significant downward revisions were made to the MSP for all types of taps and mixers. Taps and mixers produced in Germany are very high priced, which is because Germany nowadays mostly produces high quality, high priced, bathroom and kitchen fittings, whereas more economy priced products are imported from abroad. This includes German manufacturers, which often produce their premium products in their domestic production site(s), whereas lower priced ranges are sourced from foreign factories, including Poland, Hungary and Portugal. Average prices for mixers, not including any type of taps, produced in Germany have an MSP of around €70 to €80 in recent years, whereas imported models, again only referring to mixers, are significantly lower prices, with MSPs between €35 to €40 in recent years. This includes models imported by German manufacturers from their foreign production plants but also all other imported brands, such as the ones imported from e.g., China, Italy, Spain, France and Poland.

3.9.1 Historical Trends and Forecasts : Flushing Valves

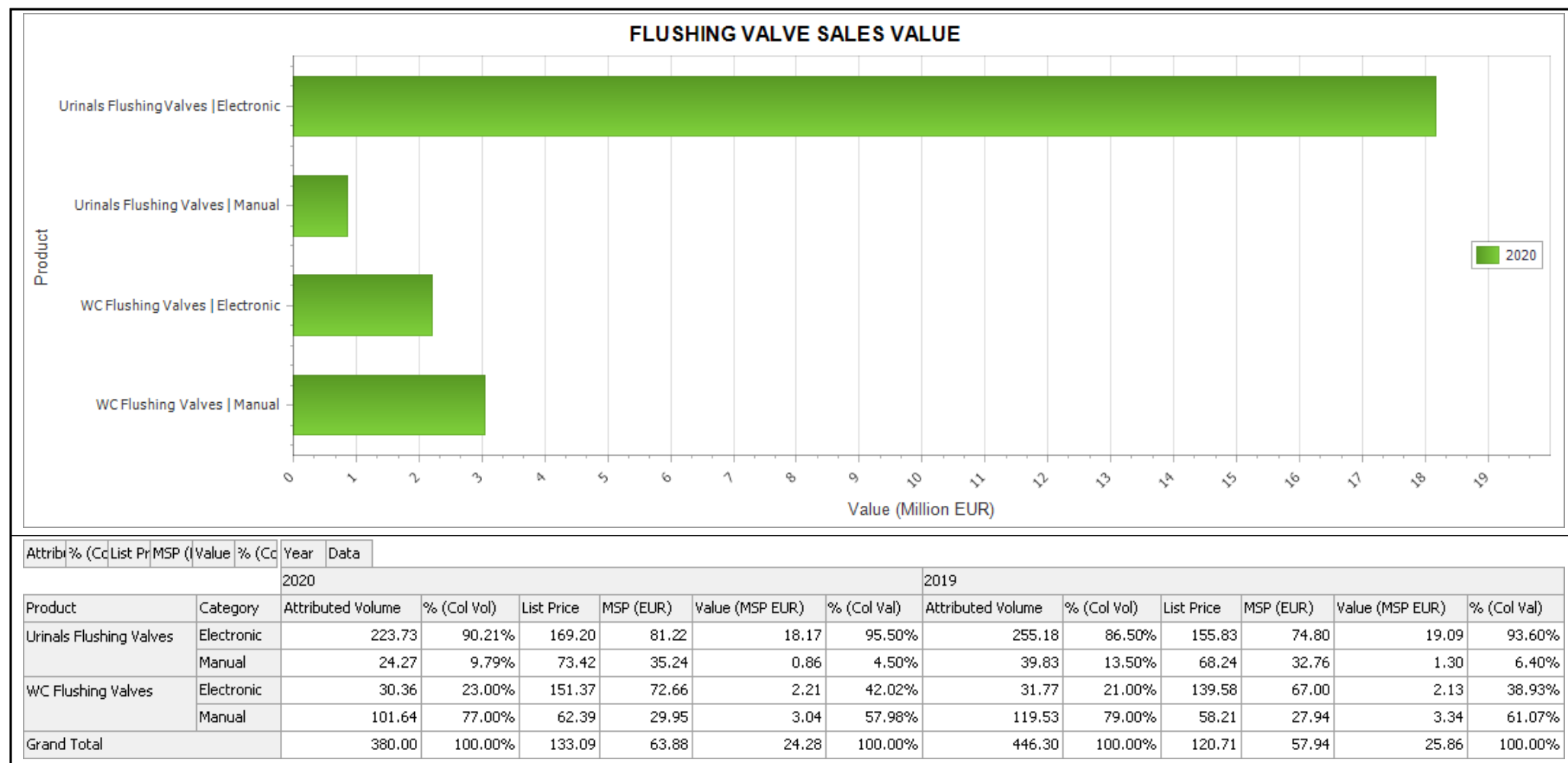
DE - SUMMARY OF TRENDS IN THE FLUSHING VALVE MARKET : HISTORICAL AND FORECASTS



Source: BRG based on industry interviews.

3.9.2 Prices and Market Values : Flushing Valves

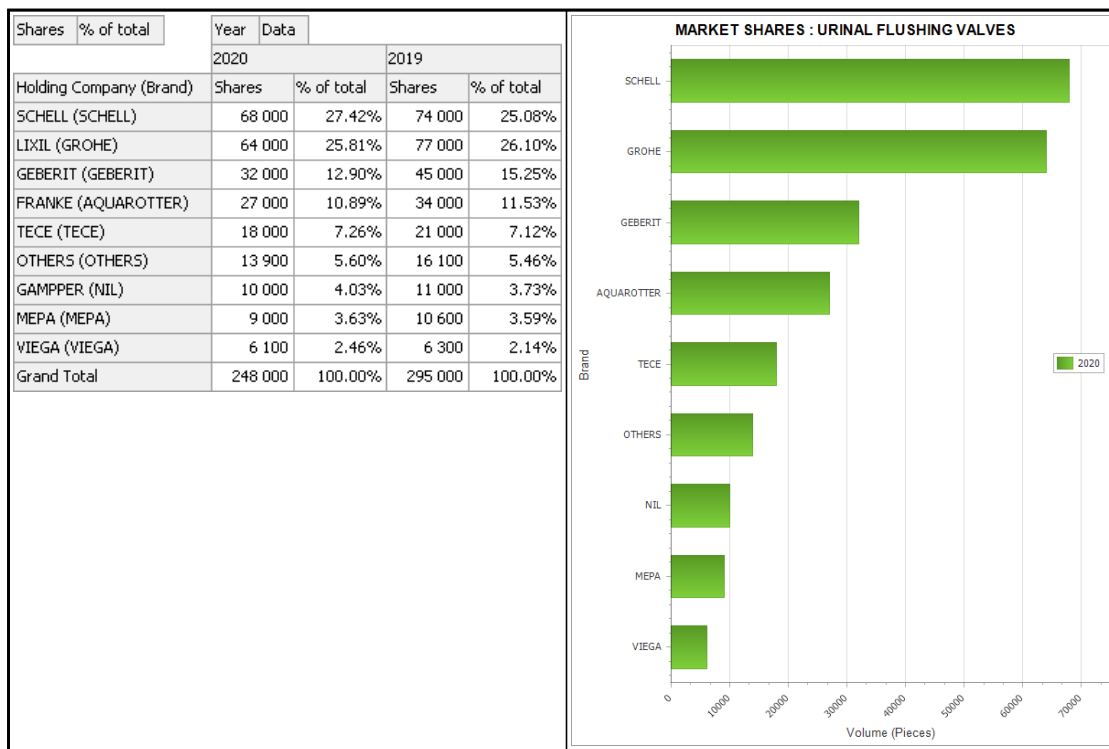
DE - ESTIMATED VALUES OF THE FLUSHING VALVE MARKET



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price)

3.9.3 Market Shares : Urinal Flushing Valves

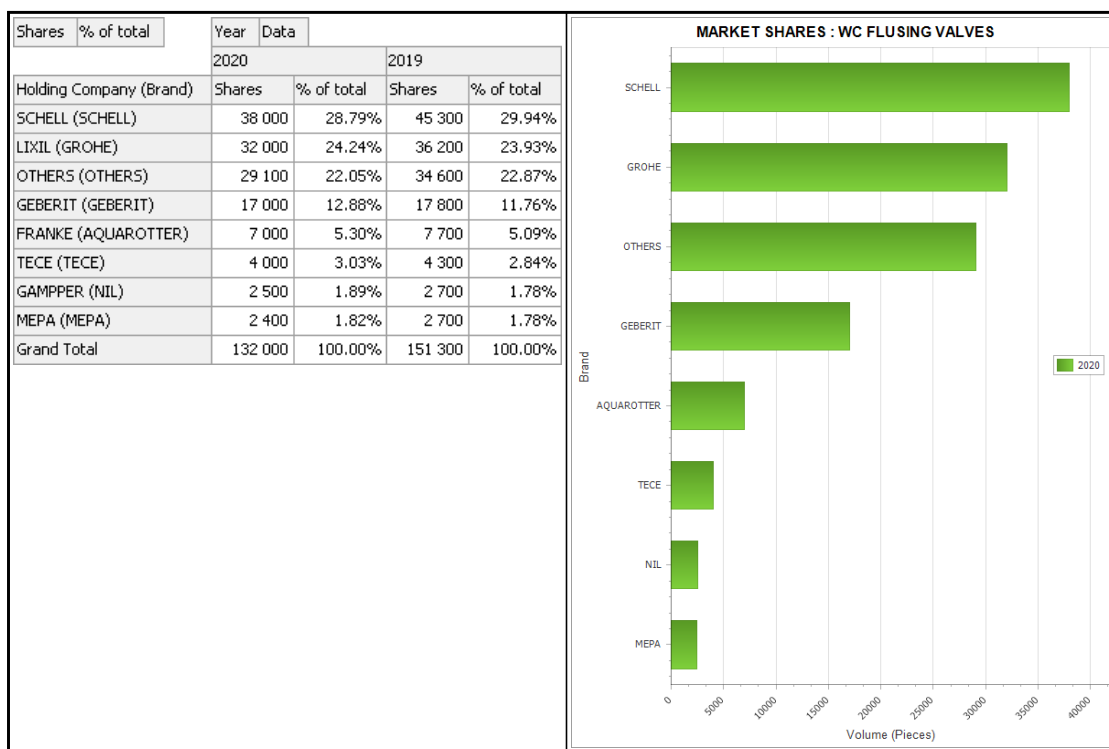
DE - ESTIMATED MARKET SHARES FOR URINAL FLUSHING VALVES



Source: BRG based on industry interviews.

3.9.4 Market Shares : WC Flushing Valves

DE - ESTIMATED MARKET SHARES FOR WC FLUSHING VALVES

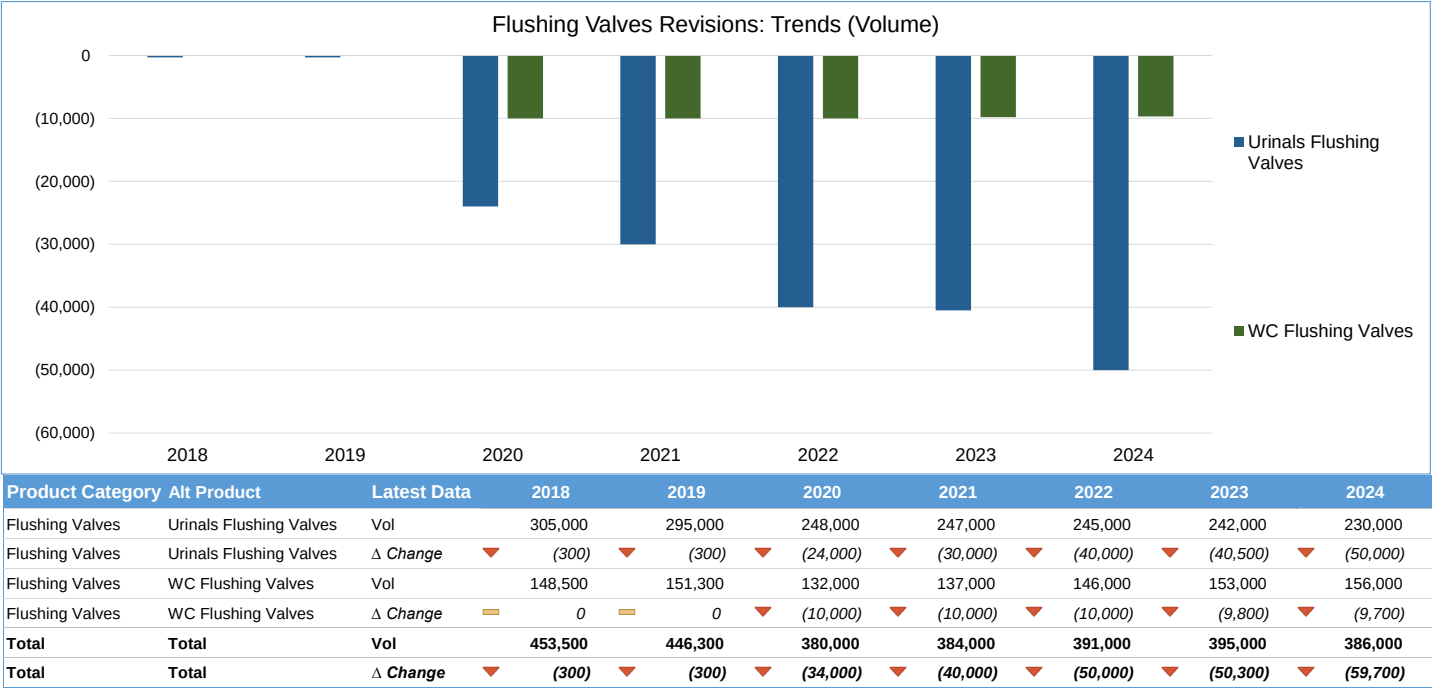


Source: BRG based on industry interviews.

3.10 Revisions in the German Flushing Valves Market (April 2021)

DE - REVISIONS IN THE GERMAN FLUSHING VALVES MARKET

Data Comparison: October 2020



No revisions have been made to historical volumes for this market.

GERMANY

SHOWER WALL

THE EUROPEAN BATHROOM & KITCHEN PRODUCT MARKETS



Market Characteristics and Technological Trends

The market for shower wall products goes back to the late 1960s when they were introduced for the first time, although growth only accelerated from the 1980s onwards. Cyclical drivers that have the biggest impact on shower wall sales are new build construction, renovation expenditure and private consumption. In general, the market for shower walls is a mature market, with no major fluctuations seen in overall sales volumes.

Over the years, most growth has come from shower enclosures, as showers have taken share from baths and, nowadays, a substantial number of dwellings in Germany is only equipped with a shower. This explains the positive development in sales of shower enclosures compared to bath screens, as baths are increasingly replaced by large shower solutions, hampering the demand for bath screens. The popularity of walk-in showers or wet shower rooms with a shower drain has been influencing demand from traditional shower wall manufacturers. A walk-in shower usually requires a single piece of glass, which can also be tailor-made by glass manufacturers and glaziers. On a local level, the traditional shower wall manufacturers face competition from window and glass installation companies. Conventional shower cubicles have nearly disappeared from the German market. In contrast to other European countries, shower enclosures and shower trays, as well as baths and bath screens, have historically not developed in unison and the relationship has not been 1:1.

In rented dwellings, as well as some low-end hotel rooms, shower curtains may be used instead of shower enclosures. Recently, the relationship between shower trays and shower enclosures has become even less pronounced as the share of wet shower rooms in the market is increasing. One of the main product trends of the past has certainly been the move from plastic to glass products. In the lower price segment, the plastics products are more common than in other segments. The low-priced glass is very thin, and some end users wrongly believe that it is too fragile. Therefore, plastic is more common in the lower price segments. The share of glass in all shower wall products has grown strongly over the last two decades, now accounting for most sales in all segments of the shower wall market. Plastic shower wall products mostly consist of polystyrol and account for the remaining share. Glass products used to be sold mainly through wholesalers but DIY stores, as well, have replaced most plastic enclosures with glass enclosures.

Hydrotherapy shower cubicles are mostly used in the fitness sector and have been developed into full wellness systems, and can include aromatherapy, connections for MP3 players, loudspeakers and coloured lighting. The combined product market segment is relatively difficult to install and rather expensive. These two factors make it a product that is less suitable to be sold in DIY stores. It remains a niche product.

Recent Market Trends

The COVID-19 pandemic shaped the developments in the year 2020 and 'business as usual' was replaced by government measures. In Germany the official lockdown, which is better understood as a 'prohibition of contact', started at the end of March 2020. Bavaria was the first state to implement measures to contain the spread of the virus, which was followed by the remaining states towards the end of March. For a few weeks the German population was advised to work from home when possible, with most non-essential stores closing for a limited time only. Measures implemented varied by state (*Bundesländer*). Most stores, including DIY stores, were allowed to open again from the beginning of April 2020 onwards, with differences from state to state in the level of restrictions and measures. In certain regions, shopping in DIY stores was limited to professionals only.

During 2020, new housing construction continued without much interruption, with only large scale project sites halted to implement protective measurements. That being said, the overall production per employee is believed to have shrunk significantly, and remains to be the case, since many employers introduced measures to protect the construction workers. The bathroom industry was declared part of the crisis infrastructure, but installers could still continue to work without many interruptions. Overall, there has been relatively little impact of the COVID-19 pandemic on the building related industry, including the bathroom market, in Germany. In some cases, end users may have postponed, or cancelled, bathroom related projects due to the uncertainty on the market, but overall demand remained relatively strong, as there was a large backlog of works from the past, benefiting sales of bathroom and kitchen equipment. At the same time, in certain instances, the COVID-19 pandemic may have contributed to additional demand. Since consumers were spending more time indoors, they had the opportunity to dedicate themselves to repair or upgrade works, whereas others might have decided to invest into their property by installing a new bathroom or kitchen due to (international) holidays being difficult because of the current pandemic.

In 2020, total sales of shower walls increased slightly stronger when compared to the previous year, being primarily driven by demand for shower enclosures, which compensated for declining sales of conventional shower cubicles and bath screens. Overall, this market profited from the increasing number of bathroom renovations and like-for-like replacement. With a large number of Germans working half time and staying at home, most people had plenty of time to dedicate themselves to DIY jobs. Demand from DIY stores increased significantly in 2020 and it is believed that it somewhat benefited sales of shower enclosures, too. Other than that, demand from the new housing and non-housing segment was negative for all the product types.

Demand for bath screens was again negative in 2020. With the trend towards showering, new baths are rarely installed to be used for showering. If a consumer opts for a bath, it is just for the purpose of having a bath. Installing a bath that is used as a shower is not contemporary because of the demographic change; for example, elderly people might struggle to get inside the bath to have a shower.

Instead, it would be more convenient to have a spacious shower. In the new house sector bathrooms tend to be bigger and German households prefer a bath without a bath screen. Walk-in showers have established themselves as a popular alternative to shower trays. The design of a walk-in shower does not require a full enclosure of the wet room. In many cases a fixed glass wall is enough to cover one side and to leave another open as an entrance. As such, walk-in showers facilitated the design of shower enclosures, so that local glass manufacturers increasingly started selling tailor-made solutions for walk-in showers. Not only can they respond faster to consumers' needs, but they are flexible to design and manufacture according to individual requirements. These advantages have led to growing demand for shower enclosures from glass manufacturers, which are not included in the BRG market share.

The shower wall market is believed to grow only marginally as long as sales of shower trays are increasing. All shower trays need a shower wall with standard sizes, which only a manufacturer can provide in large quantities. According to the 7th fundamental study of bathrooms in Germany, around 53% out of 46.2 million bathrooms in Germany have a shower wall. From all these bathrooms, there are 17.7 million original bathrooms which have not been renovated yet and about two million bathrooms which are older than 30 years. It is important to mention that about 25 million bathrooms are in owned dwellings, and around 21 million are in rented dwellings. These numbers show the huge park of showers that exist on the German market. Overall, Germans are spending more money on bathrooms and 30,000 to 40,000 euros for a complete bathroom renovation is not unusual for homeowners.

The most noticeable trend on the German market – similar to that of baths and shower trays – is towards higher quality products. The share of the upmarket price segment is growing at the expense of the middle price range. Nowadays, the material used for bath screens, conventional shower cubicles and shower enclosures is predominantly glass. Demand for plastic comes from the economy segment, which is sometimes preferred by consumers instead of very thin glass due to perceived safety attributes. Thin glass is used in lower price segments, which some consumers perceive as being more breakable; therefore, they prefer the use of plastic instead. The most common shower enclosures are with a pivoting, no door (side wall) or sliding door. Sliding doors are preferred in lower price segments with small bathrooms. There is no need for a handle and less space is required to open a door. The second most popular choice is to have no door at all, which is likely to be found in wet rooms. Five-angle and round door types hardly exist in Germany anymore. Some manufacturers have them in their portfolio for the sake of completeness.

Demand for hydrotherapy cubicles is relatively stable and the product serves an established niche market. In 2020, sales slightly decreased when compared to the previous year.

According to industry feedback there is a growing trend of manufacturers supplying large building projects directly with their products. All large shower wall

manufacturers have their own installers, who would carry out the installation in big projects. The reason is to minimise the risk of wrong calculation and installations. All works that need to be handled with care are often performed by manufacturers themselves. An installer would not have the time to learn about all the different products and brands and, therefore, the manufacturers take over all the technical and more complicated works. For example, there is a strong trend towards frameless shower walls. However, only a small percentage of installers has the skill to install such a frameless enclosure. Therefore, manufacturers selling such products would almost always carry out the installations themselves.

Prices and Market Values

Prices for all shower wall products increased in 2020 to compensate for inflation and rising production costs such as higher wages and raw material costs. Additionally, the overall market value increased as German consumers look for better quality products, which come at considerably higher prices. The level of consumer spending increased further in Germany; and homeowners tend to prefer higher-priced shower wall models. In general, prices differ significantly, not only between product segments but also between product types such as five-angled shower enclosures versus corner entry models. In more expensive price ranges, local glass manufacturers are increasing their market share and intensified the competition on the market.

The MSP for conventional shower cubicles has slightly increased when compared to 2019. This product is usually sold in the economy and lower price segment. The demand is mainly covered by 'Other' manufacturers, which produce in the Far East and which distribute mainly in DIY and online stores. Prices for hydrotherapy cubicles increased in 2020 in alignment with the wellness trend and consumers' preference for better quality. Based on the quality of hydrotherapy cubicles, prices can vary massively between different models. In general, products sold within DIY stores are cheaper than those sold via direct sales and wholesale channels, which are meant to be more exclusive. However, according to industry feedback, DIY stores have extended their product range of shower enclosures. New sales concepts have been implemented and brands, which previously were only sold via the wholesaler, are also available in DIY stores. Manufacturers are increasingly following a multi-channel approach and they hope to profit from their brand awareness among DIY consumers. The reason for this step is the disadvantage that manufacturers are facing when competing with retail brands.

New Housing

The German construction industry slightly contracted in 2020 due to the negative impact of the COVID-19 crisis. Therefore, demand for all products decreased in terms of volume when compared to the previous year. Apartments, which are built for social housing on a low budget and with limited bathroom space are more likely to have a bath with bath screens and conventional cubicles. In most other buildings, either rented or owned, showers are more common and the number of shower enclosures is increasing. Using a bath for showering is not common

anymore, unless in social housing or dwellings with limited space and the wish to have a bath. In such cases a bath can still have simple shower curtain.

Demand on the overall shower wall market is linked to developments on the residential construction market. With new constructions on a high level, sales of shower wall products, bath screens and conventional shower cubicles to the new housing end-use segment are believed to account to just under 20% of total sales. It is important to highlight the high penetration rate of walk-in showers within new housing. A higher number of walk-in showers increases the likelihood of purchasing a tailor-made shower enclosure from a glass manufacturer. Such products are not included in BRG's market share and, therefore, are a serious competitor for traditional shower wall manufacturers. In contrast, a large construction project with many new bathrooms is more likely to be equipped by the manufacturer's own installers.

Housing RMI

The German shower wall market is mature and, as such, relies strongly on replacement sales for the bulk of its sales. Demand for all products, except for bath screens and conventional shower cubicles, increased in terms of volume when compared to the previous year. However, replacement demand accounted for most sales, in particular for conventional shower cubicles, where like-for-like replacements are most common. This product fulfils a niche and is installed in a place where the lack of room does not permit the installation of larger showers or even complete bathrooms. A classic example is the installation of a shower cubicle as a second shower in a dwelling with fitness rooms, when a quick (and temporary) solution is required. In alignment with the trend of showering, many households decide to get rid of their old bath and replace it with a big shower. According to BRG research, shower drain manufacturers sell a significant amount into the housing RMI market. This gives an indication that walk-in showers are not only popular in new housing, but also in existing markets. Shower enclosure manufacturers indirectly compete with glass manufactures in existing bathrooms where walk-in showers are installed.

Non-Housing

According to EUROCONSTRUCT, non-housing construction output notably decreased in 2020 due to the COVID-19 pandemic. The only exception, which generated a slightly larger construction output in value on a year-to-year basis, was buildings for education and health. All other sectors such as industrial, hospitality, office and commercial buildings recorded a strong decline as projects got postponed or cancelled due to the economic uncertainty. Schools, universities and even hospitals are not likely to create strong demand for shower walls. Overall, demand for shower walls had a notable decline in 2020 for all types of products.

Supply Structure

The supply structure in the German market remained stable, with a moderate increase of online, DIY and direct sales. The same competitors are active in the market for shower enclosures and the market for bath screens such as SCHULTE, KERMI and SPRINZ. SCHULTE (HSK) is the market leader following a multi-channel approach. This can create conflicts for the installer, who might be selling the same shower enclosure that is also available in DIY stores. The market for shower wall products is fragmented, with domestic and international players active. When compared to other sanitary products, a relatively high share of shower wall products is sold directly to installers. This is particularly the case for manufacturers selling branded products, which are more difficult to install than the average shower enclosure.

Many manufacturers supply their products as OEM sales to wholesalers, local installers, DIY and internet shops. Branded sales of shower enclosures are mainly supplied by HUEPPE, KERMI and SPRINZ, as well as by other companies such as DUSCHOLUX and DURAVIT on a marginal scale. A great share of shower enclosures is sold by wholesalers or online shops that use private labels. The GC GROUP supplies its private label VIGOUR and is successful on a national level. It is supplied by the Italian manufacturer DUKA, among others. There are numerous private label sales and it would be out of scope to allocate sales by private labels. Therefore, the market share in the BRG report shows sales by manufacturers' brand, which also include private labels sold through wholesalers.

The share of DIY products is high, with most products being manufactured in China. On a local or regional level, traditional manufacturers increasingly face competition from window and glass installer companies. These sales are not included in the BRG market share. Conventional shower cubicles either come as high-end products or very inexpensive products. Low-end products are available in DIY stores and food discounters such as ALDI and LIDL. Supply of hydrotherapy cubicles comes, among others, from specialist hydrotherapy product producers. In this case, the product is more likely to be of superior quality with prices that by far exceed the hydrotherapy cubicles that are offered online and in DIY stores.

Forecasts

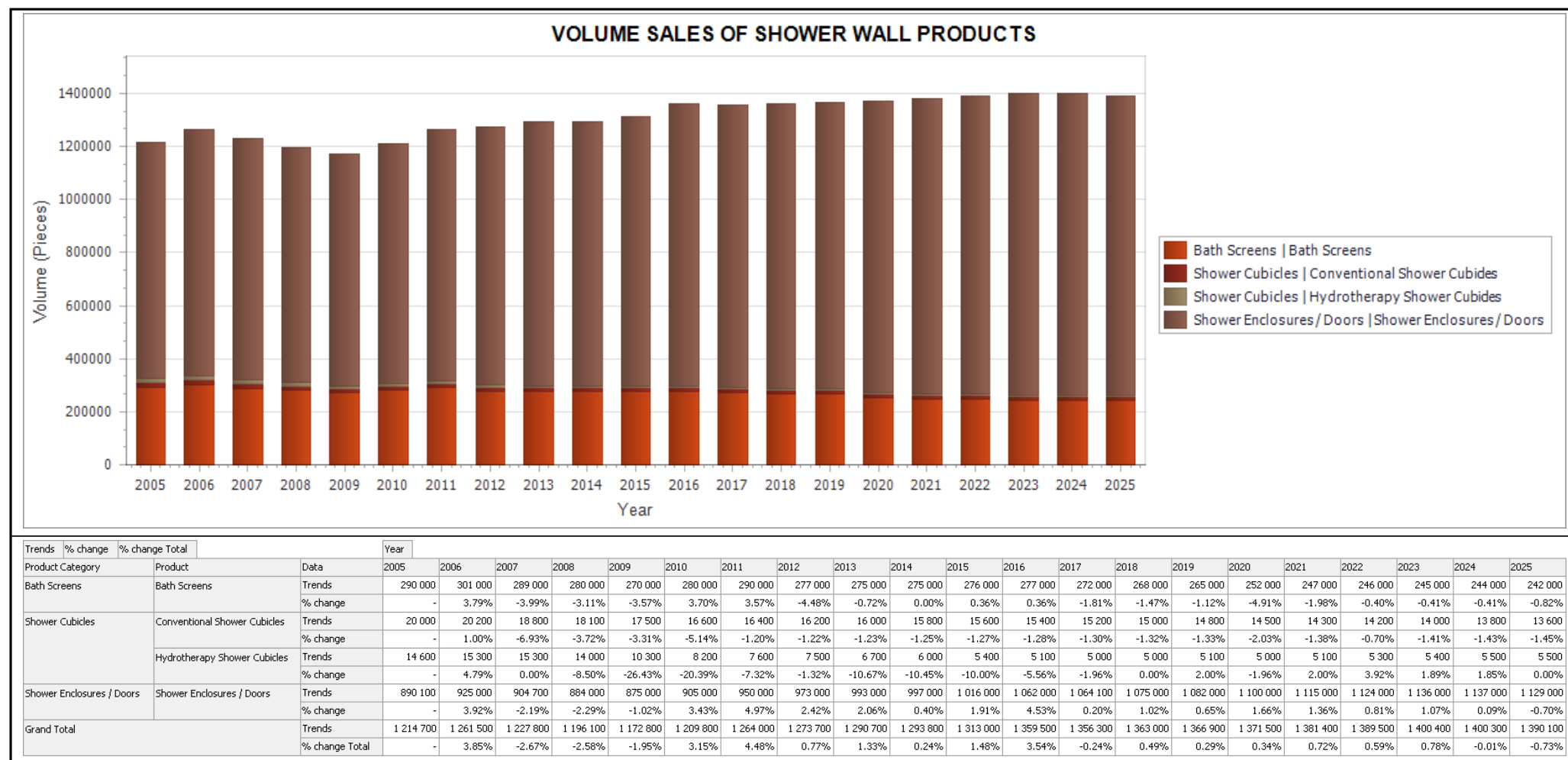
It is believed that the increase in residential construction, which is expected from 2021 onwards, should positively affect demand for shower enclosures. Demand for shower enclosures is estimated to grow on a small scale until 2024, before turning slightly negative in 2025. This development is linked to demand from the new housing segment, which accounts for a significant share of the total market. At the same time demand from RMI is believed to stay positive, although with a slightly smaller growth rate when compared to 2020. Demand from non-housing is believed to remain flat during the forecast period and sales are believed to fluctuate marginally. Bath screens and conventional shower cubicles are likely to remain declining markets during the forecast period.

Most demand is expected to come from the RMI sector, on which the German shower wall market highly depends. RMI activities are expected to remain positive throughout the forecast period. The need for new bathrooms and the conversion of old baths into larger showers is likely to be on high level and their demand is expected to fuel sales in the shower wall market. The expected increase of sales in the shower wall market is driven by only one product, which is the shower enclosure. The decreasing penetration of baths in new and existing buildings is expected to negatively influence the sales of bath screens, in favour of shower enclosures. This is true because baths are no longer used for showering, which makes bath screen sales decrease for both RMI and new housing. As far as conventional shower cubicles are concerned, their demand is expected to decrease further towards the end of the forecast period. The reason for this is that they are outdated products which are only used if there is no other option such as shower trays or walk-in showers. The popularity of the walk-in shower continues to rise in the new housing and replacement segment, although at a slower growth rate. This continuous trend accounts for a large share of all shower installations in new bathrooms in Germany. Additionally, there is a clear trend towards extra-large showers, which require modern designs for shower enclosures. Manufacturers must keep in mind these trends to come up with solutions to avoid losing shares to local glass manufactures. One of the answers are frameless shower enclosures, which are likely to gain further share in near future.

It is expected that the German shower wall market is likely to benefit from the baby boom generation from the 1950s who are in the process of retiring and who are more likely to make investments. Demand for hydrotherapy shower cubicles is believed to have reached its turning point, slowly growing sales are expected with a shift towards the upper and luxury price segments.

4.2 Historical Trends and Forecasts : Shower Wall

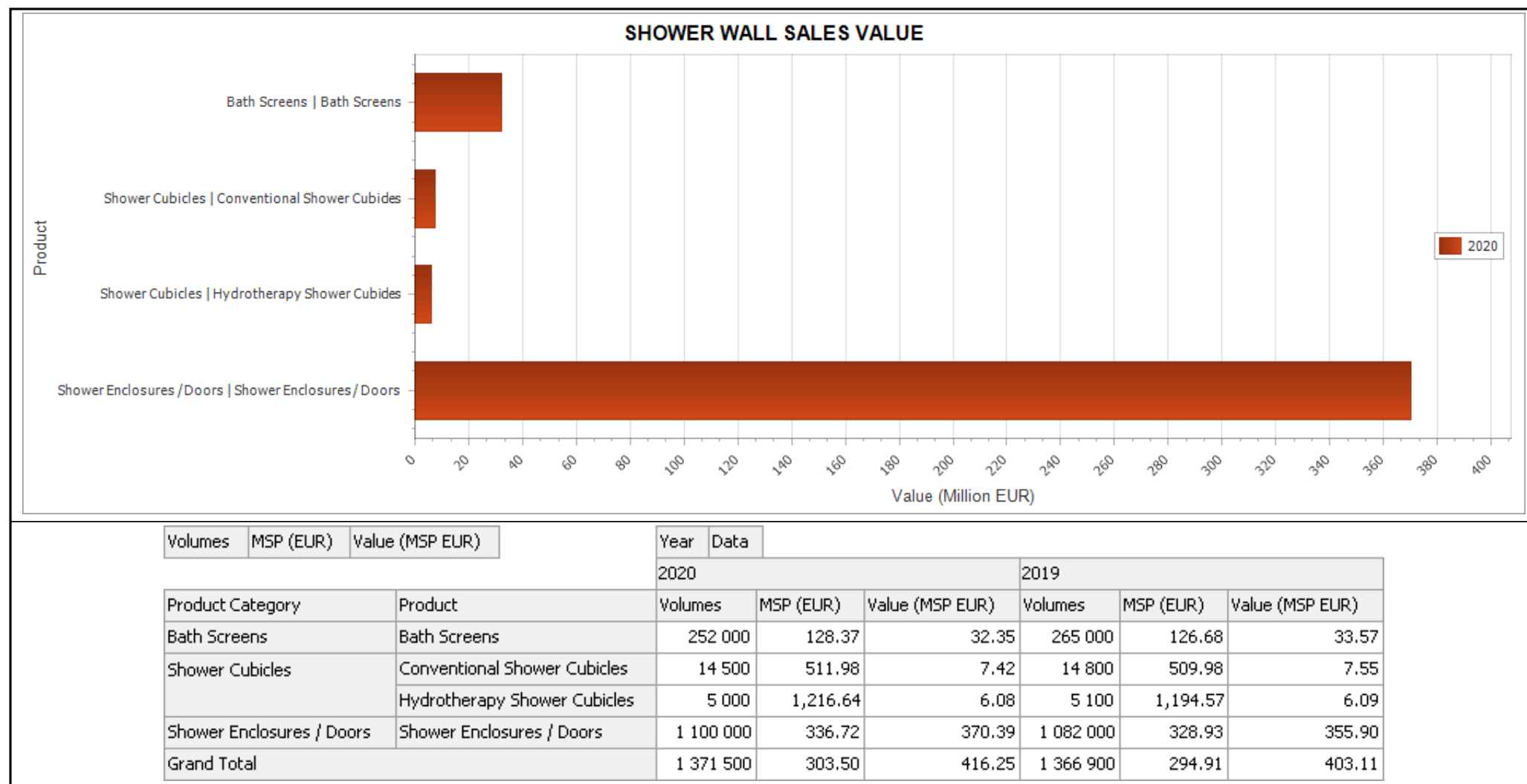
DE - SUMMARY OF TRENDS IN THE SHOWER WALL MARKET : HISTORICAL AND FORECASTS



Source: BRG based on industry interviews.

4.3.1 Prices and Market Values : Shower Wall

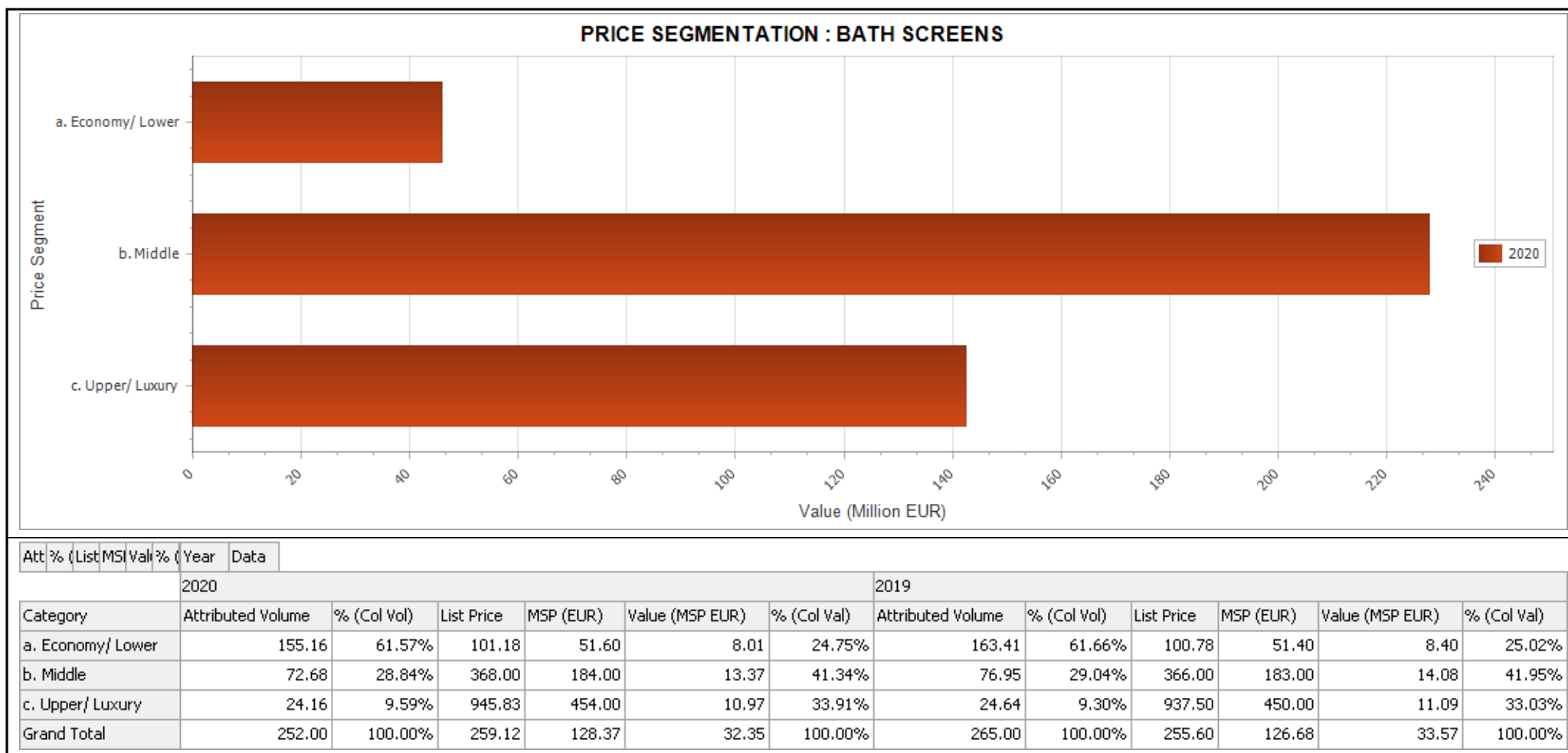
DE - ESTIMATED VALUES OF THE SHOWER WALL MARKET



Source: BRG based on industry interviews. Values Shown in Millions (at MSP = Manufacturers' Selling Price)

4.3.2 Price Segmentation : Bath Screens

DE - ESTIMATED PRICE SEGMENTATION FOR BATH SCREENS

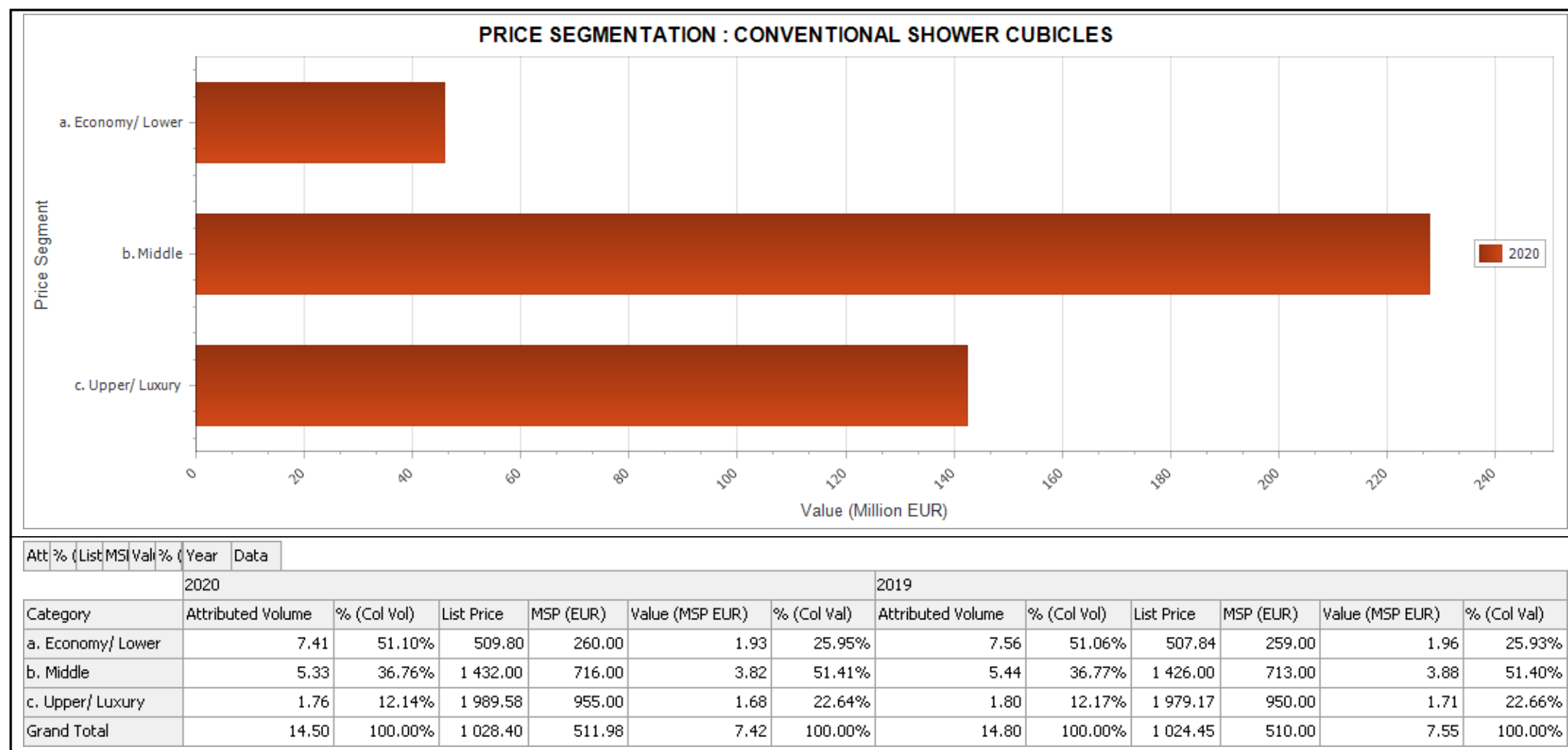


Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€125, Middle €125-€250, Upper/ Luxury >€250

4.3.3 Price Segmentation : Conventional Shower Cubicles

DE - ESTIMATED PRICE SEGMENTATION FOR CONVENTIONAL SHOWER CUBICLES

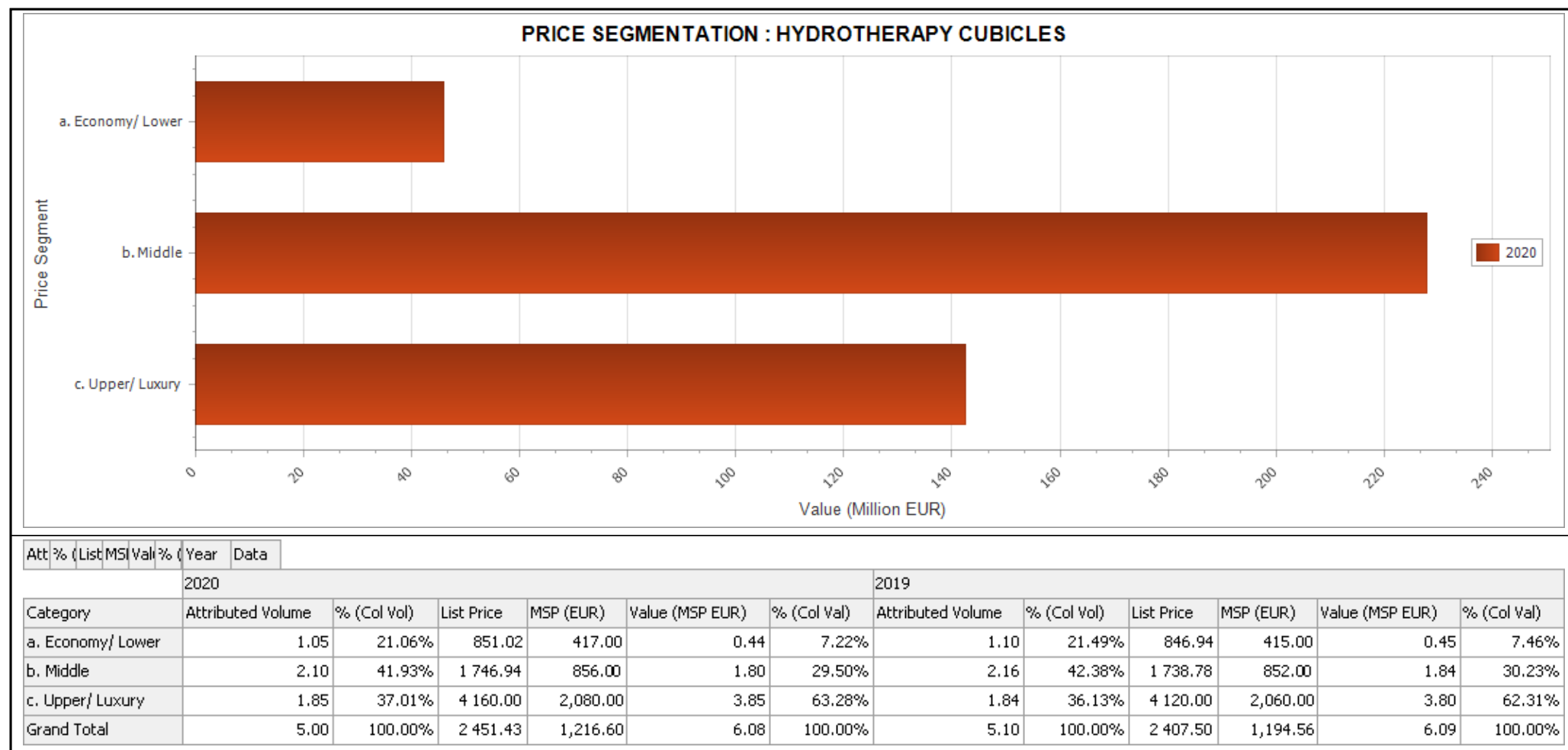


Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€500, Middle €500-€900, Upper/ Luxury >€900

4.3.4 Price Segmentation : Hydrotherapy Shower Cubicles

DE - ESTIMATED PRICE SEGMENTATION FOR HYDROTHERAPY CUBICLES

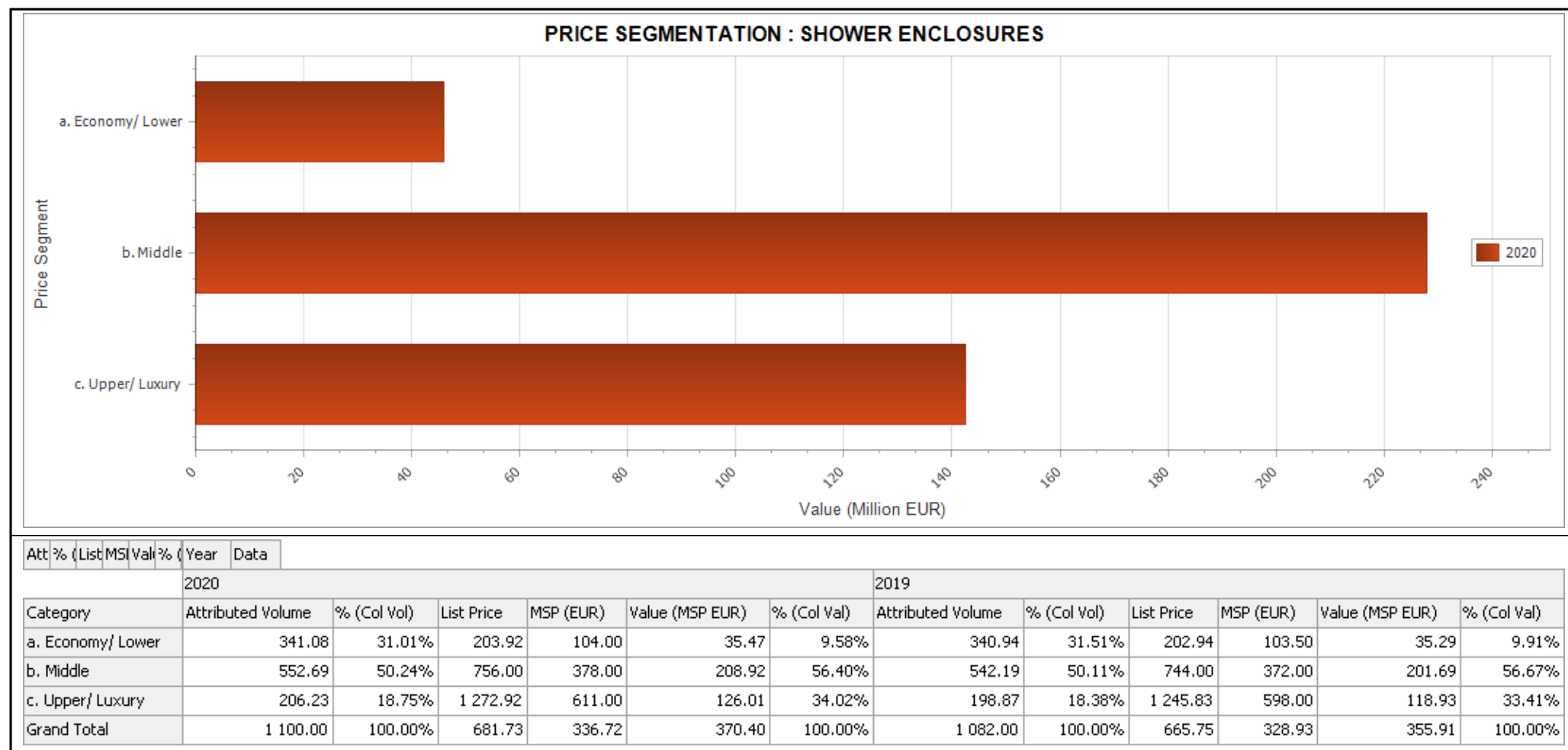


Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€600, Middle €600-€1,200, Upper/ Luxury >€1,200

4.3.5 Price Segmentation : Shower Enclosures

DE - ESTIMATED PRICE SEGMENTATION FOR SHOWER ENCLOSURES

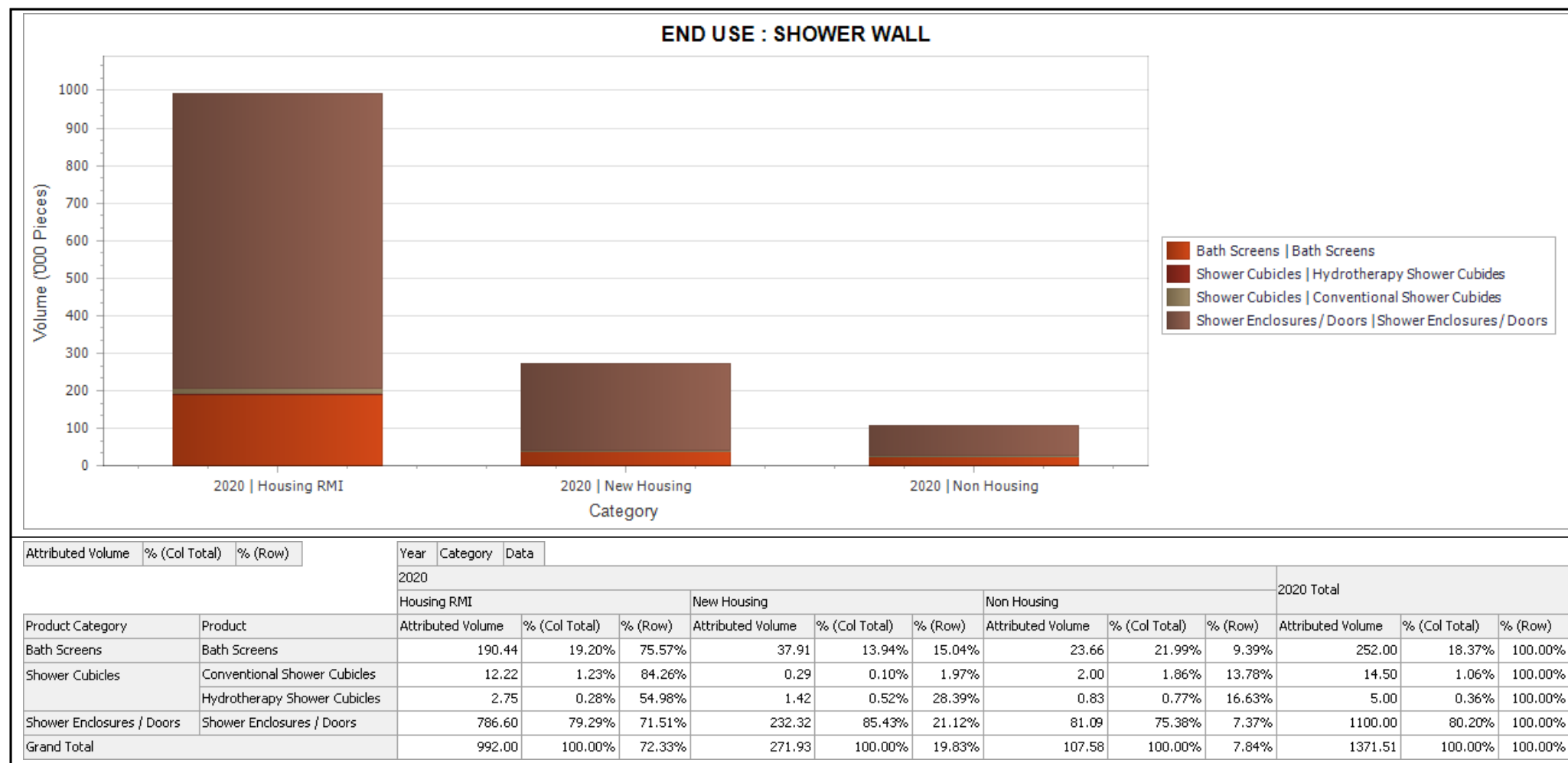


Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€250, Middle €250-€500, Upper/ Luxury >€500

4.4.1 End Use : Base Year

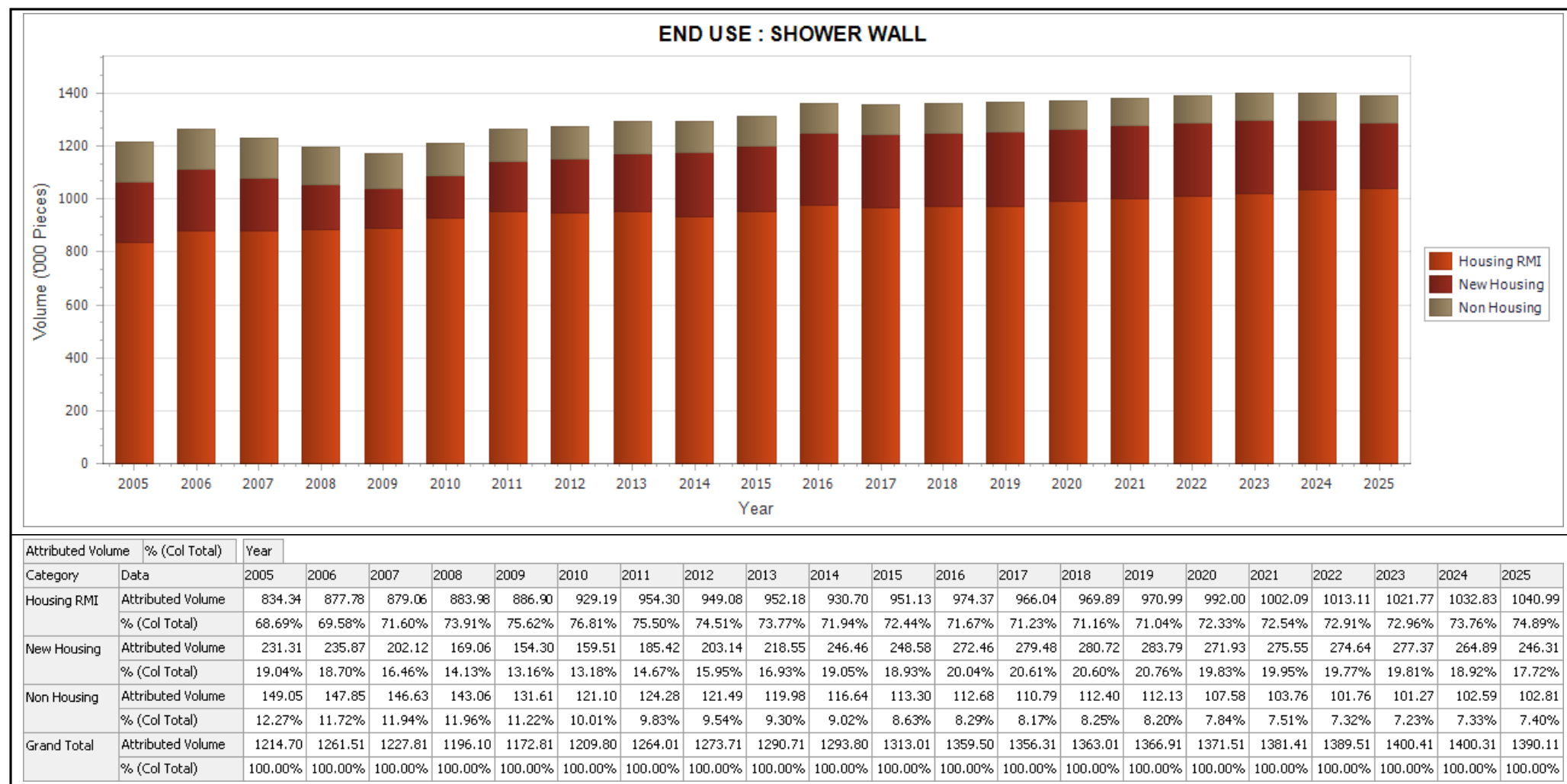
DE - SALES OF SHOWER WALL BY TYPE AND END USE SEGMENT ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

4.4.2 End Use : Historical and Forecasts

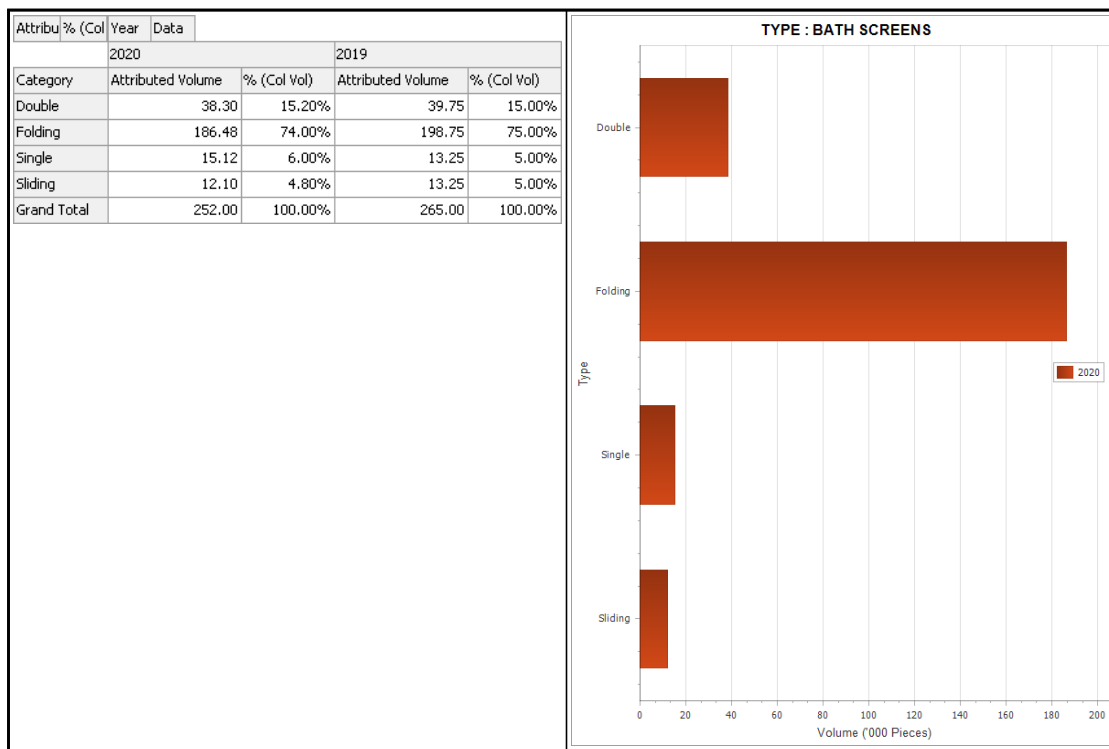
DE - SALES OF SHOWER WALL PRODUCTS BY END USE SEGMENT ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

4.4.3 Sales by Type : Bath Screens

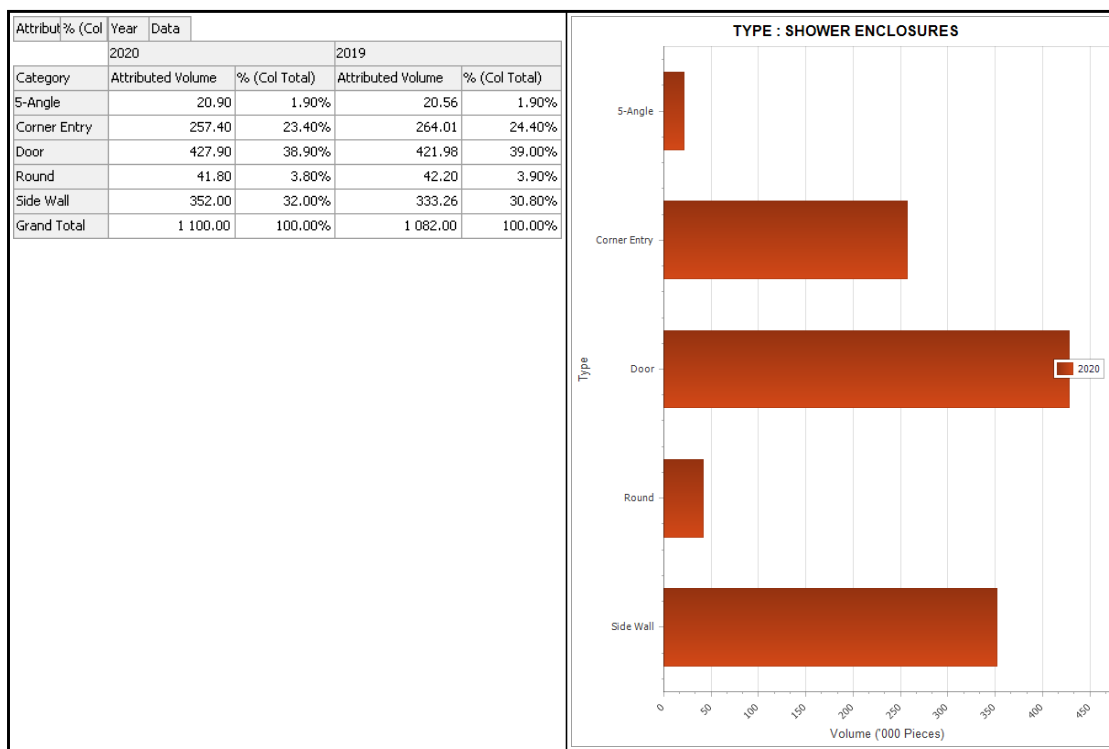
DE - BATH SCREEN SALES BY TYPE ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

4.4.4 Sales by Type : Shower Enclosures

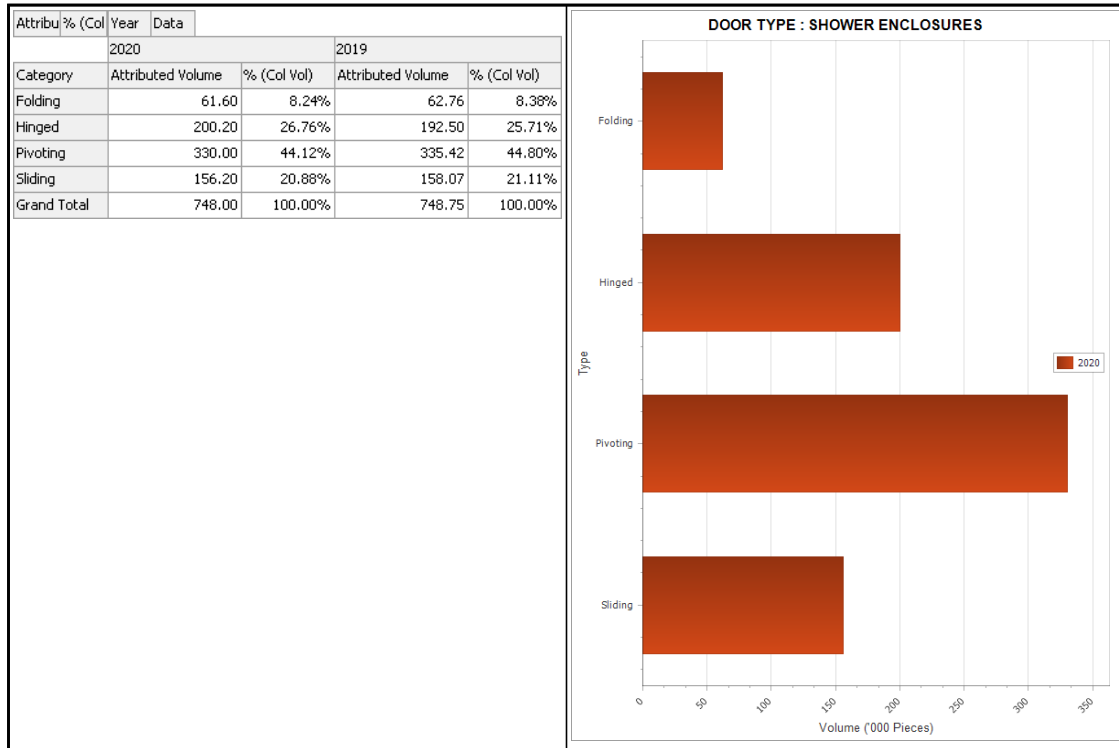
DE - SHOWER ENCLOSURE SALES BY TYPE ('000 PIECES)



Source: BRG based on industry interviews

4.4.5 Sales by Type of Door : Shower Enclosures

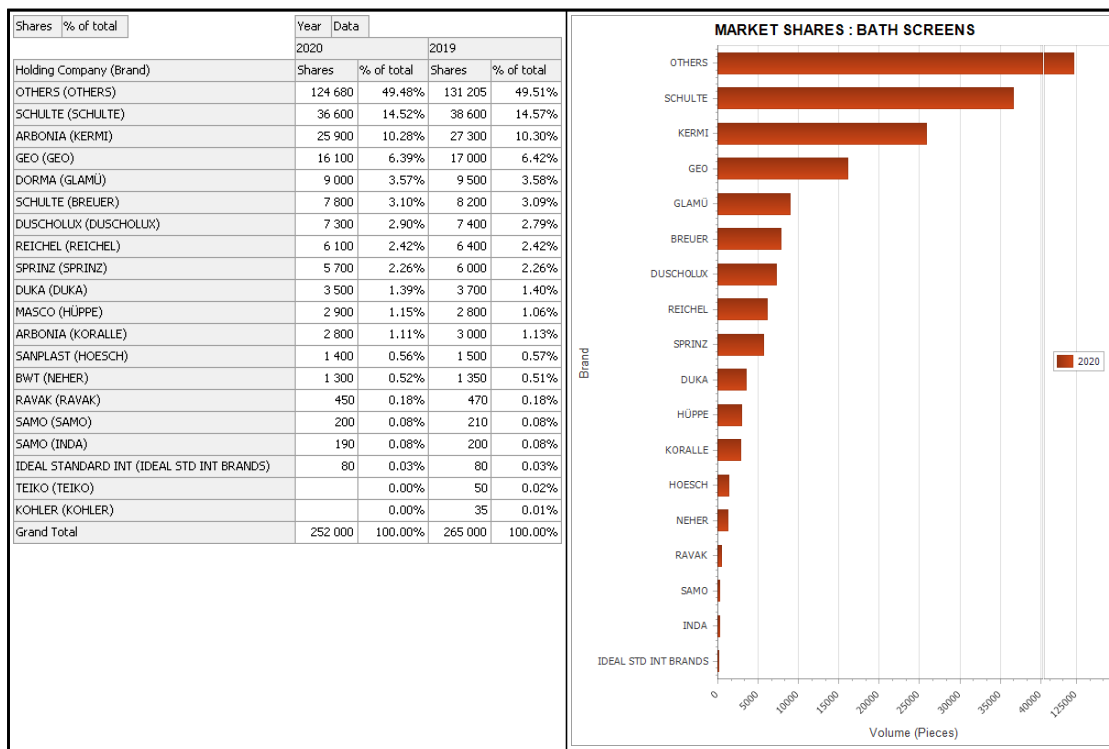
DE - SHOWER ENCLOSURE SALES BY DOOR TYPE ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

4.5.1 Market Shares : Bath Screens

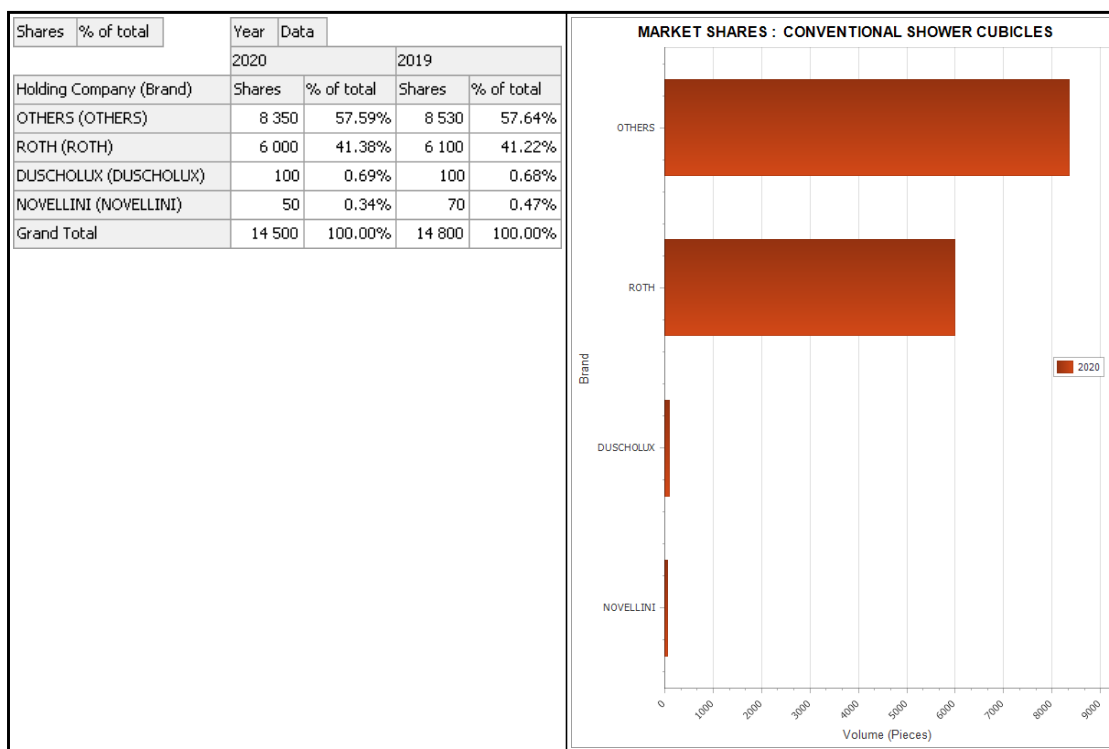
DE - ESTIMATED MARKET SHARES FOR BATH SCREENS



Source: BRG based on industry interviews.

4.5.2 Market Shares : Conventional Shower Cubicles

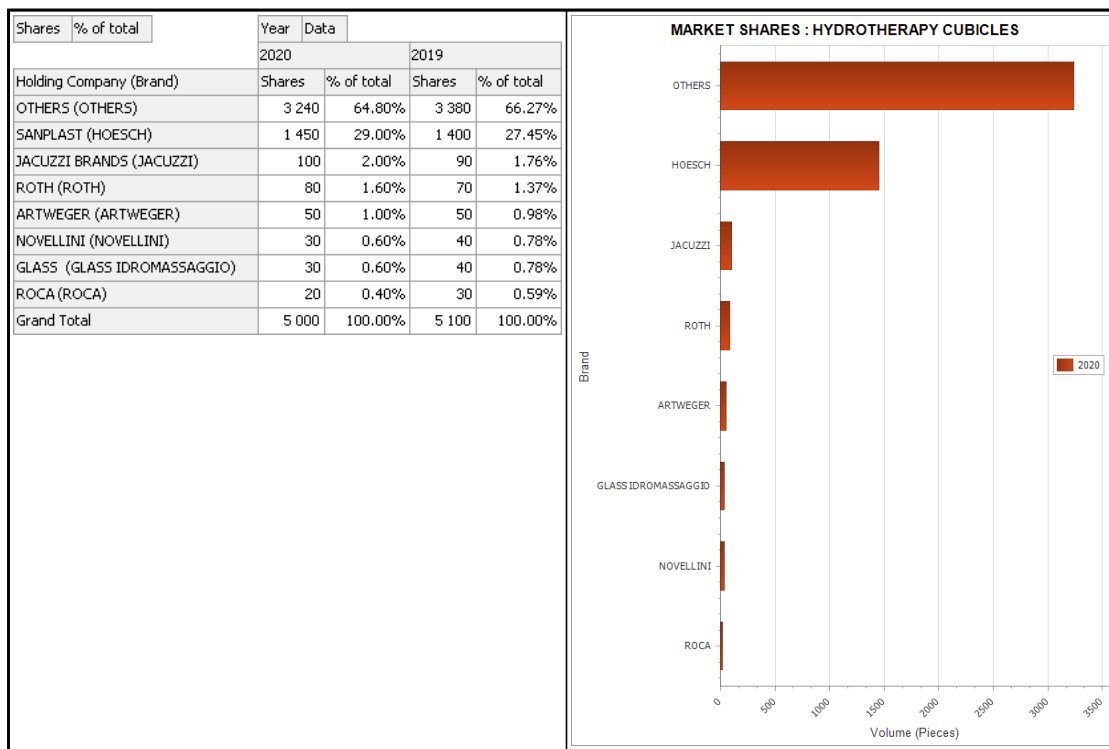
DE - ESTIMATED MARKET SHARES FOR CONVENTIONAL SHOWER CUBICLES



Source: BRG based on industry interviews.

4.5.3 Market Shares : Hydrotherapy Shower Cubicles

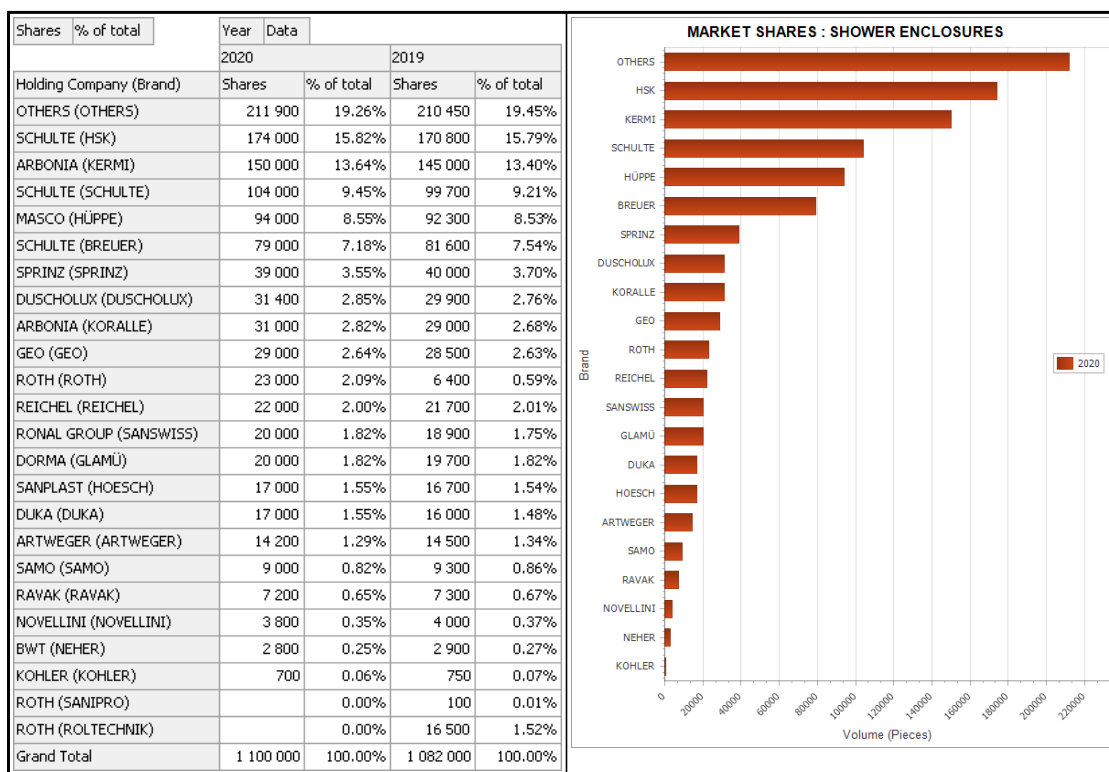
DE - ESTIMATED MARKET SHARES FOR HYDROTHERAPY CUBICLES



Source: BRG based on industry interviews.

4.5.4 Market Shares : Shower Enclosures

DE - ESTIMATED MARKET SHARES FOR SHOWER ENCLOSURES



Source: BRG based on industry interviews.

4.6.1 Route to Market: Shower Wall
DE - ESTIMATED DISTRIBUTION FOR SHOWER WALL: 2020

Manufacturer / Importer	Manufacturers / Importers
	100.0%

Wholesale Level	53.1%
	Wholesalers / Merchants
	53.1%

Retail Level	Manufacturer	Wholesaler	Manufacturer	Wholesaler	Manufacturer	Wholesaler
	7.3%	3.6%	9.1%	7.0%	1.8%	0.2%
	Internet Retailers		DIY Retailers		Specialist Retailers	
	10.9%		16.1%		2.0%	

Manufacturer	Wholesaler	Internet	DIY	Specialist
17.1%	42.0%	3.0%	1.2%	0.1%
Installers				
63.4%				

End Use Level	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer
	8.6%	0.2%	6.9%	14.9%	1.8%	59.8%
	Residential					
	92.2%					
	New Build					

19.8%		Residential RMI		72.3%	
Project	Non-Project	Project	Non-Project	Project	Non-Project
8.0%	11.8%	2.0%	70.3%	2.0%	70.3%

Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer
3.1%	0.1%	1.0%	0.0%	0.1%	3.6%
Non-Residential					
7.8%					
New Build					

2.8%		Non-Residential RMI		5.0%	
Project	Non-Project	Project	Non-Project	Project	Non-Project
2.0%	0.8%	2.7%	2.3%	2.7%	2.3%

4.6.2 Route to Market: Shower Wall
DE - ESTIMATED DISTRIBUTION FOR SHOWER WALL: 2025

Manufacturer / Importer	Manufacturers / Importers
	100.0%

Wholesale Level	52.3%
	Wholesalers / Merchants
	52.3%

Retail Level	Manufacturer	Wholesaler	Manufacturer	Wholesaler	Manufacturer	Wholesaler
	8.0%	5.0%	6.0%	6.5%	1.7%	0.1%
	Internet Retailers		DIY Retailers		Specialist Retailers	
	13.0%		12.5%		1.8%	

Manufacturer	Wholesaler	Internet	DIY	Specialist
16.4%	40.3%	4.2%	1.0%	0.2%
Installers				
62.1%				

End Use Level	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer
	11.6%	0.3%	7.4%	11.5%	1.6%	60.2%
	Residential					
	92.6%					

Source: BRG Based on Industry Interviews. - Totals may vary due to roundings

4.6.3 Route to Market - Consolidated View : Shower Wall

DE - ESTIMATED DISTRIBUTION FOR SHOWER WALL : 2020

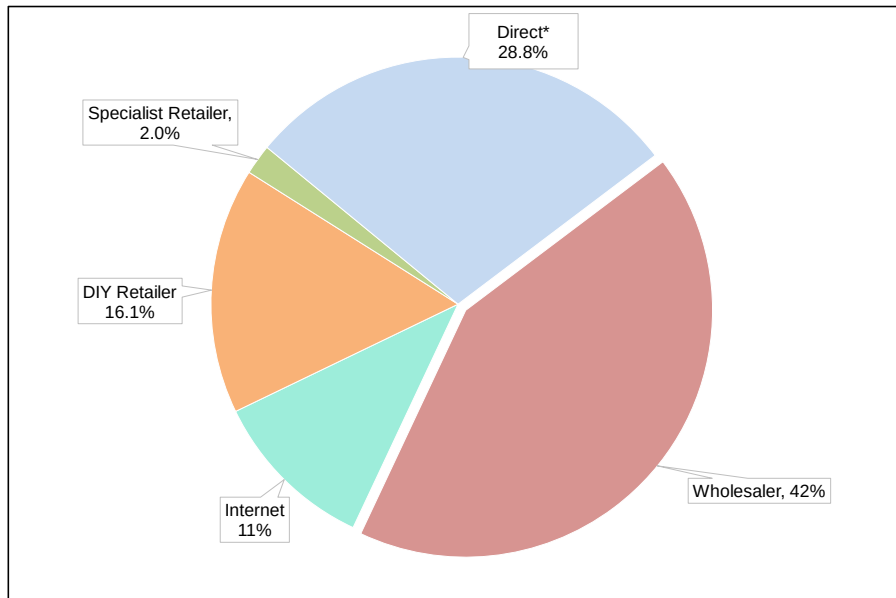
Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
42.3%	10.9%	16.1%	2.0%	28.8%	10.0%	4.7%	82.2%	3.1%
100%					14.7%		85.3%	

4.6.4 Route to Market - Consolidated View : Shower Wall

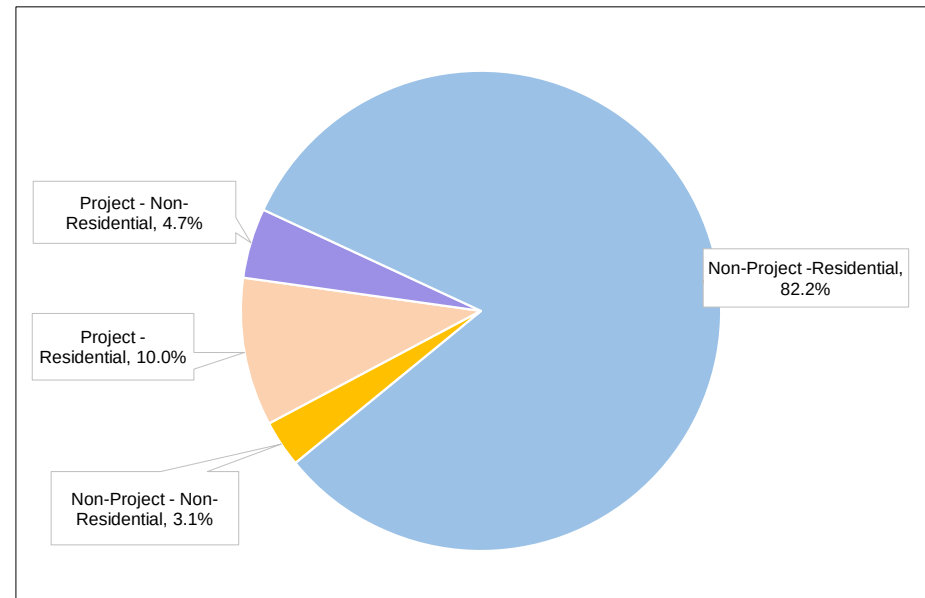
DE - ESTIMATED DISTRIBUTION FOR SHOWER WALL : 2025

Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
40.7%	13.0%	12.5%	1.8%	32.0%	10.0%	4.0%	82.6%	3.4%
100%					14.0%		86.0%	

DE - SALES OF SHOWER WALL BY DISTRIBUTION CHANNEL: 2020



DE - SALES OF SHOWER WALL BY POINT OF DECISION AND END-USE SECTOR: 2020



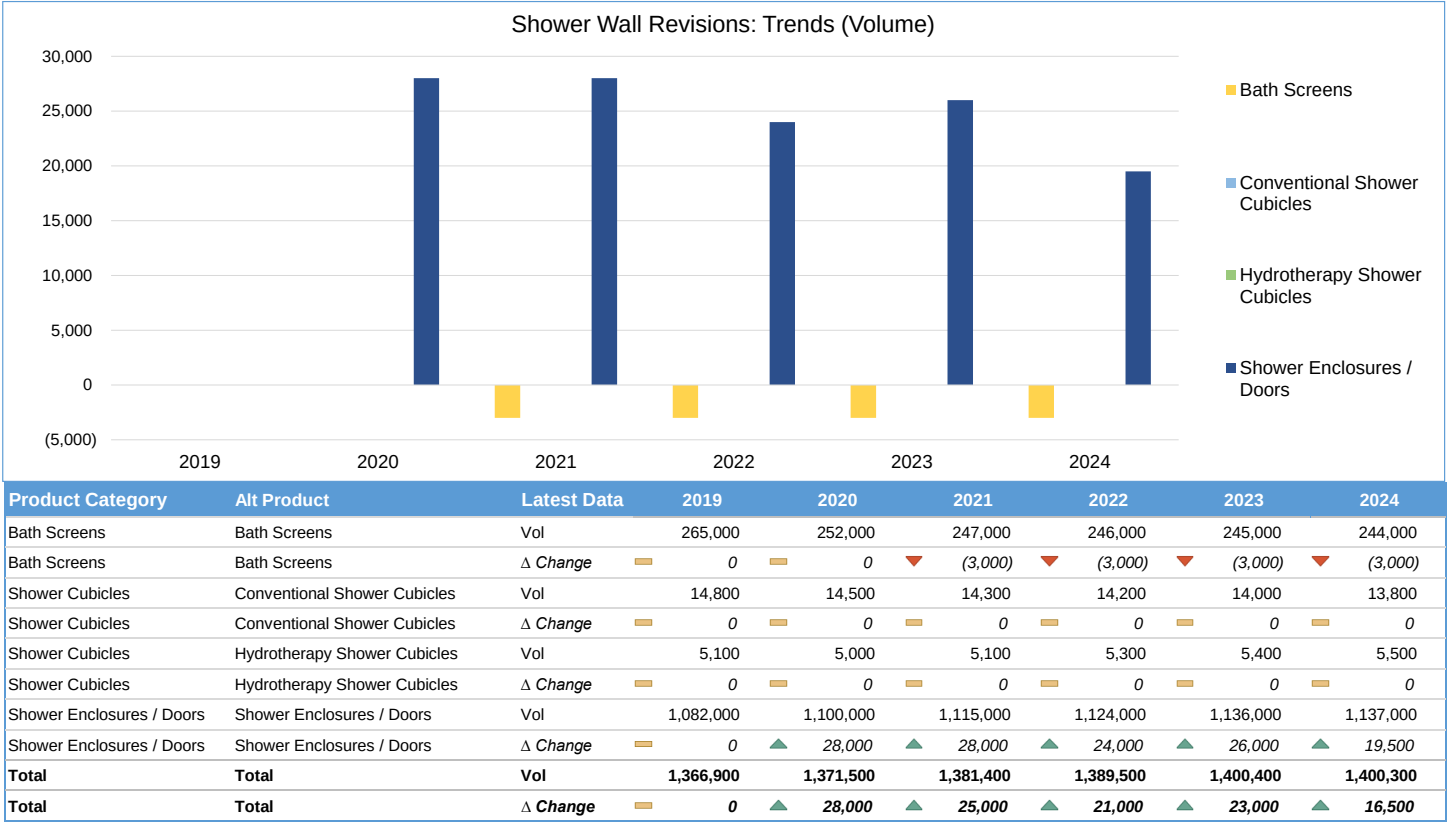
Source: BRG Based on Industry Interviews.

*Direct includes sales going directly from manufacturers to installers, contractors and private customers.

4.7 Revisions in the German Shower Wall Market (April 2021)

DE - REVISIONS IN THE GERMAN SHOWER WALL MARKET

Data Comparison: October 2020



No revisions have been made to historical volumes for this market.

GERMANY

BATHROOM FURNITURE

THE EUROPEAN BATHROOM & KITCHEN PRODUCT MARKETS



Market Characteristics and Technological Trends

Initially a purely functional space, without room for creativity when compared to the living room for example, the bathroom has gradually become a showpiece of the house. Nowadays, the bathroom represents the flagship of quality and a high lifestyle for not only houses and, in a minor part, flats, but also for hotels, restaurants and other non-residential buildings. Bathroom furniture products play a central role in this process, due to their characteristic of strong eye-catching impact and the fact that they are more keenly replaced than other bathroom products such as ceramic sanitary ware, which are distinguished by a long average lifespan. The overall bathroom furniture market is heavily dependent on replacement sales. Even if the German bathroom furniture market had started to develop already in the late 1970s, it initially showed only small growth potential, which accelerated in the late 1980s and early 1990s, and is also connected to the underlying trend spreading in German bathrooms of having base units with storage space instead of old wall hung washbasin models and pedestals.

To take advantage of the recent trend among German consumers to invest in their bathroom, manufacturers of bathroom furniture focus carefully on product innovations that lead to increasing demand, with some of them establishing a market, e.g. improved lighting solutions inside drawers. Moreover, an increasing number of mirrors and mirror cabinets nowadays are equipped with sophisticated lighting solutions, such as LED lighting inside the glass. Another noteworthy growing trend is bringing technology into the bathroom, to improve the whole 'bathroom experience' with, for example, integrated touchscreen radio or Bluetooth. Demand for all types of bathroom furniture is growing and it underlines the trend of having more storage space in the bathroom. German clients prefer spacious solutions with a monobloc washbasin unit for their main bathroom, while base units with a small washbasin are mostly installed in guest toilets. The reason for the indisputable popularity of the combination of base units sold with a monobloc countertop is due to the fear that, long term, water may damage the wood of the furniture. A relevant trend present in the mirror and mirror cabinet market sees the share of mirror cabinets, which are built into the wall, increasing further. These models, which are counted by BRG as mirror cabinets, have the advantage of having the same shape as a mirror but, at the same time, offering storage space. In some modern bathrooms stylish openings are made in the wall to be used for storage space. Such storage solutions are likely not to replace bathroom furniture and instead might reduce the need for bathrooms accessories only.

Recent Market Trends

The COVID-19 pandemic shaped the developments in the year 2020 and 'business as usual' was replaced by government measures. In Germany the official lockdown, which is better understood as a 'prohibition of contact', started at the end of March 2020. Bavaria was the first state to implement measures to contain the spread of the virus, which was followed by the remaining states towards the end of

March. For a few weeks the German population was advised to work from home when possible, with most non-essential stores closing for a limited time only. Measures implemented varied by state (*Bundesländer*). Most stores, including DIY stores, were allowed to open again from the beginning of April 2020 onwards, with differences from state to state in the level of restrictions and measures. In certain regions, shopping in DIY stores was limited to professionals only.

During 2020, new housing construction continued without much interruption, with only large scale project sites halted to implement protective measures. That being said, the overall production per employee is believed to have shrunk significantly, and remains to be the case, since many employers introduced measures to protect the construction workers. The bathroom industry was declared part of the crisis infrastructure, but installers could still continue to work without many interruptions. Overall, there has been relatively little impact of the COVID-19 pandemic on the building related industry, including the bathroom market, in Germany. In some cases, end users may have postponed, or cancelled, bathroom related projects due to the uncertainty on the market, but overall demand remained relatively strong, as there was a large backlog of works from the past, benefiting sales of bathroom and kitchen equipment. At the same time, in certain instances, the COVID-19 pandemic may have contributed to additional demand. Since consumers were spending more time indoors, they had the opportunity to dedicate themselves to repair or upgrade works, whereas others might have decided to invest into their property by installing a new bathroom or kitchen due to (international) holidays being difficult because of the current pandemic.

The German bathroom furniture market increased in 2020 with a slightly stronger growth rate when compared to the previous year. Demand was significantly driven by the RMI segment, which compensated for negative new and non-housing sales. The residential RMI segment benefited from the COVID-19 pandemic in a way that Germans were spending more time at home while still being financially secured. As a result, greater demand for bathroom furniture was recorded from the DIY segment and e-commerce. Germans invested into their homes and the drive towards more wellness and space in the bathroom benefited overall demand.

Base units recorded the stronger increase, followed by separate cabinets and mirrors and mirror cabinets. Separate cabinets are not considered to be an essential item in a bathroom due to lack of space and, therefore, they have slightly lower growth rates than base units. There is a trend towards base units with mineral cast counter tops, which are gaining share at the expense of ceramic washbasins. Mineral cast counter tops are usually offered by manufacturers who do not produce their own ceramic sanitary ware. They are found in DIY and online shops and compete with lower prices when compared to ceramic products. Most bathroom furniture is attached to the wall to facilitate cleaning from underneath. German consumers prefer a simple design that is mainly white in colour with polished surfaces. Regarding the colours of bathroom furniture, there is an increasing demand for models with oak design, which is a natural wood colour in different tones. There are different designs and the trend is more towards a rustic feel, with a natural-like texture on the furniture surface. According to industry

feedback, about half of the furniture is white, followed by different tones of black and grey and then oak with natural colours.

Regarding the type of washbasins, there are mainly three different solutions on the Germany bathroom market. Most hotels use a countertop basin, which is a washbasin that sits on top of a counter, which is usually made from wood. This style has found its way to the residential market and modern bathrooms are increasingly equipped with a countertop basin. This development is taking away share from the classical monobloc countertop, which by far has the highest share. The third most popular solution is the washbasin only. The total area at the top of the base unit is taken up by a washbasin.

Prices and Market Values

Prices in the bathroom furniture market increased across all products in 2020. The price increase was driven by higher raw material prices, which were consequently passed on to consumers. There is a trend of German end users to move away from the middle-priced segment towards the economy and upper price range. In the past years there has been a strong increase of market players in Germany from manufacturers that mainly focus on housing furniture. They started to include bathroom furniture into their portfolio. German end users who can afford it, invest more into good quality products. These products are less expensive than luxury lines that are presented at trade fairs, such as chic real wood editions. Demand for luxury products is only limited and consumers tend to spend a little bit less for a high-quality product.

At the same time, there is an increasing number of manufacturers that introduced bathroom furniture at the economy and lower price range. With a growing supply outside the traditional wholesale level, end users no longer need to buy bathroom furniture from professional installers. Instead, they can make use of alternative distribution channels such as e-commerce and DIY stores to look for cheaper solutions. Therefore, the share of the economy/lower price segment increased more in 2020. An exception is separate base units, which are more likely to be installed in larger bathrooms, where homeowners would prefer to invest into quality products. The MSP for mirrors and mirror cabinets increased the strongest as end users prefer modern large mirrors with built-in LED lights. These products are gaining market shares and are replacing simple mirrors and traditional mirror cabinets.

New Housing

In 2020, the new build market developed slightly negatively for residential apartment construction and for one- and two-family homes. Sales in new housing have decreased accordingly across all the categories of the bathroom furniture market. In terms of the product mix, separate cabinets are less likely to be part of a bathroom and they have a smaller share of total bathroom furniture sales in new housing. The reason for this is that empty floor space is needed to fit a separate cabinet and most bathrooms lack available space. According to the 7th

fundamental bathroom study the average size of a German bathroom with WC increased from 7.8 m² in 2006 to 9.1 m² in 2017. Rented apartments with 8.2 m² average size are the smallest, followed by apartment owners with 8.6 m² and house owners with 10.3 m². This development towards larger bathrooms is likely to benefit the bathroom furniture market.

According to industry feedback, not only is the main bathroom getting larger, there is a tendency to have two bathrooms installed in a dwelling. In addition of having one main bathroom, a smaller en-suite bathroom is added to the master bedroom, or a guest toilet is replaced with a smaller bathroom. In both cases, it is likely that bathroom furniture will be installed. Base units account for a similar volume as mirrors and medicine cabinets. Demand for all bathroom furniture products decreased in alignment with fewer new housing completions. From 2021 until 2023, demand from new housing is expected to grow, before turning negative in 2024 and 2025.

Housing RMI

The largest part of demand for bathroom furniture products comes from the RMI market. In 2020, sales of all bathroom furniture products were positive. Most German consumers had more free time, while financial support during the COVID-19 pandemic was secured by government funds. Overall, demand from alternative distribution channels such as DIY and the Internet had a stronger than average performance. Mirrors and mirror/medicine cabinets had the biggest growth rate, followed by base units and separate cabinets. The renovation segment tends to favour sales of higher priced products and there is a trend to purchase high quality products on the Internet. At the same time, there is a growing supply of bathroom furniture outside the traditional sanitary industry and end users can choose from different distributions channels. There is a greater choice of more affordable products available outside the traditional three-step distribution. Installers are less likely to be interested in selling bathroom furniture. They might have only a limited selection of products as it is not their priority business. Therefore, they might just install everything behind the wall and the end user buys the furniture elsewhere. For example, furniture shops have a greater variety and they are cheaper, too. Moreover, furniture shops are not affected by lack of professionals and can provide installation service once the bathroom has been finalised by the installer. Overall, demand for bathroom furniture is expected to remain positive throughout the forecast period.

Non-Housing

Total non-residential construction output strongly decreased in 2020 as a result of the COVID-19 pandemic. The only exception is investments into buildings for education and health, which had a slightly positive development. All other sectors such as industrial, hospitality, office and commercial buildings had a strong decline as projects got postponed or cancelled due to economic uncertainty. This development had a negative effect on the German bathroom furniture market. Demand for non-housing bathroom furniture mainly originates in the recreational

sector in places such as hotels, healthcare buildings, nursing homes and spa and wellness facilities. In 2020, these sectors were hit hardest by the lockdown restrictions and any new or upcoming renovation projects were likely to be put on hold. Demand for base units, mirror cabinets and separate cabinets was negative in 2020. The non-housing segment only plays a minor role and generates only a small share of total demand. For example, demand for bathroom furniture is more likely to come from privately owned hospitals, which represent only one third of total hospitals. Hospitals under public ownership have a limited disposable budget and their bathrooms would be equipped with only the necessary. In contrast, private hospitals and medical wellness facilities often have well-equipped rooms with bathrooms that make you feel like home.

Supply Structure

The main players are domestic manufacturers such as PELIPAL, FACKELMANN and STEINHOF (PURIS). The German bathroom furniture market is very fragmented, with many active players. The competition is high with new participants trying to compete for a limited pool of consumers. For example, wholesalers use their local presence to promote OEM products that they supply under private labels. With the growing share of alternative distribution channels, the market share of manufacturers that supply through wholesalers has been decreasing. Few companies are bathroom furniture specialists, e.g. DURAVIT and BURGBAD. Instead, most companies are also involved in the manufacturing of kitchen furniture, e.g. FACKELMANN and NOBILIA or in home furniture, e.g. VOGEL AUER and SCHIEBE MEIER. Some manufacturers design their base units to fit a brand of ceramic sanitary ware, for example. They collaborate with other bathroom manufacturers to provide a complete set. With a tightening bottleneck of installers, local carpenters saw their opportunity to supply bathroom furniture. Like glass manufacturers competing with shower wall suppliers, carpenters gained market share on a marginal scale. Another example in the mirror and mirror cabinet market are specialists for mirrors that are not producing other bathroom furniture items but are gradually increasing their penetration in the market.

According to industry feedback, furniture shops are losing slightly their share of the total bathroom furniture market. The reason is that they start saving on personnel that is taking care of bathroom furnishing sales. Bathroom furniture only makes up about 3% of the total furniture shop sales. Therefore, bathroom furniture is not important to them and they try to cut costs by focusing on the main products only. In contrast, DIY stores and e-commerce are gaining share disproportionately. Within the DIY stores, it is mainly the ones doing well which have competent personnel for consulting end users, e.g. GLOBUS and OBI. If a DIY lacks good service and product presentation, then sales of bathroom furniture remain low.

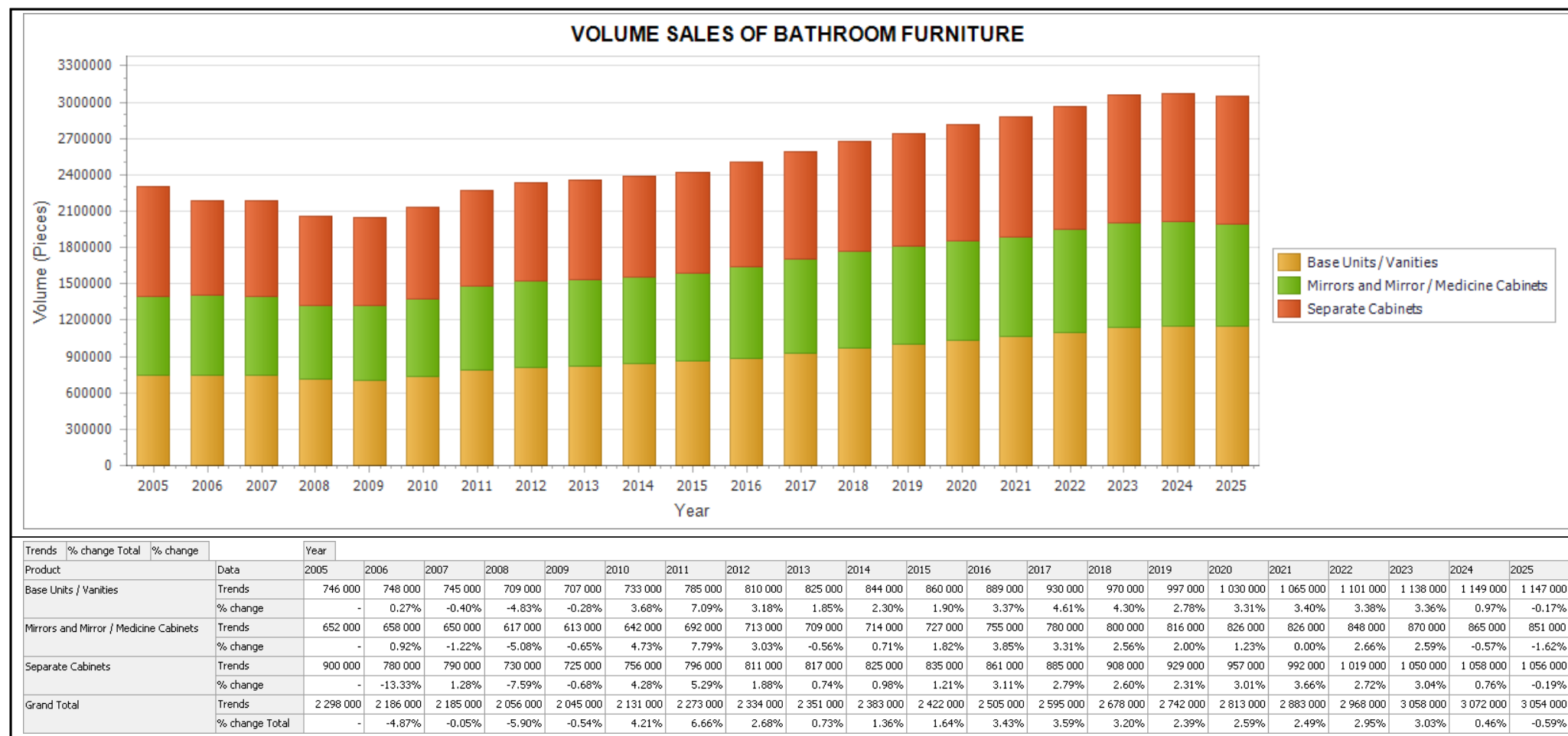
Forecasts

Demand for bathroom furniture is expected to remain positive during the forecast period, with the only exception in the year 2025. Market growth is expected to be mainly driven by new housing completions and RMI demand. Towards the end of

the forecast period new housing demand is expected to turn negative and demand from the RMI segment is likely to weaken. In terms of volume, the German bathroom furniture market highly depends on product replacements and bathroom renovations. In most new bathrooms and renovated bathrooms there is the need for storage space. At the same time there is an increasing need for wellness and beauty products, which need to be stored away. End users add furniture to their bathroom to make it homelier. For example, the use of a separate cabinet not only adds an object to your bathroom, it also provides space for candles, incense, or a souvenir.

5.2 Historical Trends & Forecasts : Bathroom Furniture

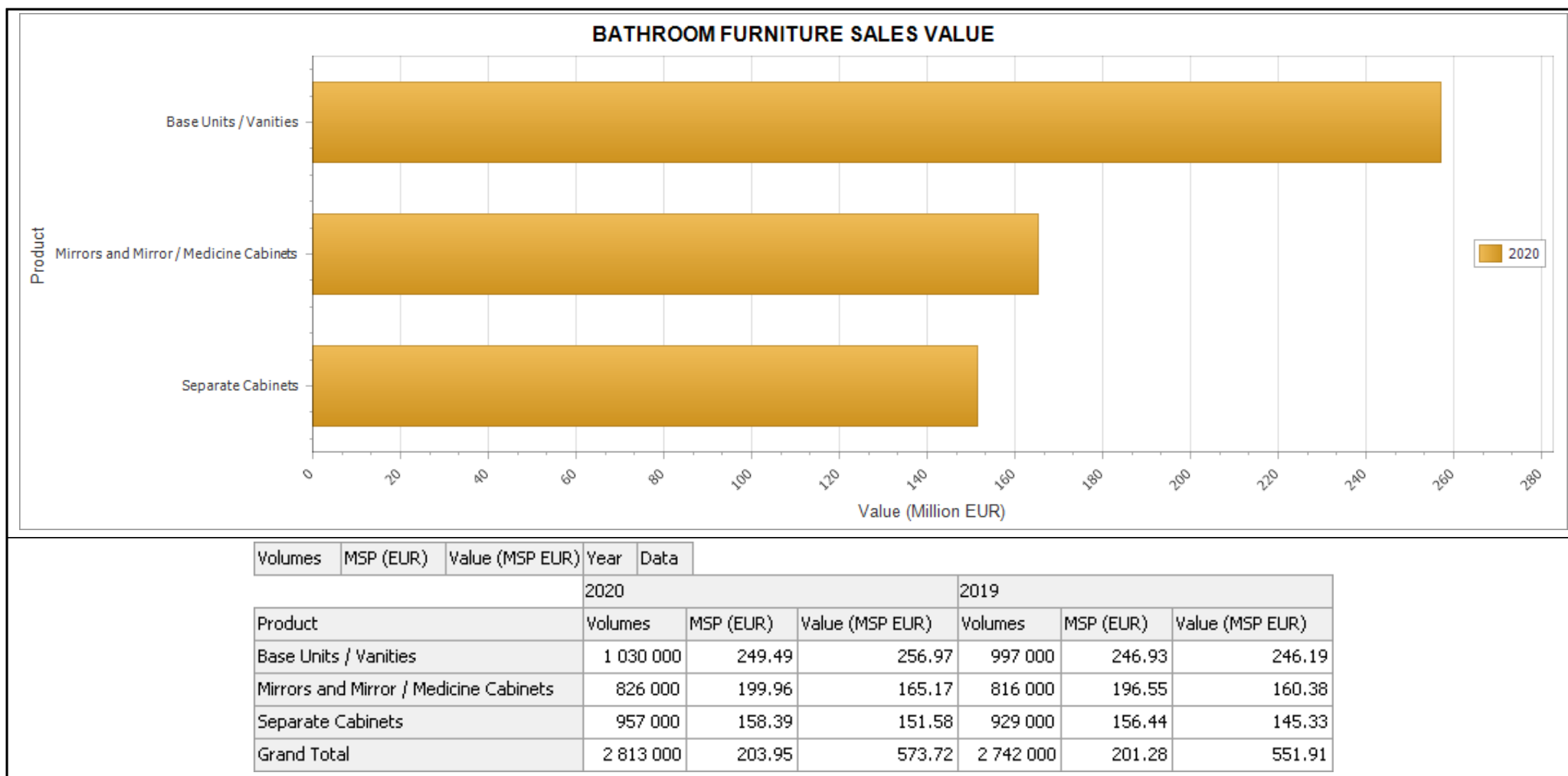
DE - SUMMARY OF TRENDS IN THE BATHROOM FURNITURE MARKET : HISTORICAL AND FORECASTS



Source: BRG based on industry interviews.

5.3.1 Prices and Market Values : Bathroom Furniture

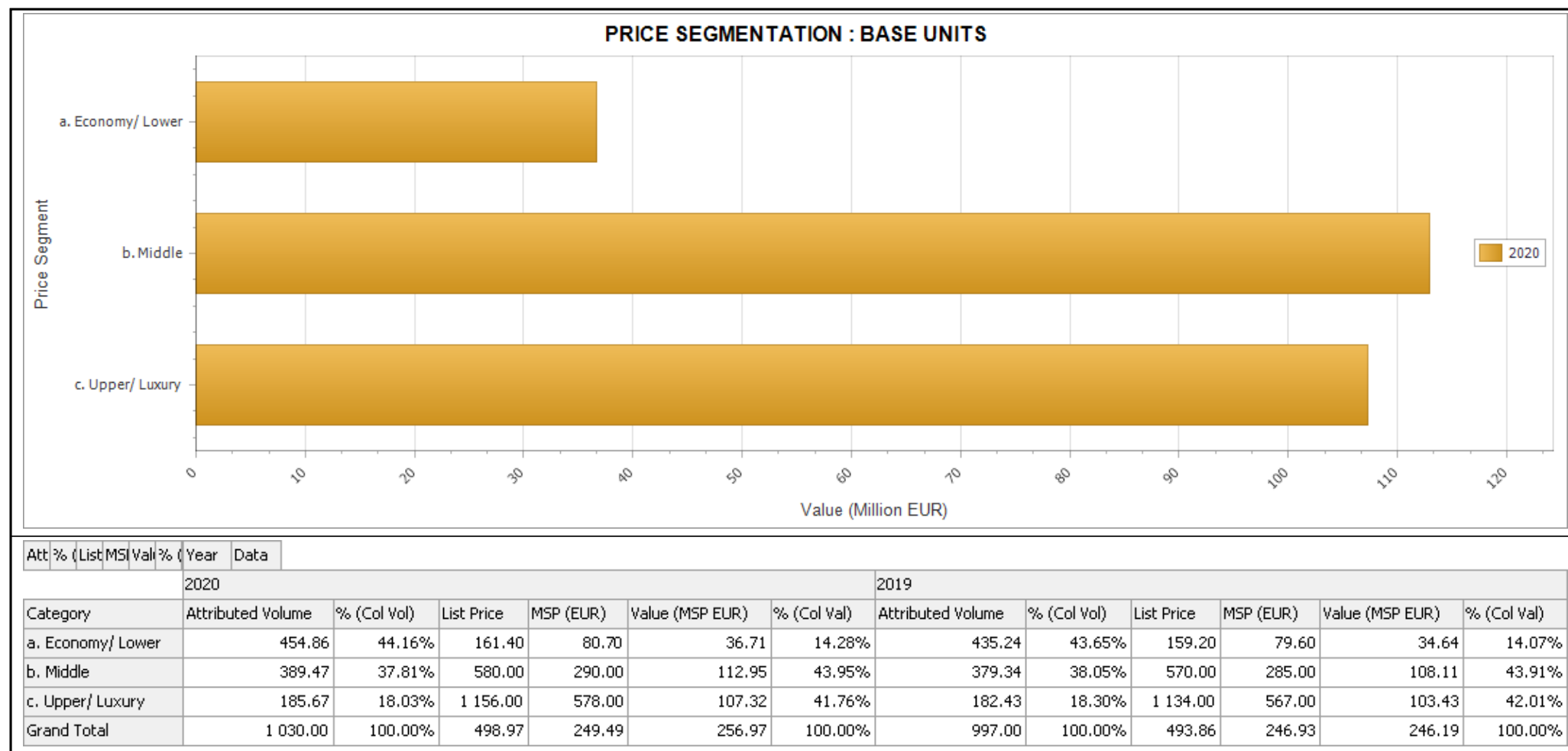
DE - ESTIMATED VALUES OF THE BATHROOM FURNITURE MARKET



Source: BRG based on industry interviews. Values Shown in Millions (at MSP = Manufacturers' Selling Price)

5.3.2 Price Segmentation : Base Units

DE - ESTIMATED PRICE SEGMENTATION FOR BASE UNITS

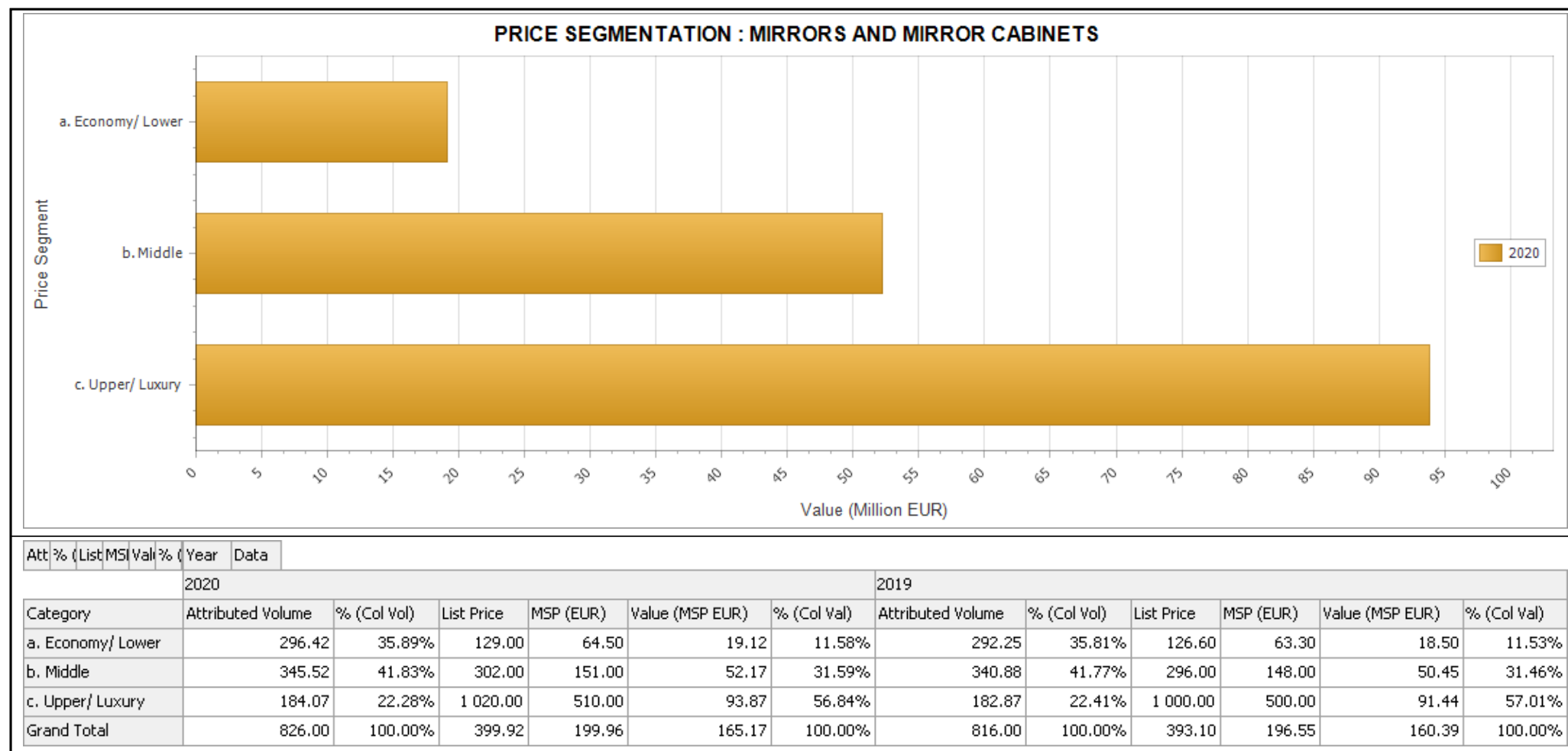


Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€200, Middle €200-€400, Upper/ Luxury >€400

5.3.3 Price Segmentation : Mirrors and Mirror Cabinets

DE - ESTIMATED PRICE SEGMENTATION FOR MIRRORS AND MIRROR CABINETS

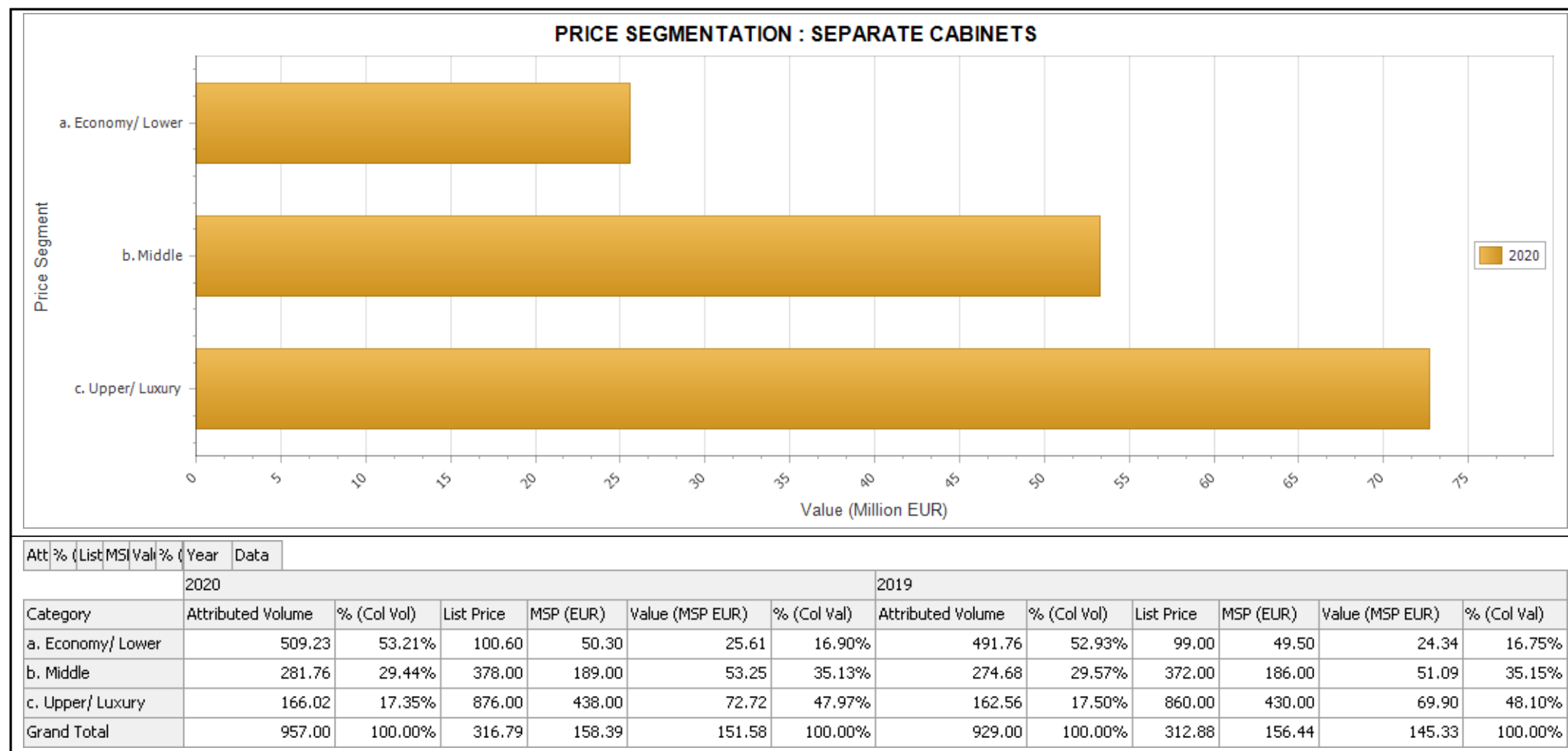


Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€100, Middle €100-€200, Upper/ Luxury >€200

5.3.4 Price Segmentation : Separate Cabinets

DE - ESTIMATED PRICE SEGMENTATION FOR SEPARATE CABINETS

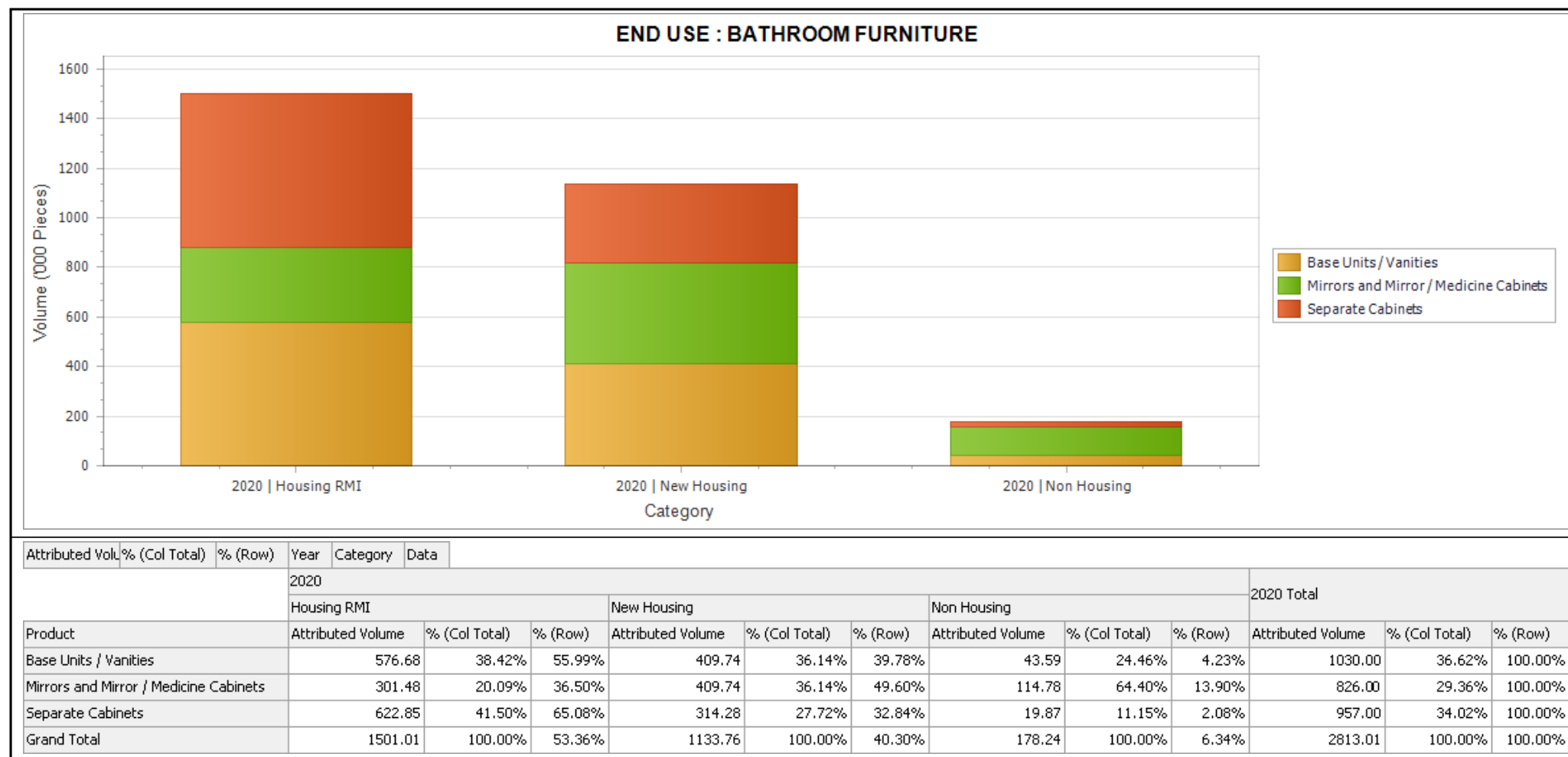


Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€125, Middle €125-€250, Upper/ Luxury >€250

5.4.1 End Use : Base Year

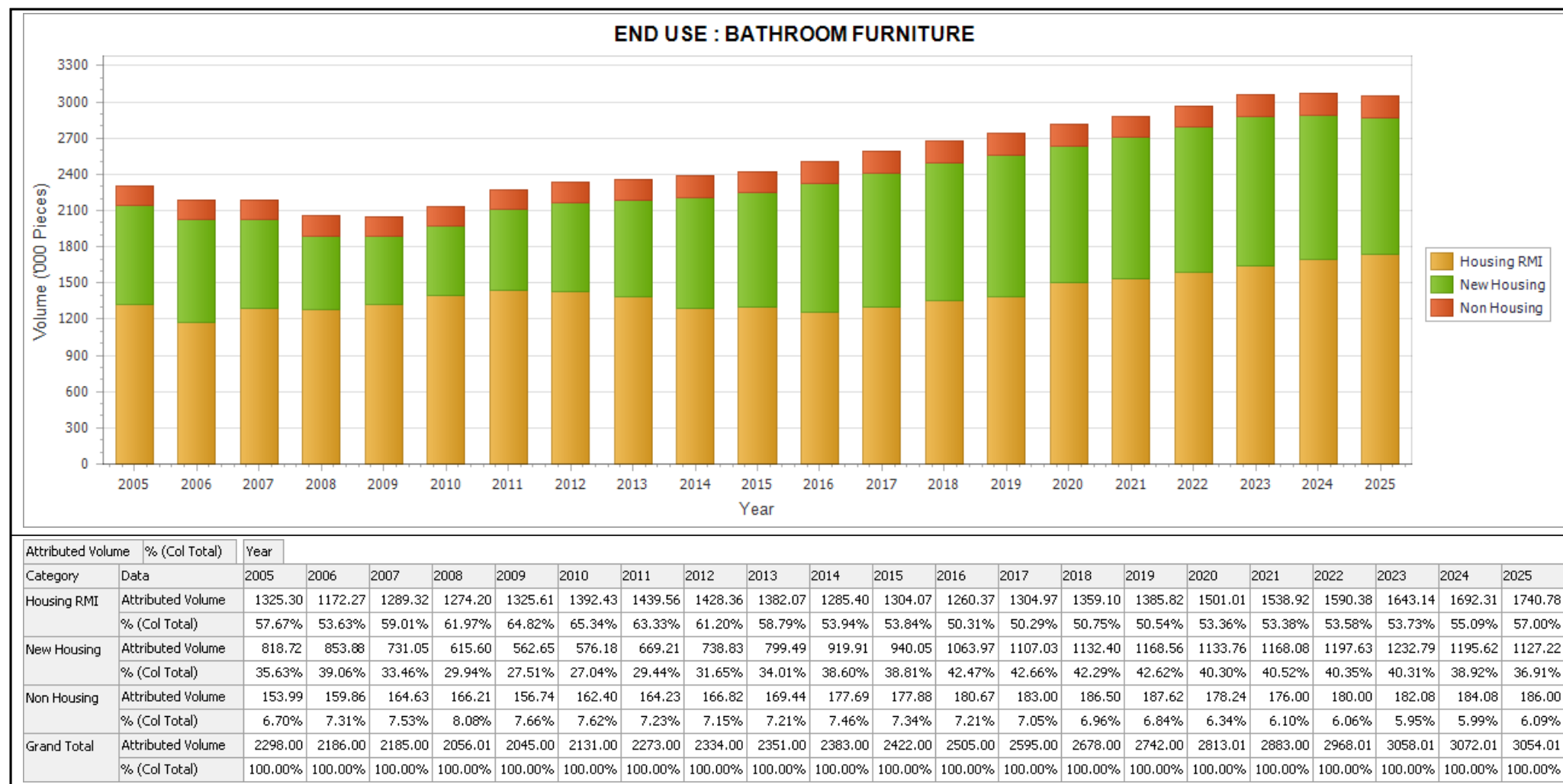
DE - SALES OF BATHROOM FURNITURE BY TYPE AND END USE SEGMENT ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

5.4.2 End Use : Historical and Forecasts

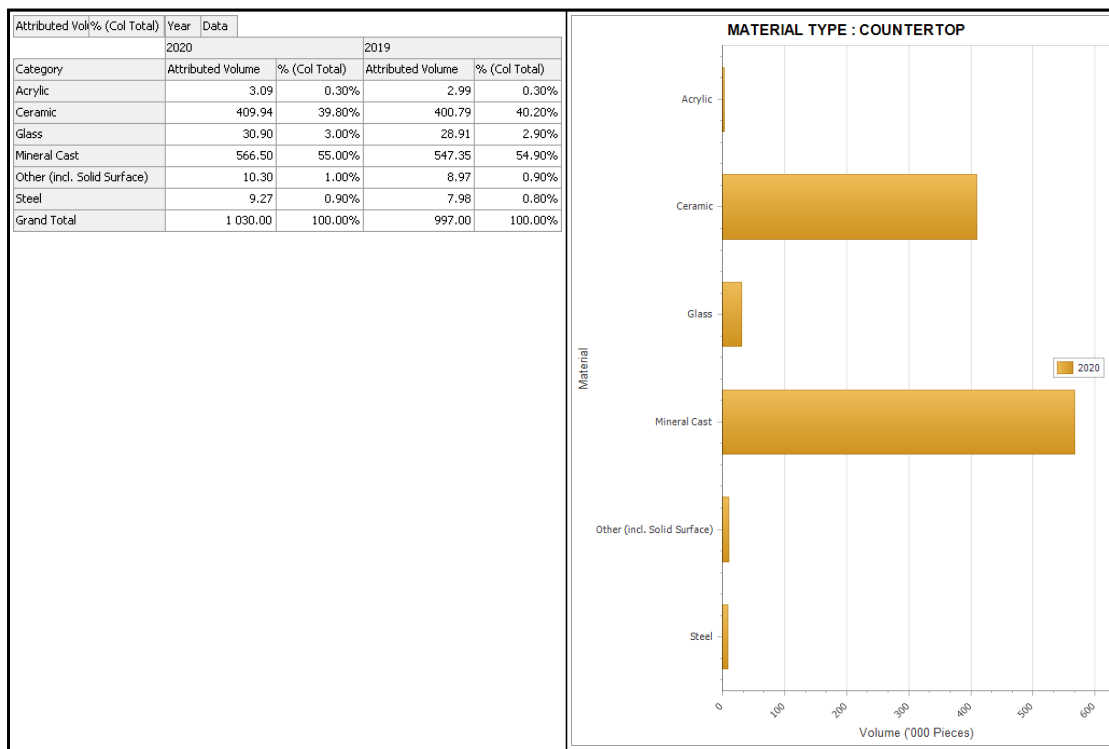
DE - SALES OF BATHROOM FURNITURE BY END USE SEGMENT ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

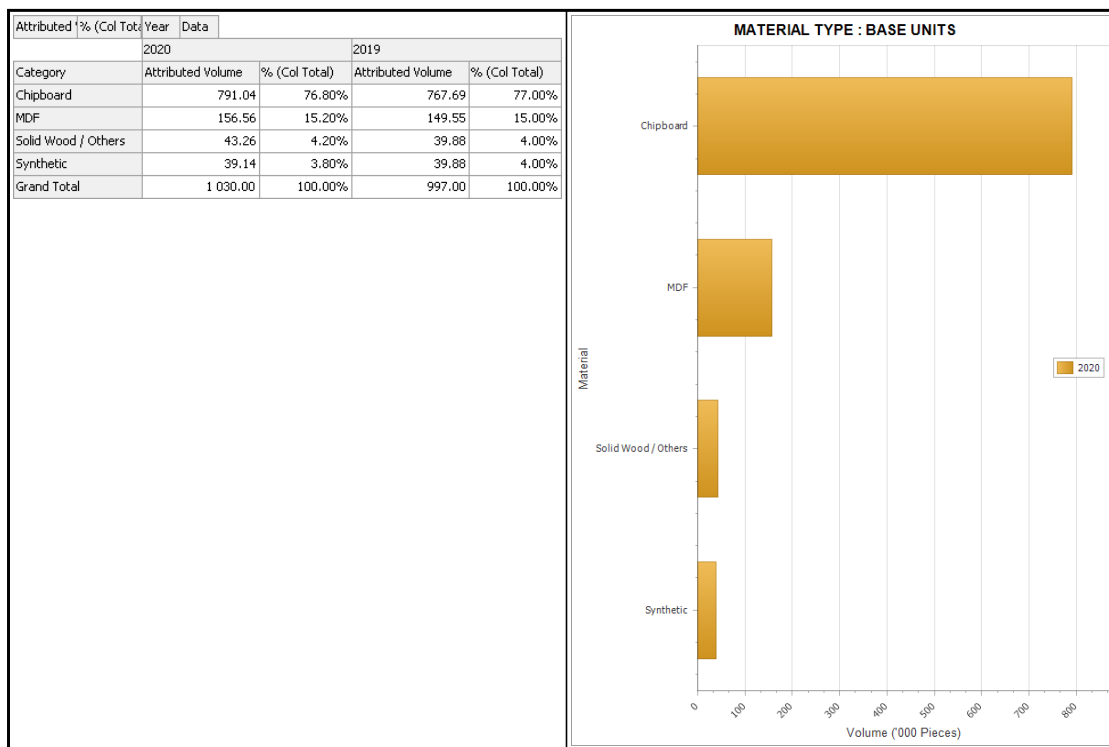
5.4.3 Sales by Type of Washbasin Material : Base Units

DE - BASE UNIT SALES BY WASHBASIN MATERIAL TYPE ('000 PIECES)



5.4.4 Sales by Type of Material : Base Units

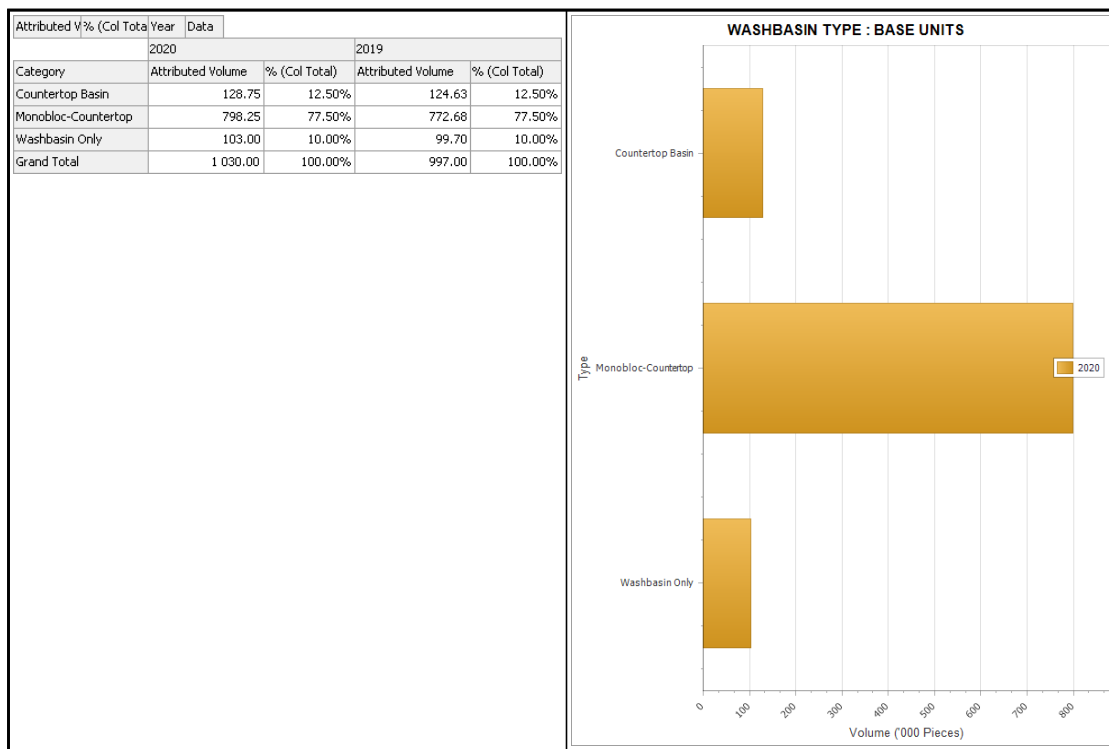
DE - BASE UNIT SALES BY MATERIAL TYPE ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

5.4.5 Sales by Type of Washbasin : Base Units

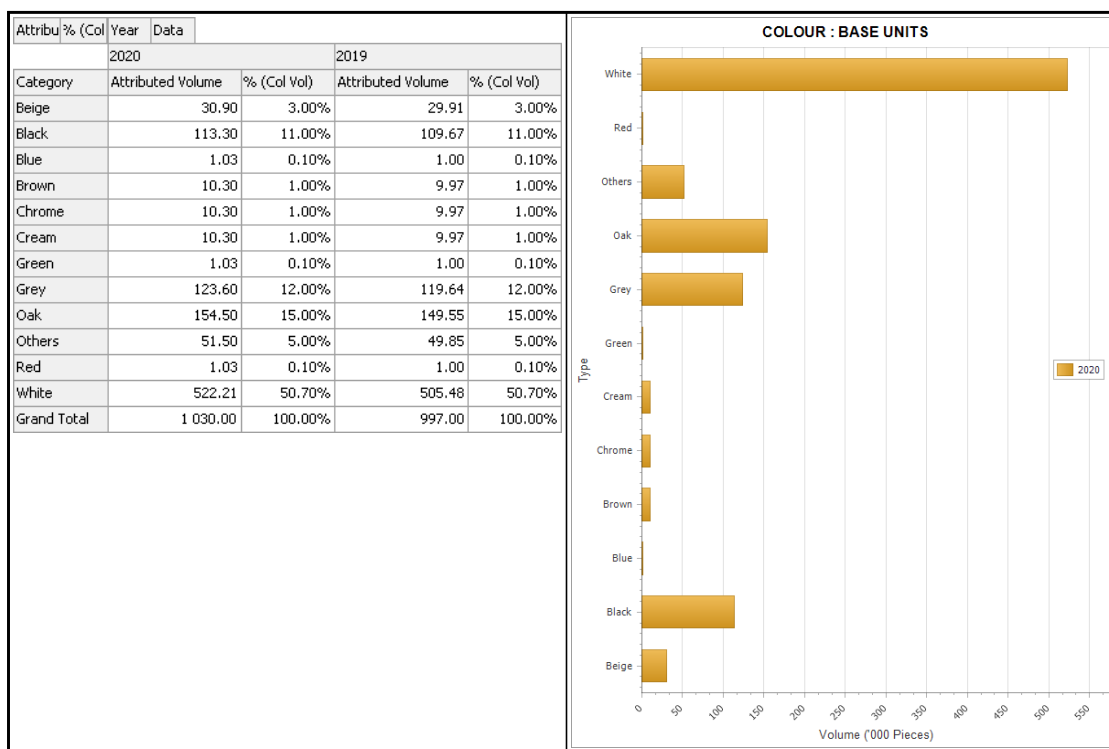
DE - BASE UNIT SALES BY WASHBASIN TYPE ('000 PIECES)



N.B. Washbasins only (non-countertop) are included in the above volumes for purposes of completeness.

5.4.6 Sales by Type of Colour : Base Units

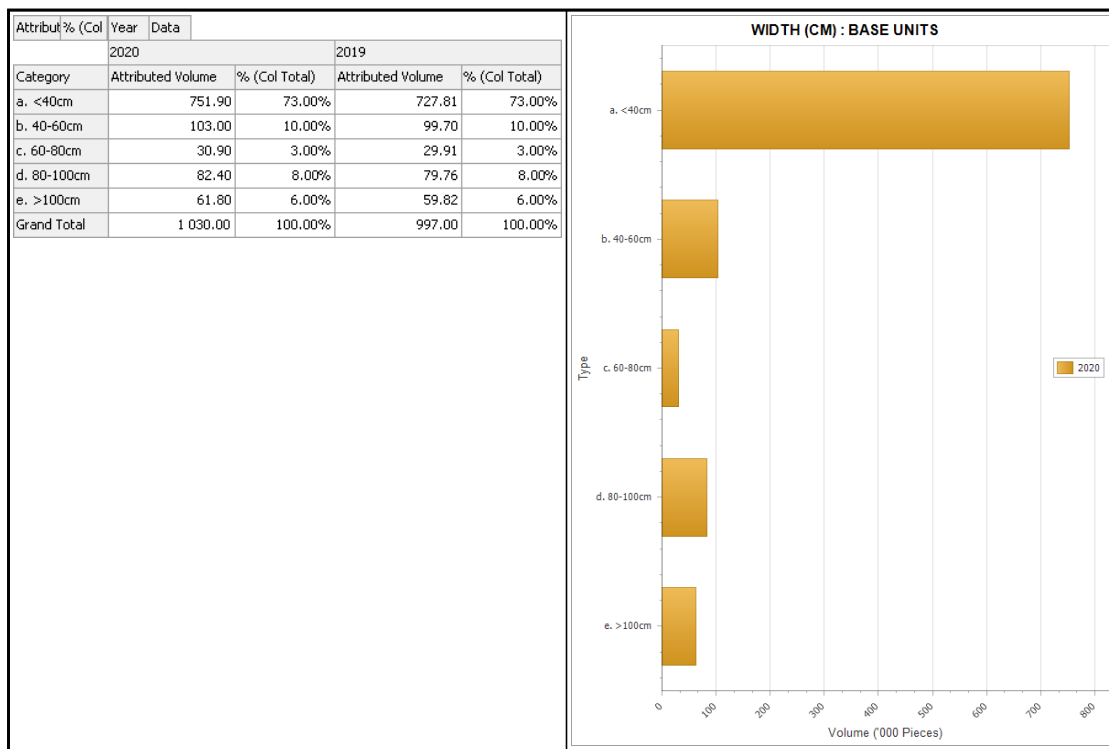
DE - BASE UNIT SALES BY COLOUR ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

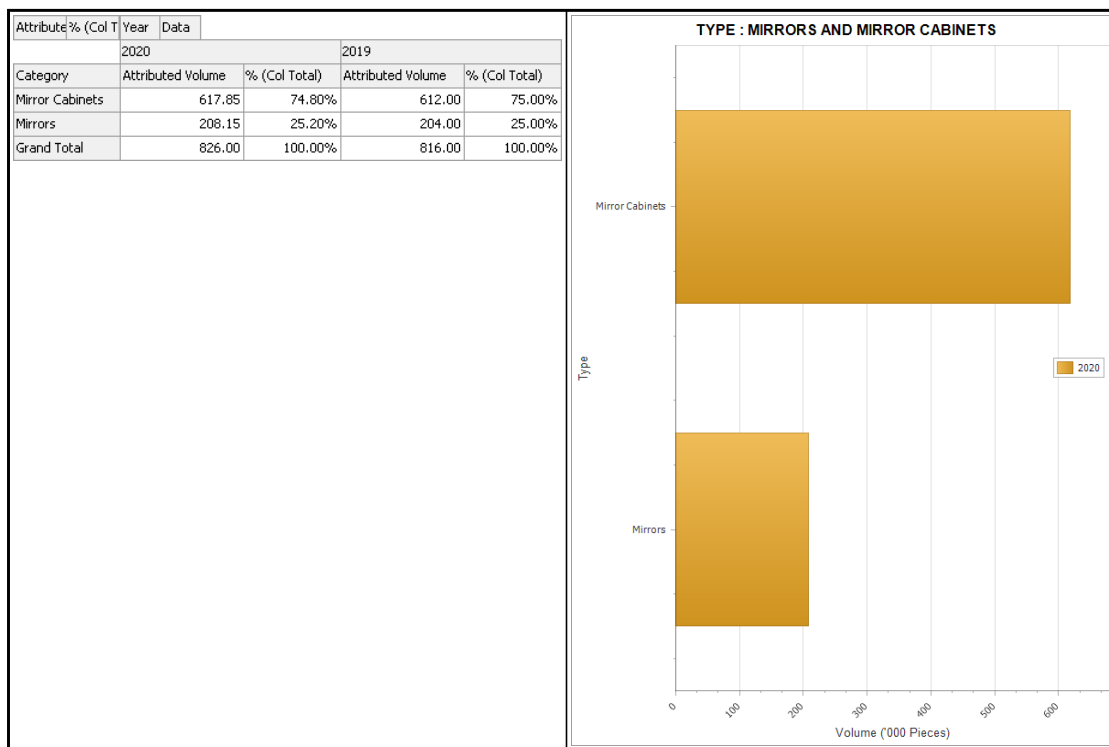
5.4.7 Sales by Width : Base Units

DE - BASE UNIT SALES BY WIDTH ('000 PIECES)



5.4.8 Sales by Type : Mirror and Mirror Cabinets

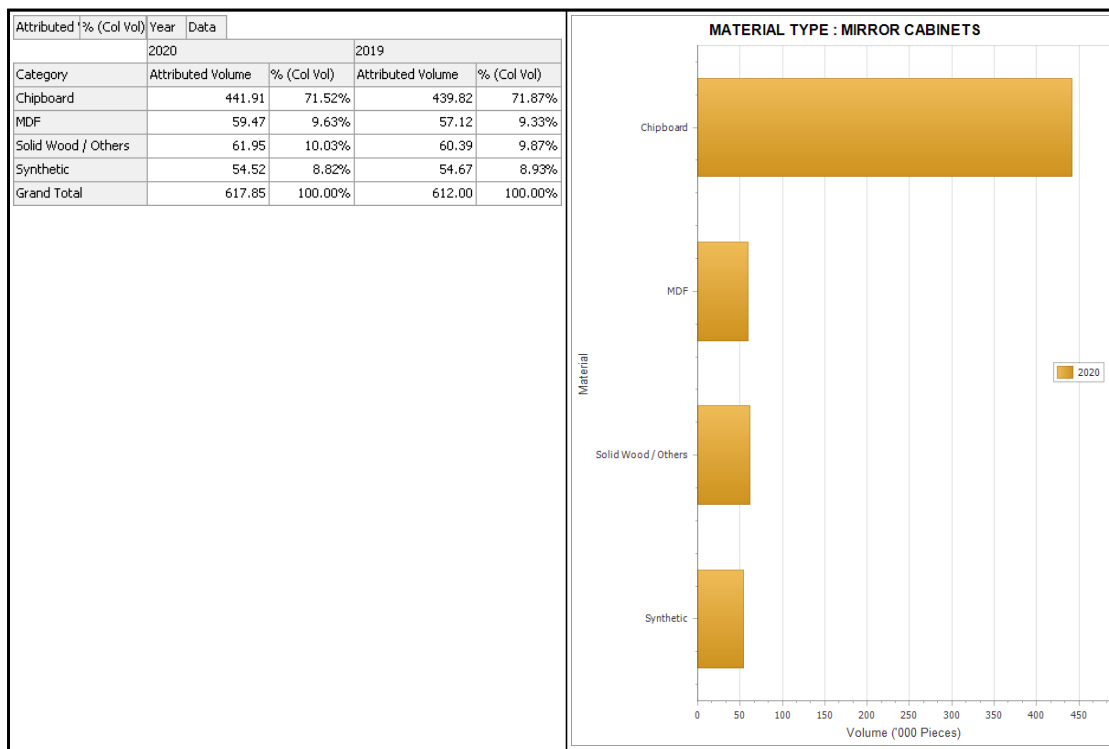
DE - MIRROR AND MIRROR CABINET SALES BY TYPE ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

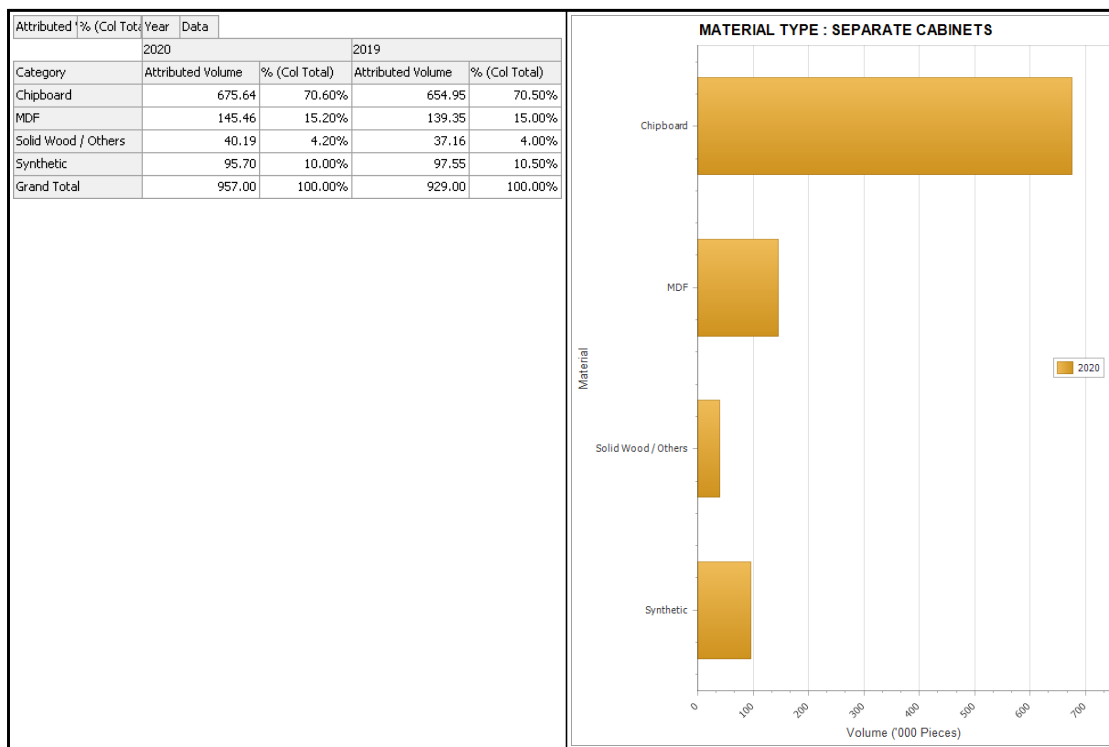
5.4.9 Sales by Type of Material : Mirror Cabinets

DE - MIRROR CABINET SALES BY MATERIAL TYPE ('000 PIECES)



5.4.10 Sales by Type of Material : Separate Cabinets

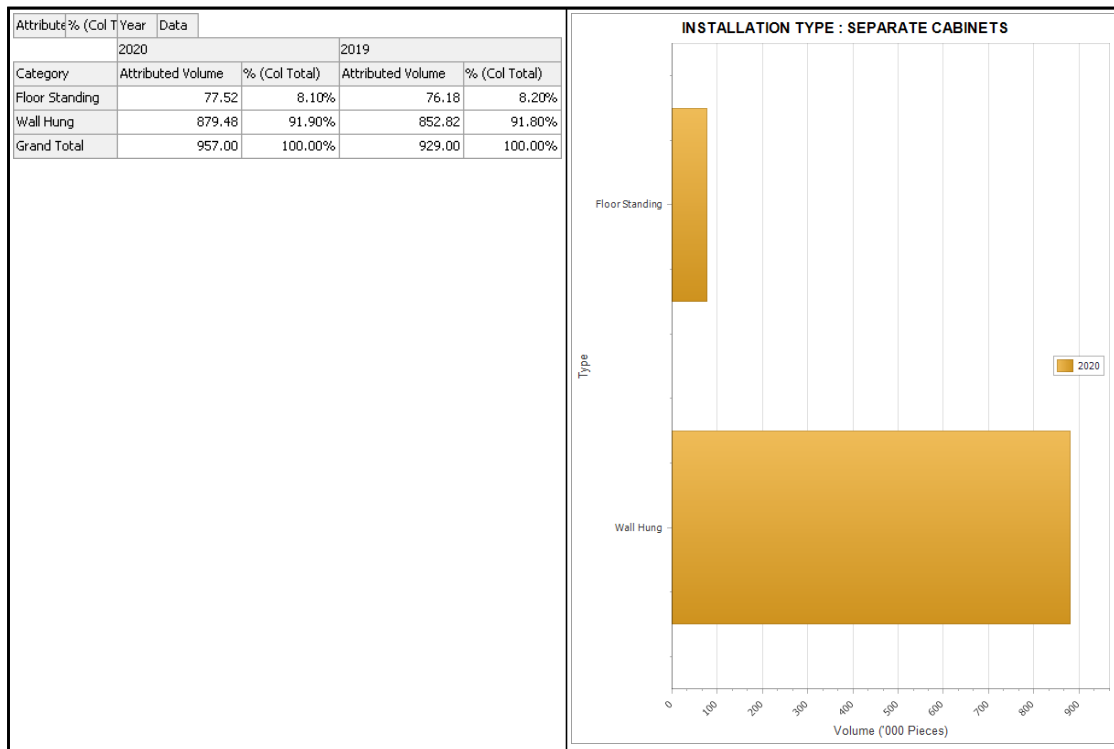
DE - SEPARATE CABINET SALES BY MATERIAL TYPE ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

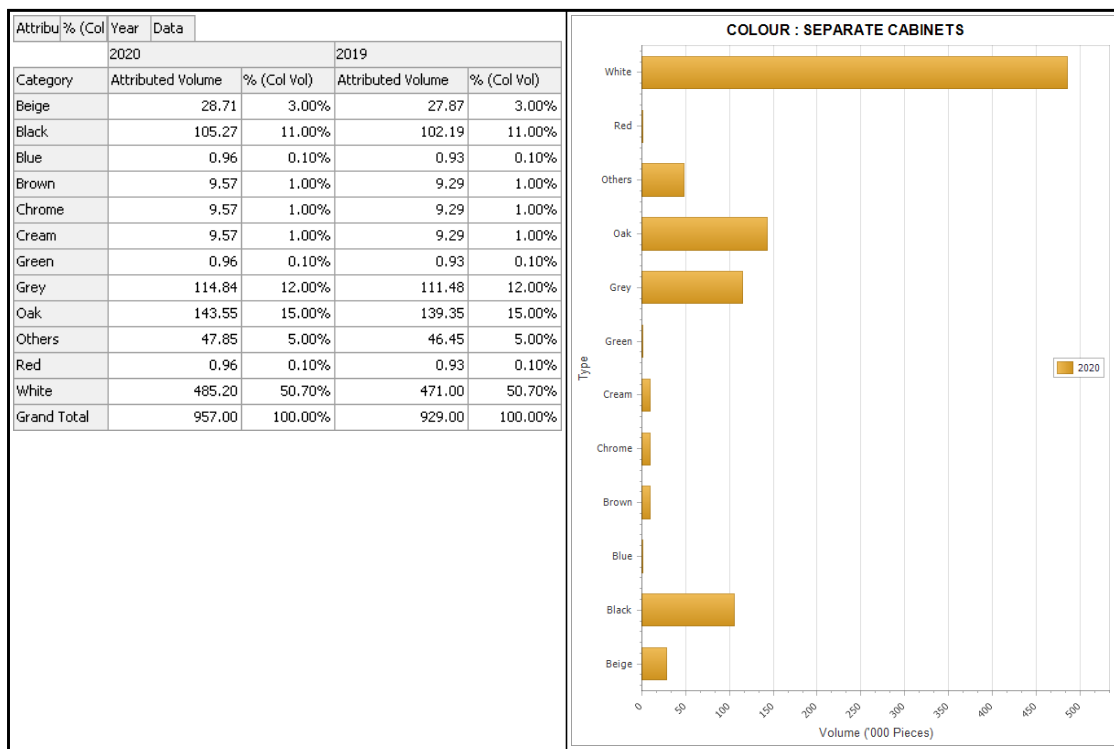
5.4.11 Sales by Type of Installation : Separate Cabinets

DE - SEPARATE CABINET SALES BY INSTALLATION TYPE ('000 PIECES)



5.4.12 Sales by Type of Colour : Separate Cabinets

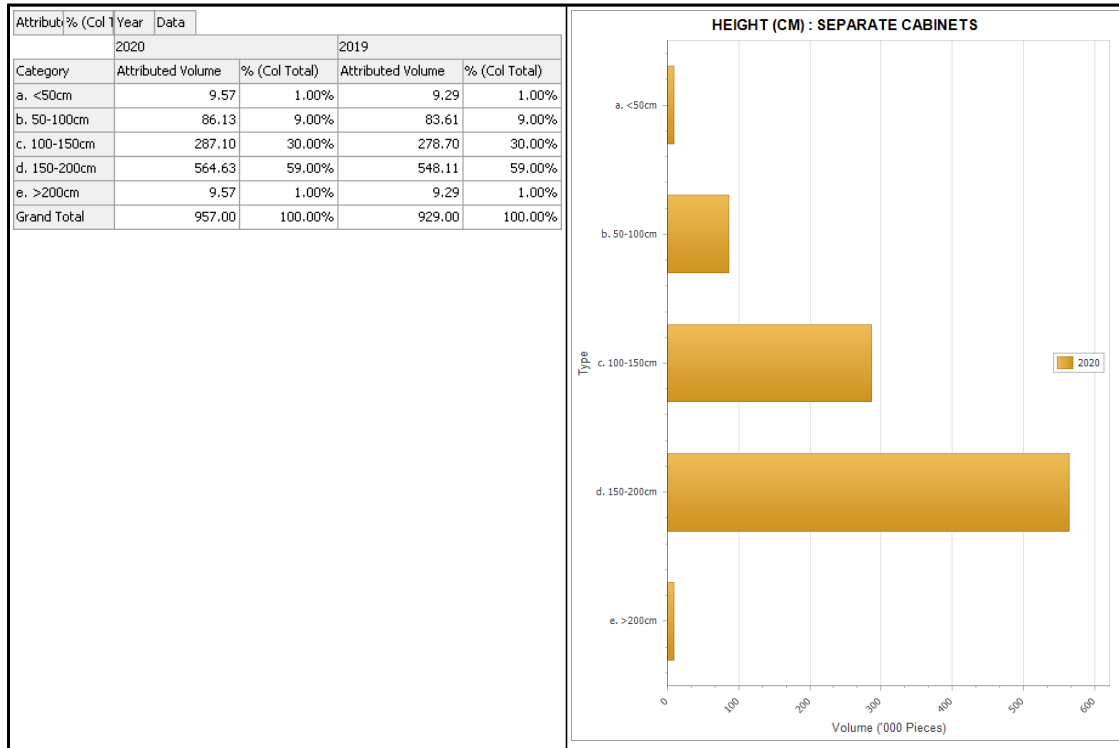
DE - SEPARATE CABINET SALES BY COLOUR ('000 PIECES)



Source: BRG based on industry interviews

5.4.13 Sales by Height : Separate Cabinets

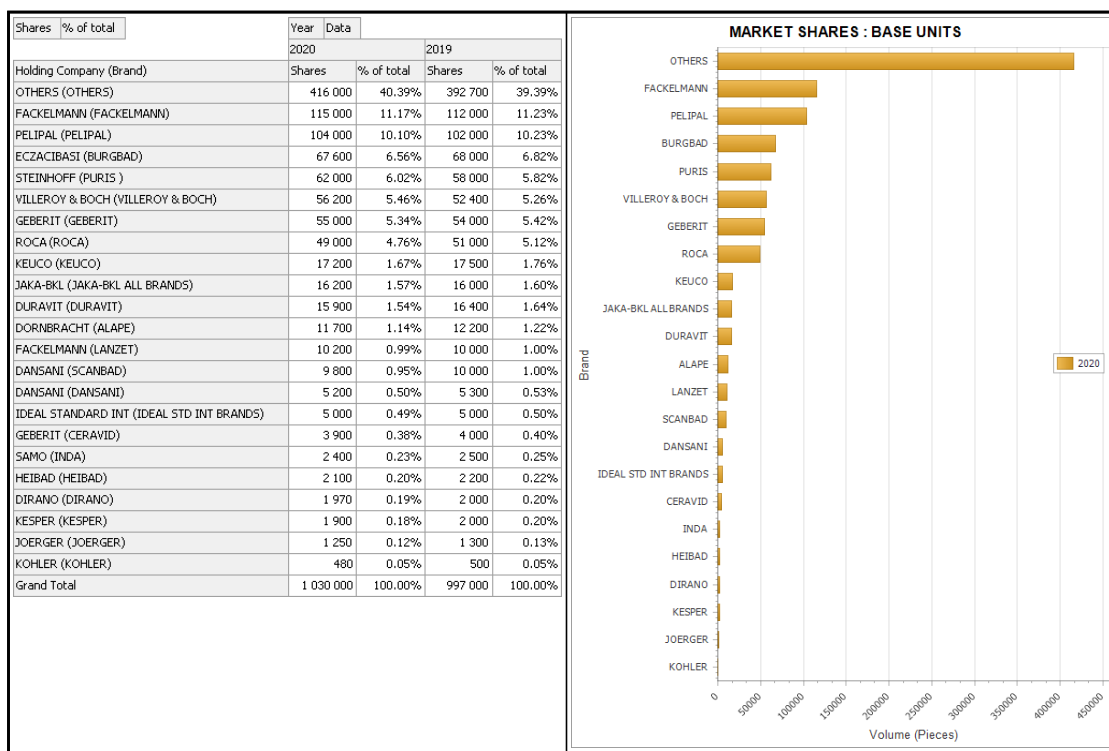
DE - SEPARATE CABINET SALES BY HEIGHT ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

5.5.1 Market Shares : Base Units

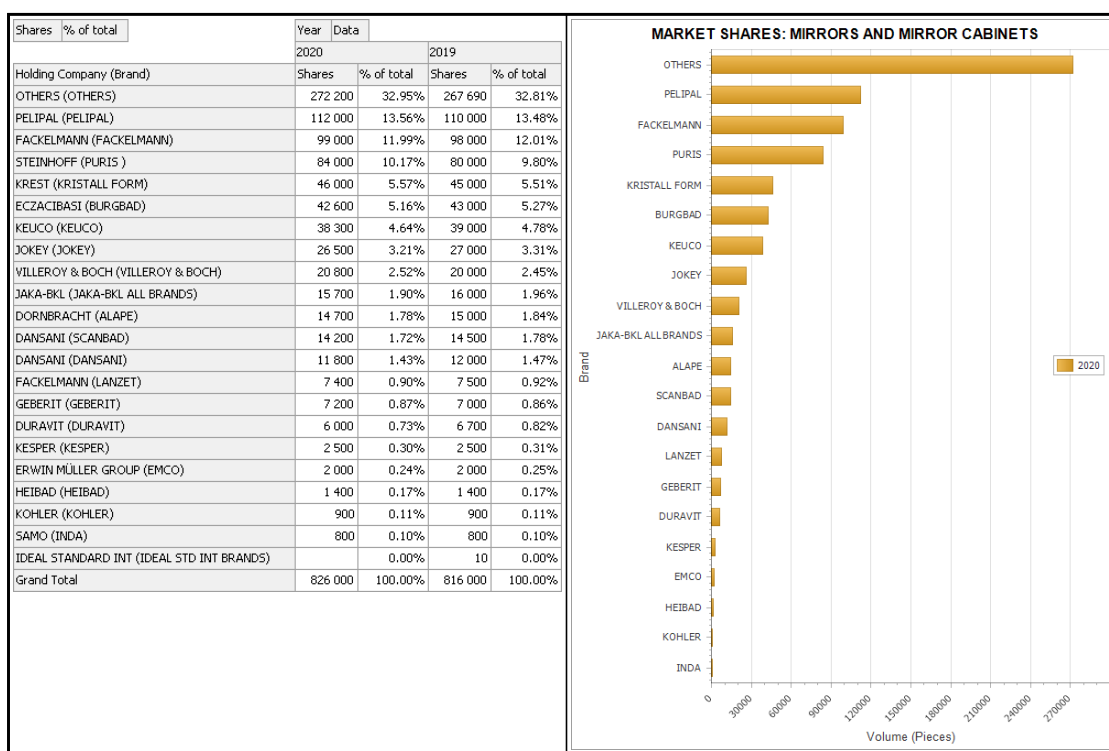
DE - ESTIMATED MARKET SHARES FOR BASE UNITS



Source: BRG based on industry interviews.

5.5.2 Market Shares : Mirror and Mirror Cabinets

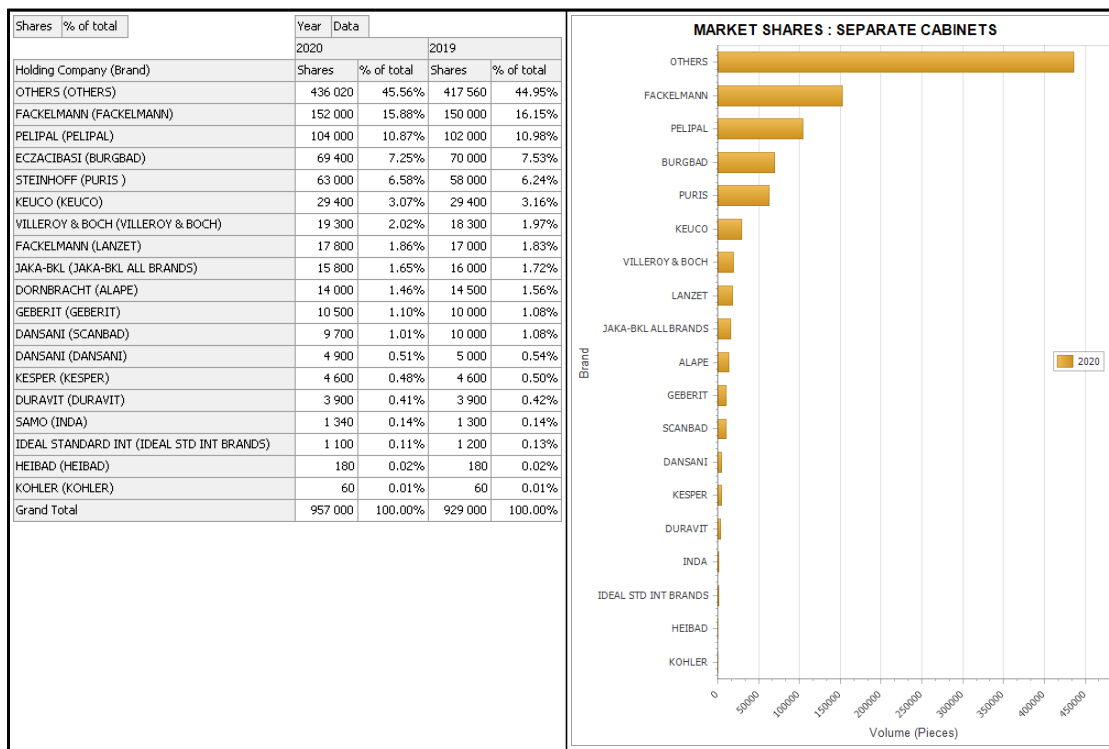
DE - ESTIMATED MARKET SHARES FOR MIRRORS AND MIRROR CABINETS



Source: BRG based on industry interviews.

5.5.3 Market Shares : Separate Cabinets

DE - ESTIMATED MARKET SHARES FOR SEPARATE CABINETS



Source: BRG based on industry interviews.

5.6.1

Route to Market: Bathroom Furniture
DE - ESTIMATED DISTRIBUTION FOR BATHROOM FURNITURE: 2020

Manufacturer / Importer	Manufacturers / Importers
	100.0%

Wholesale Level	34.3%
	Wholesalers / Merchants
	34.3%

Retail Level	Manufacturer 5.4%	Wholesaler 4.2%	Manufacturer 20.4%	Wholesaler 4.0%	Manufacturer 30.4%	Wholesaler 3.0%
	Internet Retailers		DIY Retailers		Specialist Retailers	
	9.6%		24.4%		33.4%	

Manufacturer	Wholesaler	Internet	DIY	Specialist
6.8%	23.0%	2.0%	1.2%	1.6%
Installers				
34.6%				

End Use Level	Manufacturer 0.0%	Wholesaler 0.0%	Internet 7.6%	DIY 23.2%	Specialist 31.0%	Installer 31.8%
	Residential					
	93.7%					
	New Build			Residential RMI		
	40.3%			53.4%		
	Project		Non-Project	Project		Non-Project
	8.0%		32.3%	1.0%		52.4%
	2.6%	0.1%	0.0%	0.0%	0.8%	2.8%
	Non-Residential					
	6.3%					
	New Build			Non-Residential RMI		
	2.1%			4.2%		
	Project		Non-Project	Project		Non-Project
	0.8%		1.3%	1.8%		2.4%

5.6.2

Route to Market: Bathroom Furniture
DE - ESTIMATED DISTRIBUTION FOR BATHROOM FURNITURE: 2025

Manufacturer / Importer	Manufacturers / Importers
	100.0%

Wholesale Level	32.4%
	Wholesalers / Merchants
	32.4%

Retail Level	Manufacturer 6.6%	Wholesaler 6.0%	Manufacturer 20.9%	Wholesaler 4.0%	Manufacturer 30.0%	Wholesaler 3.0%
	Internet Retailers		DIY Retailers		Specialist Retailers	
	12.6%		24.9%		33.0%	

Manufacturer	Wholesaler	Internet	DIY	Specialist
7.0%	19.0%	5.0%	1.4%	1.4%
Installers				
33.8%				

End Use Level	Manufacturer 0.1%	Wholesaler 0.3%	Internet 7.6%	DIY 23.5%	Specialist 30.6%	Installer 31.8%
	Residential					
	93.9%					
	New Build			Residential RMI		
	36.9%			57.0%		
	Project		Non-Project	Project		Non-Project
	7.8%		29.1%	2.0%		55.0%
	3.0%	0.1%	0.0%	0.0%	1.0%	2.0%
	Non-Residential					
	6.1%					
	New Build			Non-Residential RMI		
	2.0%			4.1%		
	Project		Non-Project	Project		Non-Project
	0.9%		1.1%	1.9%		2.2%

Source: BRG Based on Industry Interviews. - Totals may vary due to roundings

5.6.3 Route to Market - Consolidated View : Bathroom Furniture

DE - ESTIMATED DISTRIBUTION FOR BATHROOM FURNITURE : 2020

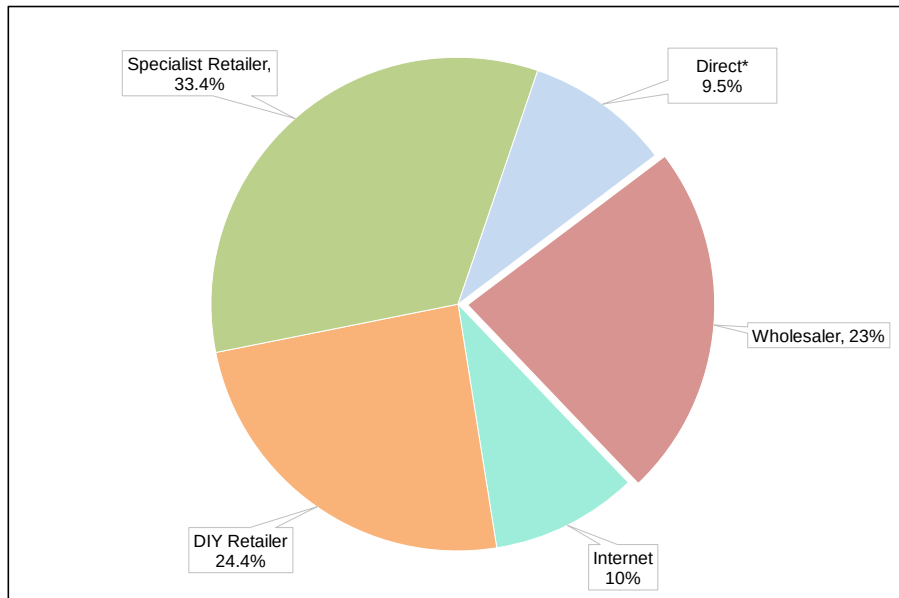
Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
23.1%	9.6%	24.4%	33.4%	9.5%	9.0%	2.6%	84.7%	3.7%
100%					11.6%		88.4%	

5.6.4 Route to Market - Consolidated View : Bathroom Furniture

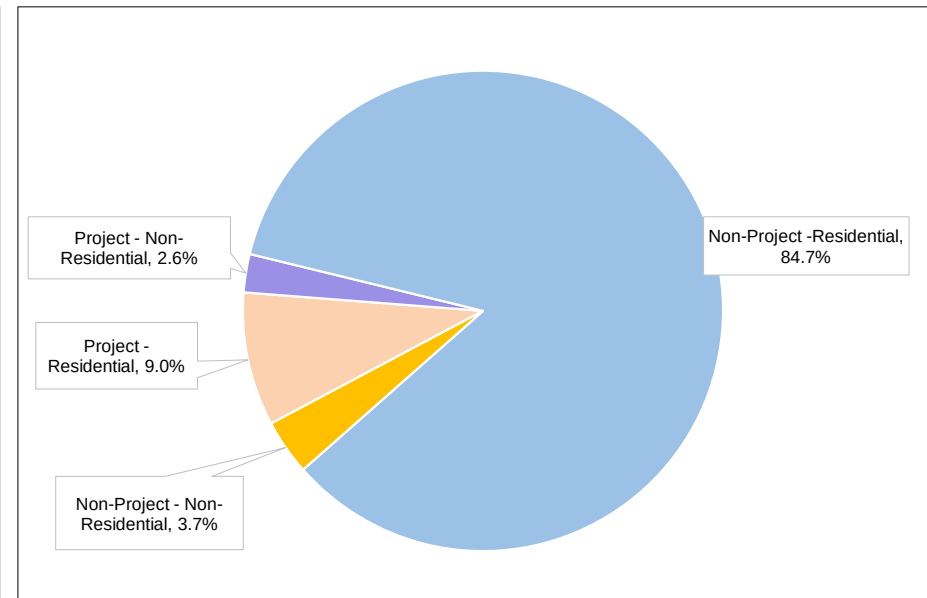
DE - ESTIMATED DISTRIBUTION FOR BATHROOM FURNITURE : 2025

Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
19.4%	12.6%	24.9%	33.0%	10.1%	9.8%	2.8%	84.1%	3.3%
100%					12.6%		87.4%	

DE - SALES OF BATHROOM FURNITURE BY DISTRIBUTION CHANNEL: 2020



DE - SALES OF BATHROOM FURNITURE BY POINT OF DECISION AND END-USE SECTOR: 2020



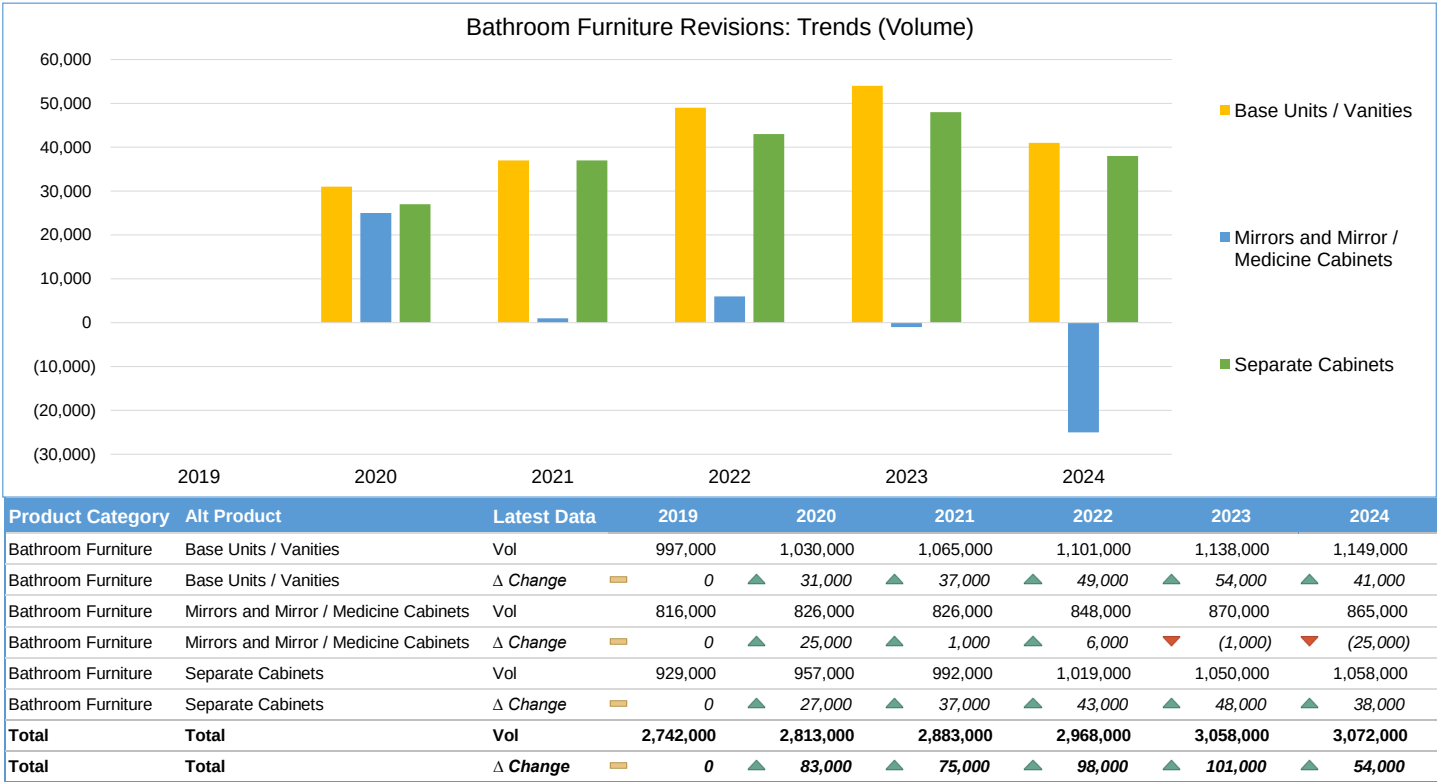
Source: BRG Based on Industry Interviews.

*Direct includes sales going directly from manufacturers to installers, contractors and private customers.

5.7 Revisions in the German Bathroom Furniture Market (April 2021)

DE - REVISIONS IN THE GERMAN BATHROOM FURNITURE MARKET

Data Comparison: October 2020



No revisions have been made to historical volumes for this market.

GERMANY

BATHROOM ACCESSORIES

THE EUROPEAN BATHROOM & KITCHEN PRODUCT MARKETS



Market Characteristics and Technological Trends

The market for bathroom accessories in Germany can be split into two broad categories. The first category includes products that are installed during the building phase and are then hardly ever replaced. Replacement usually happens only if the entire bathroom is renovated; for example, to adapt to the needs of an elderly person. Therefore, sales numbers are low in relation to other products. The products in the second category have limited durability and are replaced on a regular basis because they have a short lifespan or should be replaced for hygienic reasons. Numbers of units sold are usually high for these products. Across these products list prices range from very low to high, depending on the brand and distribution channel.

The German bathroom accessories market is a mature market, driven largely by replacement demand and the developments in the construction industry. This market has experienced strong fluctuations in value since 2000, being in line with the development of the construction industry. The overall market did not experience the effects of the economic crisis and has been growing again since 2010, which has been in line with Germany's economic recovery, as well as developments seen on the housing market. Sales are a zero-sum game, i.e. market shares gained by one company are lost by another.

Dispensers for liquid soaps have gained in popularity and have started to replace soap holders. They are usually made of metal, glass, or plastics. An increasing number of manufacturers offer wall mounted soap dispensers, wall mounted toothbrush holders and tumblers. They are usually sold as a set. The most replaced bathroom products are individual toilet brushes, which are low in value but high in turnover. Toilet brushes have the highest sales in volume, followed by towel and bathrobe hooks. The most common material in the market is metal and its share is constantly increasing. However, quality and prices of metal can vary considerably and substitution processes between different metals exist. Compound materials are increasing in importance, especially for freestanding bathroom applications. Wood is basically non-existent in the German market, mainly because it is regarded to be unsuitable for wet rooms. In the lower-priced segments, the most common materials are plastics, ceramics, glass and steel. In the medium-priced segments, brass, zinc and steel are the most common materials. In the luxury segment, brass, at times refined with gold and crystals, is used.

In recent years, there has been a trend towards bathroom accessories that can be glued or sucked onto the wall. Drilling a hole is avoided because it can damage tiles, which nowadays are larger with thin joints. German consumers prefer not to damage their expensive tiles and opt for a gluing or suction solution. The company NIE WIEDER BORHEN, which is owned by TESA since 2017, launched bathroom accessories products together with a patented technique, which makes it possible to glue all types of bathroom accessories and to remove them without any trace. Another looming trend on the bathroom accessories market is 'away from the wall'.

About twenty years ago, nearly every bathroom in Germany was equipped with wall attached, drilled, towel, toothbrush and toilet paper holders.

Nowadays, Gers tend to store their towels in a cupboard, or on a freestanding towel holder, or an electric towel warmer. Fewer accessories are mounted on a wall and this leads to decreasing demand of traditional products. A growing number of consumers buys WC brushes made from silicone, which prevents dirt from sticking onto them and, consequently, prolonging the lifespan of this product significantly. According to industry feedback, on average private end users buy five to six accessories for a new bathroom. Most end users, buy three units of bathroom accessories per one trip to DIY stores. A large share of end users buys five to six units when carrying out a full bathroom renovation. Many buy only one piece to replace a broken one.

Recent Market Trends

The COVID-19 pandemic shaped the developments in the year 2020 and 'business as usual' was replaced by government measures. In Germany the official lockdown, which is better understood as a 'prohibition of contact', started at the end of March 2020. Bavaria was the first state to implement measures to contain the spread of the virus, which was followed by the remaining states towards the end of March. For a few weeks the German population was advised to work from home when possible, with most non-essential stores closing for a limited time only. Measures implemented varied by state (*Bundesländer*). Most stores, including DIY stores, were allowed to open again from the beginning of April 2020 onwards, with differences from state to state in the level of restrictions and measures. In certain regions, shopping in DIY stores was limited to professionals only.

During 2020, new housing construction continued without much interruption, with only large scale project sites halted to implement protective measures. That being said, the overall production per employee is believed to have shrunk significantly, and remains to be the case, since many employers introduced measures to protect the construction workers. The bathroom industry was declared part of the crisis infrastructure, but installers could still continue to work without many interruptions. Overall, there has been relatively little impact of the COVID-19 pandemic on the building related industry, including the bathroom market, in Germany. In some cases, end users may have postponed, or cancelled, bathroom related projects due to the uncertainty on the market, but overall demand remained relatively strong, as there was a large backlog of works from the past, benefiting sales of bathroom and kitchen equipment. At the same time, in certain instances, the COVID-19 pandemic may have contributed to additional demand. Since consumers were spending more time indoors, they had the opportunity to dedicate themselves to repair or upgrade works, whereas others might have decided to invest into their property by installing a new bathroom or kitchen due to (international) holidays being difficult because of the current pandemic.

In 2020, overall bathroom accessories sales strongly increased when compared to the previous year. Total market value grew with a strong growth rate. This positive

development can be explained by German consumers spending more time at home. At the same time Germans invested more into their home by making everyday life more convenient. The Internet was used to purchase products that can easily contribute to more wellness and wellbeing in the bathroom. The topic for hygiene started to play a more significant role and bathroom accessories linked to more hygiene such as soap dispensers and towel holders significantly increased in demand as a result of the COVID-19 pandemic. There is a trend away from drilling and products that are fixed to the wall such as toothbrush holders are more likely to decline because German consumers prefer not to damage expensive tiles.

Bathroom accessories are a simple way to refurbish a bathroom quickly. Demand is mainly driven by private consumption, which recorded a significantly stronger growth in 2020 when compared to the previous year. Bathroom accessories sales in volume and in value rose accordingly. The average price increased marginally in 2020. End users spent more on products that made them feel better and safer during the lockdown. The share of the upper price segment slightly increased in 2020 when compared to previous years. At the same time, the economy and lower price segments increased, too. This can be explained by the growing trend of buying through alternative distribution channels, such as the Internet, supermarkets and furniture stores. Overall, it can be said that there is a trend towards the more extreme price segments at the expense of the middle segment.

An overall trend within the bathroom accessories market can be observed for products that can be easily fixed to the wall. This is the case for small items such as towel holders, but also for bigger items such as soap dispensers, tumbler holder, towel rails and even shower supports. The gluing technology by BRAVAT and NIE WIEDER BORHEN can hold weights of up to 400 kg and does not leave any marks after being removed. Demand for such products is believed to increase as it provides a drilling-free solution.

Prices and Market Values

The bathroom accessories market is highly competitive with foreign manufacturers, mainly from the Far East, reaching out to consumers directly. The price competition between manufacturers and distributors is increasing and it has an impact on the MSPs. Price increases for traditional products are limited and it is the innovative products that achieve a higher margin. It is estimated that around 90% of all bathroom accessories products are produced in China. This high percentage underlines the fact, that selling prices are justified by distribution channels. With a shift towards online purchases, end users are likely to scout for the cheapest products available. Since the installation of bathroom accessories products does not require an installer, end users are likely to install these products themselves. Germans are changing their shopping habits and the use of the Internet has gained great importance. As a result, there has been a slow shift away from the middle price segment towards products in the economy-priced ranges. At the same time, the trend to avoid drilling creates demand for alternative products, which are likely to be more expensive.

A significant shift towards innovative products has compensated the growing share of cheaper e-commerce sales and led to a marginal increase of the MSP. Bathroom accessories products with a longer lifespan, like towel rings and shelves, are usually higher in value, but lower in volume, while products with a short lifespan and the need to be replaced on a regular basis, such as toilet brushes, represent low values but high volumes. Rising labour costs in China and a strengthened Chinese currency cause the production costs to increase. An increase in Chinese production cost might only be small, however, due to the intense competition, it is difficult to ask for higher prices. Therefore, some manufacturers such as FACKELMANN abandoned the production in China and transferred it back to Germany. Instead of employing thousands of people, sophisticated production sites in Germany have become more financially viable.

New Housing

Since a new bathroom needs to be fully equipped with several bathroom accessories, sales to the new build residential segment represent an important share in the overall performance of the total bathroom accessories market. The new housing segment share of total sales had a small decrease when compared to the previous year. In a new bathroom a harmonious design is of importance for the end user, who chooses bathroom accessories to match design of taps and mixers. Major manufactures, such as GROHE, DURAVIT and IDEAL STANDARD, recognised this trend, especially in the upper price segments, and expanded their product range with bathroom accessories that match their portfolio in terms of style and accent lines. The intention is to benefit from added value by offering a one-stop solution for a perfectly designed bathroom. End users who can afford it are likely to spend the extra costs, whereas price sensitive end users are likely to look for cheaper accessories elsewhere. Since the end user no longer just buys a functional accessory but rather an additional component of the entire bathroom solution, products made from higher quality are more likely to be accepted. The overall bathroom accessories market did not benefit from the high number of new build completions like in previous years. Instead, demand from new dwellings declined due to fewer new housing completions in 2020. The new housing sector is expected to grow again from 2021 onwards, before turning negative in 2024 and 2025.

Housing RMI

Replacement demand accounts by far for most sales in the bathroom accessories market. The relatively short lifespan of several bathroom accessories fuels the need for periodical replacements to a certain extent. Given the different nature of the products being part of the bathroom accessories market, lifespans can differ greatly. While some products have a shorter lifespan than other bathroom products, such as toothbrush holders compared to baths and shower trays, some others, like shelves, can resist up to several years without being replaced. Frequently heavy bathroom replacement activities also include the substitution of old bathroom accessories with new ones that generally are better suited to the new

style of the renovated bathroom, so that a total renewal of five or six units on average happens at once. Overall, RMI activity strongly increased in 2020 when compared to the previous year. This growth is linked to the trend towards more hygiene among the large German population. At the same time end users spent more time at home and used the Internet and DIY stores to improve their homes. It is likely that this change in consumer behaviour benefited sales of bathroom accessories as they can easily be replaced.

Non-Housing

Demand for bathroom accessories in the German non-housing market notably declined in terms of value in 2020. Since public and commercial places were restricted to limited use only, bathrooms were less frequently used. New construction and refurbishment projects have been put on hold due to the COVID-19 crisis and unnecessary investments were avoided. The hotel and leisure industry were hit hard by the lockdown and most businesses struggled to remain in business. Other places might have invested into more hygienic materials and technology to ensure a more hygienic and healthy experience for their guests. Such products would only be the exception and most likely limited to the upper and luxury product range. Replacement demand from public areas such as offices, sports venues, educational institutions and transportation facilities is likely to have been limited due to restricted use.

Bathroom accessories in hotels and offices need to be both sturdy and pleasing to the eye. In the hotel lobby toilets are equipped with high-quality products that are long lasting. The style remains the same and the appearance is of great importance. In the accommodation bathroom accessories are of lower quality as they are changed every two to three years. They must look new and modern and can be cheaper due to a much shorter life cycle. Hospitals and nursing homes require certain characteristics to meet usability, quality standards and health regulations. The standard material in some non-housing facilities such as kitchens is stainless steel due to its antibacterial properties, ease of cleaning and long lifespan. In 2020, investments into all sectors such as hospitality, offices and commercial buildings recorded a notable decline. The only exception were buildings for education and health, which developed slightly positively in 2020. The situation is expected to slightly improve in 2021 and from 2022 sales into the non-housing segment are believed to start growing again.

Supply Structure

The market for bathroom accessories is fragmented and characterised by many small to medium-sized companies, the share for which is negligible in comparison to the major companies. Companies in the market include bathroom specialists, which offer bathroom taps and mixers, bathroom furniture and bathroom accessories. Most taps and mixers manufacturers have extended their product spectrum over the years to include bathroom accessories. Additionally, many high street retailers, such as TCHIBO, also offer bathroom accessories. The upper priced segment is dominated by German companies. These include KEUCO,

HEWI, the ERWIN MÜLLER GROUP (EMCO) and LIXL (GROHE). KEUCO produces in Germany, in Hemer and in Gütersloh. EMCO is in Lingen, HEWI is in Mengerlinghausen, and GROHE produces in Germany, Portugal, Thailand and Canada.

It is estimated that between 10-20% of bathroom accessories in terms of volume are purchased directly from China. Moreover, most German bathroom accessories are produced in China. Traditional DIY suppliers are CORAM NV (TIGER) and FACKELMANN. DIY stores and wholesalers supply private labels, which are performing positively. They source large numbers of products from the Far East. The products that are found in high street retailers are often sourced from the Far East, too. Specialists in the distribution flow include supermarkets, department stores and furniture retailers, such as IKEA.

In 2016, both the production companies SAM SCHULTE and SAF ARMATUREN, which are part of the SAM GROUP, went into insolvency. The SAM VERTRIEBS GMBH & CO. KG, as well as the companies in Austria and Switzerland, were not affected by the insolvency application. Around three months after the application, the SAM SCHULTE GROUP was bought by the international holding KINEN. The investors from Europe, Hong Kong and China have had previous exposure to the sanitary industry and aim to secure the operations of the German manufacturers. According to industry feedback the branch in Menden Germany was completely dissolved and the company SAM VERTRIEBS GmbH was closed. The new founded company SAM DESIGN GmbH is trying to get back into the market. However, they can only recover two product lines at the most and most customers lost their trust in the company and sales plunged in 2019 to an insignificant volume.

At the beginning of 2017, TESA acquired NIE WIEDER BOHREN, which was founded in 2005. Both companies have been working closely together since 2010. NIE WIEDER BOHREN owns several patents for gluing techniques that can be removed without any stains.

Forecasts

BRG expects the bathroom accessories market to continue to perform strongly during the forecast period. Following the trend towards more hygiene and investments into residential bathrooms, demand is likely to remain strong from private consumers. Demand from commercial use is likely to recover slowly and the pace depends on the development of the COVID-19 pandemic. Demand from the new housing segment is likely to decline towards the end of the forecast in alignment with fewer new housing completions. Although the growth rate of housing completions is expected to weaken, demand from new housing is likely to remain on a high level.

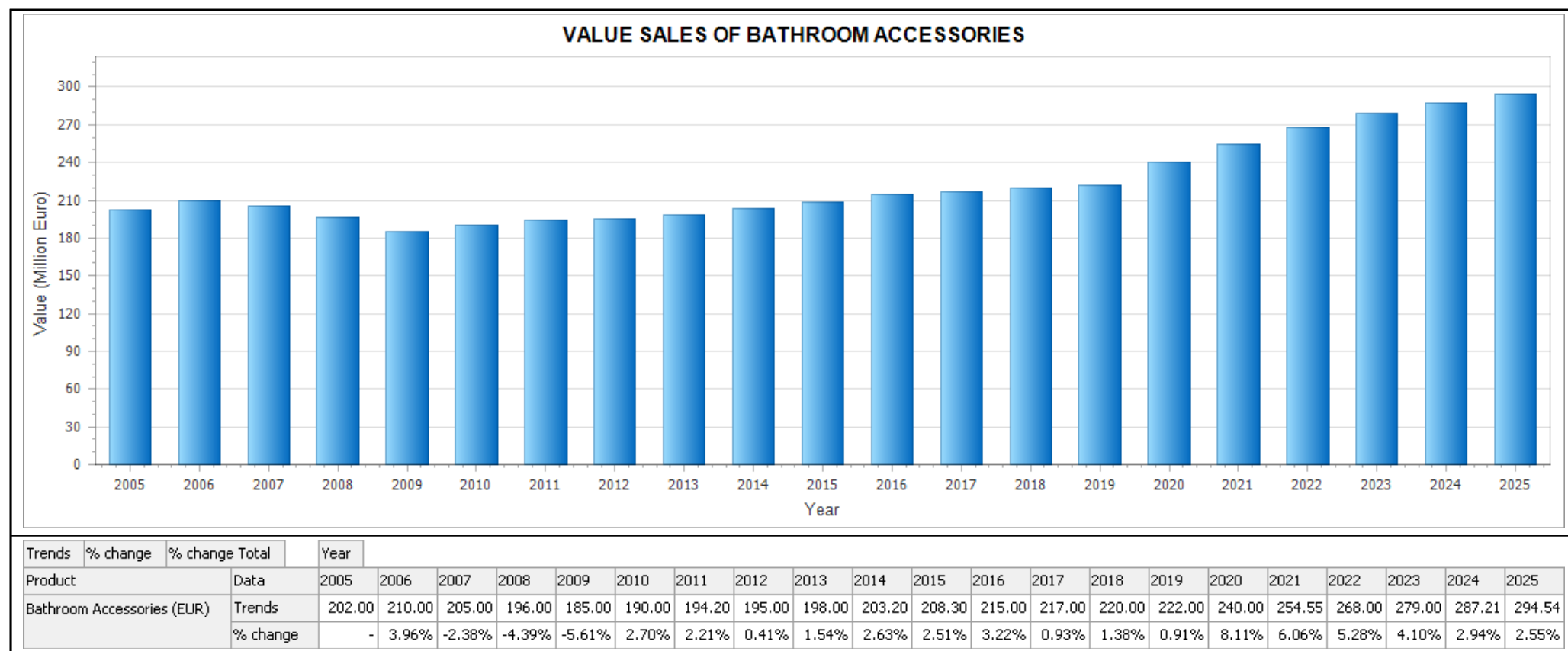
Furthermore, sales of bathroom accessories products are believed to continue growing in the replacement segment in accordance with the high demand of bathroom renovations. According to the fundamental study of Germany's bathroom industry by VDS from 2017, it is estimated that around 17.7 million bathrooms have

not been renovated since their construction or moving in. Their average age accounts to approximately 19.5 years, and around two million are believed to be older than 30 years. Potential for the bathroom accessories market does exist and may be favourably influenced by the trend of German consumers to appreciate their bathroom as a modern and comfortable living space as opposed to a purely functional one.

It is believed that changes in consumer behaviour will impact the distribution of bathroom accessories. This is an ongoing trend that has been observed by manufacturers claiming that sales through wholesalers are weakening to the benefit of other distribution channels, mainly DIY stores and e-commerce. Innovative products make the installation much easier and a broader range of end users can install them themselves. Not only prices, but also design and practicality play an increasing importance. Consumers have an increasing number of products to choose from, while manufacturers are facing more competition.

6.2 Historical Trends and Forecasts

DE - SUMMARY OF TRENDS IN THE BATHROOM ACCESSORIES MARKET : HISTORICAL AND FORECASTS (MILLION EURO)



Source: BRG based on industry interviews.

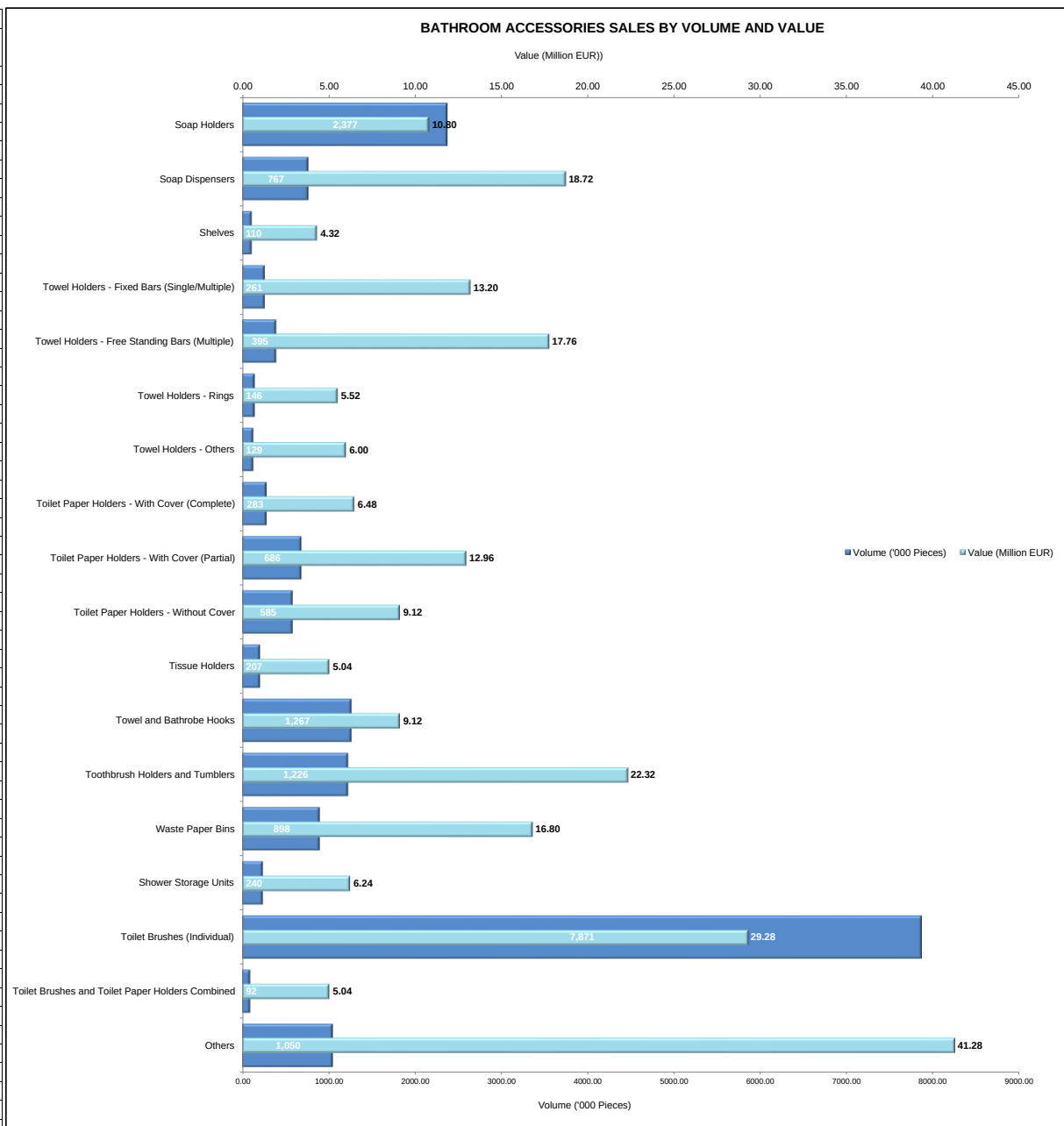
6.3 Detailed Prices and Market Values : Bathroom Accessories

DE - ESTIMATED VALUES OF THE BATHROOM ACCESSORIES MARKET BY TYPE OF PRODUCT

Market Value		2020	
		Volume ('000 Pieces)	Value (Million Euro)
Soap Holders	Volume / Value	2 376.8	10.8
	Share	12.8%	4.5%
	Average Price (EUR)		4.5
Soap Dispensers	Volume / Value	767.2	18.7
	Share	4.1%	7.8%
	Average Price (EUR)		24.4
Shelves	Volume / Value	110.5	4.3
	Share	0.6%	1.8%
	Average Price (EUR)		39.1
Towel Holders: Fixed Bars (Single/Multiple)	Volume / Value	261.4	13.2
	Share	1.4%	5.5%
	Average Price (EUR)		50.5
Towel Holders: Free Standing Bars (Multiple)	Volume / Value	394.7	17.8
	Share	2.1%	7.4%
	Average Price (EUR)		45.0
Towel Holders: Rings	Volume / Value	146.4	5.5
	Share	0.8%	2.3%
	Average Price (EUR)		37.7
Towel Holders: Others	Volume / Value	129.0	6.0
	Share	0.7%	2.5%
	Average Price (EUR)		46.5
Toilet Paper Holders: With Cover (Complete)	Volume / Value	283.0	6.5
	Share	1.5%	2.7%
	Average Price (EUR)		22.9
Toilet Paper Holders: With Cover (Partial)	Volume / Value	685.7	13.0
	Share	3.7%	5.4%
	Average Price (EUR)		18.9
Toilet Paper Holders: Without Cover	Volume / Value	584.6	9.1
	Share	3.1%	3.8%
	Average Price (EUR)		15.6
Tissue Holders	Volume / Value	207.4	5.0
	Share	1.1%	2.1%
	Average Price (EUR)		24.3
Towel and Bathrobe Hooks	Volume / Value	1 266.7	9.1
	Share	6.8%	3.8%
	Average Price (EUR)		7.2
Toothbrush Holders and Tumblers	Volume / Value	1 226.4	22.3
	Share	6.6%	9.3%
	Average Price (EUR)		18.2
Waste Paper Bins	Volume / Value	898.4	16.8
	Share	4.8%	7.0%
	Average Price (EUR)		18.7
Shower Storage Units	Volume / Value	240.0	6.2
	Share	1.3%	2.6%
	Average Price (EUR)		26.0
Toilet Brushes (Individual)	Volume / Value	7 871.0	29.3
	Share	42.3%	12.2%
	Average Price (EUR)		3.7
Toilet Brushes and Toilet Paper Holders Combined	Volume / Value	92.5	5.0
	Share	0.5%	2.1%
	Average Price (EUR)		54.5
Others	Volume / Value	1 050.4	41.3
	Share	5.6%	17.2%
	Average Price (EUR)		39.3
TOTAL	Volume in '000/Value in Million	18 591.9	240.0
	Share	100.0%	100.0%
	Average Price (EUR)		12.9

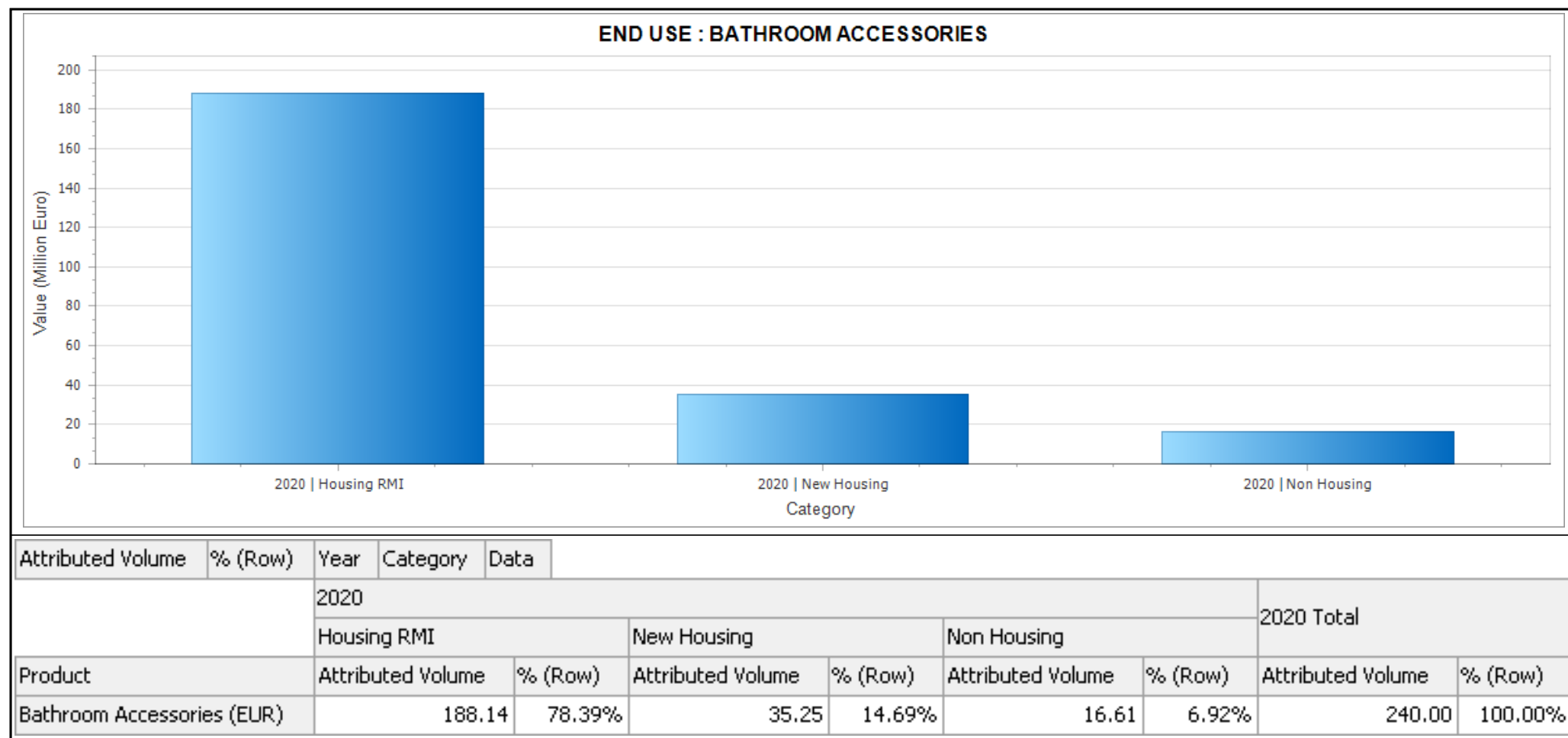
Source: BRG based on industry interviews

N.B. Prices Shown at MSP (= Manufacturers' Selling Price).



6.4.1 End Use : Base Year

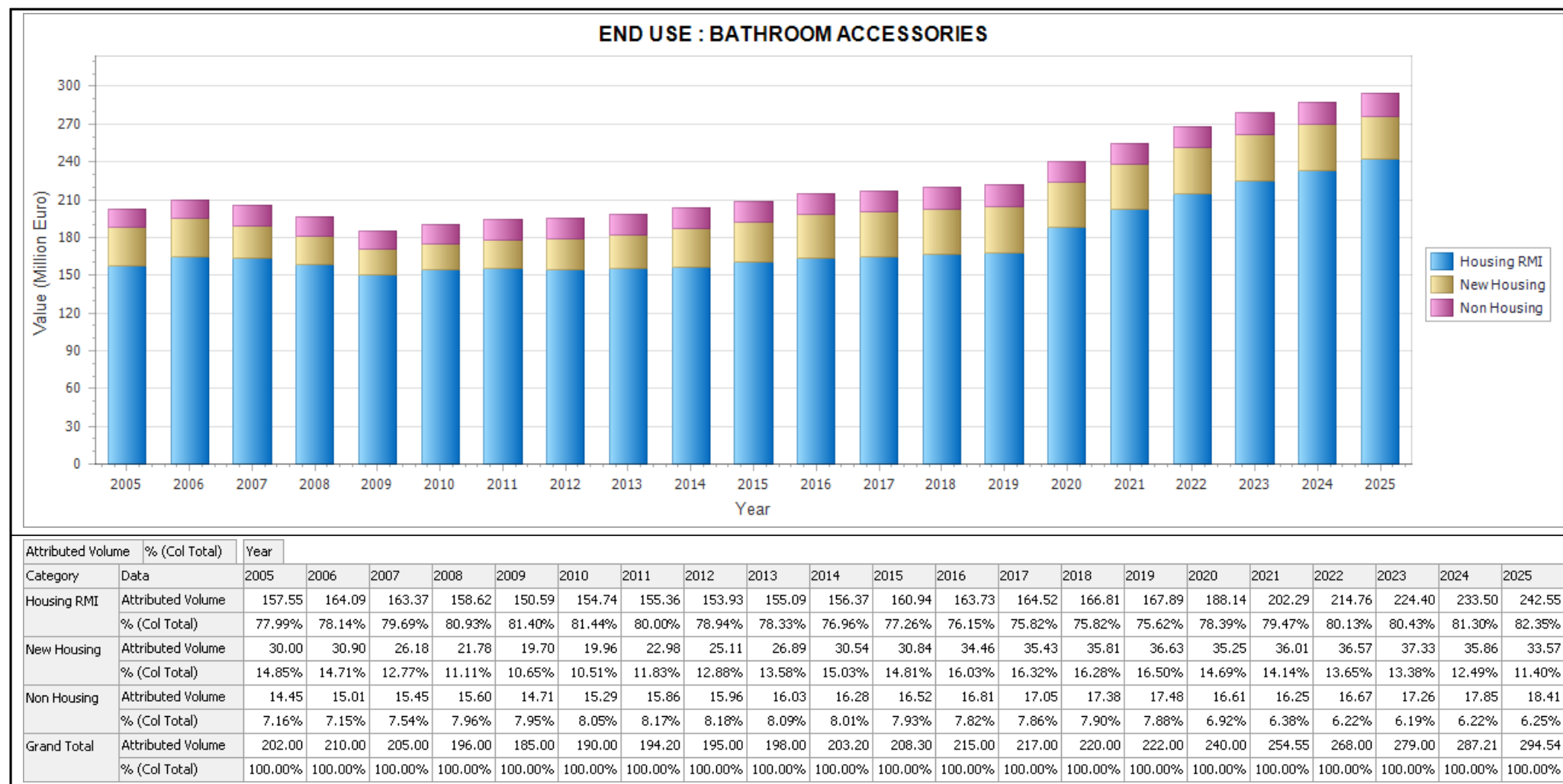
DE - SALES OF BATHROOM ACCESSORIES BY END USE SEGMENT (MILLION EURO)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

6.4.2 End Use : Historical and Forecasts

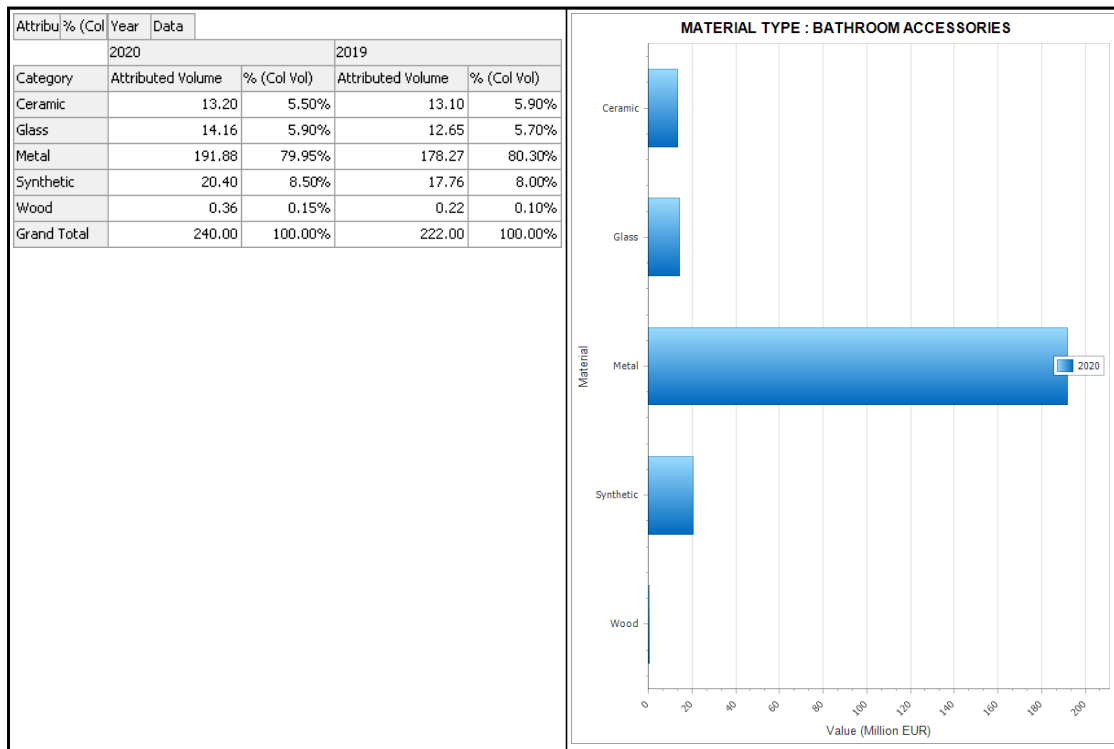
DE - SALES OF BATHROOM ACCESSORIES BY END USE SEGMENT (MILLION EURO)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

6.4.3 Sales by Type of Material : Bathroom Accessories

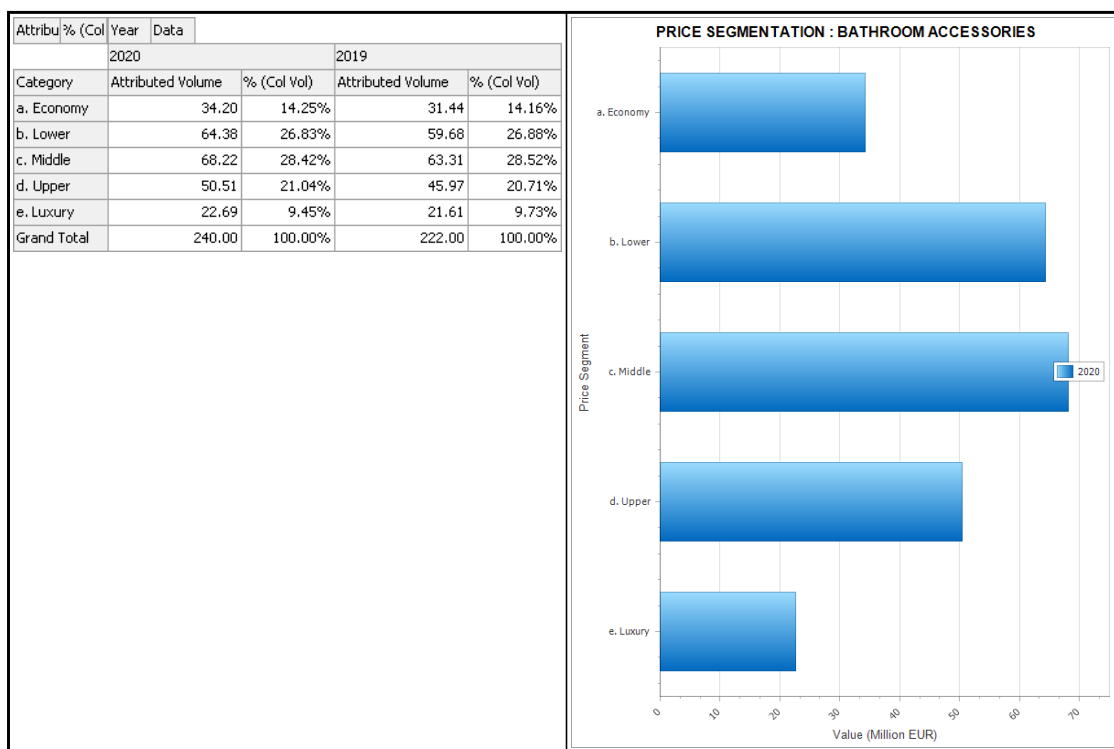
DE - BATHROOM ACCESSORY SALES BY MAIN MATERIAL TYPE (MILLION EUR)



Source: BRG based on industry interviews

6.4.4 Sales by Price Segment : Bathroom Accessories

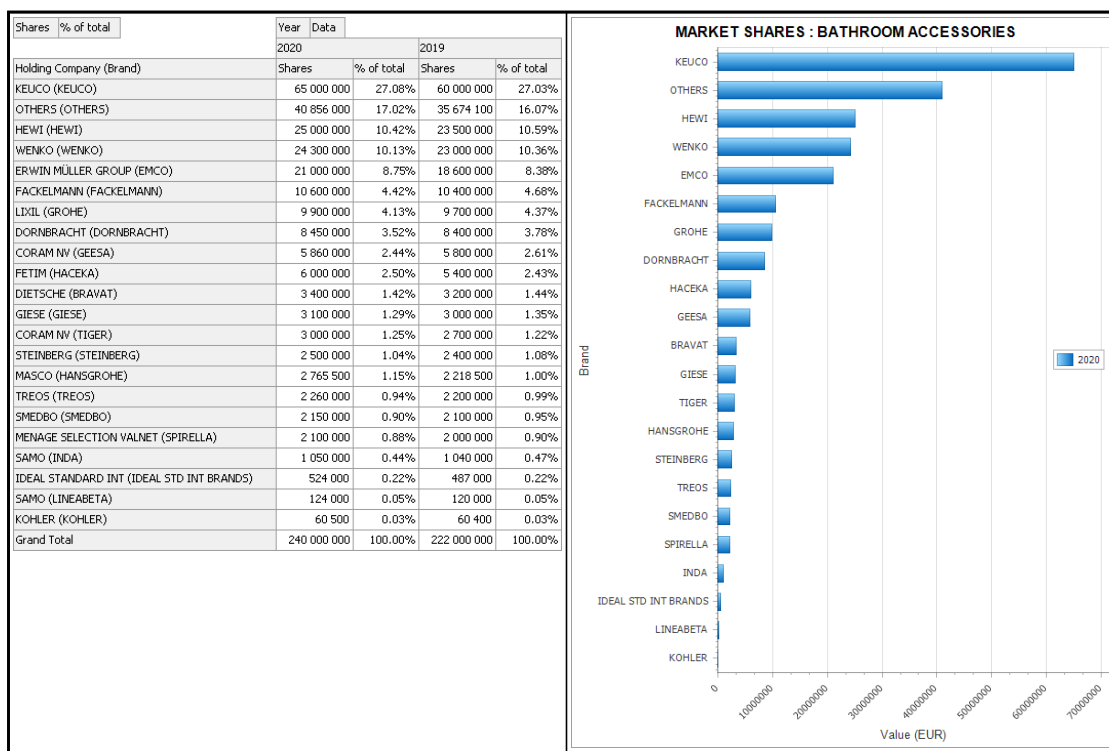
DE - BATHROOM ACCESSORY SALES BY PRICE SEGMENT (MILLION EUR)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

6.5 Market Shares : Bathroom Accessories

DE - ESTIMATED MARKET SHARES FOR BATHROOM ACCESSORIES (EUR)



Source: BRG based on industry interviews.

6.6.1 Route to Market: Bathroom Accessories (EUR)
DE - ESTIMATED DISTRIBUTION FOR BATHROOM ACCESSORIES (EUR): 2020

Manufacturer / Importer	Manufacturers / Importers
	100.0%

Wholesale Level	56.6%
	Wholesalers / Merchants
	56.6%

Retail Level	Manufacturer	Wholesaler	Manufacturer	Wholesaler	Manufacturer	Wholesaler
	9.2%	5.0%	18.6%	11.0%	10.9%	12.0%
	Internet Retailers		DIY Retailers		Specialist Retailers	
	14.2%		29.6%		22.9%	

Manufacturer	Wholesaler	Internet	DIY	Specialist
2.0%	28.2%	2.0%	1.6%	3.3%
Installers				
37.0%				

End Use Level	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer
	0.5%	0.2%	11.5%	27.0%	19.0%	34.8%
	Residential					
	93.1%					
	New Build			Residential RMI		
	14.7%			78.4%		
	Project	Non-Project		Project	Non-Project	
	3.0%	11.7%		1.0%	77.4%	

Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer	
2.2%	0.2%	0.7%	1.0%	0.6%	2.2%	
Non-Residential						
6.9%						
	New Build			Non-Residential RMI		
	2.3%			4.6%		
	Project	Non-Project		Project	Non-Project	
	1.0%	1.3%		1.4%	3.2%	

6.6.2 Route to Market: Bathroom Accessories (EUR)
DE - ESTIMATED DISTRIBUTION FOR BATHROOM ACCESSORIES (EUR): 2025

Manufacturer / Importer	Manufacturers / Importers
	100.0%

Wholesale Level	55.0%
	Wholesalers / Merchants
	55.0%

Retail Level	Manufacturer	Wholesaler	Manufacturer	Wholesaler	Manufacturer	Wholesaler
	11.0%	7.0%	18.0%	10.0%	11.0%	11.0%
	Internet Retailers		DIY Retailers		Specialist Retailers	
	18.0%		28.0%		22.0%	

Manufacturer	Wholesaler	Internet	DIY	Specialist
2.0%	26.5%	3.1%	1.7%	1.5%
Installers				
34.8%				

End Use Level	Manufacturer		Wholesaler		Internet		DIY		Specialist		Installer	
	0.5%		0.3%		13.9%		25.3%		19.7%		34.0%	
	Residential											
	93.7%											
	New Build						Residential RMI					
	11.4%						82.3%					
	Project			Non-Project			Project			Non-Project		
	2.3%			9.1%			1.6%			80.7%		

Manufacturer		Wholesaler		Internet		DIY		Specialist		Installer		
2.5%		0.2%		1.0%		1.0%		0.8%		0.8%		
Non-Residential												
6.3%												
	New Build						Non-Residential RMI					
	2.0%						4.3%					
	Project			Non-Project			Project			Non-Project		
	0.9%			1.1%			1.0%			3.3%		

Source: BRG Based on Industry Interviews. - Totals may vary due to roundings

6.6.3 Route to Market - Consolidated View : Bathroom Accessories

DE - ESTIMATED DISTRIBUTION FOR BATHROOM ACCESSORIES : 2020

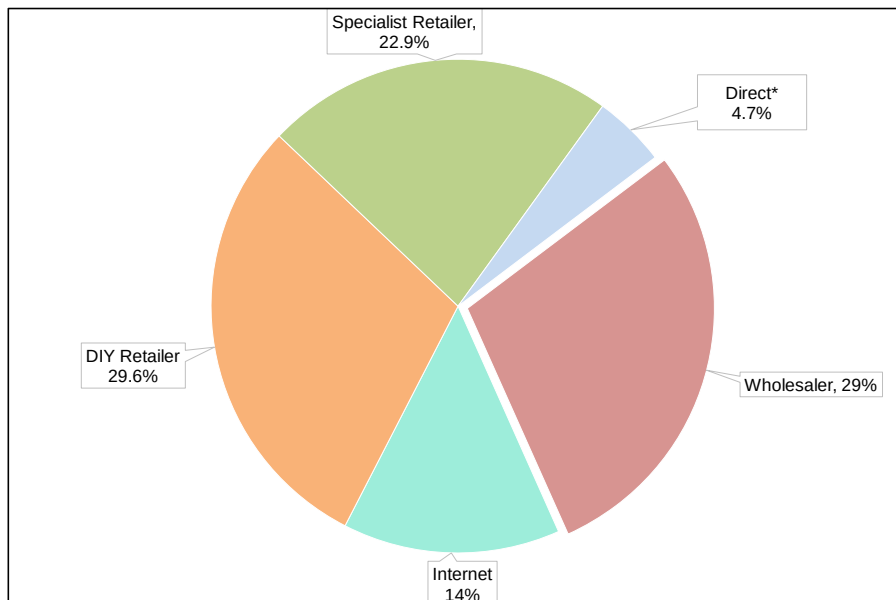
Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
28.6%	14.2%	29.6%	22.9%	4.7%	4.0%	2.4%	89.1%	4.5%
100%					6.4%		93.6%	

6.6.4 Route to Market - Consolidated View : Bathroom Accessories

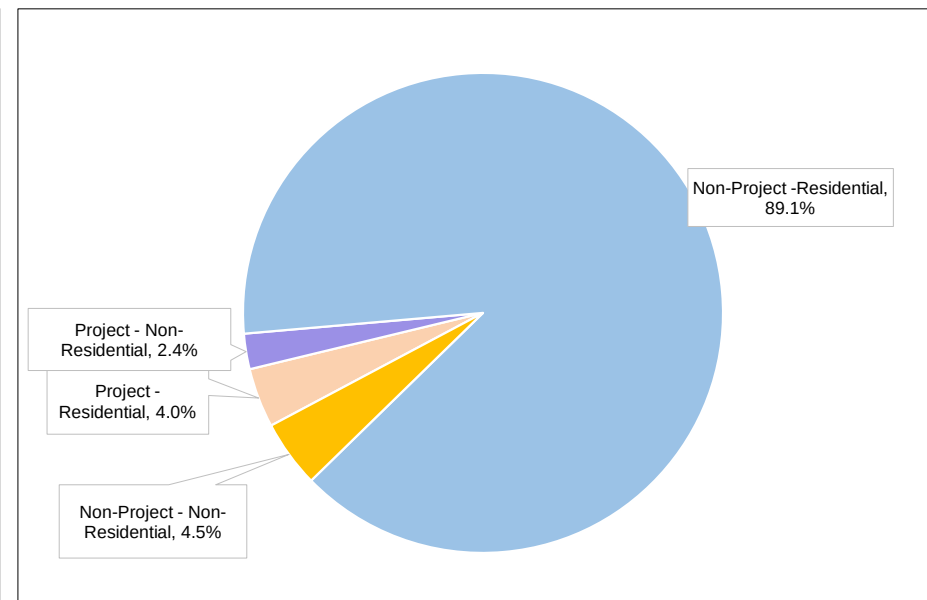
DE - ESTIMATED DISTRIBUTION FOR BATHROOM ACCESSORIES : 2025

Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
27.0%	18.0%	28.0%	22.0%	5.0%	3.9%	1.9%	89.8%	4.4%
100%					5.8%		94.2%	

DE - SALES OF BATHROOM ACCESSORIES BY DISTRIBUTION CHANNEL: 2020



DE - SALES OF BATHROOM ACCESSORIES BY POINT OF DECISION AND END-USE SECTOR: 2020



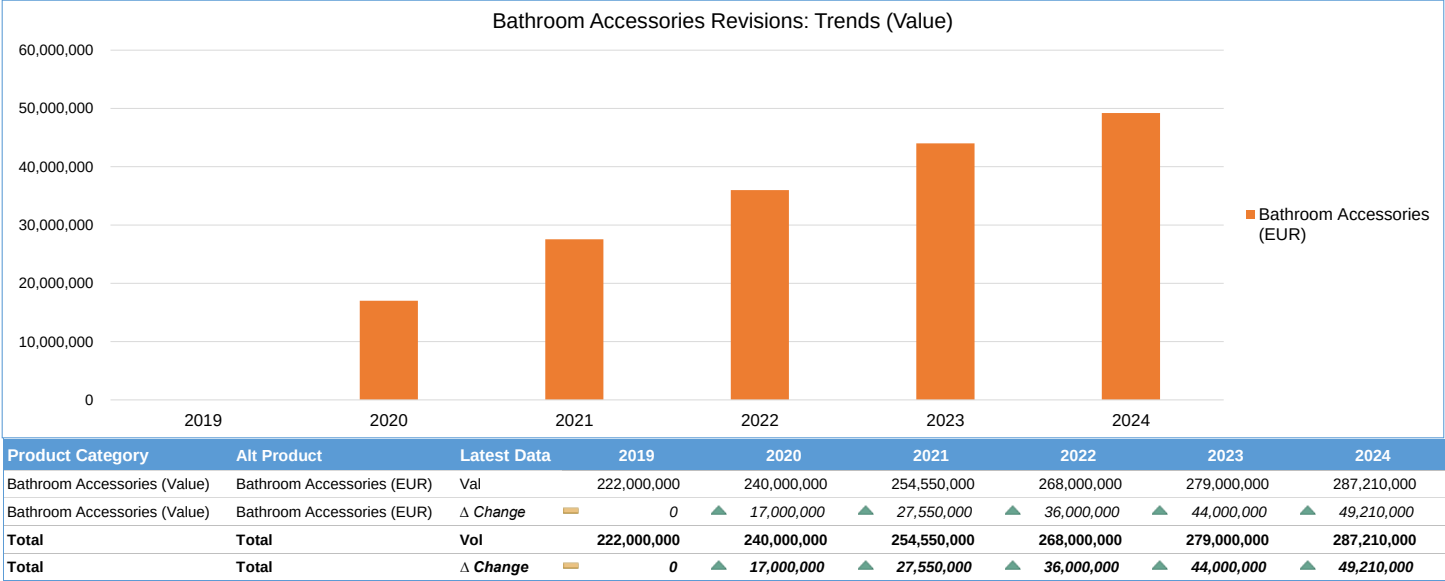
Source: BRG Based on Industry Interviews.

*Direct includes sales going directly from manufacturers to installers, contractors and private customers.

6.7 Revisions in the German Bathroom Accessories Market (April 2021)

DE - REVISIONS IN THE GERMAN BATHROOM ACCESSORIES MARKET

Data Comparison: October 2020



No revisions have been made to historical volumes for this market.

GERMANY

SHOWER ACCESSORIES

THE EUROPEAN BATHROOM & KITCHEN PRODUCT MARKETS



Market Characteristics and Technological Trends

The German shower accessories market has benefited from the trend away from bathing towards showering. Over the last decade the market has performed well with growing sales on a yearly basis. A wide variety of shower heads is available with established German manufacturers, such as GROHE, HANSGROHE and DORNBACHT, offering high-priced models. Low-end alternatives are sold through non-traditional channels, such as retail shops, such as TCHIBO, or supermarkets and food discounters, e.g. ALDI, LIDL and REAL, with occasional sales.

Shower heads are one of the bathroom items that are replaced regularly, depending on the quality of the water in a geographic area. Most common are shower heads that are adjustable in height and they are replacing traditional shower heads with a single function, as they provide more comfort for the consumer. In recent years, fixed heads, in particular so-called rain showers, have increasingly taken away market shares from adjustable shower heads. A fixed shower head is almost always sold together with a hand shower. Least common are fixed side jets, which failed to establish themselves on the market. Fixed shower heads are common in public showers at swimming pools or sports centres. The latest trend are shower pipes, which have been rapidly gaining popularity. Shower pipes were introduced by HANSGROHE in the mid-2000s and sales have been increasing ever since. Nowadays, end users not only can buy high-priced shower pipes, but also low-priced products have become easily available in DIY and online stores. According to BRG definition, a shower pipe consists of a system that includes a fixed shower head that is mounted onto the mixer using a pipe, hand shower head and a flexible hose, shower support and a shower mixer, which can be thermostatic or non-thermostatic.

The German market for shower accessories has a high share of flexible hoses, which are either made of metal or plastic usually in a metallic colour. The share of plastic flexible hoses, which look like metal, is slowly increasing and its share is estimated at more than 54%. Demand for flexible hoses is growing as they are commonly installed in larger showers in combination with a fixed shower head. Shower supports include bars as well as holders. Holders that are mounted on the wall are rarely sold in the German market, and bars account for most sales. Shower bars used to be the shower item with the longest lifespan and were replaced as part of a complete bathroom refurbishment.

Nowadays, it has become easier to replace shower bars by using existing drillings or non-drilling techniques for new installations. Sales of shower supports are expected to be negatively affected by growing popularity of shower pipes, rain and hand showers. Shower sets and shower mixers are either first time installations in new buildings (often concealed) or installations as part of a bathroom refurbishment. These are being increasingly sold, as they provide the end users with a certain convenience in buying a matching set of products. The most common

set consists of a shower head, hose and bar without a mixer. This is followed by the complete set, which also includes a mixer. Closely after is the shower set that consists of a shower head and hose only without a mixer. This is a typical replacement product in case the shower head or hose needs replacing.

Despite the popularity of shower sets among end users, single item replacements also do happen. Several product innovations, like integrated wireless speakers and colourful lights in the shower head, have been introduced to the market and have established their market niche. Shower panels consist of body jets and multiple showerheads, which can replace standard shower sets. Sales started to take off when HANSGROHE and the French manufacturer VALENTIN introduced their units. Sales, however, have started to decline following the introduction of shower pipes (also by HANSGROHE) and fixed shower heads with hand showers. These comparably simple and less expensive products have experienced a surge in popularity. Demand for shower panels peaked more than a decade ago. Since then demand has been declining on a yearly basis.

Recent Market Trends

The COVID-19 pandemic shaped the developments in the year 2020 and 'business as usual' was replaced by government measures. In Germany the official lockdown, which is better understood as a 'prohibition of contact', started at the end of March 2020. Bavaria was the first state to implement measures to contain the spread of the virus, which was followed by the remaining states towards the end of March. For a few weeks the German population was advised to work from home when possible, with most non-essential stores closing for a limited time only. Measures implemented varied by state (*Bundesländer*). Most stores, including DIY stores, were allowed to open again from the beginning of April 2020 onwards, with differences from state to state in the level of restrictions and measures. In certain regions, shopping in DIY stores was limited to professionals only.

During 2020, new housing construction continued without much interruption, with only large scale project sites halted to implement protective measures. That being said, the overall production per employee is believed to have shrunk significantly, and remains to be the case, since many employers introduced measures to protect the construction workers. The bathroom industry was declared part of the crisis infrastructure, but installers could still continue to work without many interruptions. Overall, there has been relatively little impact of the COVID-19 pandemic on the building related industry, including the bathroom market, in Germany. In some cases, end users may have postponed, or cancelled, bathroom related projects due to the uncertainty on the market, but overall demand remained relatively strong, as there was a large backlog of works from the past, benefiting sales of bathroom and kitchen equipment. At the same time, in certain instances, the COVID-19 pandemic may have contributed to additional demand. Since consumers were spending more time indoors, they had the opportunity to dedicate themselves to repair or upgrade works, whereas others might have decided to invest into their property by installing a new bathroom or kitchen due to (international) holidays being difficult because of the current pandemic.

Following the positive path of the past years, the German market for shower accessories grew also in 2020, at a slightly stronger rate when compared to the previous year. This development can be ascribed to the stronger performance of the RMI segment, which compensated for a decline from new and non-housing sales. The impact of the COVID-19 pandemic caused most Germans to spend more time at home, which was used to invest into homes. More time was available and planned projects could be carried out. With DIY shops staying open during most of the year in addition to a great variety of products available on the Internet, consumers were faced with favourable conditions to improve their showers.

Like with other bathroom products, there is a shift towards high quality products, which are mainly boosted by the popularity of large rain showers. On the other hand, there is a trend towards cheaper solutions with a certain quality standard. German consumers tend to either buy cheap shower heads (with or without a thermostat) or upper market products such as fixed rain showers together with a hand shower. For example, concealed thermostatic mixers that are covered with glass, introduced by HANSGROHE, have received positive feedback among consumers. Moreover, demand for fixed shower heads with a concealed connection to a thermostatic mixer is growing. In almost all cases a hand shower is installed together with a fixed shower head.

Demand for shower panels stabilised in 2020 due to additional demand from consumers interested in more wellness. Due to lockdown restrictions consumers were more likely to invest into a hydrotherapy product at home. It is believed that this marginal demand compensated for the overall declining trend. Shower panels have been suffering from the rising preference towards shower pipes, and this continues to affect sales demand. There is a significant demand that comes mainly from the economy/lower price level, whilst the upper/luxury level has almost disappeared.

The shower pipe market has undergone the strongest growth rate in 2020, excelling the performance of the previous year. The main reason for the success is the experience of a rain shower, which end users are increasingly fond of. At the same time, more manufacturers are offering shower pipes for more affordable prices. In the past it was difficult to buy a shower pipe for less than EUR 400, whilst, nowadays, consumers can buy models even for EUR 100. In contrast to the other shower accessories products, the share of shower pipes sold in the middle and lower price segments has increased in 2020. This shift did not result in a lower MSP, because it is believed that the share of shower pipes with a thermostatic mixer is increasing. They are believed to be significantly more expensive and, therefore, the overall MSP increased.

Prices and Market Values

The development of the average MSP for shower accessories differs greatly from one product to the other. With the gaining popularity of rain showers, end users are willing to spend more on a superior product. Since rain showers are more

expensive than a standard shower head, the average price of shower heads increased the strongest.

Most manufacturers have increased their prices to account for inflation and production costs linked to higher wages and raw material costs. This development is reflected by the average increase of flexible hose sales.

The MSP of shower pipes, especially in middle and upper price segments, contributed to a notable increase. With growing competition on the market, manufacturers find it difficult to justify strong price increases. At the same time good quality products become more readily available in DIY stores and online. This further puts pressure on the competition and end users are able to spend a bit more to afford a branded shower pipe with a thermostatic mixer.

In contrast to previous years, the average MSP of shower panels slightly increased in 2020 as the share of products sold in the economy price segment dropped. At the same time the number of products sold in the upper and middle price range increased. The shift towards more expensive products and the marginal increase of the MSP led to an overall increase.

New Housing

New housing demand for all products was negative in 2020 when compared to the previous year due to a lower number of new dwelling completions. The only exception are shower pipes, for which demand slightly increased in new housing. This can be explained by the growing popularity for this product, which is taking away share from more traditional solutions. It can be considered that shower pipes are the standard when installing a large shower and only concealed options, which are more likely in the upper price segments, are preventing sales from growing.

According to industry feedback, the number of bathrooms per dwelling is increasing. It is becoming more common to have two bathrooms in a standard 100 m² apartment. The trend is to either convert the guest toilet into a smaller second bathroom, or to equip the master bedroom with an en-suite bathroom. This development benefits the shower accessories market; however, only shower pipes profited from it in 2020 due to the overall decline in new housing completions. It is believed that the share of shower pipes is going to increase to the expense of traditional shower sets.

The share of concealed shower installations is increasing in this segment as it is easier to install when building a new bathroom. Especially in the upper price segment, concealed systems are gaining popularity. In this case a large rain shower is fixed to the ceiling and a hand shower with a flexible hose is attached to the wall. Modern concealed mixers or thermostat mixers have only buttons to adjust water temperature and pressure; however, there are also more affordable solutions with a tap.

The replacement market represents the most significant segment for shower accessories, because of the shorter lifespan of these products when compared to ceramic sanitary ware, for example. Flexible hoses tend to break when used and moved regularly and shower heads are likely to become useless from limescale corrosion. The lifespan of a flexible hose is estimated to be about 15 years or longer and depends on its use. KOTTMANN, for example, gives a seven year warranty on their flexible hose. If a flexible hose breaks outside the warranty, then end users go to DIY stores to buy replacements there. If it breaks within the warranty time, then it is replaced professionally by the company. Demand for flexible hoses was positive in 2020 with a slightly bigger growth rate when compared to the previous year. Flexible hoses are either made of stainless steel or synthetics, with a trend towards the latter. The reason for this is that steel products may start to rust, while synthetic hoses are visually almost identical to steel – and less expensive. It is estimated that the overall market for flexible hoses is dominated by plastic hoses.

The number of shower head sales in the replacement segment generate the bulk of sales due to their shorter product longevity. Replacement sales not only refer to substitutions of a single shower accessory, but also include sales of complete shower sets, often installed at the time when a bathroom refurbishment takes place. Demand for shower heads is believed to be a growing market, which benefits from the overall trend towards showering. Demand for shower supports in the RMI segment was slightly positive in 2020 when compared to the previous year. This product is mostly sold in a set together with a shower head and a flexible hose. End users prefer to buy a complete set to make sure that all individual products match each other. Additionally, it is an easy and affordable way to give a newer look to the shower by replacing weary parts.

Demand for shower panels in the replacement segment slightly increased in 2020. The product is outdated and is more likely to be replaced by a rain shower, which is nowadays associated with more wellness. However, a significant volume of shower panel sales in the replacement market remains for like-for-like replacements of products that were installed in the past. These products are of lower quality and usually need regular replacement when exposed to hard water and frequent use. It is believed that the lockdown restrictions contributed to some growth for this product as end users invested into more wellness products at home.

Shower pipes are the most popular product for bathroom renovations and have become a common solution to enhance the showering experience. While prices used to be high in the past, nowadays, there is a great variety of good quality products for an affordable price. This development mirrors the rapidly growing demand for this product, especially in the replacement segment. Looking ahead, demand is likely to remain high and shower pipes are likely to grow their share in bathroom renovations. The product can be easily fitted to an existing shower.

Non-Housing

In 2020, the non-housing segment underwent a significantly negative development in both renovation and new non-residential construction completions. The investments into buildings for education and health were the only exception, which had a slightly positive development. All other sectors such as industrial, hospitality, office and commercial buildings had a strong decline as projects got postponed or cancelled due to the COVID-19 pandemic. As a result almost all shower accessories products sales decreased in 2020. The only exception are shower pipes, which recorded a small increase that year. This can be explained by a growing share at the expense of more traditional solutions.

Most sales within the non-housing segment go to the commercial field, such as sports centres, hotels, nursing homes and, in a minor part, to private hospitals. Demand for shower pipes in the non-housing segment has played a significant share in the past. Before finding its way to residential bathrooms, the product was popular in hotel bathrooms. Nowadays, shower pipes are increasingly installed in residential bathrooms and the non-housing segment continues losing share when compared to the total market volume.

Overall demand in the non-housing segment is closely linked to the development of the non-housing construction market. Demand for 2021 and beyond is expected to continue to be negative, before starting to recover from 2024 onwards. Since the share of total shower accessories sales into non-housing is limited by buildings that have a shower, demand is likely to stay small in comparison to the total market size.

Supply Structure

The main market players for shower accessories are HANSGROHE, LIXL (GROHE) and SEE (SCHÜTTE). In addition, there are other German manufacturers such as DORNBRACHT, KEUCO and KLUDI competing in a mature market. In contrast to the other manufacturers, SEE (SCHÜTTE) focuses on other distribution channels than the wholesale. Several players from outside the industry supply products through different distribution channels, too. Like for bathroom accessories and furniture, retailers such as TCHIBO and food discounters such as LIDL, PENNY, NETTO, NORMA, MARKTKAUF and HERKULES, generate large sales volumes. They tackle the lower-priced segment of the market with inexpensive imports from the Far East. In particular, the flexible hose market is characterised by a high share of OEM manufacturing – with retailers competing on prices with inexpensive imported products.

An important OEM manufacturer is MHK SYSTEMS (former KOTTMANN), which supplies major companies with high quality products, as well as DIY stores that sell it under private labels. MHK SYSTEMS acquired the license rights for the KOTTMANN brand and the machine park for shower hose production from the company in 2019. Since then, KOTTMANN branded sales were stopped in Germany and KOTTMANN branded flexible hoses are mainly sold to Arabic

countries. Its total sales by material can be divided into about one third metal and two thirds plastic.

Many of the large bathroom manufacturers such as DUSCHOLUX, DUSAR and LIXIL (GROHE) stopped supplying hydrotherapy products such as shower panels in the past. SEE (SCHÜTTE) is the market leader for this product and sells most of their products to the replacement segment.

Regarding shower pipes, MASCO (HANSGROHE) is the market leader and, as a pioneer, launched the first shower pipes in 2008. Other manufacturers were quickly to follow and demand for shower pipes has been growing significantly. LIXIL (GROHE) is almost equally successful with its shower pipes now and it is believed that both manufacturers together have a market share over 50%. Other significant manufacturers are ORAS (HANSA) and SEE (SCHÜTTE). SEE (SCHÜTTE) introduced a shower pipe in black colour, which is sold through many different distribution channels.

Forecasts

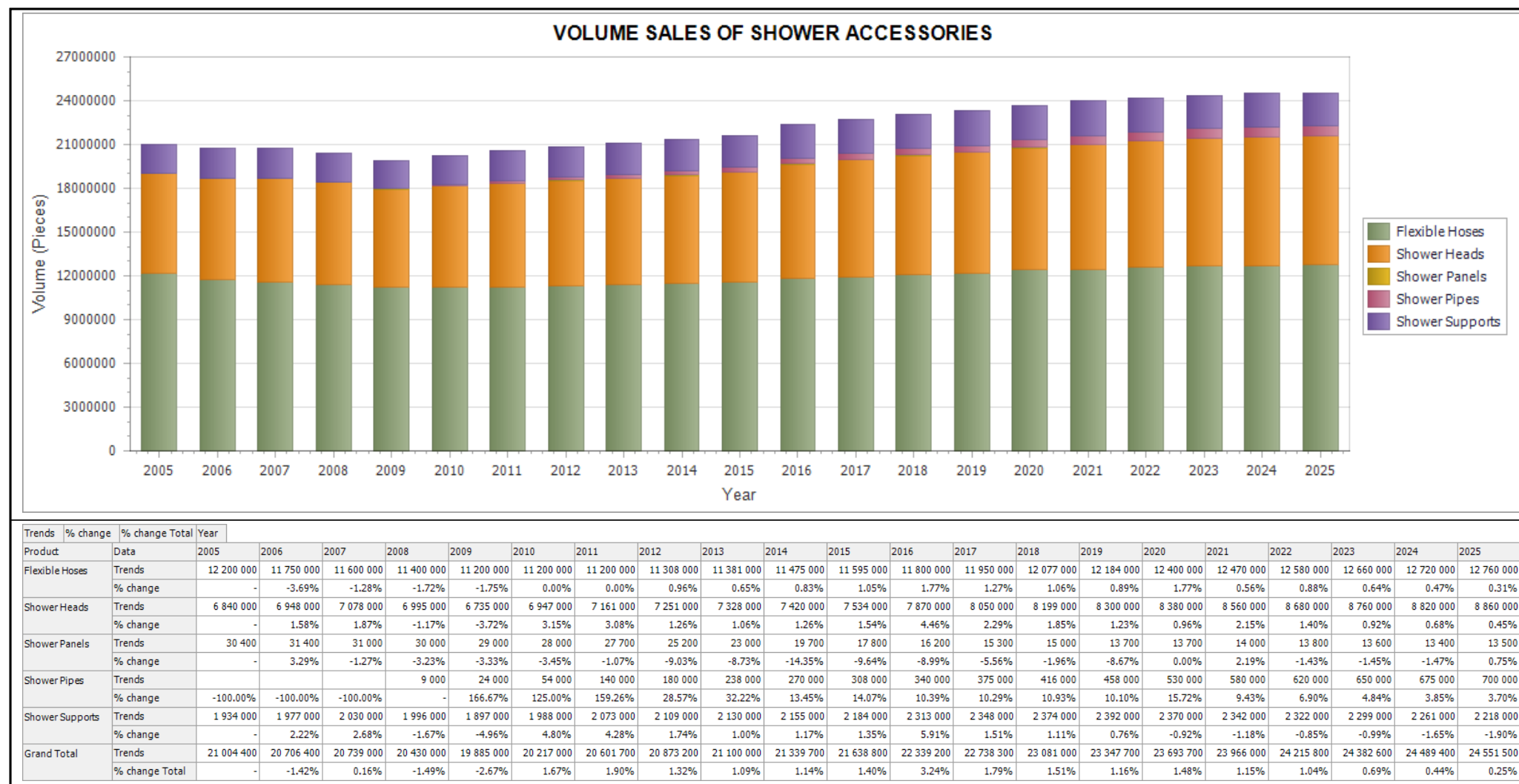
The German shower accessories market is expected to develop positively during the forecast period. The shower accessories market, as a mature market, depends mainly on replacement sales. Demand from the RMI segment is expected to remain positive during the forecast period. Like-for-like sales and demand from bathroom refurbishments are believed to compensate for declining new residential and non-housing sales. More traditional shower accessories are believed to give way to shower pipes, which are expected to gain share in new bathrooms and sales are expected to grow strongly during the forecast period.

Demand for flexible hoses is likely to grow stronger over the forecast period as most new showers are equipped with a hand shower. Moreover, shower head sales should profit from this development, too. In contrast, demand for shower supports is expected to decline marginally due to the strengthening trend towards shower pipes. Demand for hydrotherapy shower panels is expected to decline again from 2022 onwards. While the share of the upper price segment is likely to shrink further in the long term again, the share of the price entry segment is expected to be sustained from like-to-like replacement sales.

The shower pipe market is expected to have the strongest growth rates towards the end of the forecast period. This relatively new product is gaining market share and popularity in the German shower market. With manufacturers launching new models on the market, end users have a growing choice of colours, price ranges and styles to choose from. This development is likely to stimulate a further growth, while the MSP is expected to continue declining.

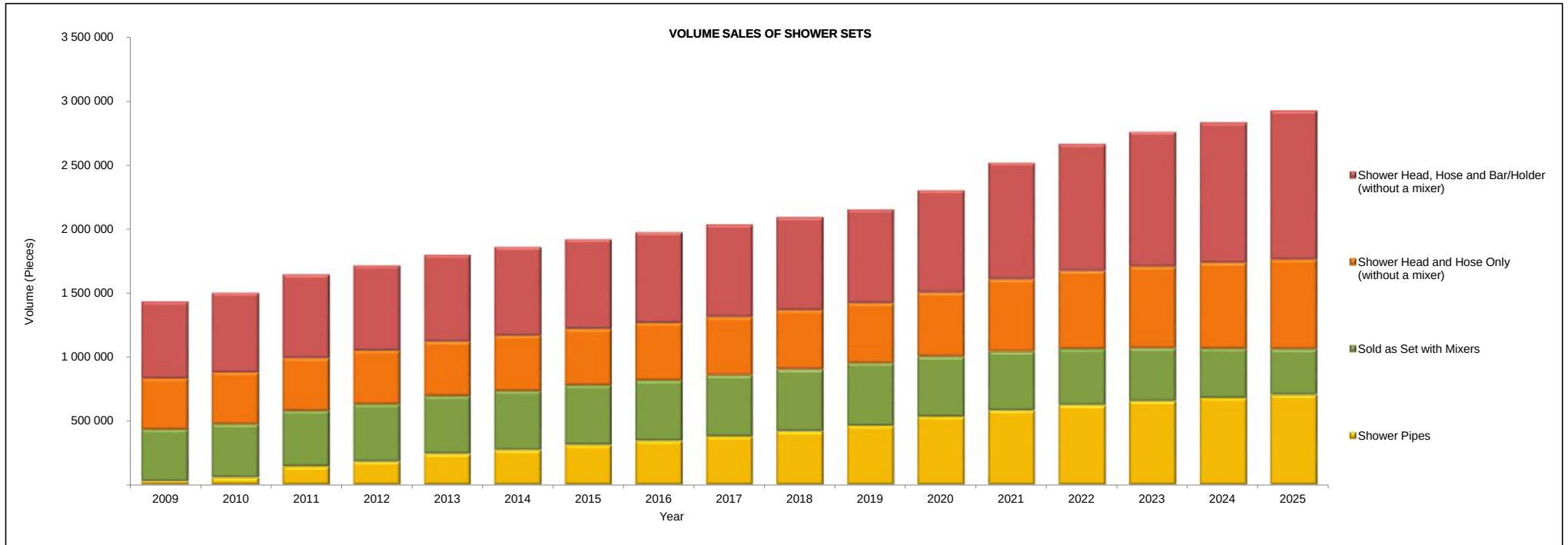
7.2.1 Historical Trends & Forecasts : Shower Accessories

DE - SUMMARY OF TRENDS IN THE SHOWER ACCESSORIES MARKET : HISTORICAL AND FORECASTS



Source: BRG based on industry interviews.

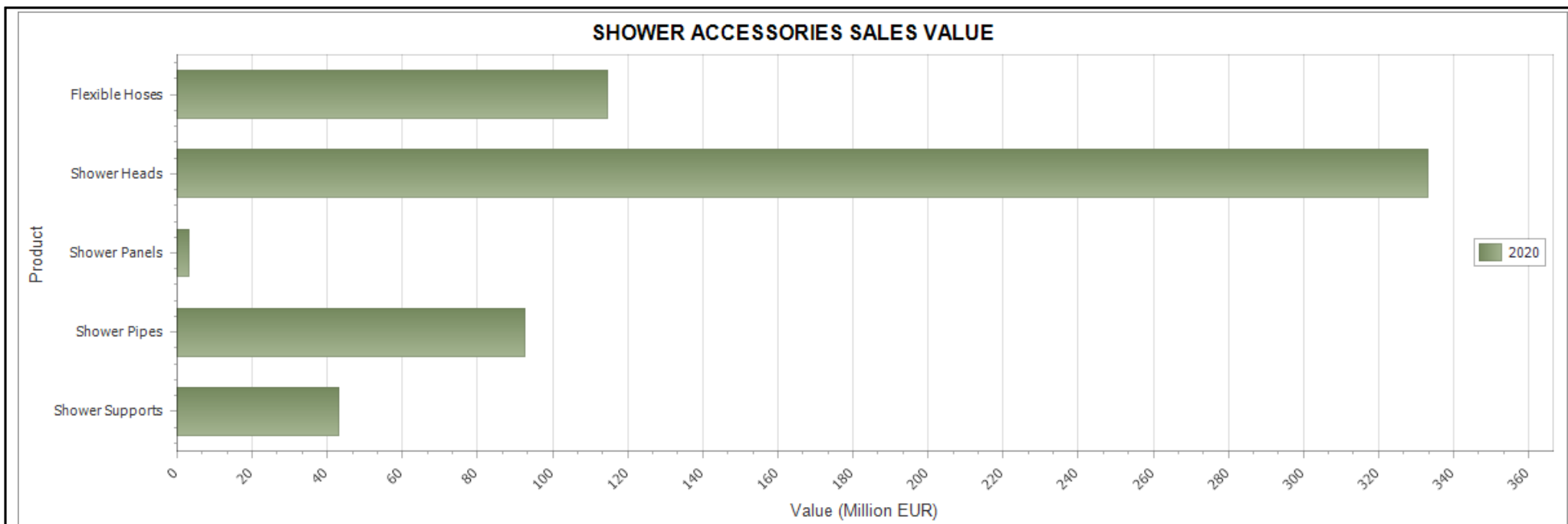
7.2.2 Historical Trends and Forecasts : Shower Sets
DE - SUMMARY OF TRENDS IN THE SHOWER SETS MARKET : HISTORICAL AND FORECASTS



Product Type	Product	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Shower Sets	Shower Head, Hose and Bar/Holder (without a mixer)	600 000	620 000	651 000	666 000	678 000	690 000	700 000	709 000	720 000	725 000	732 000	797 000	909 000	989 000	1 048 000	1 093 000	1 162 500
	% Change	-	3.33%	5.00%	2.30%	1.80%	1.77%	1.45%	1.29%	1.55%	0.69%	0.97%	8.88%	14.05%	8.80%	5.97%	4.29%	6.36%
	Shower Head and Hose Only (without a mixer)	400 000	407 000	414 000	421 000	428 000	435 000	442 000	449 000	456 000	463 000	470 000	503 000	565 000	610 000	638 000	670 000	699 333
	% Change	-	1.75%	1.72%	1.69%	1.66%	1.64%	1.61%	1.58%	1.56%	1.54%	1.51%	7.02%	12.33%	7.96%	4.59%	5.02%	4.38%
	Sold as Set with Mixers	405 000	417 000	436 000	445 000	452 000	460 000	467 000	473 000	478 000	483 000	488 000	468 000	458 000	438 000	415 000	386 000	357 000
	% Change	-	2.96%	4.56%	2.06%	1.57%	1.77%	1.52%	1.28%	1.06%	1.05%	1.04%	-4.10%	-2.14%	-4.37%	-5.25%	-6.99%	-7.51%
	Shower Pipes	24 000	54 000	140 000	180 000	238 000	270 000	308 000	340 000	375 000	416 000	458 000	530 000	580 000	620 000	650 000	675 000	700 000
	% Change	-	125.00%	159.26%	28.57%	32.22%	13.45%	14.07%	10.39%	10.29%	10.93%	10.10%	15.72%	9.43%	6.90%	4.84%	3.85%	3.70%
Grand Total		1 429 000	1 498 000	1 641 000	1 712 000	1 796 000	1 855 000	1 917 000	1 971 000	2 029 000	2 087 000	2 148 000	2 298 000	2 512 000	2 657 000	2 751 000	2 824 000	2 918 833
% Change Total		-	4.83%	9.55%	4.33%	4.91%	3.29%	3.34%	2.82%	2.94%	2.86%	2.92%	6.98%	9.31%	5.77%	3.54%	2.65%	3.36%

Source: BRG based on Industry Interviews.

7.3.1 Prices and Market Values : Shower Accessories
DE - ESTIMATED VALUES OF THE SHOWER ACCESSORIES MARKET

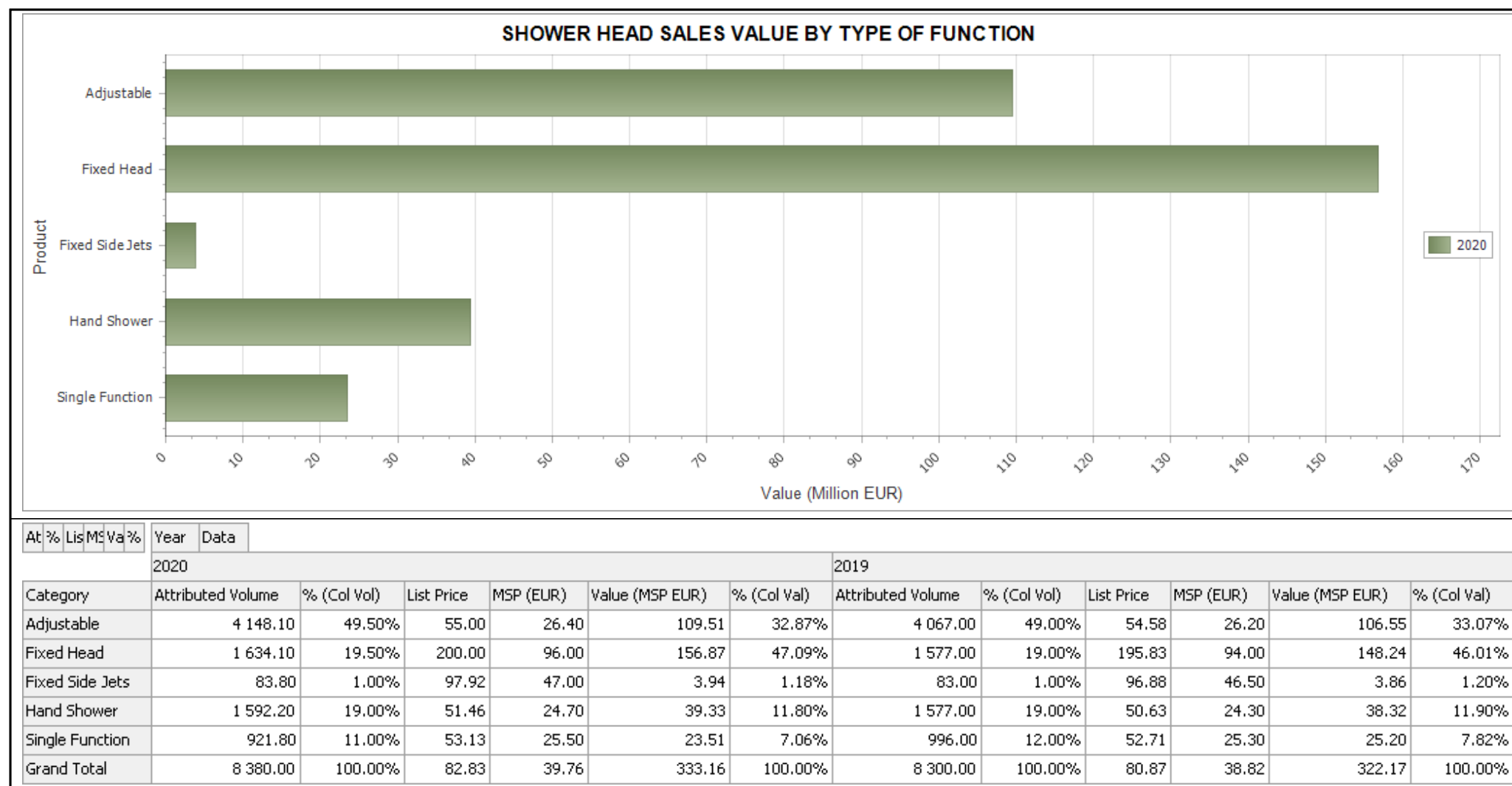


Volum	MSP (l	Value	Year	Data			
				2020	2019		
Product	Volumes	MSP (EUR)	Value (MSP EUR)		Volumes	MSP (EUR)	Value (MSP EUR)
Flexible Hoses	12 400 000	9.24	114.58		12 184 000	9.13	111.24
Shower Heads	8 380 000	39.76	333.19		8 300 000	38.82	322.21
Shower Panels	13 700	216.50	2.97		13 700	211.55	2.90
Shower Pipes	530 000	174.78	92.63		458 000	172.40	78.96
Shower Supports	2 370 000	18.17	43.06		2 392 000	18.07	43.22
Grand Total	23 693 700	24.75	586.43		23 347 700	23.92	558.53

Source: BRG based on industry interviews. Values Shown in Millions (at MSP = Manufacturers' Selling Price)

7.3.2 Prices and Market Values by Type : Shower Heads

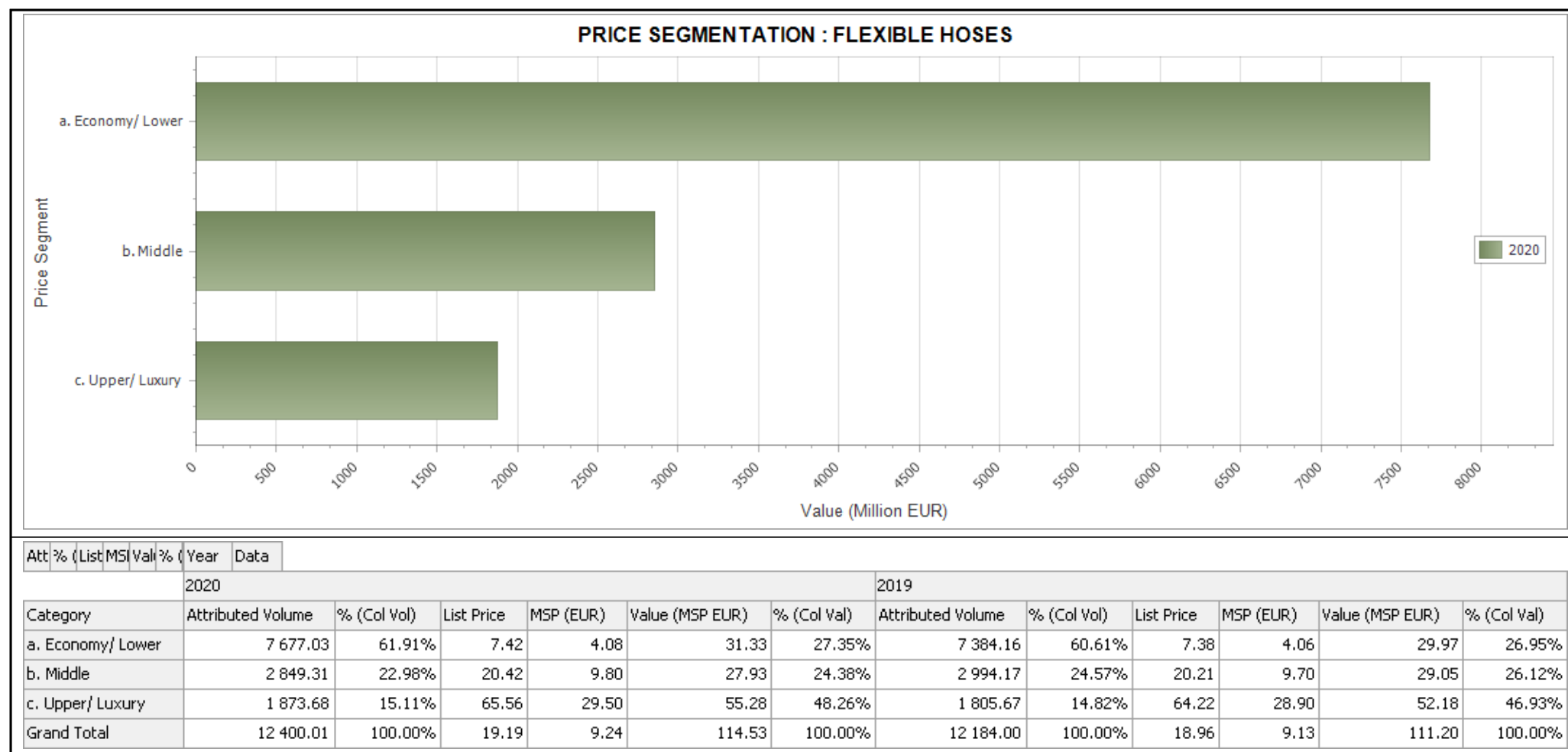
DE - ESTIMATED VALUES OF THE SHOWER HEAD MARKET BY TYPE OF FUNCTION



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price)

7.3.3 Price Segmentation : Flexible Hoses

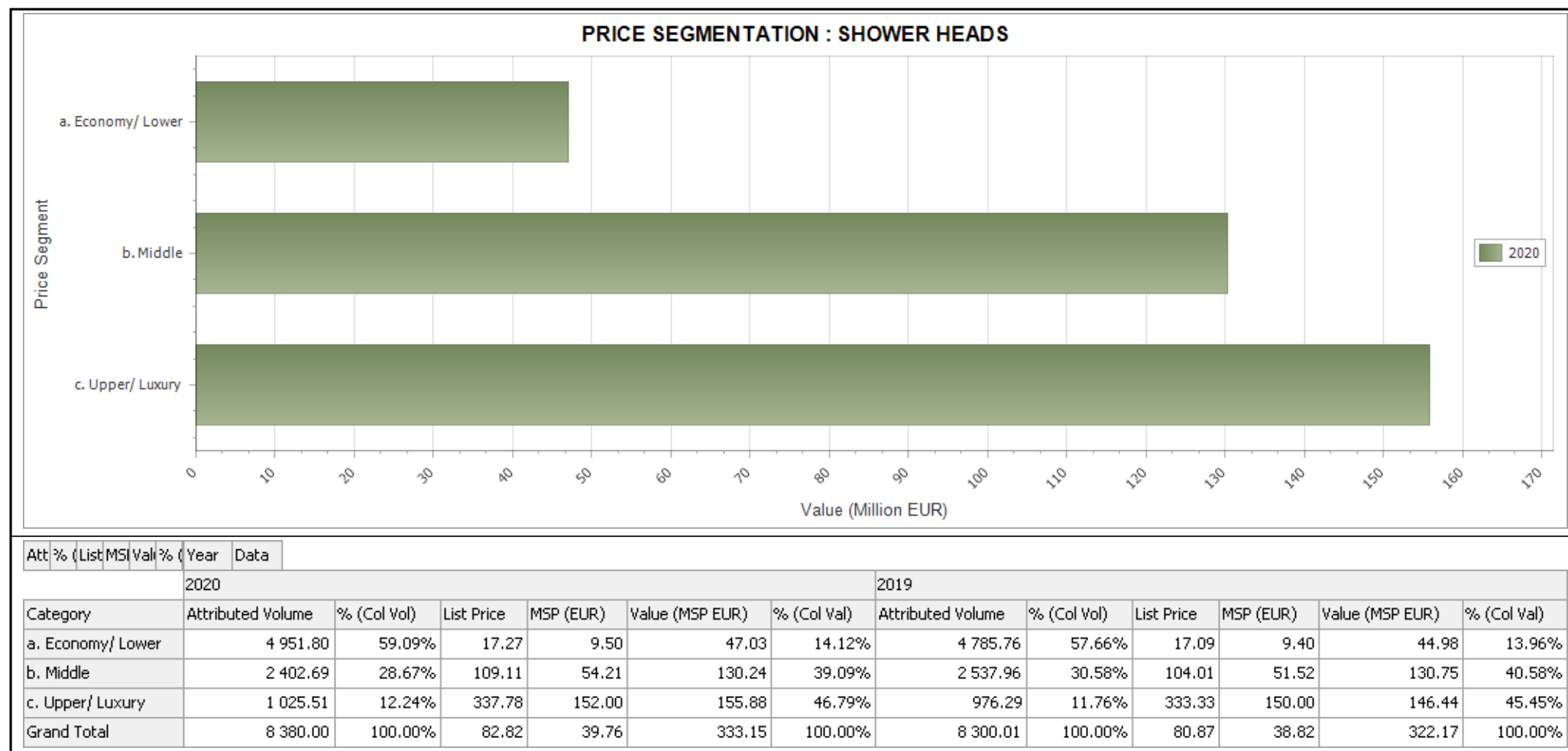
DE - ESTIMATED PRICE SEGMENTATION FOR FLEXIBLE HOSES



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€5, Middle €5-€15, Upper/ Luxury >€15

7.3.4 Price Segmentation : Shower Heads

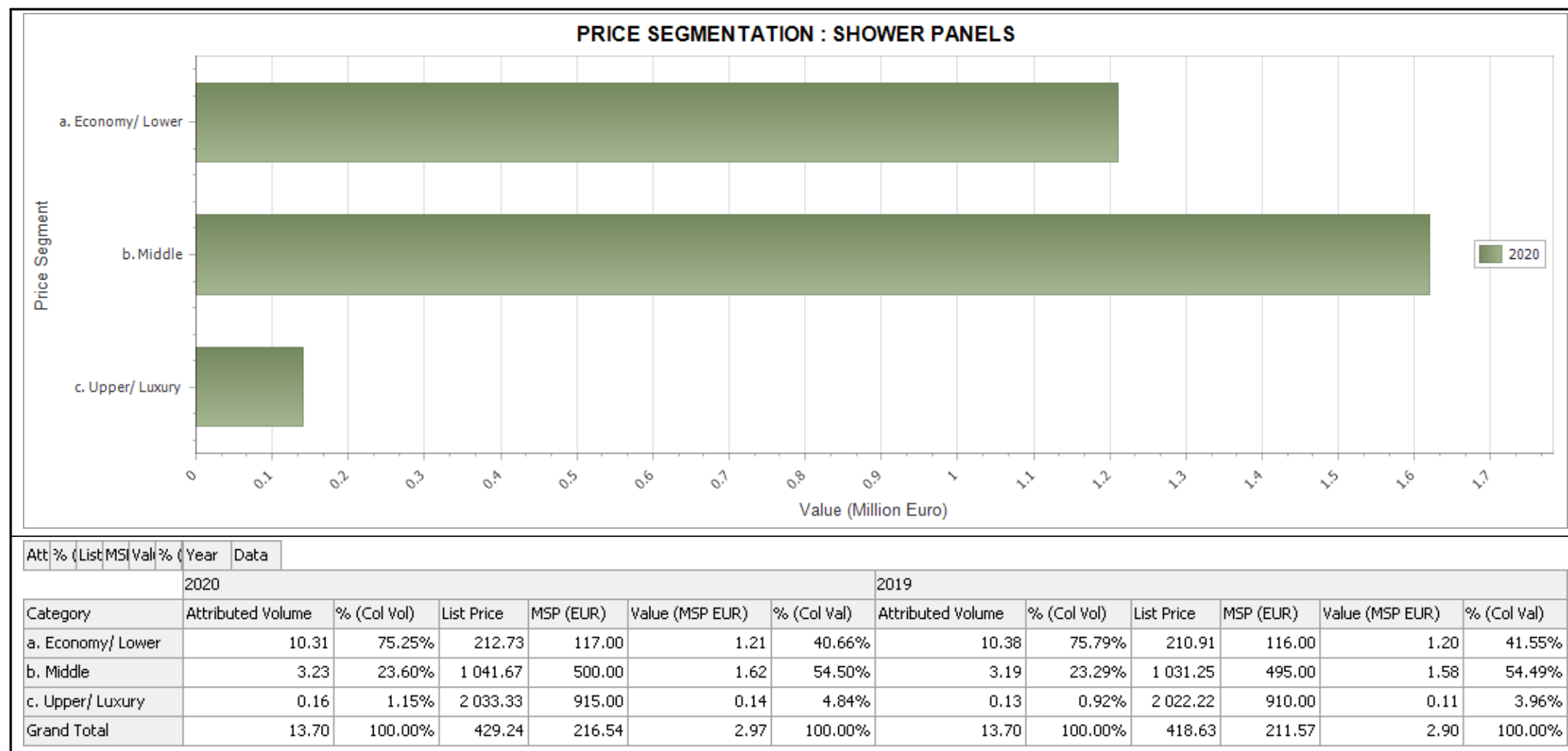
DE - ESTIMATED PRICE SEGMENTATION FOR SHOWER HEADS



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€40, Middle €40-€80, Upper/ Luxury >€80

7.3.5 Price Segmentation : Shower Panels

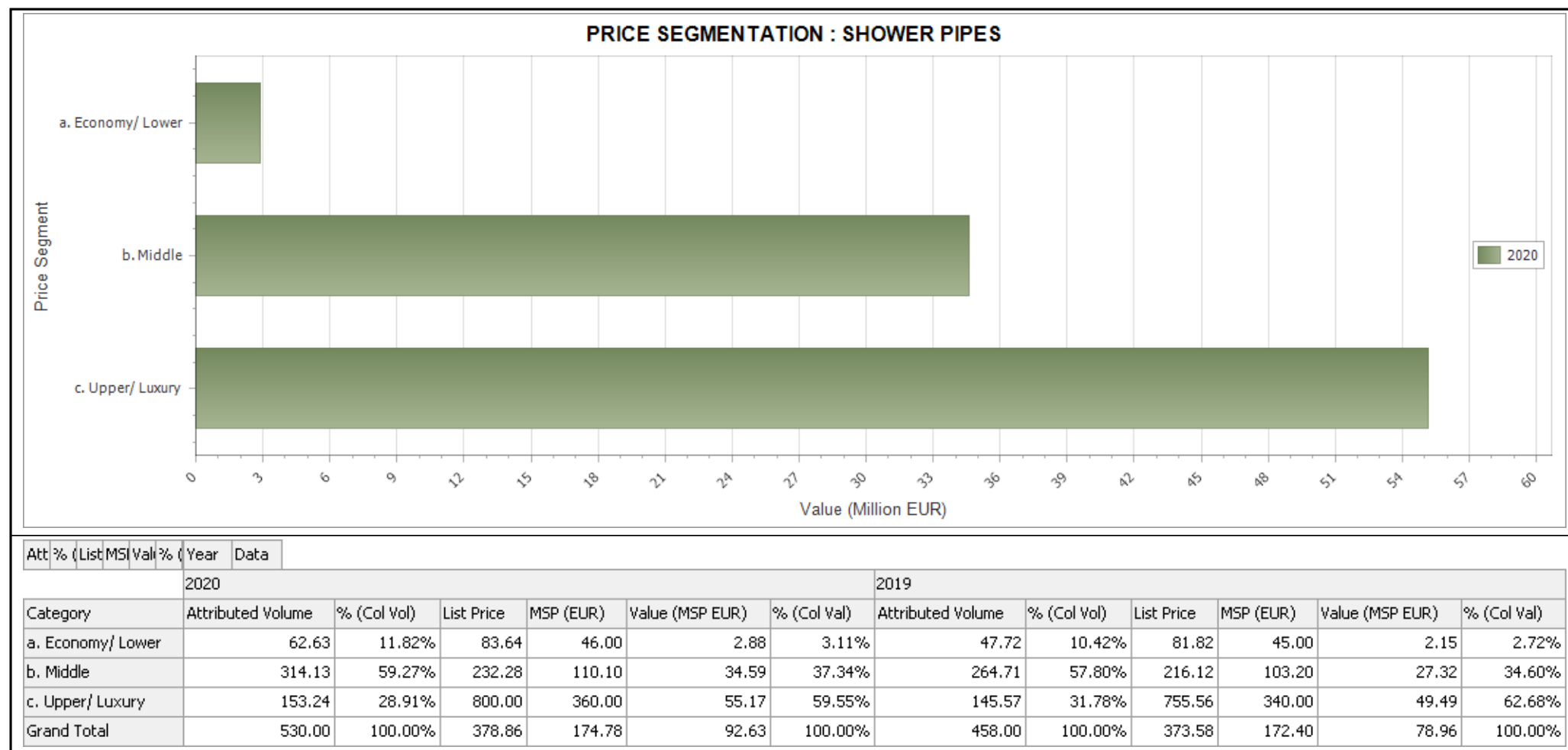
DE - ESTIMATED PRICE SEGMENTATION FOR SHOWER PANELS



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€300, Middle €300-€600, Upper/ Luxury >€600

7.3.6 Price Segmentation : Shower Pipes

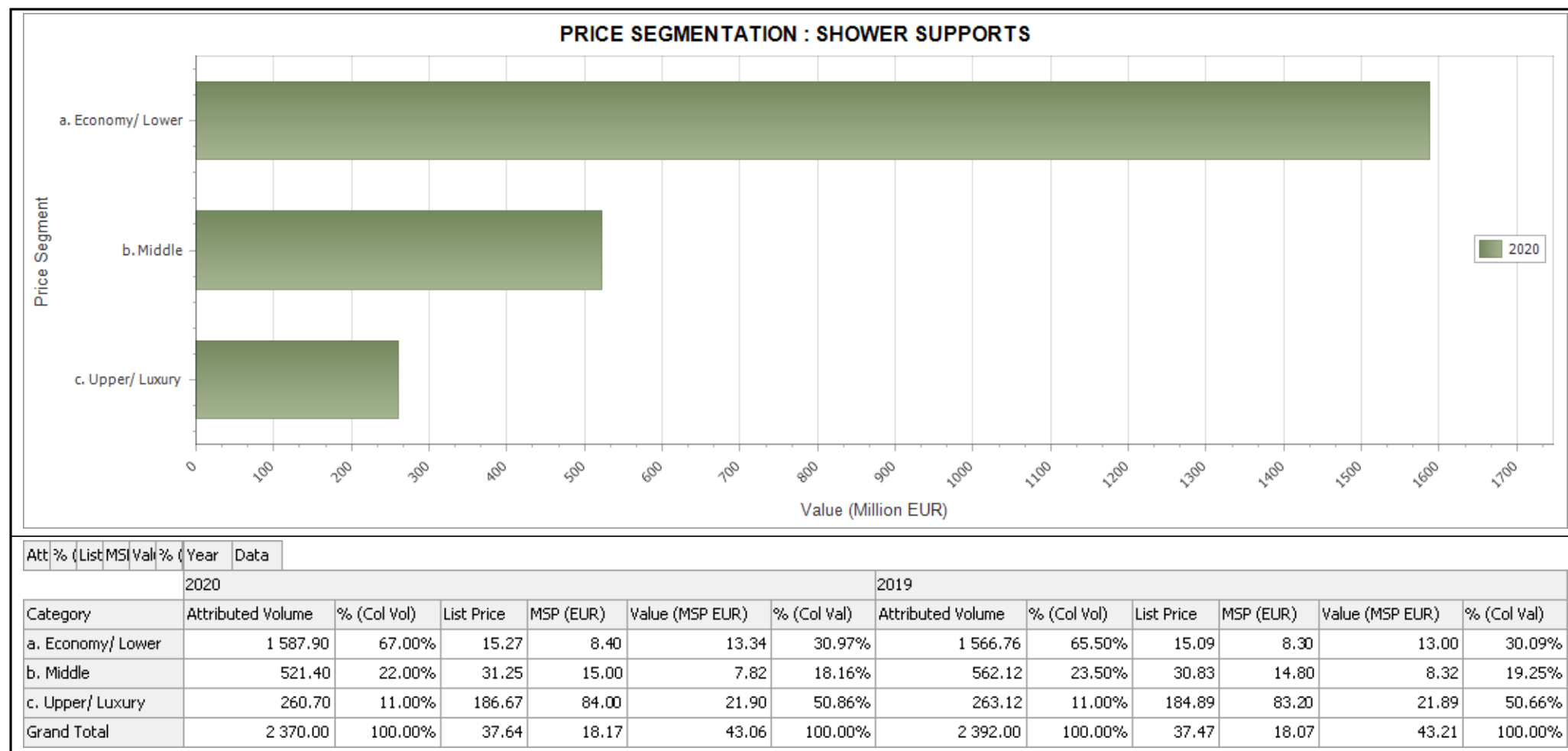
DE - ESTIMATED PRICE SEGMENTATION FOR SHOWER PIPES



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€75, Middle €75-€150, Upper/ Luxury >€150

7.3.7 Price Segmentation : Shower Supports

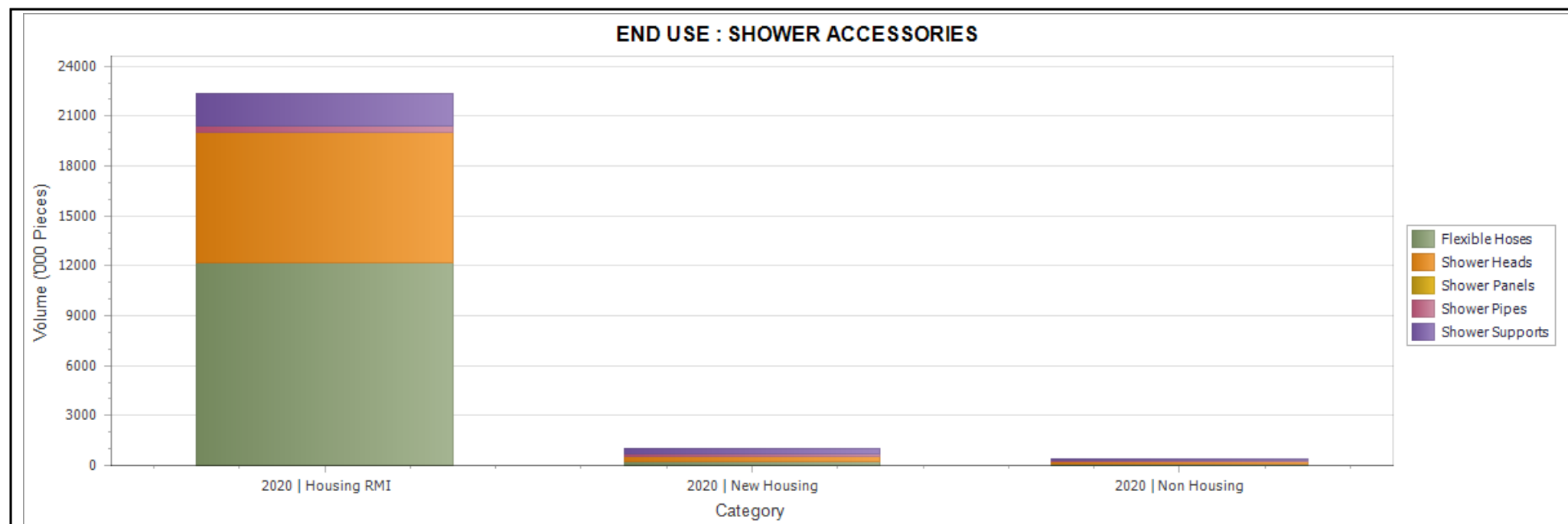
DE - ESTIMATED PRICE SEGMENTATION FOR SHOWER SUPPORTS



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€10, Middle €10-€30, Upper/ Luxury >€30

7.4.1 End Use : Base Year

DE - SALES OF SHOWER ACCESSORIES BY TYPE AND END USE SEGMENT ('000 PIECES)

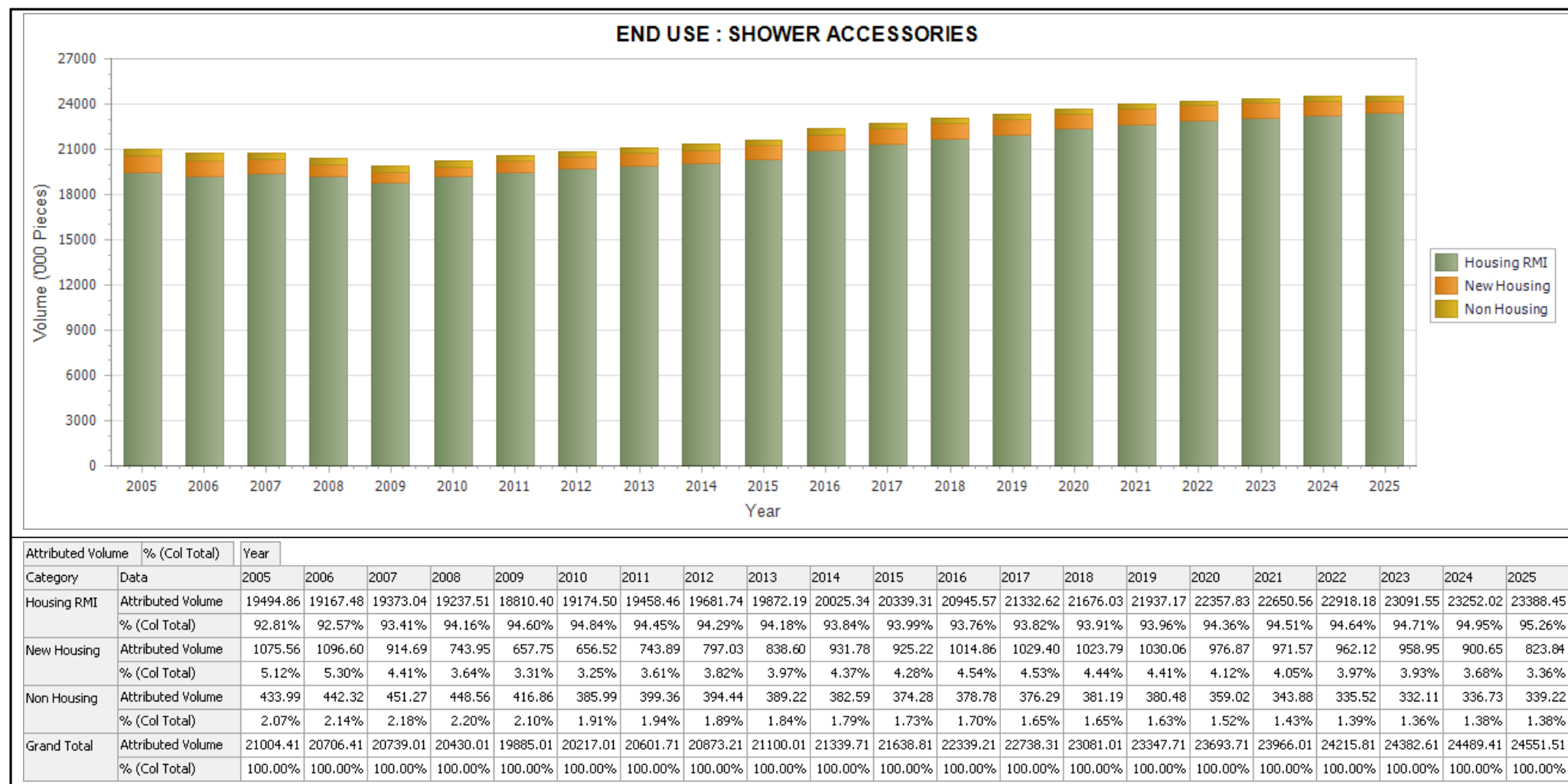


Attribu	% (Co	% (Ro	Year	Category	Data							
2020										2020 Total		
Housing RMI				New Housing			Non Housing					
Product	Attributed Volume	% (Col Total)	% (Row)	Attributed Volume	% (Col Total)	% (Row)	Attributed Volume	% (Col Total)	% (Row)	Attributed Volume	% (Col Total)	% (Row)
Flexible Hoses	12152.38	54.35%	98.00%	201.06	20.58%	1.62%	46.56	12.97%	0.38%	12400.00	52.33%	100.00%
Shower Heads	7852.88	35.12%	93.71%	359.95	36.85%	4.30%	167.17	46.56%	1.99%	8380.00	35.37%	100.00%
Shower Panels	10.85	0.05%	79.19%	1.80	0.18%	13.11%	1.06	0.29%	7.70%	13.70	0.06%	100.00%
Shower Pipes	353.57	1.58%	66.71%	106.87	10.94%	20.16%	69.56	19.37%	13.12%	530.00	2.24%	100.00%
Shower Supports	1988.14	8.89%	83.89%	307.19	31.45%	12.96%	74.67	20.80%	3.15%	2370.00	10.00%	100.00%
Grand Total	22357.83	100.00%	94.36%	976.87	100.00%	4.12%	359.02	100.00%	1.52%	23693.71	100.00%	100.00%

Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

7.4.2 End Use : Historical and Forecasts

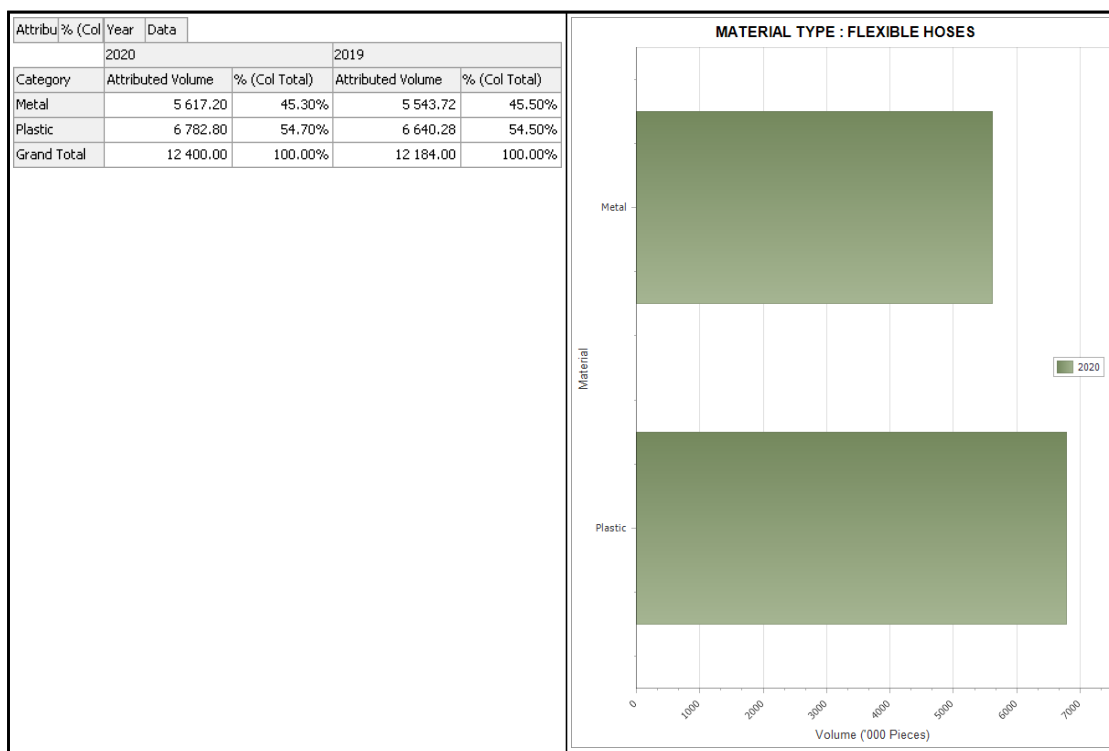
DE - SALES OF SHOWER ACCESSORIES BY END USE SEGMENT ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

7.4.3 Sales By Type of Material : Flexible Hoses

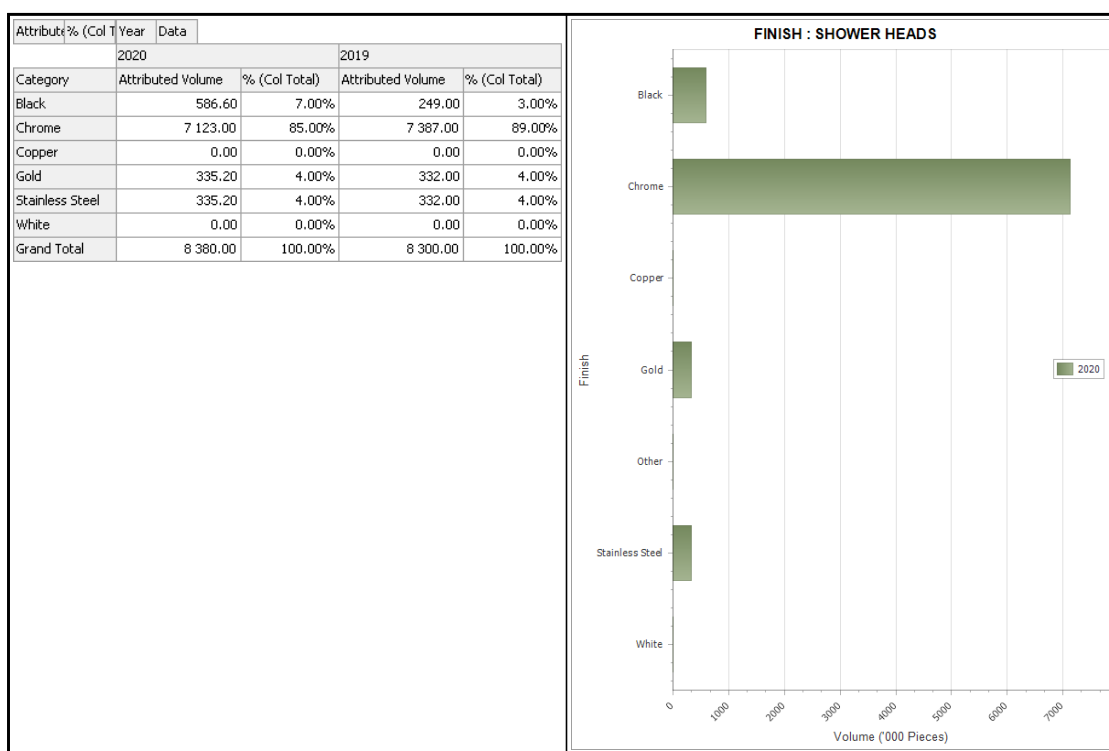
DE - FLEXIBLE HOSE SALES BY MATERIAL TYPE ('000 PIECES)



Source: BRG based on industry interviews

7.4.4 Sales by Type of Finish : Shower Heads

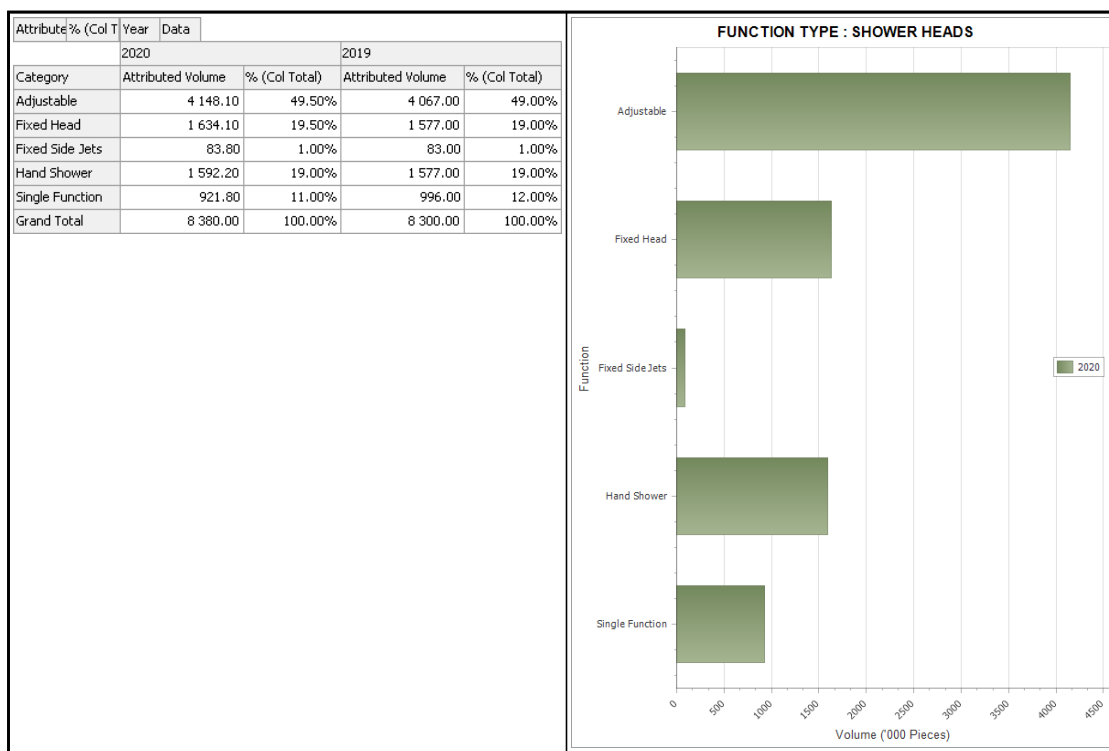
DE - SHOWER HEAD SALES BY FINISH TYPE ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

7.4.5 Sales By Type of Function : Shower Heads

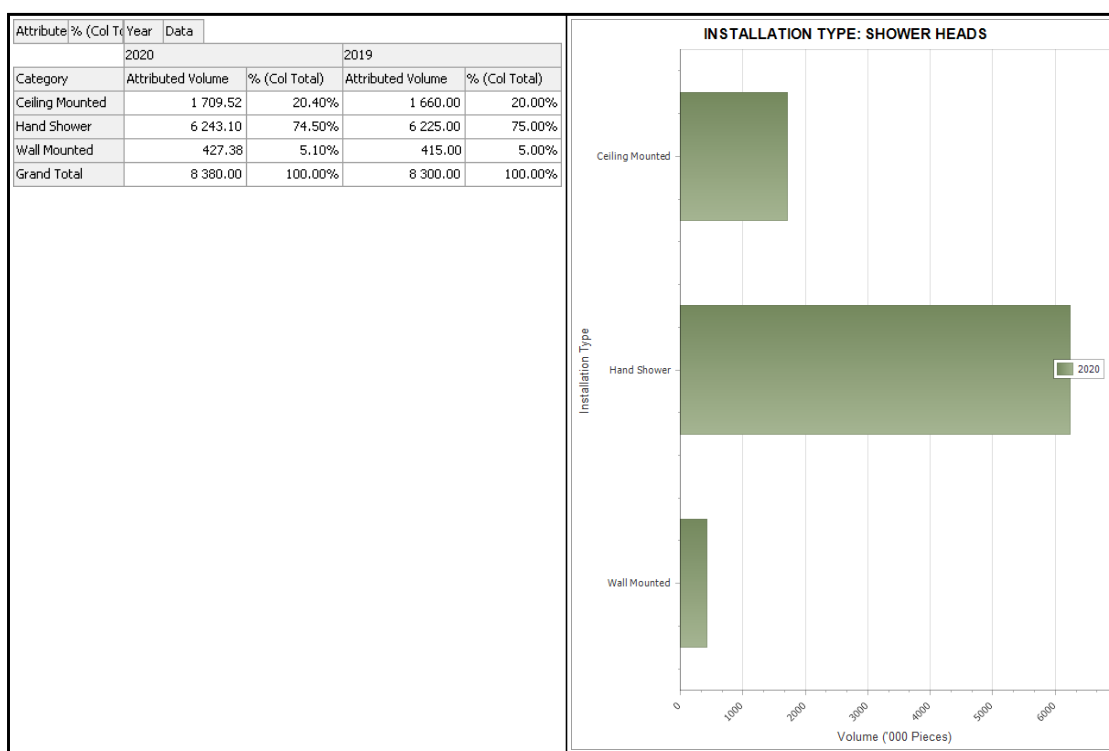
DE - SHOWER HEAD SALES BY FUNCTION TYPE ('000 PIECES)



Source: BRG based on industry interviews

7.4.6 Sales by Type of Installation : Shower Heads

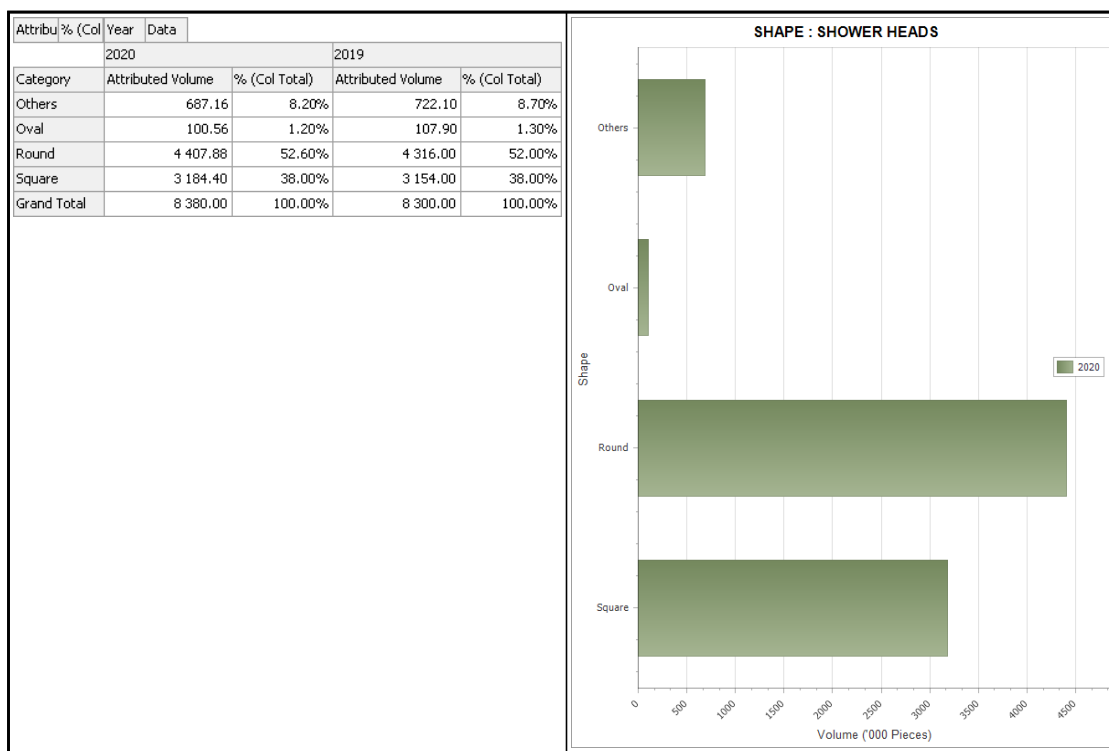
DE - SHOWER HEAD SALES BY INSTALLATION TYPE ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

7.4.7 Sales By Type of Shape : Shower Heads

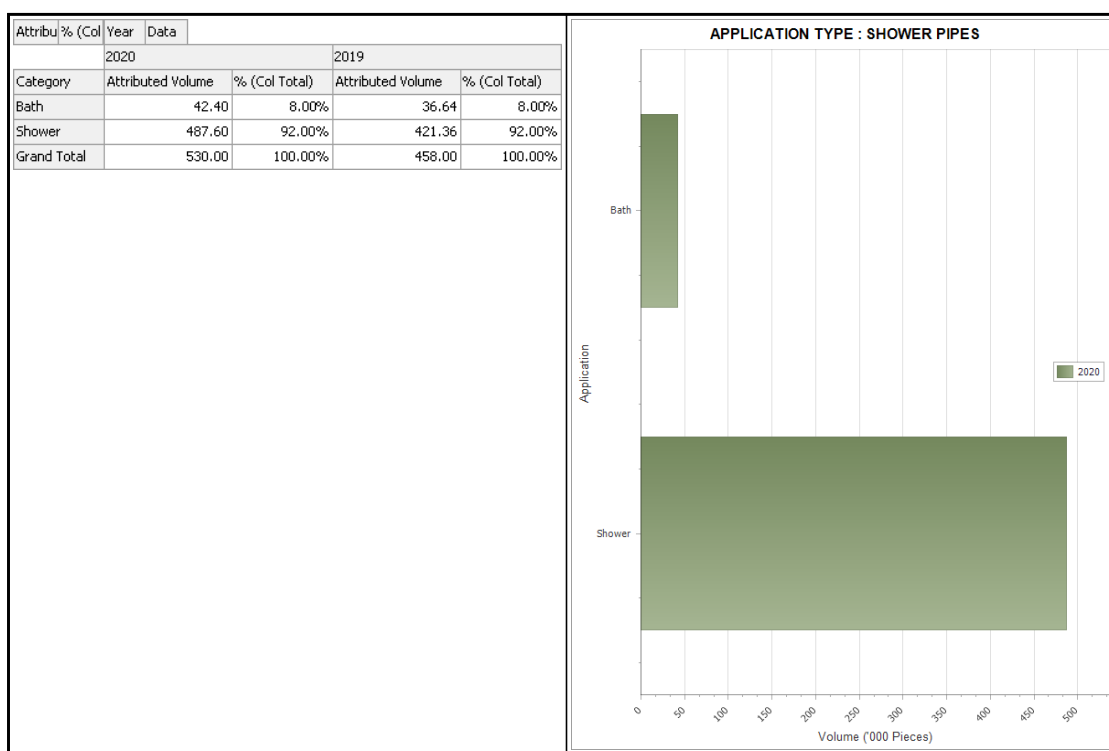
DE - SHOWER HEAD SALES BY SHAPE ('000 PIECES)



Source: BRG based on industry interviews

7.4.8 Sales by Type of Application : Shower Pipes

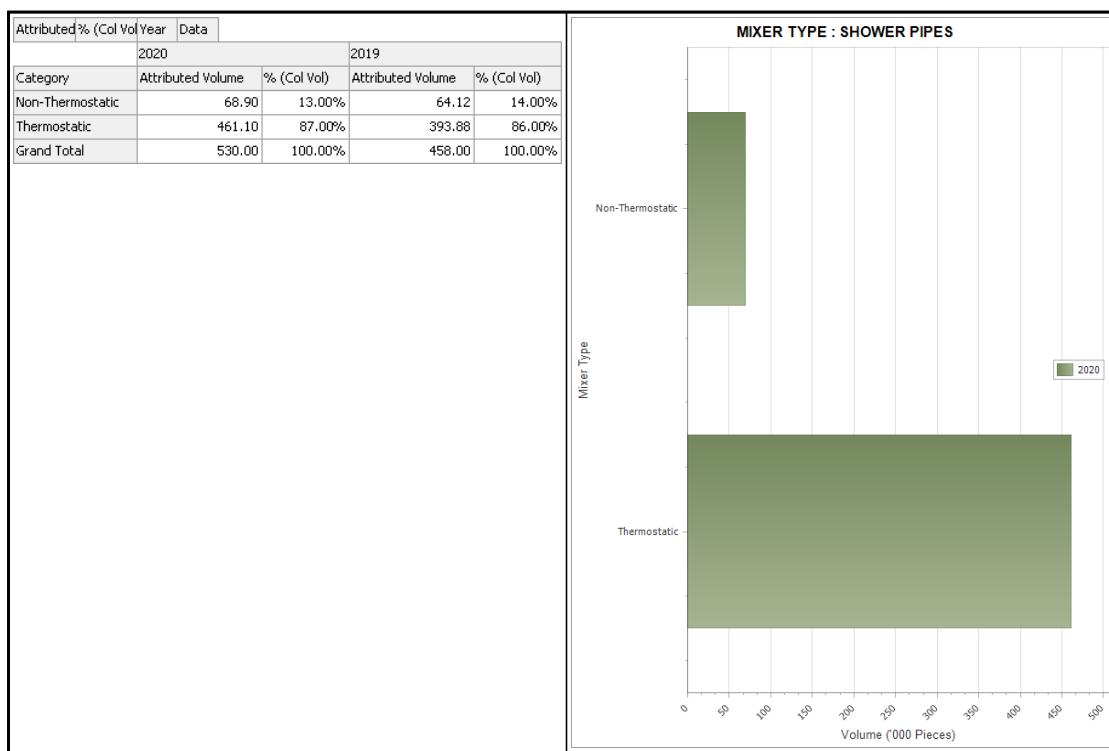
DE - SHOWER PIPE SALES BY APPLICATION TYPE ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

7.4.9 Sales by Type of Mixer : Shower Pipes

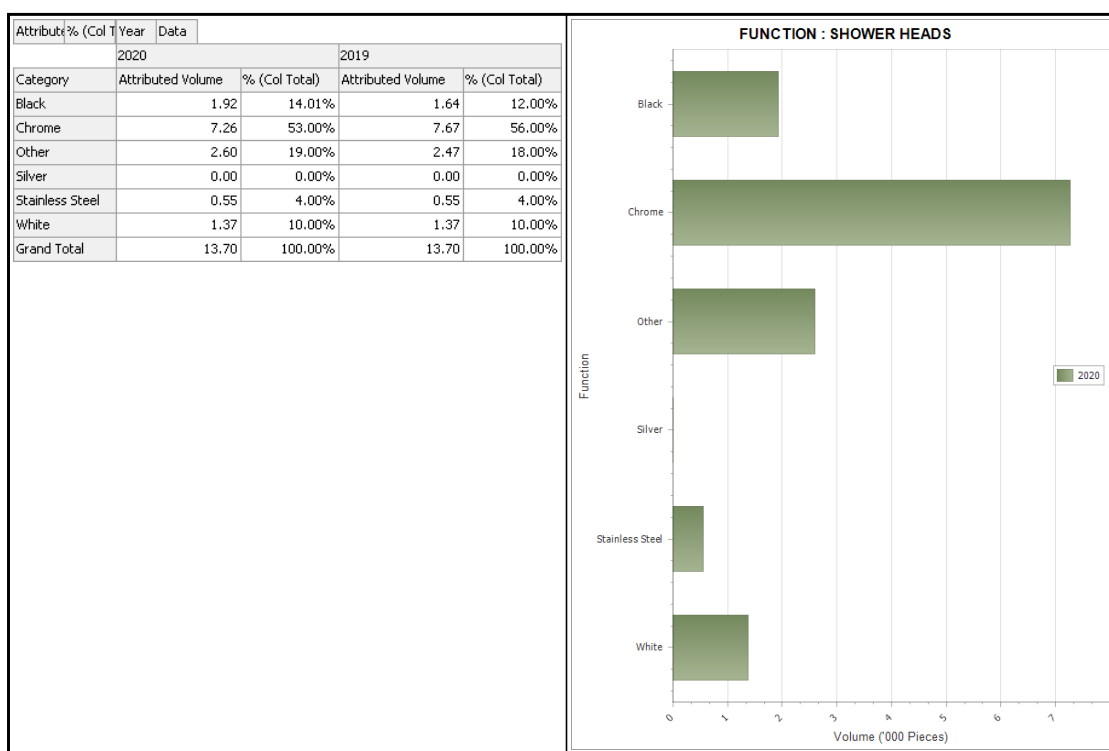
DE - SHOWER PIPE SALES BY MIXER TYPE ('000 PIECES)



Source: BRG based on industry interviews

7.4.10 Sales by Type of Finish: Shower Panels

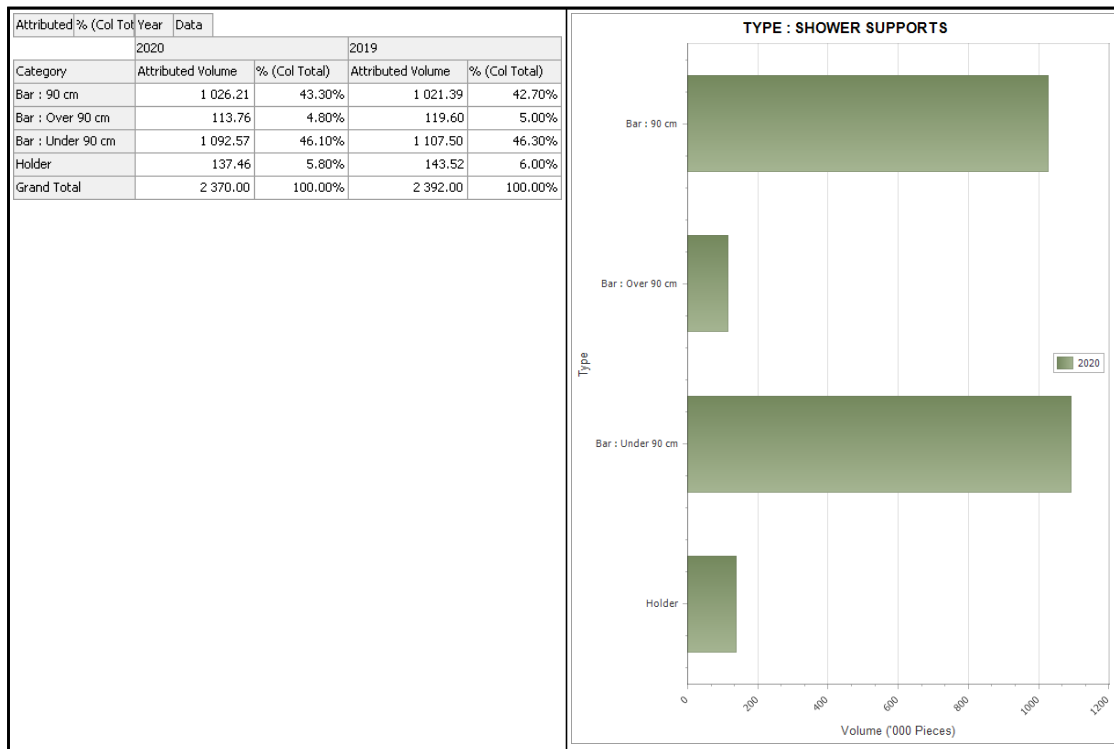
DE - SHOWER PANEL SALES BY FINISH TYPE ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

7.4.11 Sales By Type : Shower Supports

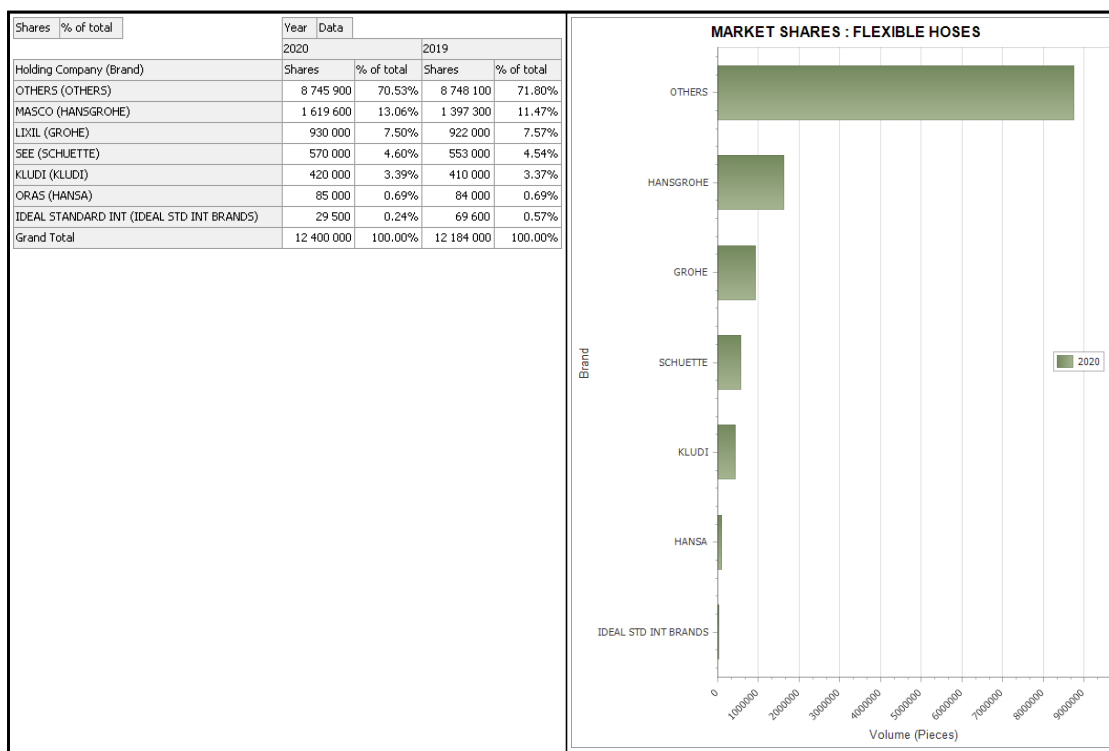
DE - SHOWER SUPPORT SALES BY TYPE ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

7.5.1 Market Shares : Flexible Hoses

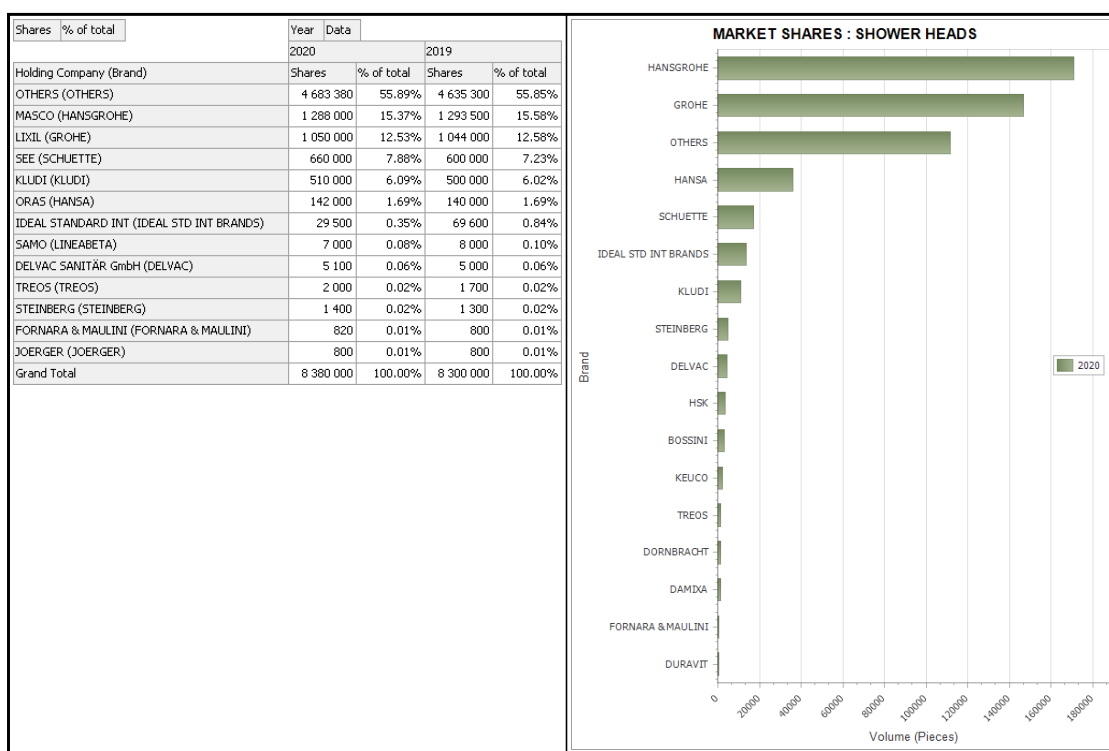
DE - ESTIMATED MARKET SHARES FOR FLEXIBLE HOSES



Source: BRG based on industry interviews.

7.5.2 Market Shares : Shower Heads

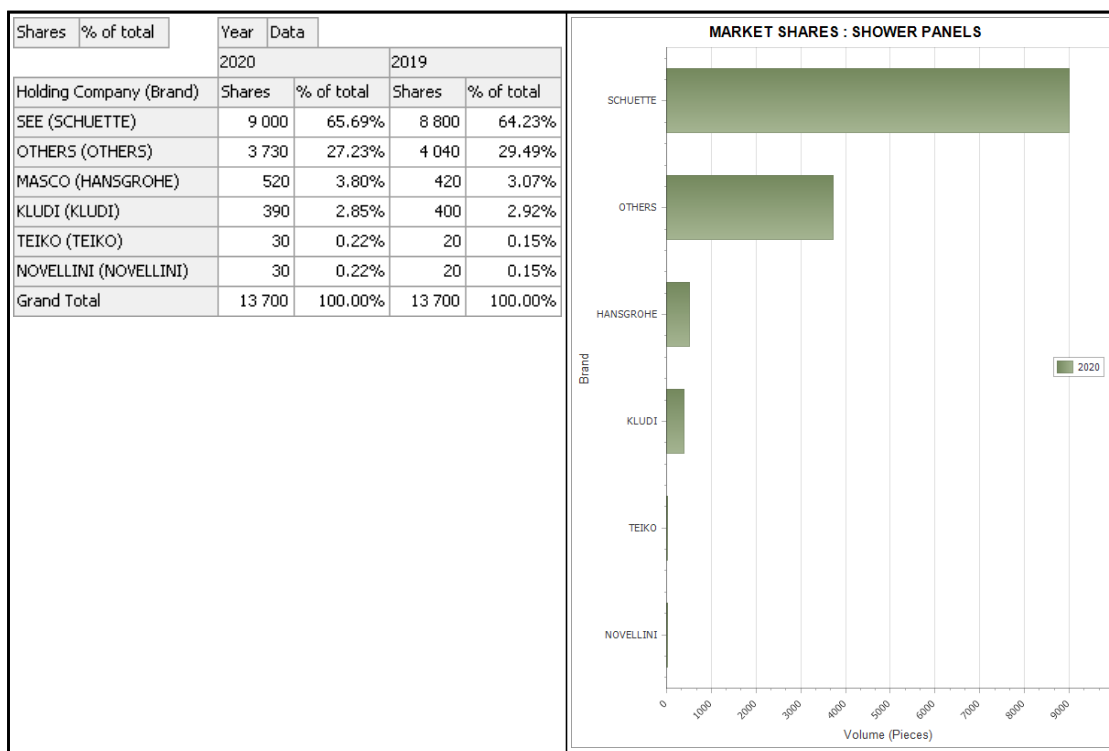
DE - ESTIMATED MARKET SHARES FOR SHOWER HEADS



Source: BRG based on industry interviews.

7.5.3 Market Shares : Shower Panels

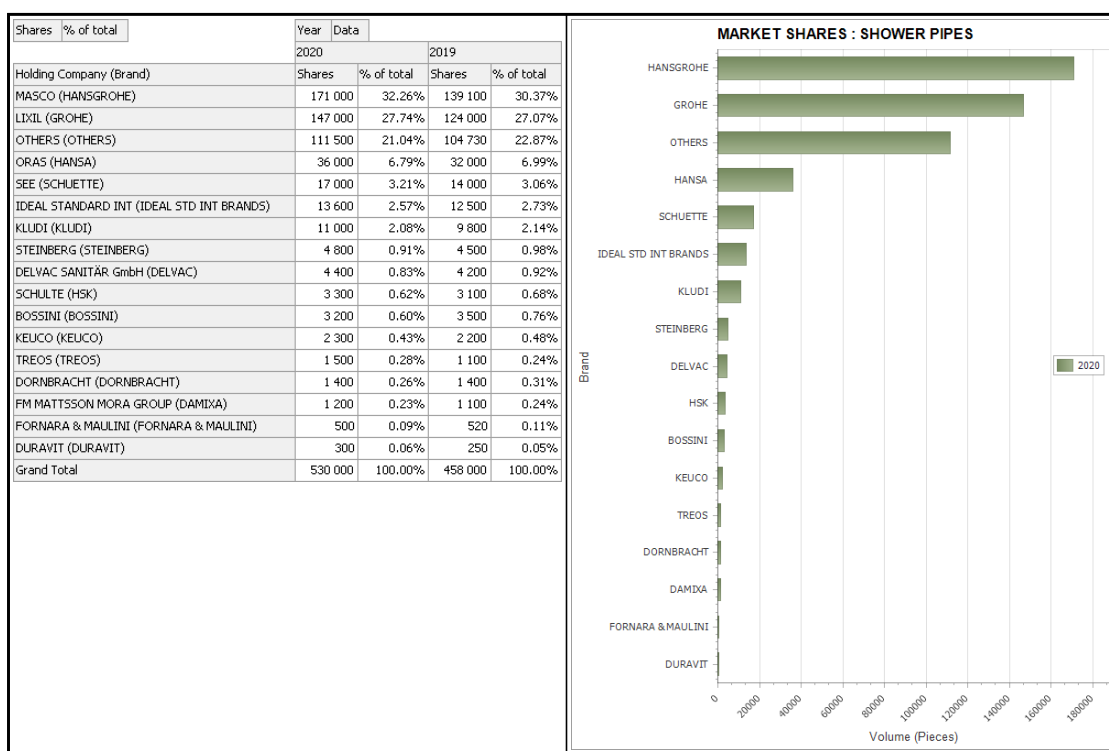
DE - ESTIMATED MARKET SHARES FOR SHOWER PANELS



Source: BRG based on industry interviews.

7.5.4 Market Shares : Shower Pipes

DE - ESTIMATED MARKET SHARES FOR SHOWER PIPES



Source: BRG based on industry interviews.

7.6.1

Route to Market: Shower Accessories
DE - ESTIMATED DISTRIBUTION FOR SHOWER ACCESSORIES: 2020

Manufacturer / Importer	Manufacturers / Importers
	100.0%

Wholesale Level	56.2%
	Wholesalers / Merchants
	56.2%

Retail Level	Manufacturer	Wholesaler	Manufacturer	Wholesaler	Manufacturer	Wholesaler
	2.0%	12.0%	25.7%	21.0%	11.8%	9.0%
	Internet Retailers		DIY Retailers		Specialist Retailers	
	14.0%		46.7%		20.8%	

Manufacturer	Wholesaler	Internet	DIY	Specialist
3.4%	14.0%	2.8%	2.0%	0.8%
Installers				
23.0%				

End Use Level	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer
	0.2%	0.1%	10.8%	44.7%	20.0%	22.6%	0.7%	0.1%	0.4%	0.0%	0.0%	0.3%
	Residential						Non-Residential					
	98.5%						1.5%					
	New Build			Residential RMI			New Build			Non-Residential RMI		
	4.1%			94.4%			0.4%			1.1%		
	Project	Non-Project		Project	Non-Project		Project	Non-Project		Project	Non-Project	
	1.8%	2.3%		1.7%	92.7%		0.4%	0.0%		0.1%	1.0%	

7.6.2

Route to Market: Shower Accessories
DE - ESTIMATED DISTRIBUTION FOR SHOWER ACCESSORIES: 2025

Manufacturer / Importer	Manufacturers / Importers
	100.0%

Wholesale Level	56.0%
	Wholesalers / Merchants
	56.0%

Retail Level	Manufacturer	Wholesaler	Manufacturer	Wholesaler	Manufacturer	Wholesaler
	3.2%	18.0%	23.9%	17.0%	12.0%	9.0%
	Internet Retailers		DIY Retailers		Specialist Retailers	
	21.2%		40.9%		21.0%	

Manufacturer	Wholesaler	Internet	DIY	Specialist
4.0%	12.0%	3.4%	1.8%	1.0%
Installers				
22.2%				

End Use Level	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer
	0.3%	0.0%	17.4%	39.1%	20.0%	21.8%
	Residential					
	98.6%					
	New Build			Residential RMI		
	3.4%			95.3%		
	Project	Non-Project		Project	Non-Project	
	2.0%	1.4%		1.5%	93.8%	

Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer	
0.6%	0.0%	0.4%	0.0%	0.0%	0.4%	
Non-Residential						
1.4%						
	New Build			Non-Residential RMI		
	0.6%			0.8%		
	Project	Non-Project		Project	Non-Project	
	0.5%	0.1%		0.7%	0.1%	

Source: BRG Based on Industry Interviews. - Totals may vary due to roundings

7.6.3 Route to Market - Consolidated View : Shower Accessories

DE - ESTIMATED DISTRIBUTION FOR SHOWER ACCESSORIES : 2020

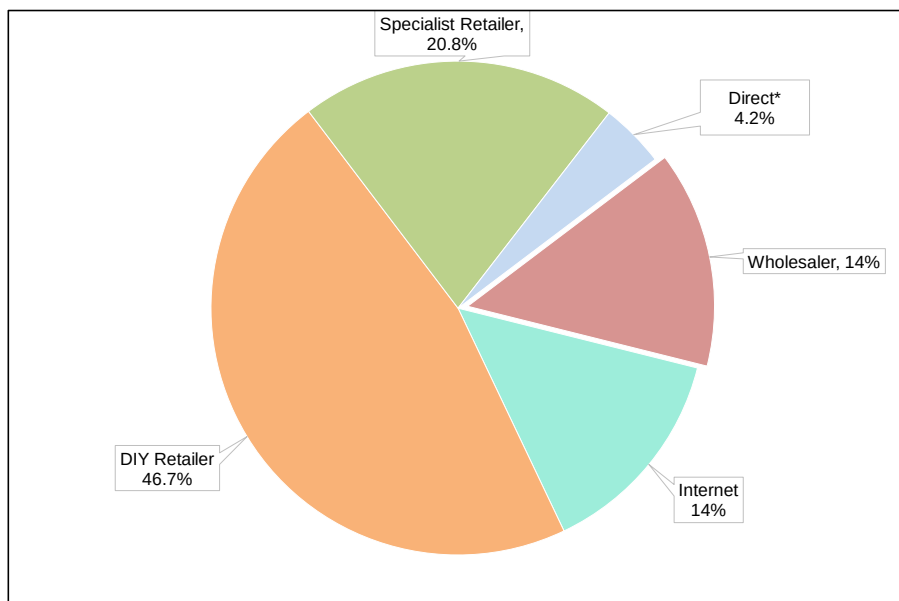
Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
14.2%	14.0%	46.7%	20.8%	4.2%	3.5%	0.5%	95.0%	1.0%
100%					4.0%		96.0%	

7.6.4 Route to Market - Consolidated View : Shower Accessories

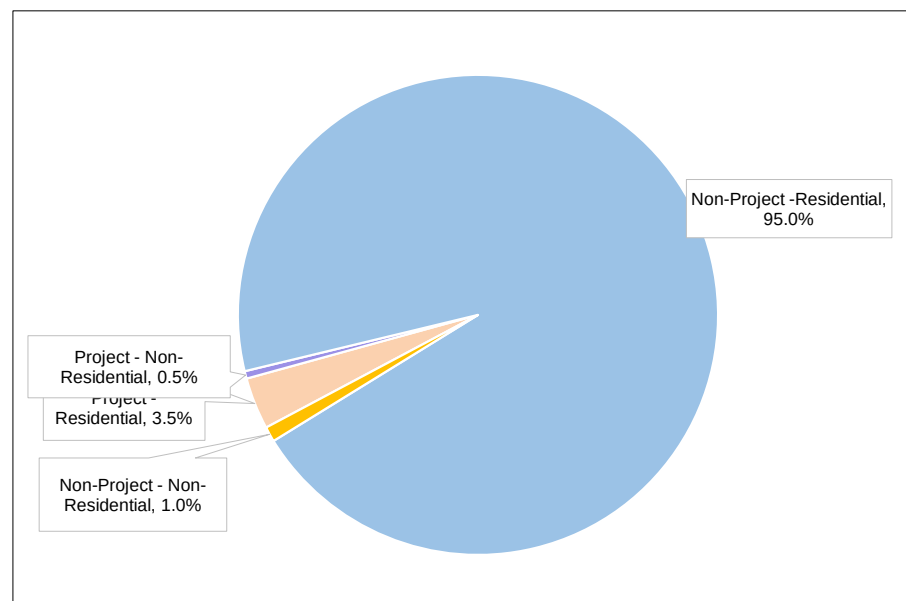
DE - ESTIMATED DISTRIBUTION FOR SHOWER ACCESSORIES : 2025

Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
12.0%	21.2%	40.9%	21.0%	4.9%	3.5%	1.2%	95.1%	0.2%
100%					4.7%		95.3%	

DE - SALES OF SHOWER ACCESSORIES BY DISTRIBUTION CHANNEL: 2020



DE - SALES OF SHOWER ACCESSORIES BY POINT OF DECISION AND END-USE SECTOR: 2020



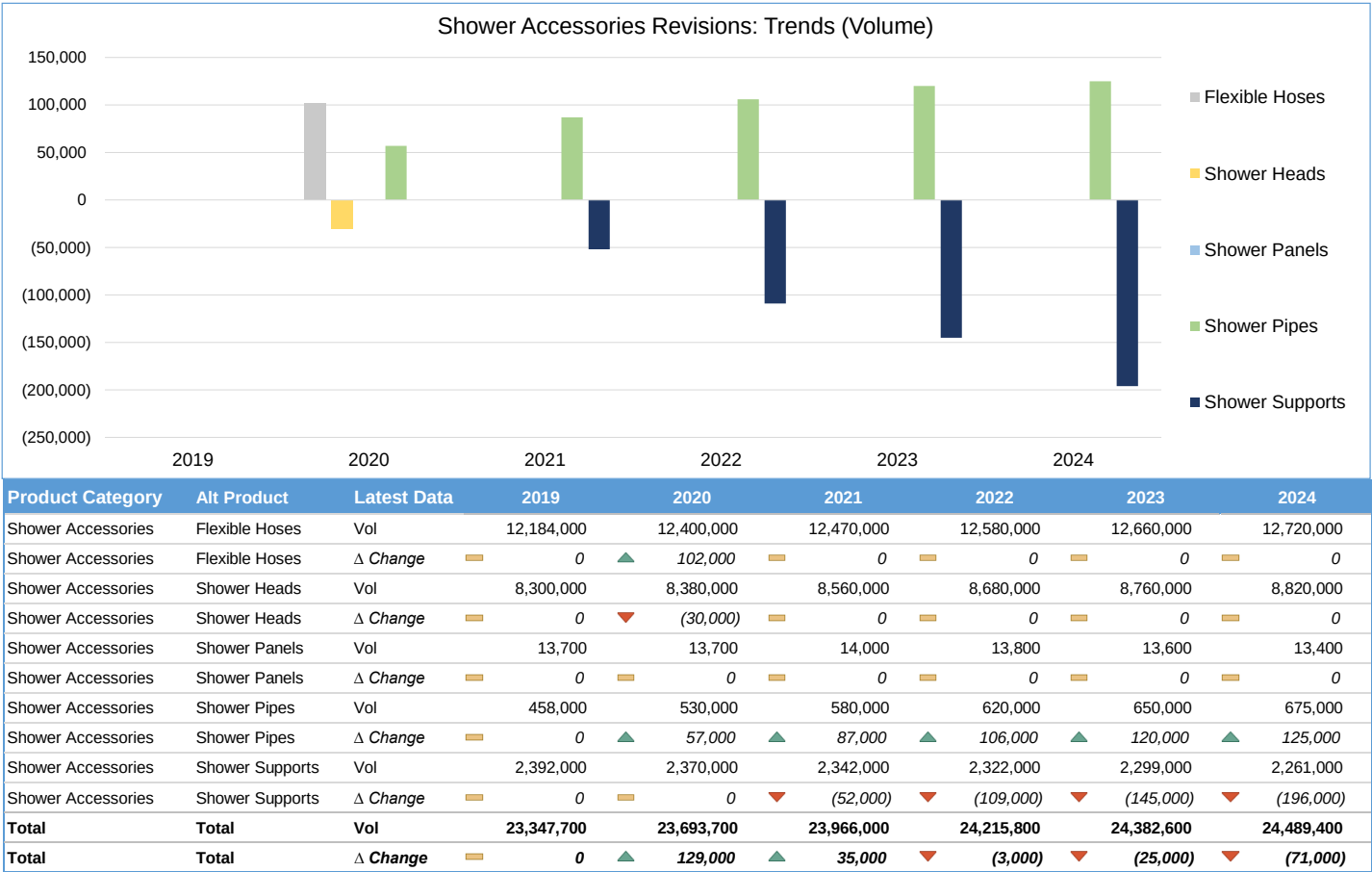
Source: BRG Based on Industry Interviews.

*Direct includes sales going directly from manufacturers to installers, contractors and private customers.

7.7 Revisions in the German Shower Accessories Market (April 2021)

DE - REVISIONS IN THE GERMAN SHOWER ACCESSORIES MARKET

Data Comparison: October 2020



No revisions have been made to historical volumes for this market.

The MSPs of shower supports and shower heads were revised down for the years 2019 and 2020. In response to the growing popularity of shower pipes, shower heads and shower supports have lost share in the upper price segments. This led to an overall decline of the MSP in recent years.

GERMANY

KITCHEN SINKS

THE EUROPEAN BATHROOM & KITCHEN PRODUCT MARKETS



Market Characteristics and Technological Trends

Sales of kitchen sinks are highly related to the overall kitchen market, as sinks are often bought as part of a new kitchen rather than as a single replacement item. Kitchens in Germany are mostly equipped with stainless steel sinks, synthetic kitchen sinks or ceramic kitchen sinks, in that order. The average kitchen in Germany is replaced every 20 years, which indicates the long lifespan of kitchen sinks. Kitchen taps and mixers are correlated less with the overall kitchen market, as these have a shorter average lifespan.

Synthetic sinks, in particular composite sinks, have gained in popularity over the last few years, especially in the DIY segment. Synthetic sinks were fashionable in the 1980s and early 1990s. After this peak, they suffered from image problems regarding the durability of the material. An improvement in the materials of composite kitchen sinks, led to better quality. Manufacturers were again able to compete with stainless steel kitchen sinks. The new type of materials made the surface more durable, easy to clean and water resistant to prevent stains. Nevertheless, stainless steel sinks still remain dominant on the German market as end users perceive these models to be the most appropriate material for a kitchen sink, in a private and commercial kitchen. The market consists largely of inset sinks that remain the most cost-efficient option, being highly durable, lightweight, having the widest scope in terms of design, and due to their simple installation.

The ceramic kitchen sink market is a niche market, accounting for the residual share of the total market. It is a high-quality market and sinks are made with vitreous china, an enamel coating to make the ceramic tougher, denser and shinier. Ceramic kitchen sinks are restricted in terms of volume, due to their look and price. The revival of the synthetic kitchen sink market brought back a wider variety of colours. Currently, black is the colour in the highest demand, especially when the sink is positioned next to CERAN cooktops, which are black, too. Other preferred colours are white, grey and, increasingly, beige. Some manufacturers are also trying bold colours such as pink, green and light blue, in line with having coloured cupboard units, fridges and kitchen appliances. However, demand is limited and German end users prefer classic colours.

In terms of innovations in the shape of the sinks, a few new models have been adapted to modern needs, such as small sinks, because of the increasing number of single households. In recent years, German end users have been spending more money for kitchen sinks. The reason is that the kitchen has gained importance as a living space. Having a modern and sophisticated kitchen is perceived to be a status symbol. Backed up by Germany's positive economic climate, Germans are investing more in a modern kitchen. The kitchen is turning into a sociable living space away from a merely functional room. End users are looking for products that contribute to an enhanced living standard to meet their individual needs.

Recent Market Trends

The COVID-19 pandemic shaped the developments in the year 2020 and 'business as usual' was replaced by government measures. In Germany the official lockdown, which is better understood as a 'prohibition of contact', started at the end of March 2020. Bavaria was the first state to implement measures to contain the spread of the virus, which was followed by the remaining states towards the end of March. For a few weeks the German population was advised to work from home when possible, with most non-essential stores closing for a limited time only. Measures implemented varied by state (*Bundesländer*). Most stores, including DIY stores, were allowed to open again from the beginning of April 2020 onwards, with differences from state to state in the level of restrictions and measures. In certain regions, shopping in DIY stores was limited to professionals only.

During 2020, new housing construction continued without much interruption, with only large scale project sites halted to implement protective measures. That being said, the overall production per employee is believed to have shrunk significantly, and remains to be the case, since many employers introduced measures to protect the construction workers. The bathroom industry was declared part of the crisis infrastructure, but installers could still continue to work without many interruptions. Overall, there has been relatively little impact of the COVID-19 pandemic on the building related industry, including the bathroom market, in Germany. In some cases, end-users may have postponed, or cancelled, bathroom related projects due to the uncertainty on the market, but overall demand remained relatively strong, as there was a large backlog of works from the past, benefiting sales of bathroom and kitchen equipment. At the same time, in certain instances, the COVID-19 pandemic may have contributed to additional demand. Since consumers were spending more time indoors, they had the opportunity to dedicate themselves to repair or upgrade works, whereas others might have decided to invest into their property by installing a new bathroom or kitchen due to (international) holidays being difficult because of the current pandemic.

In 2020, sales of all types of kitchen sinks, except for enamelled steel, increased in terms of volume with a smaller growth rate when compared to the previous year. Demand from the RMI segment grew at a lower rate and sales in the new residential and the non-housing segment declined. This resulted in an overall smaller growth for the kitchen sink market compared to the previous year. Synthetic kitchen sinks saw the biggest growth volume-wise and demand for this product continued to be strong again. Sales of stainless steel kitchen sinks account for almost two thirds of all sinks. Stainless products saw the second strongest growth after synthetic sinks in terms of volume. Stainless steel sinks generate the highest volume on the market, whereas synthetic sinks are the fastest growing type of sinks. Since kitchen sinks are mainly sold together with a new kitchen through specialist kitchen stores and furniture shops, the dependency on professional installers is less when compared to most bathroom products. Hence, the installation of kitchen sinks is not hampered by a shortage of professionals. Demand for ceramic sinks notably increased, while demand for enamelled steel

kitchen sinks continued to decline. These two products are a niche and their sales account for a small share of the market. In total, sales of overall kitchen sinks in 2020 increased strongly in volume terms when compared to 2019.

The construction industry in Germany has been on a high level and end users have been exposed to long waiting times, sometimes up to one year, to get a new bathroom installed. This situation benefits demand for new kitchens as end users in need of a new kitchen and bathroom would renovate their kitchen first. Most kitchen sinks in Germany have a one bowl configuration, followed by a 1.5 bowl. Although slightly increasing in share, two bowl configurations are rare. 1.5 bowl configurations are becoming increasingly popular and end users are willing to pay more for this product. When looking at the installation type, inset is by far dominating, only losing to undermount, which is slightly gaining share. Undermount kitchen sinks are becoming more popular due to aesthetic reasons. Nevertheless, it remains a niche as it can only be installed with certain countertops and the supply is limited.

Prices and Market Values

In 2020, average prices for all types of kitchen sinks, at manufacturer level, increased to compensate for inflation and higher production costs. At the same time, German end users increasingly invest more money for a new kitchen and are looking for high-quality products. This trend affects the share of the upper/luxury price segments for ceramic, stainless steel and synthetic kitchen sinks. These developments contributed to a notable increase of MSP in particular for stainless steel and synthetic kitchen sinks. It is worthwhile mentioning, that the overall market is on a relatively high price level when compared to Germany's neighbouring countries.

The increasing popularity of the Internet used as a purchasing channel, and an easy way to compare prices and models prior to buying decisions, further benefits German end users by making the market more transparent. However, sinks are rarely bought online on their own. Instead, they are an important aspect of a kitchen, which requires prior planning and consultation with specialist stores. Therefore, the choice for a kitchen sink is more determined by its fit to a kitchen design than by its price.

New Housing

Contrary to the previous year, Germany's new housing segment decreased in 2020 and demand for kitchen sinks was smaller when compared to 2019. The kitchen sink market is closely connected to overall new housing completions, as every household requires one kitchen sink and, in some cases, more than one is installed in the same dwelling. This, however, is more likely to be in large houses rather than apartments. The decline of residential new completions negatively influenced sales of all kitchen sink types in Germany in 2020. Stainless steel kitchen sinks continued to lose importance to the benefit of synthetic kitchen sinks. Stainless steel sinks are chosen for their long lifespan and cheap price when compared to most other

materials. Synthetic sinks are preferred since limescale stains do not show easily when compared to stainless steel sinks. Moreover, synthetic sinks have further advantages such as an abundance of shapes and colours and being scratch-resistant. Most ceramic kitchen sinks are found in upper market segments.

Housing RMI

The amount of kitchen sinks that were installed in the residential RMI segment remained on a positive level in 2020 when compared to the previous year. A further growing replacement demand might have been triggered due to the long waiting listings for bathroom renovations. Moreover, most Germans spend more time at home and the choice to spend money was limited due to the lockdown. Therefore, consumers chose to invest into their homes by replacing an old kitchen for example. A closer look at the material shows that RMI demand for all types of kitchen sinks, except for enamelled steel sinks, slightly increased. Although synthetic kitchen sinks are gaining market share, stainless kitchen sinks generate the largest volumes in this segment. The reason is that like-for-like replacement is likely to happen with the same type of material.

Stainless kitchen sinks within the price entry level are more common and end users looking for a cheap replacement are likely to choose a stainless-steel product. Also, consumers are more likely to upgrade to a synthetic kitchen sink, when buying a whole new kitchen. In this case, they want to try a new material based on recommendations and out of interest for something different. Overall, no major changes have characterised the kitchen sinks product mix. Stainless steel sinks remain the most popular product also in this segment, followed by synthetic sinks. Sales of stainless-steel sinks slightly increased, although some were substituted with synthetic kitchen sinks. Although synthetic products are not cheaper in the price entry segment, they are offered in multiple designs and colours. Stainless steel sinks are regarded to be a traditional product and end users, who are looking for something new, are more likely to choose a synthetic sink. This is particularly the case when a new kitchen is installed.

Non-Housing

Demand from the non-housing construction and RMI output was clearly negative in 2020. As a result of the COVID-19 pandemic and lockdown measures, most non-housing facilities were confronted with a difficult time. Smaller businesses have been struggling and it is likely that new and renovation projects have been put on hold. This situation is likely to remain unchanged for 2021 and sales are expected to remain negative, before turning positive again from 2022 onwards. The long-term outlook for this segment is more positive, as sales to non-residential buildings are expected to remain on a stable level, after the recovery from the COVID-19 crisis. The kitchen sinks non-residential segment is mainly represented by restaurants, bars, hotels as well as by hospitals and offices. In this segment, as in the overall market, stainless steel is the dominant product.

Supply Structure

Most kitchen sinks in Germany are sold through specialised kitchen retailers, which either source the full kitchen, including the sink, from the kitchen manufacturer or, alternatively, they source just the kitchen without the sink and they pick a suitable one directly from a manufacturer. Either way, the kitchen sink is likely to be sold together with a predesigned and planned kitchen. Examples of specialist kitchen retailers are MUSTERHAUS KÜCHEN FACHGESCHÄFT, which is part of the MHK GROUP, and PLANA, which has more than 40 kitchen stores located in Southwest Germany. Another international player that needs to be mentioned is IKEA, which also offers planning and installation of tailor-made kitchens. In recent years, IKEA has experienced significant growth and it is estimated that its share is growing quickly.

The distribution landscape of kitchen sinks is diverse with few national and many strong regional players. Specialist retailers usually have their own team of installers. The supply structure on the German kitchen sink market remained stable in 2020. Supply of ceramic sinks is dominated by German manufacturers with high quality products. In contrast, there are only a few foreign players that are active in the market. The synthetic kitchen sink market is more diverse with German and foreign market players. The two main manufacturers, BLANCO and FRANKE, have together more than 40% of the total synthetic sink market share. Moreover, these two manufacturers have more than 50% of the total stainless-steel kitchen sink market share. They cover the whole price range of kitchen sinks. The market of ceramic kitchen sinks is dominated by VILLEROY & BOCH, while enamelled is almost only produced by RIEBER. In the leading product materials, stainless steel and synthetic, the top three manufacturers are FRANKE, BLANCO and RIEBER. In mid-2017, HANS GROHE introduced stainless steel and synthetic kitchen sinks to their portfolio. The expert for taps and mixers decided to produce kitchen sinks to sell them in a bundle with their own kitchen taps and mixers.

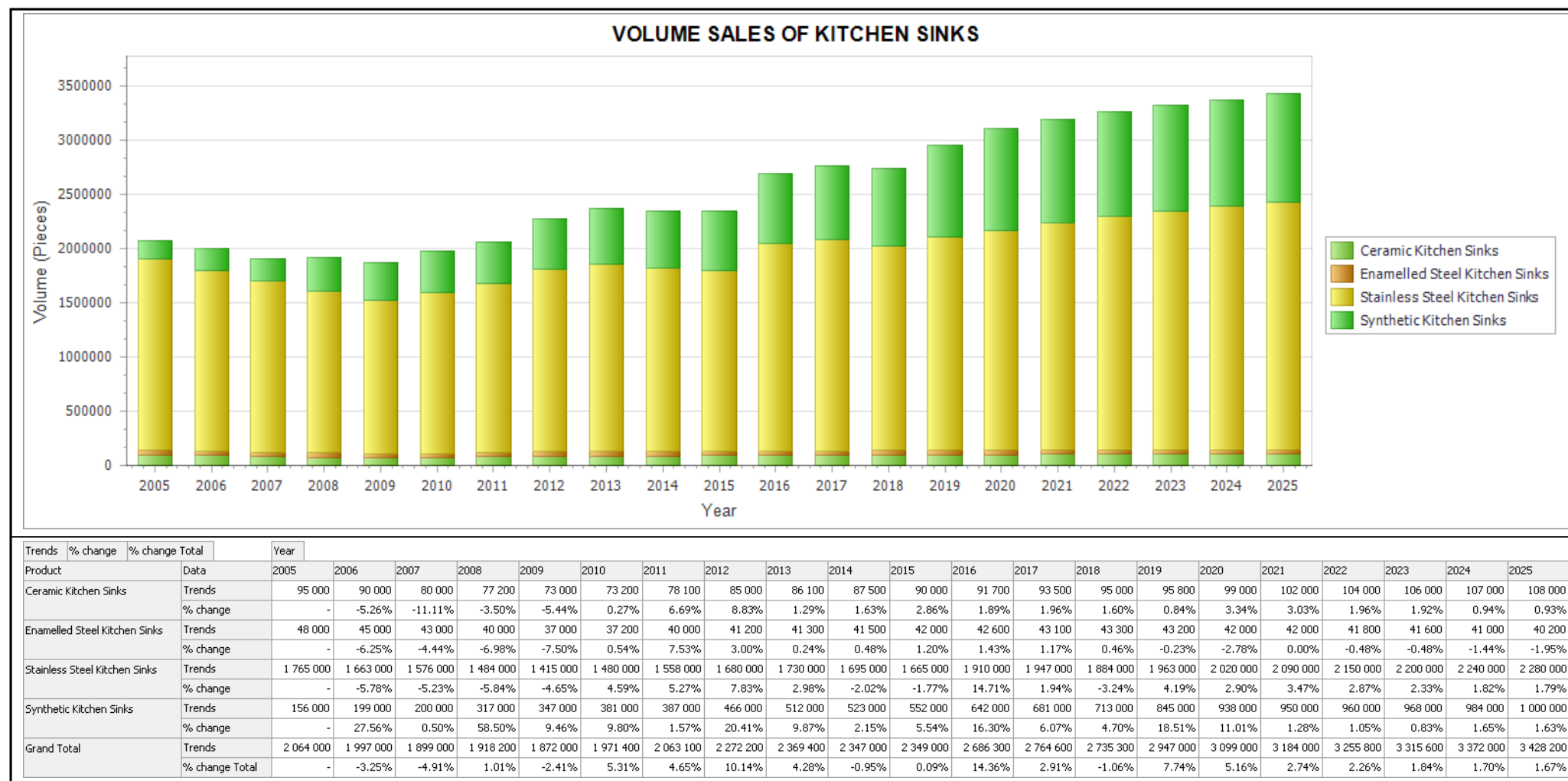
Forecasts

For the next five years the German kitchen sink market is expected to continue its growth. New construction activities are expected to recover in 2021 until 2023, before turning negative again. Overall, demand from the RMI segment is so significant in volume that it will compensate for lower sales from the non-housing and new housing segments. The impact of the COVID-19 pandemic has directed consumer spending towards their homes. Demand for new kitchens and ultimately kitchen sinks is likely to sustain the market during the forecast period. This trend is being supported by the fact that most Germans are restricted to travel and excess money is going to be invested into a new kitchen for example. Demand from non-housing, however, is expected to remain low as the leisure industry is mostly suffering from lockdown measures. Restaurants and other non-housing buildings with kitchens are likely to postpone upcoming renovations during the next couple of years. The replacement segment is responsible for most kitchen sink sales. The average lifespan of a kitchen sink is around 20 years. In most cases it is replaced together with a new kitchen. Houses built during the construction boom, which

ended around 1996, are still likely to be refurbished during the coming years.

8.2 Historical Trends and Forecasts : Kitchen Sinks

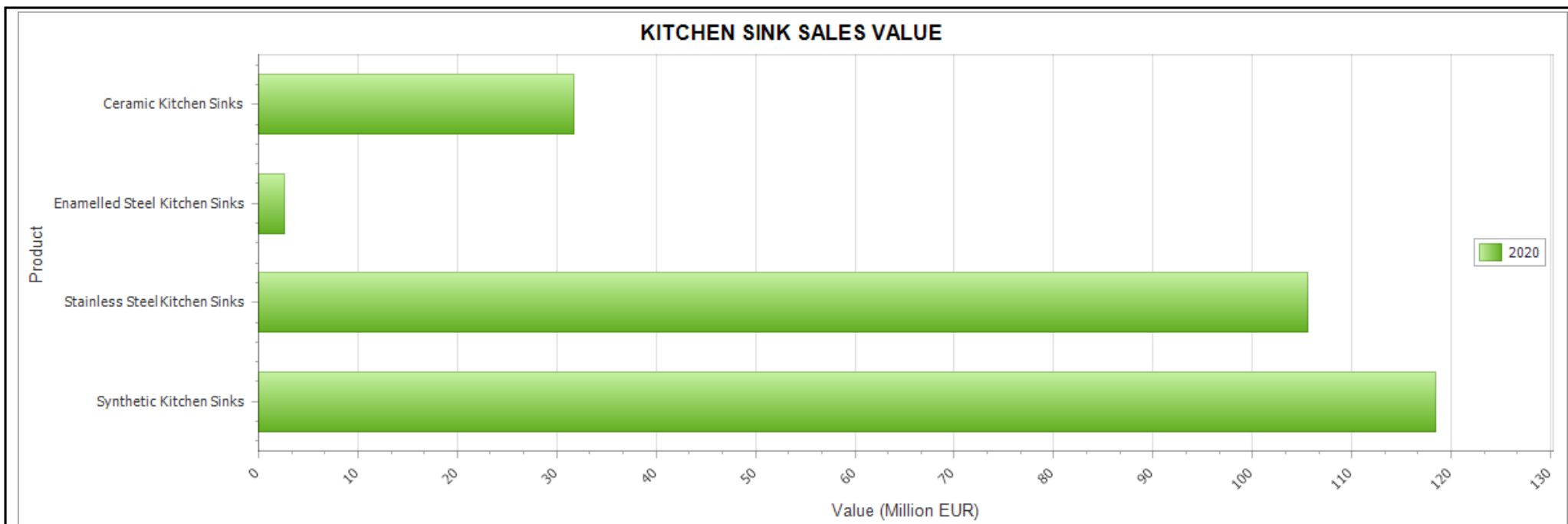
DE - SUMMARY OF TRENDS IN THE KITCHEN SINK MARKET : HISTORICAL AND FORECASTS



Source: BRG based on industry interviews.

8.3.1 Prices and Market Values : Kitchen Sinks

DE - ESTIMATED VALUES OF THE KITCHEN SINK MARKET

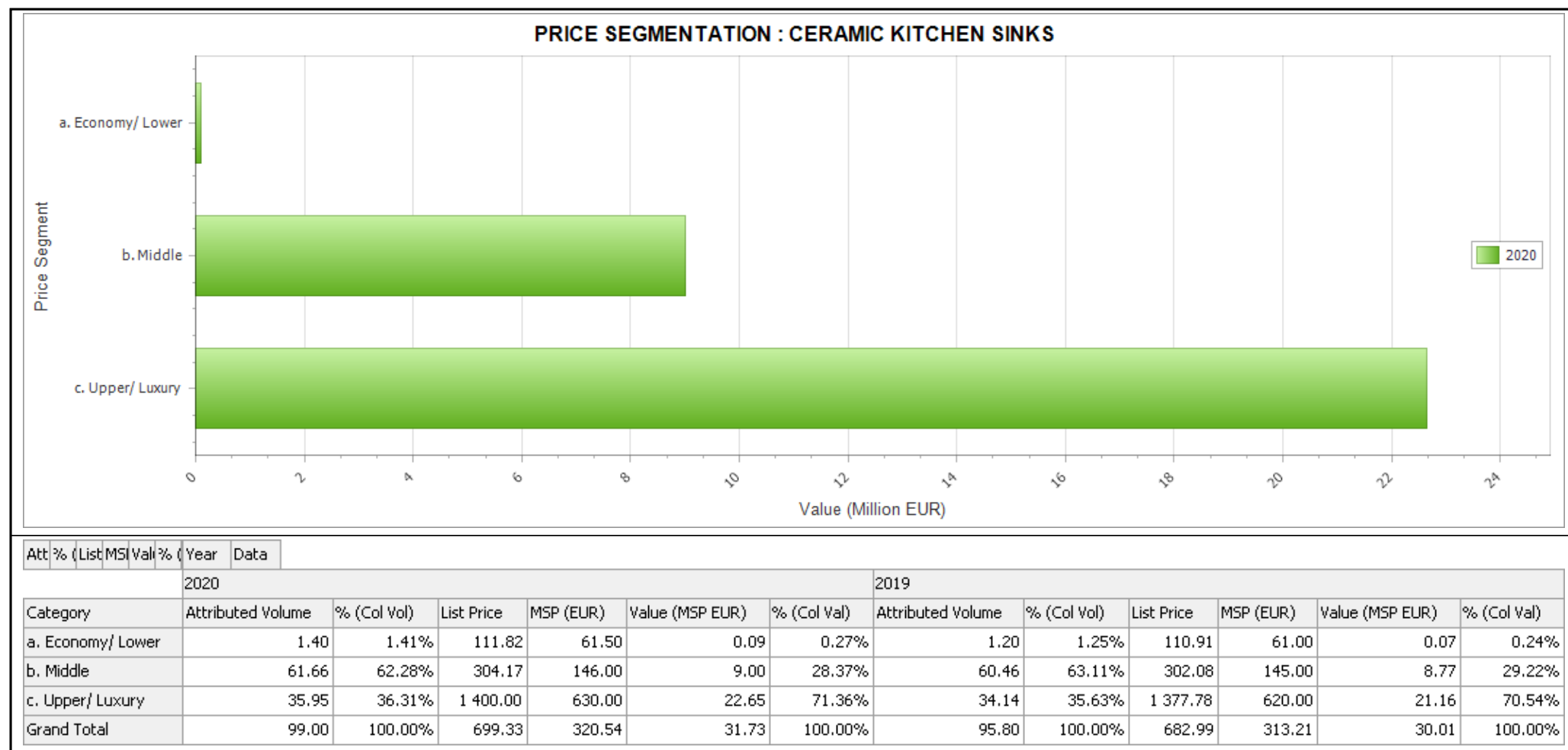


Volumes	MSP (EUR)	Value (MSP)	Year	Data			
			2020	2019			
Product	Volumes	MSP (EUR)	Value (MSP EUR)	Volumes	MSP (EUR)	Value (MSP EUR)	
Ceramic Kitchen Sinks	99 000	320.54	31.73	95 800	313.20	30.00	
Enamelled Steel Kitchen Sinks	42 000	61.68	2.59	43 200	60.58	2.62	
Stainless Steel Kitchen Sinks	2 020 000	52.23	105.50	1 963 000	50.48	99.09	
Synthetic Kitchen Sinks	938 000	126.25	118.42	845 000	122.39	103.42	
Grand Total	3 099 000	83.33	258.25	2 947 000	79.79	235.13	

Source: BRG based on industry interviews. Values Shown in Millions (at MSP = Manufacturers' Selling Price)

8.3.2 Price Segmentation : Ceramic Kitchen Sinks

DE - ESTIMATED PRICE SEGMENTATION FOR CERAMIC KITCHEN SINKS

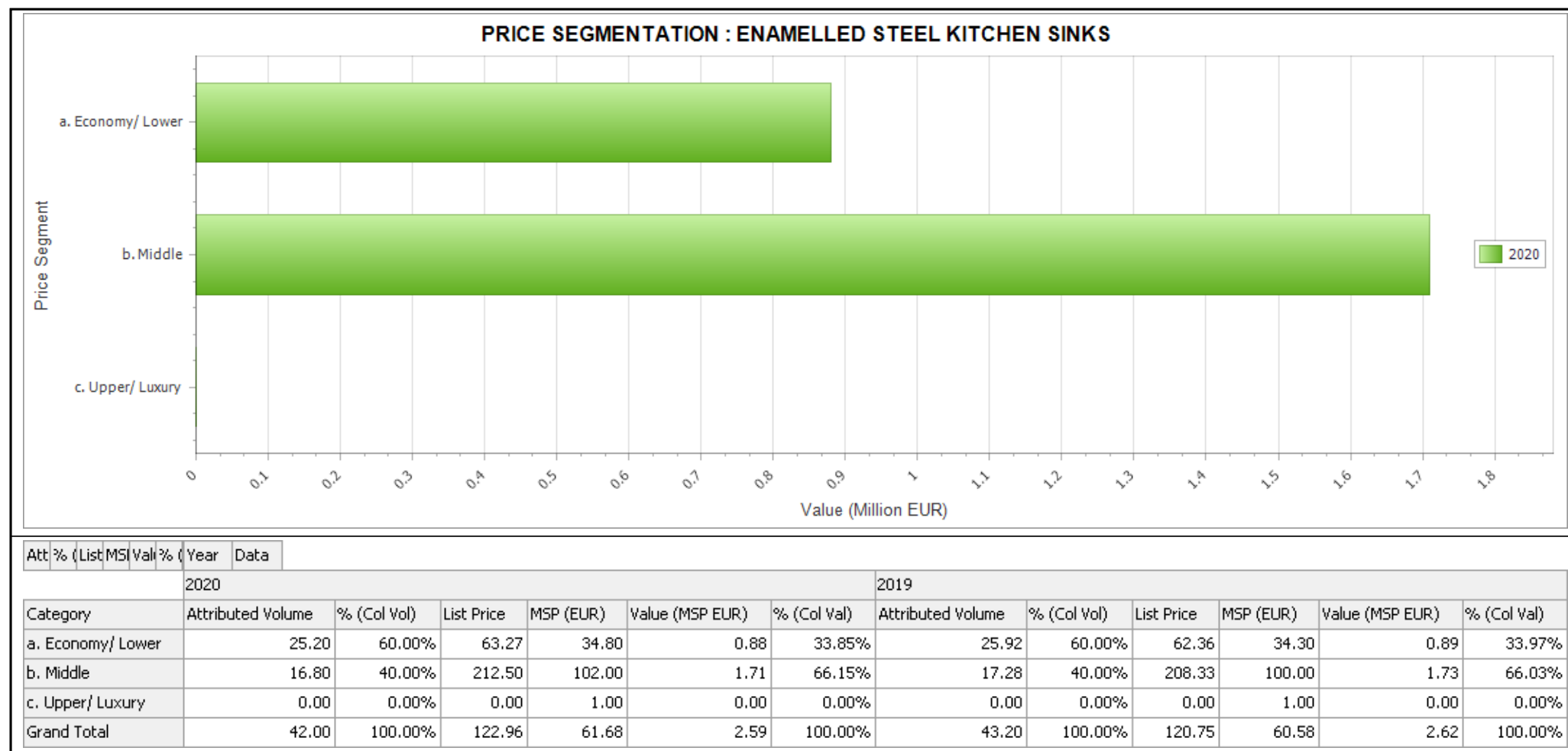


Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€100, Middle €100-€200, Upper/ Luxury >€200

8.3.3 Price Segmentation : Enamelled Steel Kitchen Sinks

DE - ESTIMATED PRICE SEGMENTATION FOR ENAMELLED STEEL KITCHEN SINKS

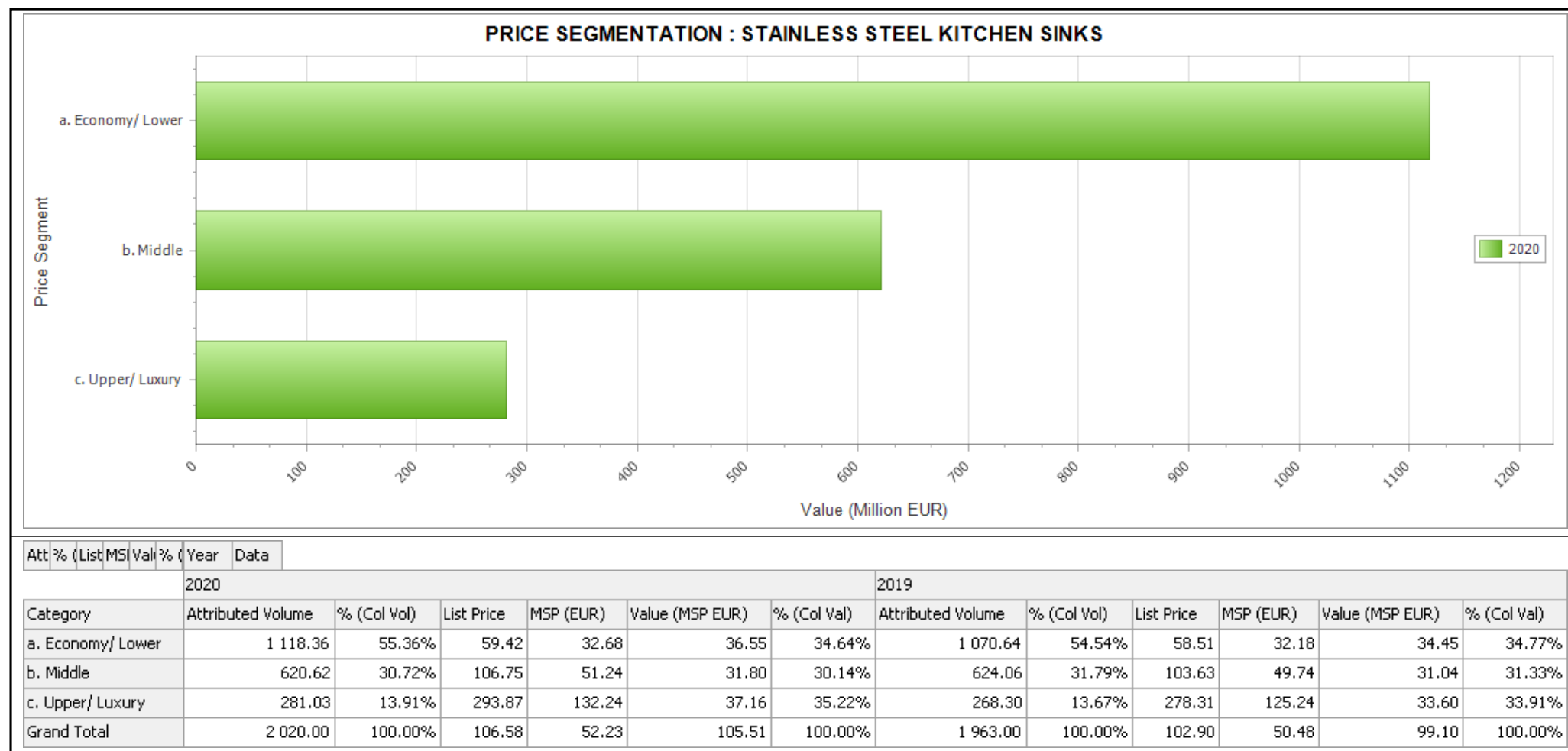


Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€75, Middle €75-€150, Upper/ Luxury >€150

8.3.4 Price Segmentation : Stainless Steel Kitchen Sinks

DE - ESTIMATED PRICE SEGMENTATION FOR STAINLESS STEEL KITCHEN SINKS

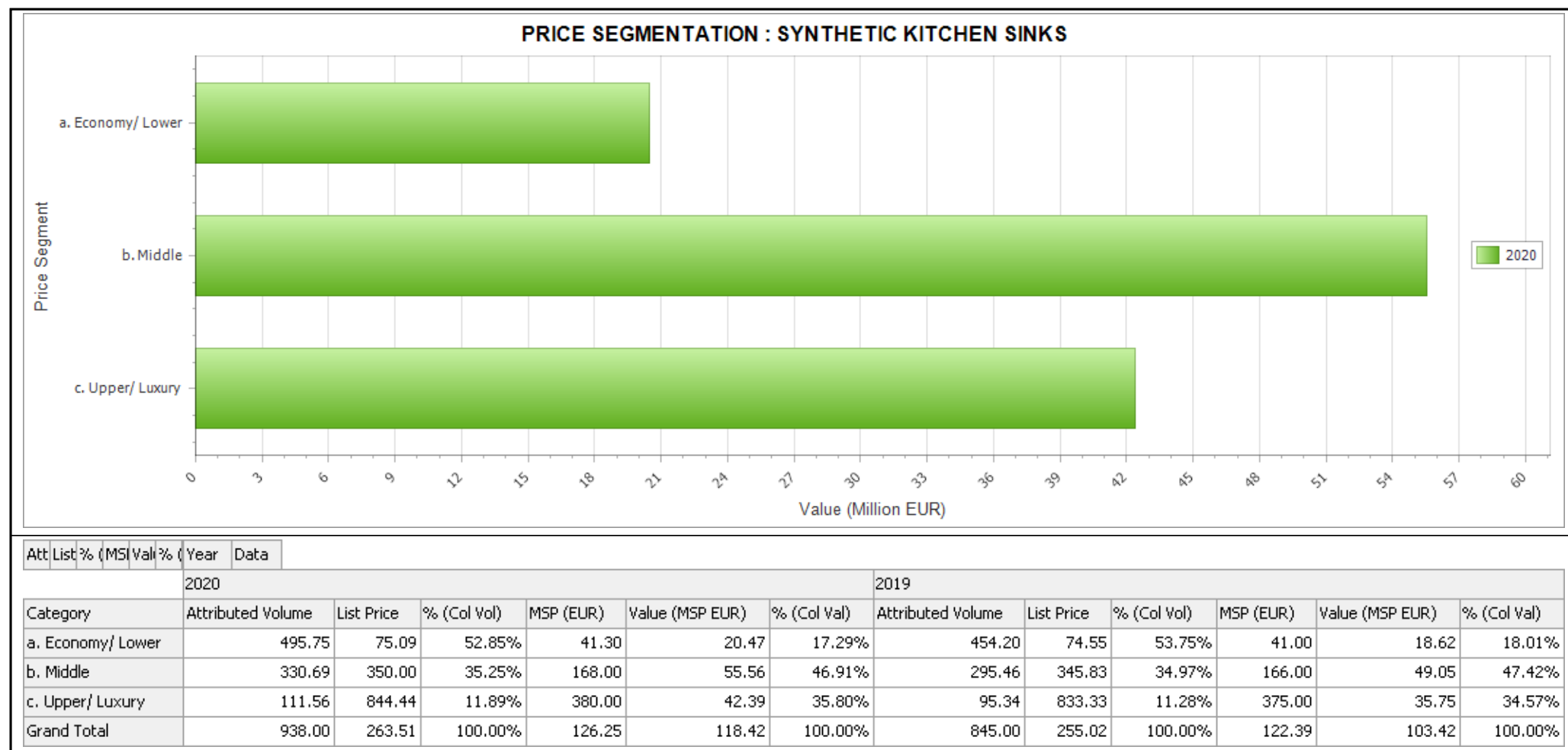


Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€40, Middle €40-€80, Upper/ Luxury >€80

8.3.5 Price Segmentation : Synthetic Kitchen Sinks

DE - ESTIMATED PRICE SEGMENTATION FOR SYNTHETIC KITCHEN SINKS

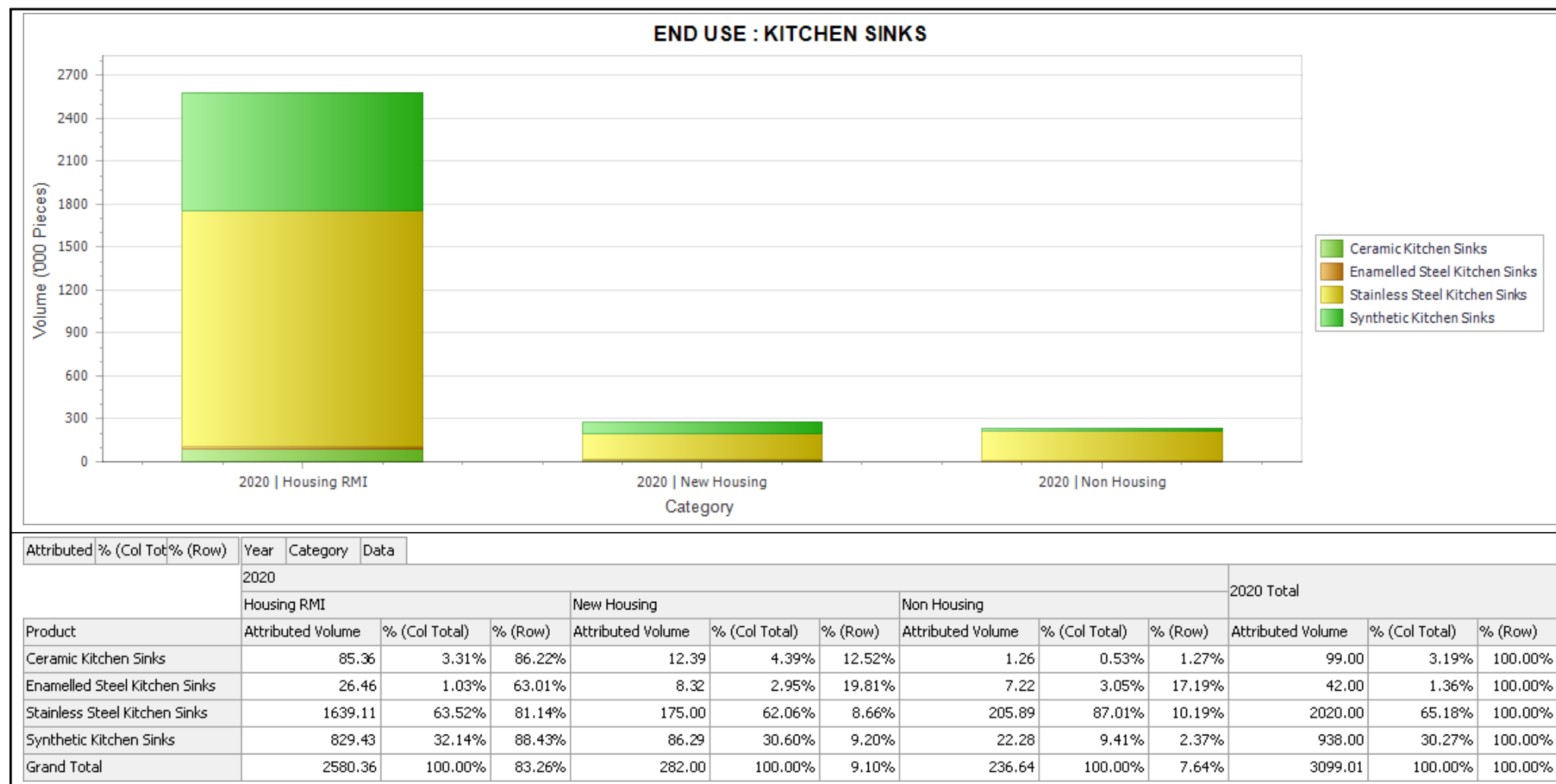


Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€125, Middle €125-€250, Upper/ Luxury >€250

8.4.1 End Use : Base Year

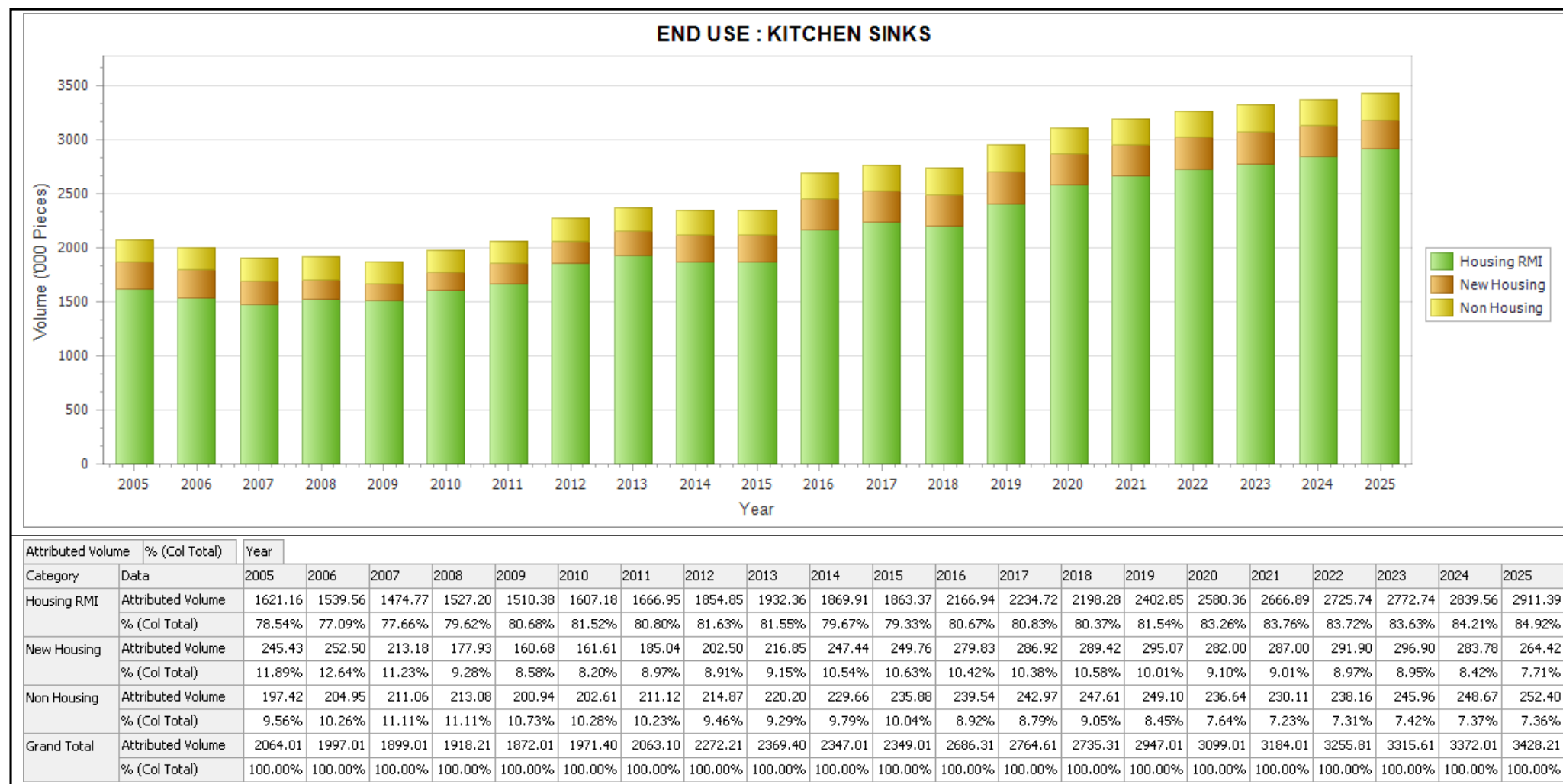
DE - SALES OF KITCHEN SINKS BY TYPE AND END USE SEGMENT ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

8.4.2 End Use : Historical and Forecasts

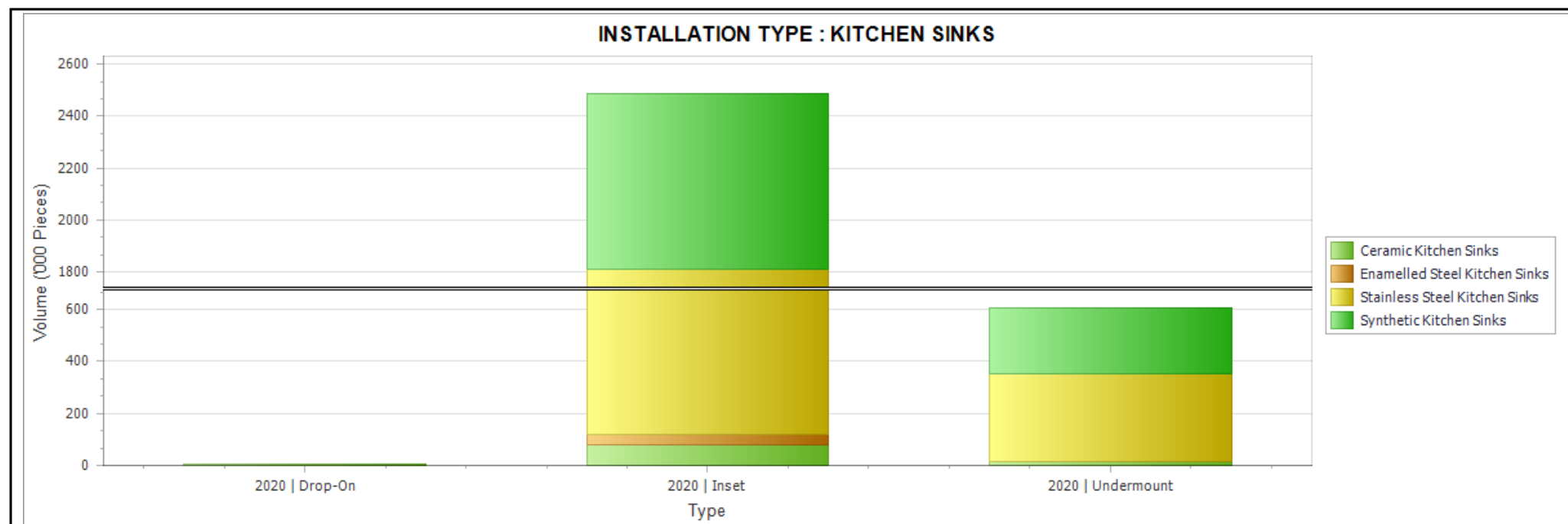
DE - SALES OF KITCHEN SINKS BY END USE SEGMENT ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

8.4.3 Sales by Type of Installation : Kitchen Sinks

DE - KITCHEN SINK SALES BY INSTALLATION TYPE ('000 PIECES)

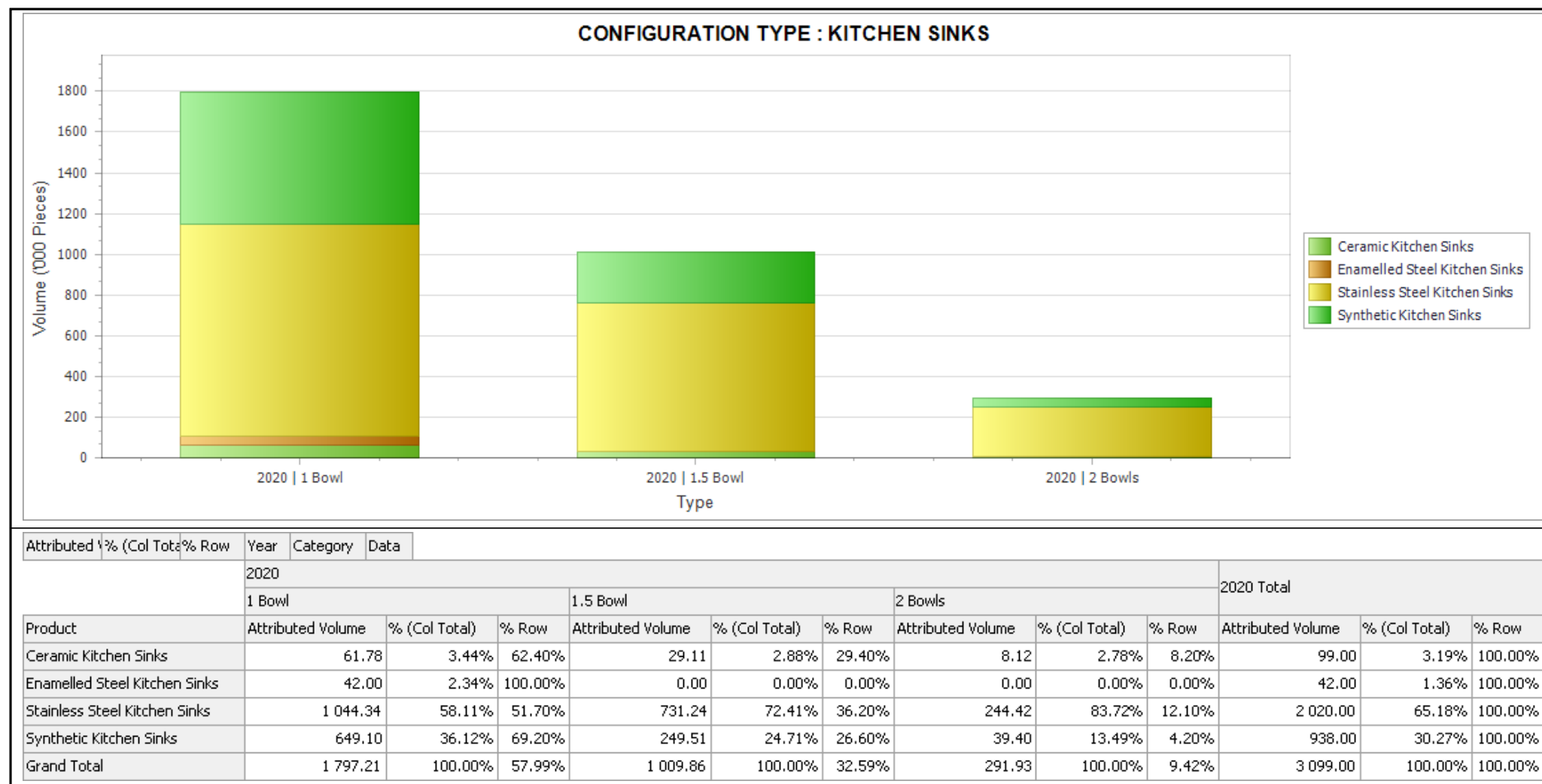


Attributed	% (Col Vol)	% Row	Year	Category	Data											
			2020													
			Drop-On			Inset			Undermount			2020 Total				
Product	Attributed Volume		% (Col Vol)	% Row	Attributed Volume		% (Col Vol)	% Row	Attributed Volume		% (Col Vol)	% Row	Attributed Volume		% (Col Vol)	% Row
Ceramic Kitchen Sinks	2.97		100.00%	3.00%	79.20		3.18%	80.00%	16.83		2.77%	17.00%	99.00		3.19%	100.00%
Enamelled Steel Kitchen Sinks	0.00		0.00%	0.00%	42.00		1.69%	100.00%	0.00		0.00%	0.00%	42.00		1.36%	100.00%
Stainless Steel Kitchen Sinks	0.00		0.00%	0.00%	1 686.70		67.77%	83.50%	333.30		54.90%	16.50%	2 020.00		65.18%	100.00%
Synthetic Kitchen Sinks	0.00		0.00%	0.00%	680.99		27.36%	72.60%	257.01		42.33%	27.40%	938.00		30.27%	100.00%
Grand Total	2.97		100.00%	0.10%	2 488.89		100.00%	80.31%	607.14		100.00%	19.59%	3 099.00		100.00%	100.00%

Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

8.4.4 Sales by Type of Configuration : Kitchen Sinks

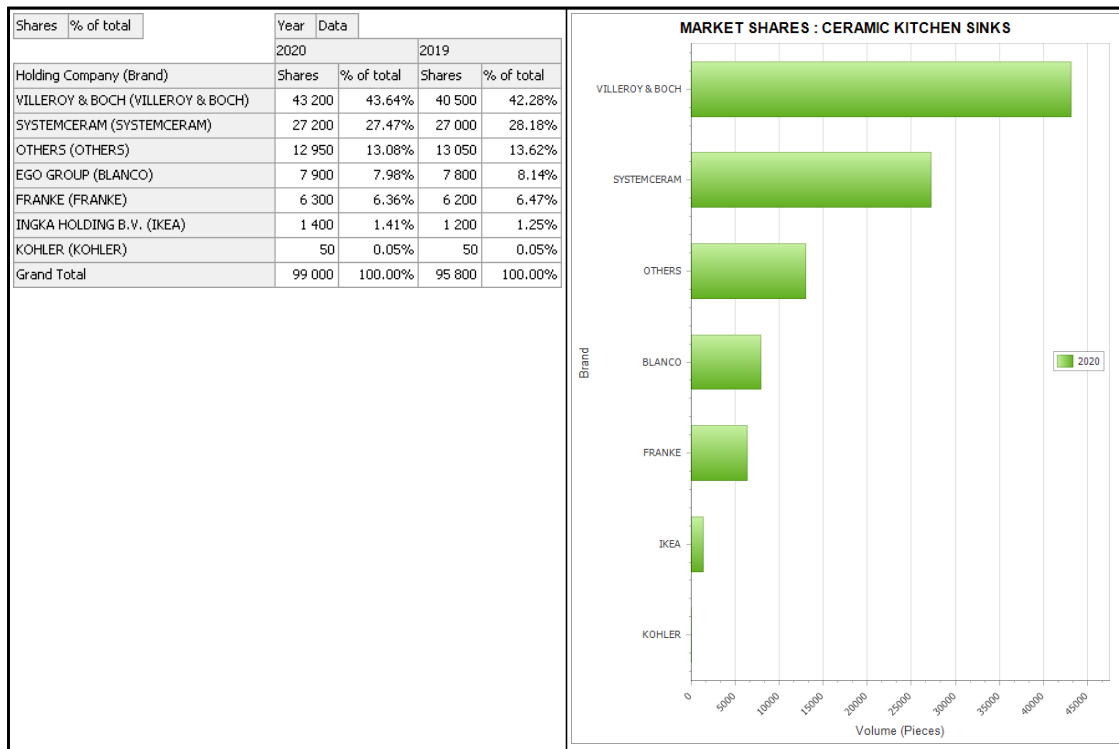
DE - KITCHEN SINK SALES BY CONFIGURATION TYPE ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

8.5.1 Market Shares : Ceramic Kitchen Sinks

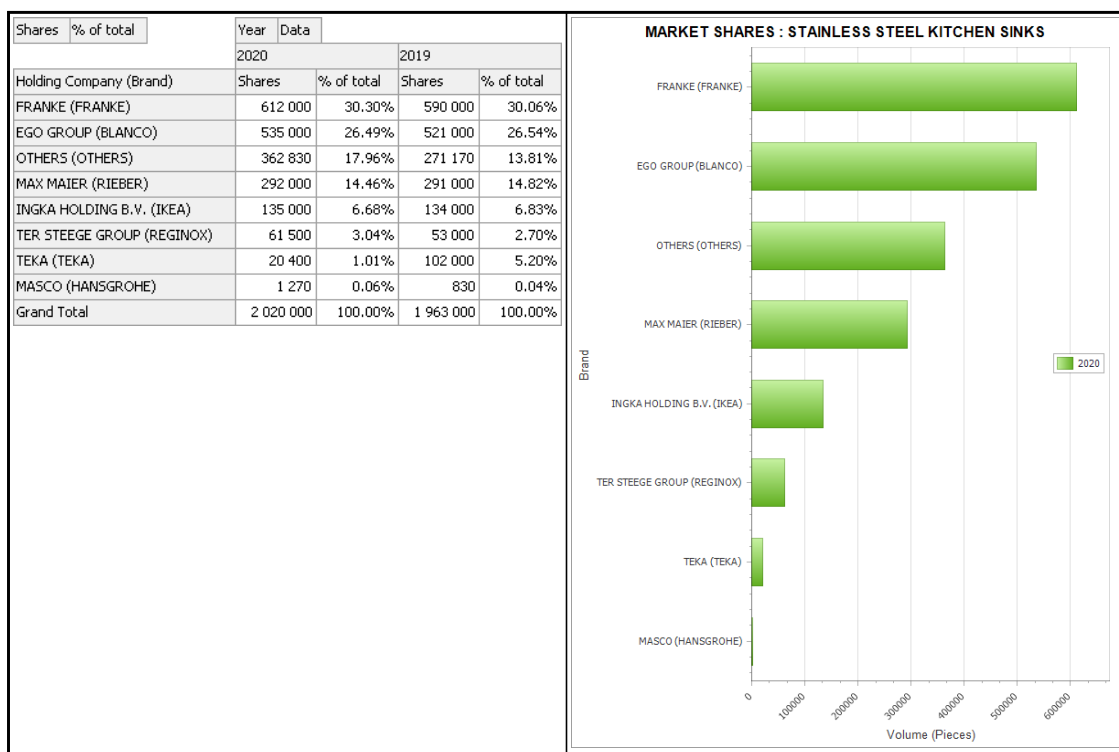
DE - ESTIMATED MARKET SHARES FOR CERAMIC KITCHEN SINKS



Source: BRG based on industry interviews

8.5.2 Market Shares : Stainless Steel Kitchen Sinks

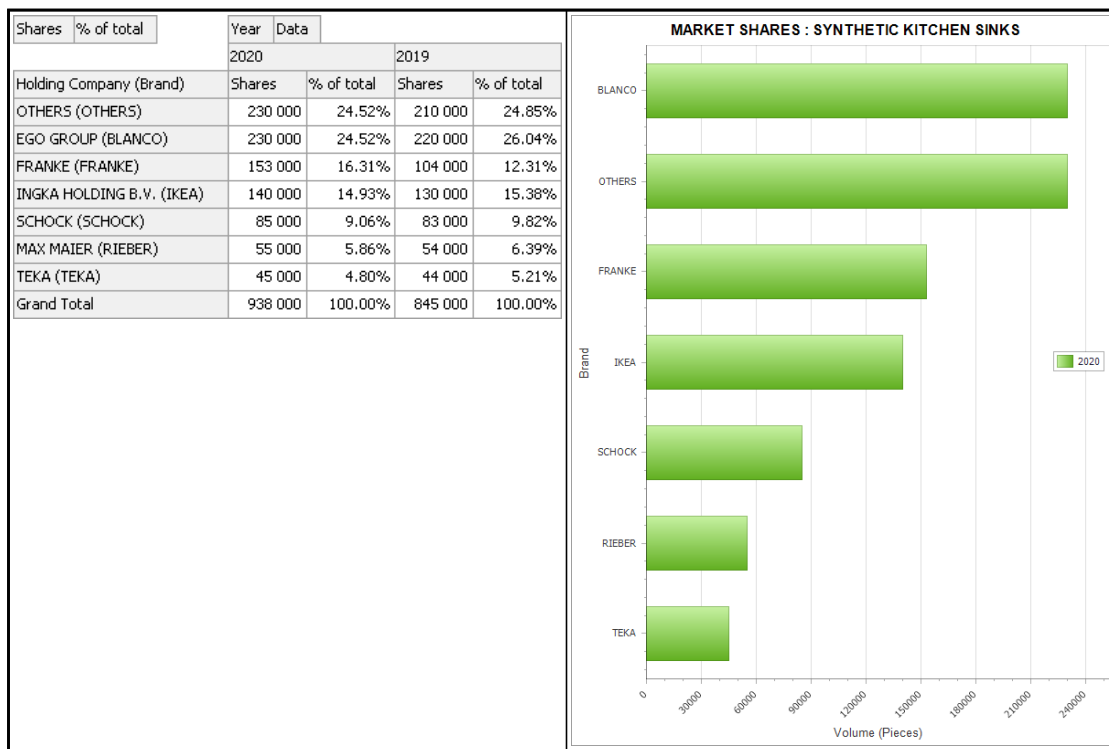
DE - ESTIMATED MARKET SHARES FOR STAINLESS STEEL KITCHEN SINKS



Source: BRG based on industry interviews.

8.5.3 Market Shares : Synthetic Sinks

DE - ESTIMATED MARKET SHARES FOR SYNTHETIC KITCHEN SINKS



Source: BRG based on industry interviews.

8.6.1

Route to Market: Kitchen Sinks
DE - ESTIMATED DISTRIBUTION FOR KITCHEN SINKS: 2020

Manufacturer / Importer

Manufacturers / Importers	
100.0%	

27.7%
Worktop Manufacturers
27.7%

Wholesale Level

Manufacturers	Worktop Manufacturer
22.1%	5.1%
Wholesalers / Merchants	
27.2%	

Retail Level

Manufacturer	Worktop Manufacturer	Wholesaler
5.2%	1.5%	1.0%
Internet Retailers		
7.7%		

Manufacturer	Worktop Manufacturer	Wholesaler
11.5%	0.0%	6.2%
DIY Retailers		
17.7%		

Manufacturer	Worktop Manufacturer	Wholesaler
28.5%	12.0%	7.0%
Specialist Retailers		
47.5%		

Manufacturer	Worktop Manufacturer	Wholesaler	Internet	DIY	Specialist
5.0%	8.0%	11.0%	0.0%	0.0%	0.0%
Installers					
24.0%					

End Use Level

Manufacturer	Worktop Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer
0.0%	0.0%	0.0%	7.1%	17.7%	45.5%	22.0%
Residential						
92.4%						
New Build				Residential RMI		
9.1%				83.3%		
Project		Non-Project		Project		Non-Project
3.7%		5.4%		9.0%		74.3%

Manufacturer	Worktop Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer
0.0%	1.1%	2.0%	0.6%	0.0%	2.0%	2.0%
Non-Residential						
7.6%						
New Build				Non-Residential RMI		
3.2%				4.4%		
Project		Non-Project		Project		Non-Project
2.0%		1.2%		2.0%		2.4%

8.6.2

Route to Market: Kitchen Sinks
DE - ESTIMATED DISTRIBUTION FOR KITCHEN SINKS: 2025

Manufacturer / Importer

Manufacturers / Importers	
100.0%	

29.0%
Worktop Manufacturers
29.0%

Wholesale Level

Manufacturers	Worktop Manufacturer
22.0%	5.0%
Wholesalers / Merchants	
27.0%	

Manufacturer	Worktop Manufacturer	Wholesaler
5.3%	1.7%	2.0%
Internet Retailers		
9.0%		

Manufacturer	Worktop Manufacturer	Wholesaler
11.2%	0.0%	6.0%
DIY Retailers		
17.2%		

Manufacturer	Worktop Manufacturer	Wholesaler
29.0%	12.4%	7.7%
Specialist Retailers		
49.1%		

Manufacturer	Worktop Manufacturer	Wholesaler	Internet	DIY	Specialist
3.5%	7.7%	10.9%	0.0%	0.0%	0.1%
Installers					
22.2%					

End Use Level

Manufacturer	Worktop Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer
0.0%	0.9%	0.2%	8.0%	17.2%	46.1%	20.2%
Residential						
92.6%						
New Build				Residential RMI		
7.7%				84.9%		
Project		Non-Project		Project		Non-Project
3.2%		4.5%		9.0%		75.9%

Manufacturer	Worktop Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer
0.0%	1.3%	0.2%	1.0%	0.0%	2.9%	2.0%
Non-Residential						
7.4%						
New Build				Non-Residential RMI		
3.0%				4.4%		
Project		Non-Project		Project		Non-Project
1.9%		1.1%		2.0%		2.4%

Source: BRG Based on Industry Interviews. - Totals may vary due to roundings

8.6.3 Route to Market - Consolidated View : Kitchen Sinks

DE - ESTIMATED DISTRIBUTION FOR KITCHEN SINKS : 2020

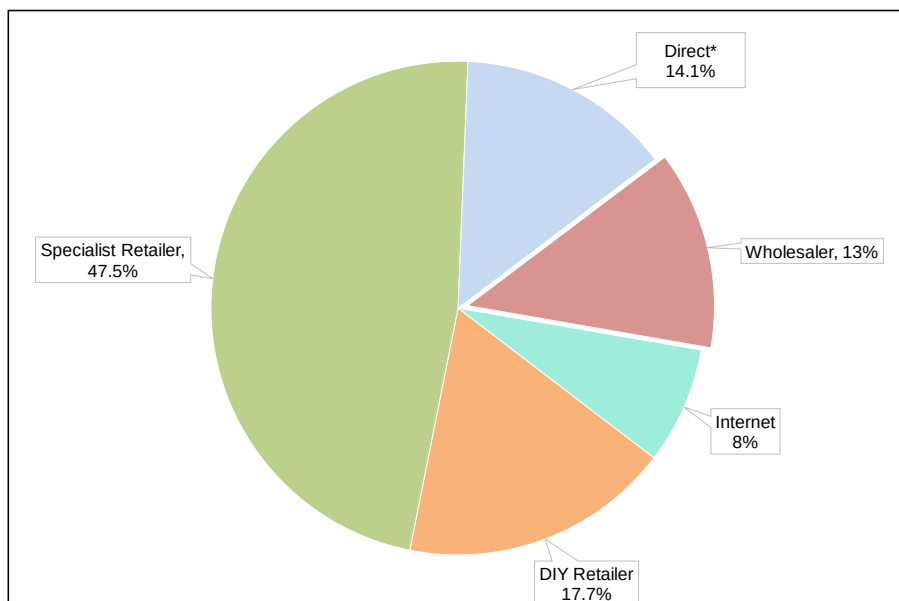
Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
13.0%	7.7%	17.7%	47.5%	14.1%	12.7%	4.0%	79.7%	3.6%
100%					16.7%		83.3%	

8.6.4 Route to Market - Consolidated View : Kitchen Sinks

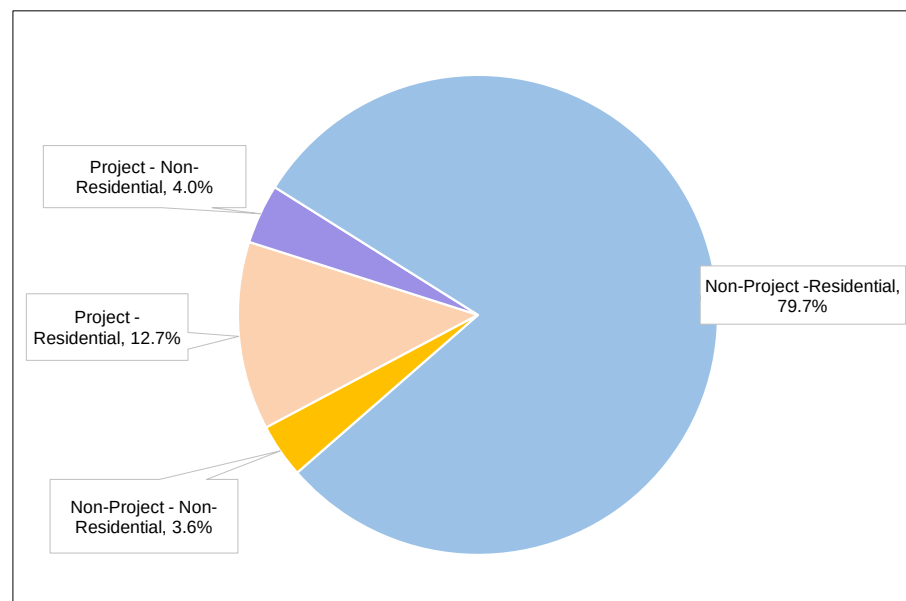
DE - ESTIMATED DISTRIBUTION FOR KITCHEN SINKS : 2025

Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
11.3%	9.0%	17.2%	49.1%	13.4%	12.2%	3.9%	80.4%	3.5%
100%					16.1%		83.9%	

DE - SALES OF KITCHEN SINKS BY DISTRIBUTION CHANNEL: 2020



DE - SALES OF KITCHEN SINKS BY POINT OF DECISION AND END-USE SECTOR: 2020



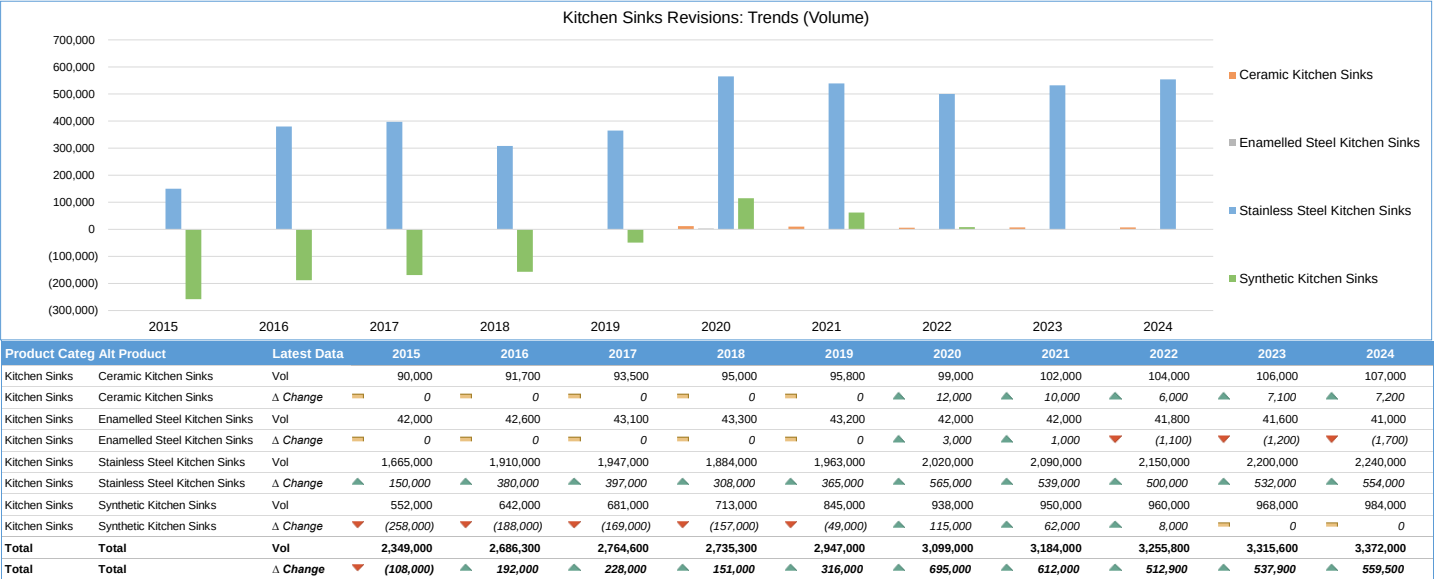
Source: BRG Based on Industry Interviews.

*Direct includes sales going directly from manufacturers to installers, contractors and private customers.

8.7 Revisions in the German Kitchen Sinks Market (April 2021)

DE - REVISIONS IN THE GERMAN KITCHEN SINKS MARKET

Data Comparison: October 2020



The market volume for stainless kitchen sinks was revised up historically due to an underestimation in domestic production, which was larger than previously estimated. Demand for synthetic kitchen sinks was revised down for the years 2005 until 2011 for the same reason. Synthetic kitchen sink sales have been growing strongly in the last two decades and this development is believed to continue during the forecast period.

The MSP of ceramic kitchen sinks was revised up for the year 2020 according to adjustments to the market share. The share of the upper price segment is bigger than previously anticipated.

GERMANY

KITCHEN TAPS AND MIXERS

THE EUROPEAN BATHROOM & KITCHEN PRODUCT MARKETS



Market Characteristics and Technological Trends

During the financial crisis, sales plunged in 2008/2009 and this was accompanied by intense price competition and a trend towards the lower-priced segment. Since 2010, the market has started to recover with solid growth, except for the year 2012. More recent developments indicate a shift of German end users towards more expensive kitchens. The kitchen has developed into a living area where people like to host and socialise with guests. The kitchen has gained importance as a status symbol and Germans are willing to spend more money for a modern kitchen. This development has been supported by the trend towards open kitchens, which are part of the living room.

The average kitchen in Germany is being replaced every 20 years. Kitchen sinks are usually replaced together with the whole kitchen. Taps and mixers correlate less with the overall kitchen market as they have a shorter average lifespan. Most sales are seen in the refurbishment sector. The German kitchen taps and mixers market is characterised by high maturity and saturation. Sales are driven by the level of private consumption in the country and the developments in the construction industry.

Several product innovations, like boiling hot water and sparkling water taps and electronic mixers for the kitchen, have been introduced to the market and established their market niche. Cleanliness and hygiene are an important topic for modern consumers, as well as comfort and design. Two head mixers are an outdated product and broken ones are likely to be replaced by one head mixers. Many companies have taken this into account and do not offer two head mixers anymore. An exception are luxury brands such as DORNBACHT, which have released newly designed two head mixers in their classic product range. Pillar taps hardly have any significance on the market for kitchen taps and mixers in Germany. DORNBACHT has a few models that are intended for filling pots and are located near the cooking stove.

One head mixers are the most popular type of mixer, and the market remains broad in terms of quality and price ranges. The shift from two head mixers towards one head mixers is also evident in the DIY market, as two head mixers do not provide a meaningful price advantage anymore. In some cases two head mixers are more expensive when compared to the average one head mixer. 'Showerettes', with a pull-out spout, or taps with flexible spouts, have increased their penetration over the last few years. Other types of kitchen taps and mixers are rather expensive if bought from well-established manufacturers. Many are characterised by an extravagant design and are fashion items. Newly launched products, such as taps provided with a switch on and off button, are expected to influence the residential and non-residential upmarket of future years.

Recent Market Trends

The COVID-19 pandemic shaped the developments in the year 2020 and 'business as usual' was replaced by government measures. In Germany the official lockdown, which is better understood as a 'prohibition of contact', started at the end of March 2020. Bavaria was the first state to implement measures to contain the spread of the virus, which was followed by the remaining states towards the end of March. For a few weeks the German population was advised to work from home when possible, with most non-essential stores closing for a limited time only. Measures implemented varied by state (*Bundesländer*). Most stores, including DIY stores, were allowed to open again from the beginning of April 2020 onwards, with differences from state to state in the level of restrictions and measures. In certain regions, shopping in DIY stores was limited to professionals only.

During 2020, new housing construction continued without much interruption, with only large scale project sites halted to implement protective measures. That being said, the overall production per employee is believed to have shrunk significantly, and remains to be the case, since many employers introduced measures to protect the construction workers. The bathroom industry was declared part of the crisis infrastructure, but installers could still continue to work without many interruptions. Overall, there has been relatively little impact of the COVID-19 pandemic on the building related industry, including the bathroom market, in Germany. In some cases, end users may have postponed, or cancelled, bathroom related projects due to the uncertainty on the market, but overall demand remained relatively strong, as there was a large backlog of works from the past, benefiting sales of bathroom and kitchen equipment. At the same time, in certain instances, the COVID-19 pandemic may have contributed to additional demand. Since consumers were spending more time indoors, they had the opportunity to dedicate themselves to repair or upgrade works, whereas others might have decided to invest into their property by installing a new bathroom or kitchen due to (international) holidays being difficult because of the current pandemic.

The German kitchen taps and mixers market performed strongly in 2020 with most demand coming from sales of one head mixers. Growth was accelerated because of positive RMI activity, which compensated for a decline from new residential and non-housing demand. With kitchens gaining importance as a status symbol, German end users are more willing to renew their kitchen. Kitchen taps are mainly sold together with a new kitchen through specialist kitchen stores. Such sales do not depend on the busy schedules of professional installers. If an end user must choose between renewing first their kitchen or bathroom, it is more likely to be the kitchen. The installation of kitchen taps and mixers is not hampered by the lack of installers. Moreover, kitchen taps and mixers are replaced three times more often than the ones in the bathroom.

Demand for kitchen taps and mixers increased for all price segments; however, the luxury and lower price segments gained stronger share. The middle and economy price segments lost shares, while the upper price segment remained unchanged in

2020. Demand from the lower segment is linked to increasing online and DIY purchases. Nowadays, these alternative distribution channels offer affordable good quality products, which end users have easy access to. Waiting months for an installer to fix a broken tap is not possible, and end users make use of faster alternatives for replacing broken taps and mixers. The growing share of the luxury price segment reflects the growing demand for electronic kitchen taps that supply hot, cold and sparkling water for example.

Prices and Market Values

The German market, when compared to its European neighbouring countries, is a relatively high-priced market. Especially in recent years, German end users have started to invest more into high quality kitchen products such as taps and mixers. In 2020, the average MSP for kitchen taps and mixers decreased, mainly due to a lower MSP of one and two head types. The market for kitchen taps and mixers is highly competitive, which makes it more difficult for manufacturers to impose extreme price increases. A significant amount of kitchen taps and mixers are imported from abroad and imports continue gaining market shares. Moreover, the consumer's trend of comparing prices on the Internet before purchasing the product increasingly obstructs remarkable price increases on the market.

A new kitchen sink is often bought together with a new tap and mixer. FRANKE and BLANCO are two dominating manufacturers, which sell their sinks together with their own taps and mixers that are produced abroad. Prices for kitchen taps and mixers range from low imports from the Far East to high well-known European brands, which are generally distributed via wholesalers and installers. Similar to the sanitary taps and mixers market, some established manufacturers, such as GROHE and HANS GROHE, offer their products through different distribution channels. The same taps and mixers might be offered at different price ranges, depending on the distribution channel. According to industry feedback, a sanitary tap found in a DIY store or online might be up to three times cheaper than the same product distributed by an installer or in a showroom.

New Housing

The decline in new housing completions in 2020 limited demand for kitchen taps and mixers. All types of kitchen taps and mixers recorded fewer sales, except for electronic taps and mixers and pillars. Electronic taps and mixers are a niche; however, they have the strongest growth rate in this sector. Demand has increased due to the trend towards modern taps that supply instant hot, cold and sparkling water. One head mixers are by far the most popular type of mixer in German kitchens. Two head mixers have benefited from more new house completions in the economic and upper price segments; however, sales figures are marginal. Sales in the new housing segment tend to follow the number of dwelling constructions very closely. Usually, one set of mixers is sold for every new build residential house. Demand is expected to grow again from 2021 until 2023, before strongly declining in 2024 and 2025.

Housing RMI

Demand for taps and mixers in the RMI segment strongly increased in 2020 when compared to the previous year. The kitchen taps and mixers market could profit from the positive trend in the renovation activities and like-for-like replacements. It is believed that the largest number of sales stems from like-for-like replacement decisions, as breakages require a quick replacement. Within this segment, the standard one head mixers account for the largest part of the sales followed by two head mixers' sales. Sales for two head mixers in the RMI segment increased slightly in 2020. However, it is believed that the share is gradually declining to the benefit of one head mixers. Demand for electronic taps and mixers is growing within the housing RMI segment, although sales remain marginal.

Non-Housing

Kitchen products have a limited demand from non-housing as not every commercial building requires a kitchen, unlike bathrooms. Most kitchen taps and mixers go into gastronomy, i.e. restaurants and hotels, as well as offices and public buildings. In 2020, demand for kitchen taps and mixers was clearly negative when compared to 2020. One head mixers are the most common item in the non-housing segment, followed by two head mixers and pillar taps. Demand from the non-housing segment is believed to remain negative in 2021, before turning positive again from 2022 onwards.

Supply Structure

The German kitchen taps and mixers market has well established players and relatively stable market shares. Like bathroom and shower accessories, the market for kitchen taps and mixers is dominated by German manufacturers. Kitchen taps and mixers are generally manufactured by companies that specialise in the bathroom sector, such as GROHE, KLUDI, DORNBRAUCH and HANS&GROHE. These manufacturers supply their product through wholesalers; however, DIY stores are increasingly playing a more important role. These companies have a strong market share in the RMI sector.

The market for new housing taps and mixers is dominated by kitchen sink manufacturers, who sell their sink together with their taps. The German company BLANCO is the clear market leader followed by the Swiss manufacturer FRANKE. SCHÜTTE is the market leader for sales outside wholesalers and distributes most of its taps and mixers through DIY stores, followed by furniture specialist stores and occasional promotions in food discount stores such as ALDI, LIDL, PENNY and NORMA. In 2011, the manufacturer SEE (SCHÜTTE) was bought by KIRCHHOFF and is part of the SEE HOLDING, which employs more than 150 employees in Wallenhorst in Germany, Salzburg in Austria, and Hong Kong. SCHÜTTE produces all its products according to German quality standards in the Far East.

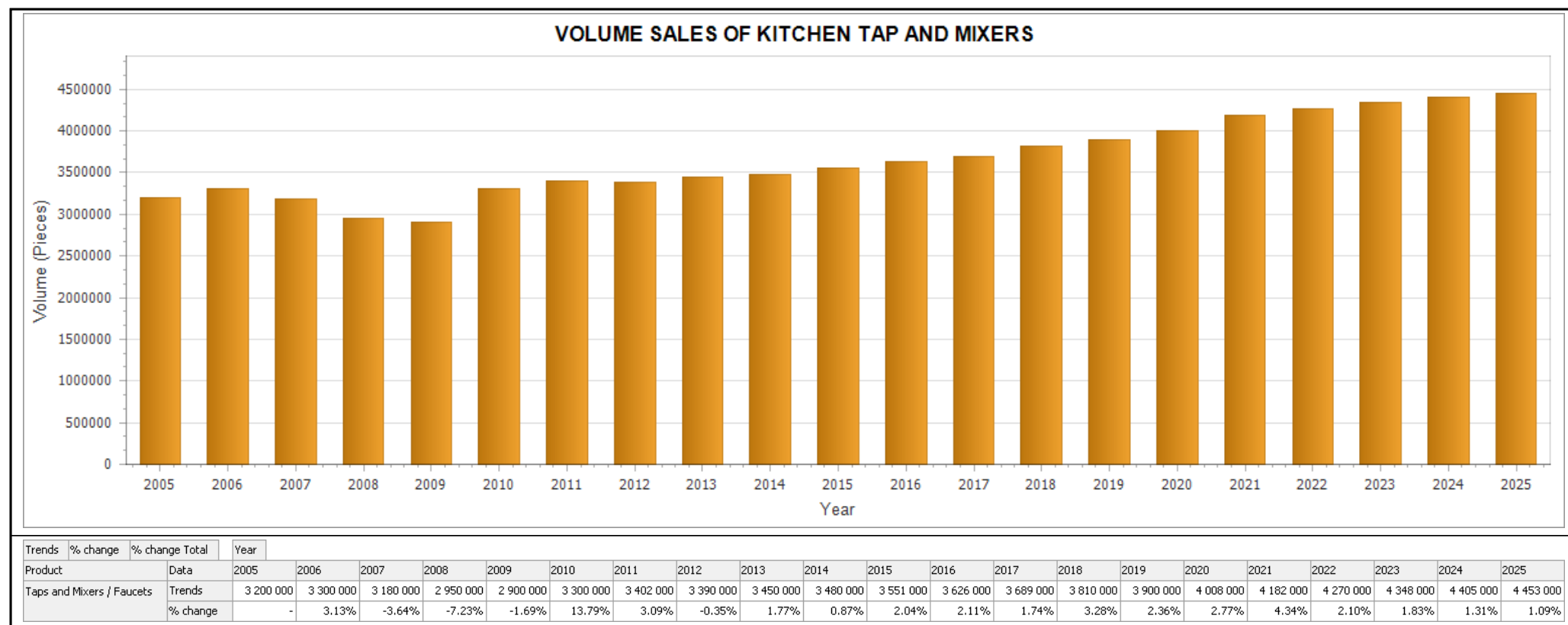
Kitchen taps and mixers are not only offered as part of special campaigns in supermarkets such as LIDL and ALDI, but they also are available in certain online shops such as REUTER. The market share of online and DIY sales is believed to increase at the expense of the traditional distribution channels. The product spectrum of kitchen sanitary taps and mixers includes residential and commercial products, such as special taps and mixers for institutional kitchens or laboratories. These products are offered by manufacturers that specialise in commercial kitchens.

Forecasts

Demand for kitchen taps and mixers is believed to remain positive during the forecast period, with growth expected to gradually decline from 2022 onwards. Sales are anticipated to slow down mainly due to a slower RMI growth rate. Instead, new housing is believed to contribute to some more sales in the years 2021 until 2023, before turning negative in 2024 and 2025 again. Most demand is expected to come from kitchen renovations and like-for-like replacements. BRG expects that overall demand is likely to weaken towards the end of the forecast period. This is due to the long-term impact of the COVID-19 crisis and its economic impact, which might limit consumer spending for new kitchens. In terms of product mix it is expected that sales of all product types, i.e. pillar, one and two head taps and mixers are going to gain in volume during the forecast period.

9.2 Historical Trends and Forecasts : Kitchen TM

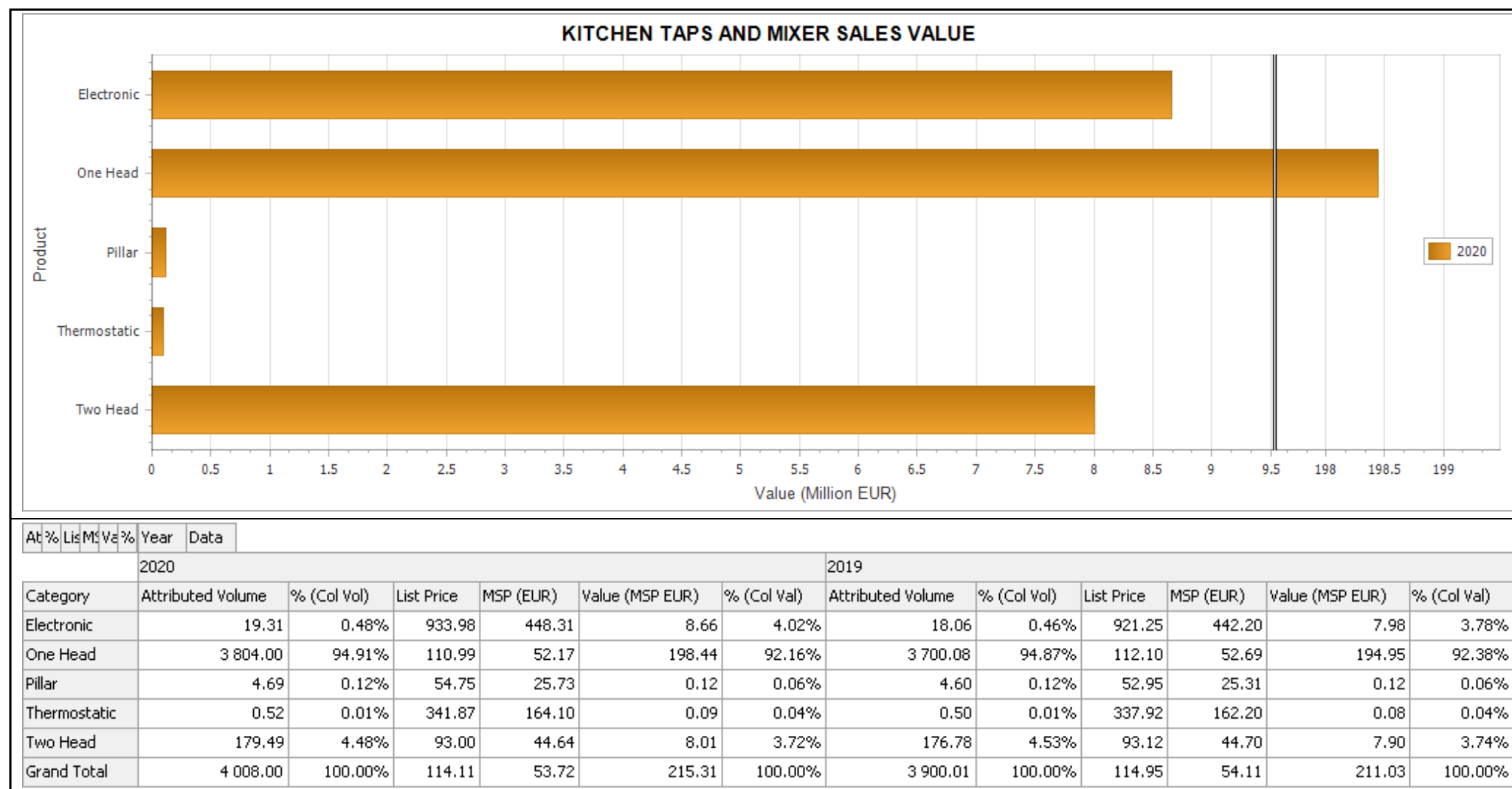
DE - SUMMARY OF TRENDS IN THE KITCHEN TAP AND MIXER MARKET : HISTORICAL AND FORECASTS



Source: BRG based on industry interviews.

9.3.1 Prices and Market Values : Kitchen Taps & Mixers

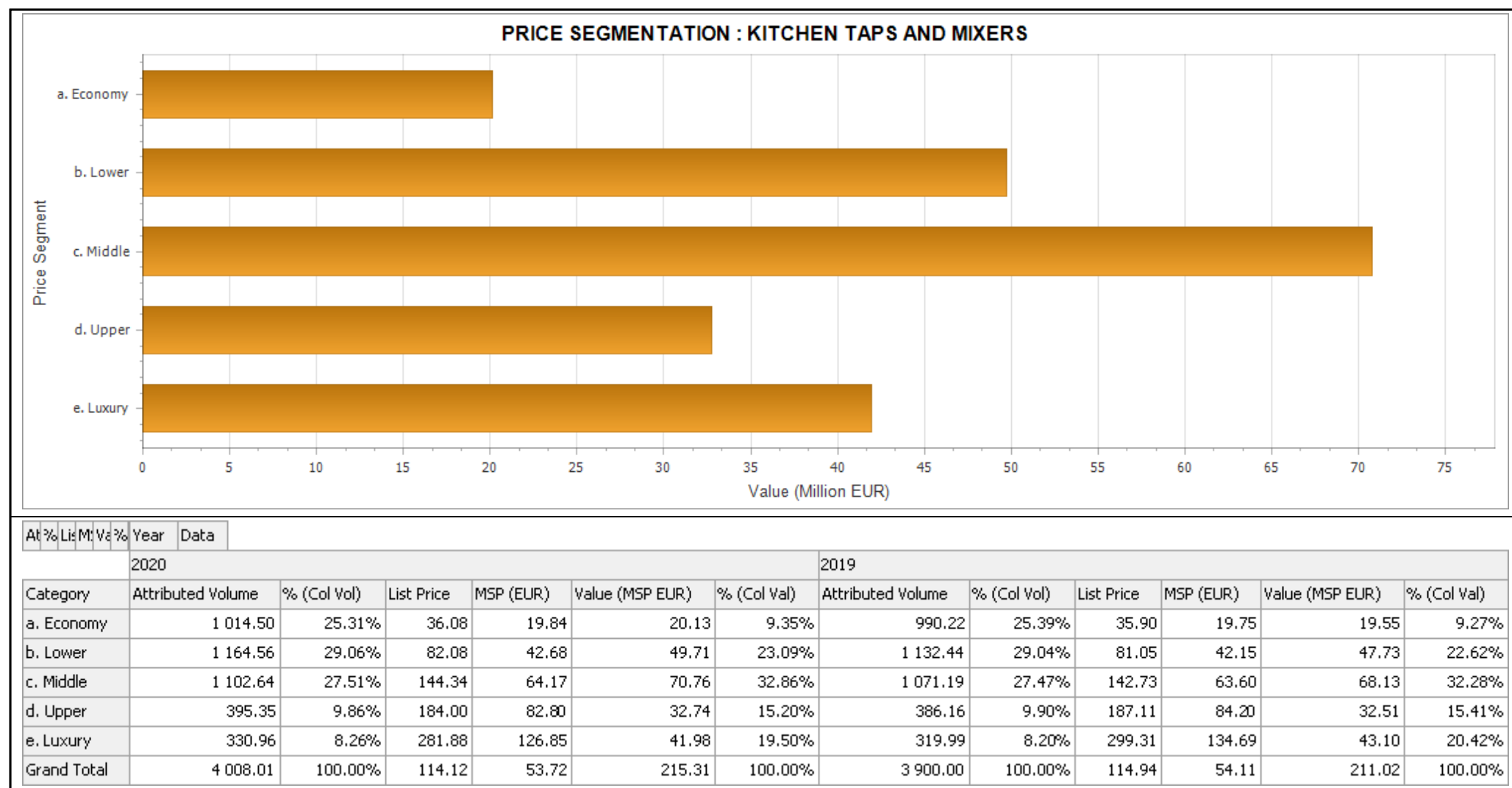
DE - ESTIMATED VALUES OF THE KITCHEN TAP AND MIXER MARKET



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price)

9.3.2 Price Segmentation : Kitchen Taps & Mixers

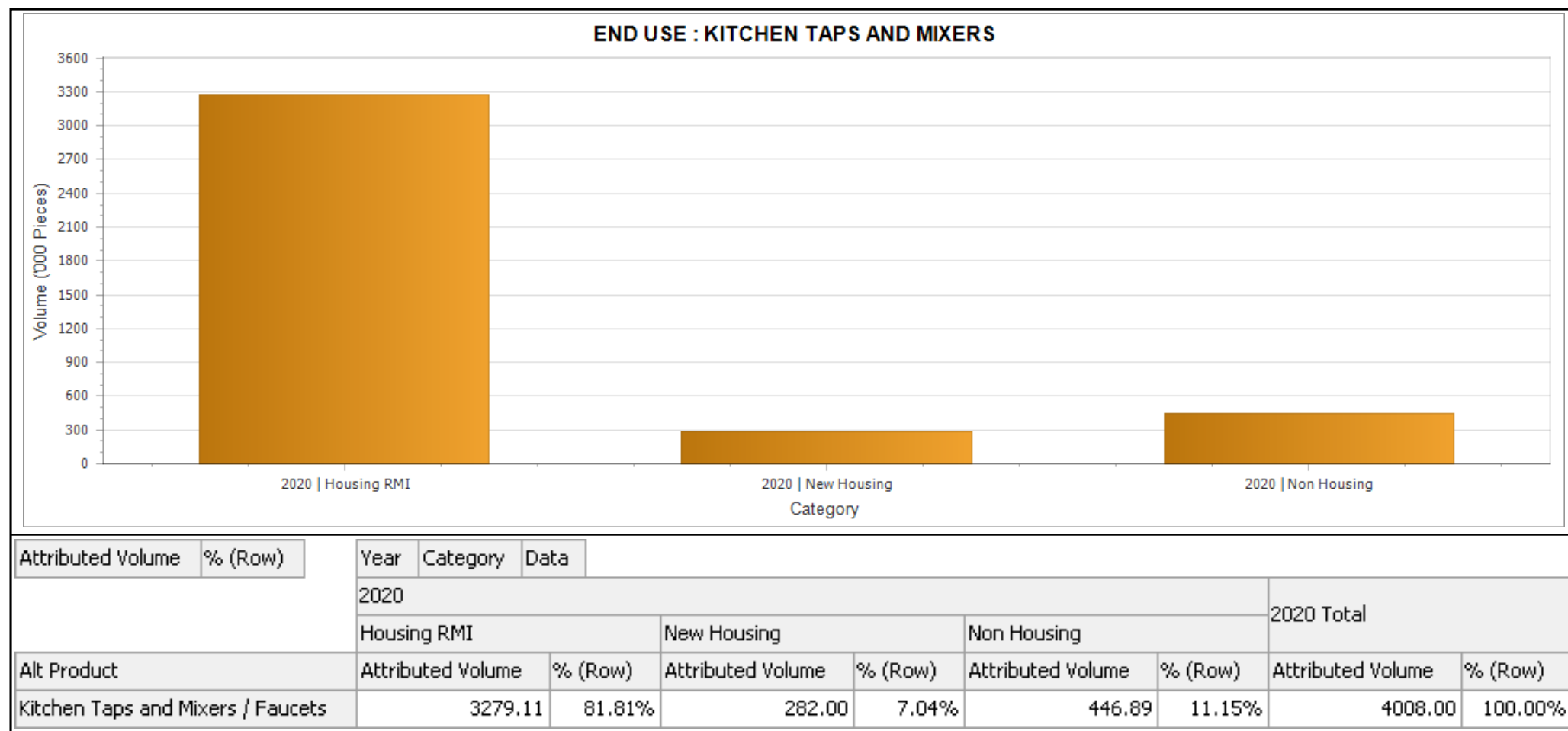
DE - ESTIMATED PRICE SEGMENTATION FOR KITCHEN TAPS AND MIXERS



Source: BRG based on industry interviews. Total may show slight variance compared to market data due to rounding. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy <€35, Lower €35-€50, Middle €50-€70, Upper €70-€110 and Luxury >€110

9.4.1 End Use : Base Year

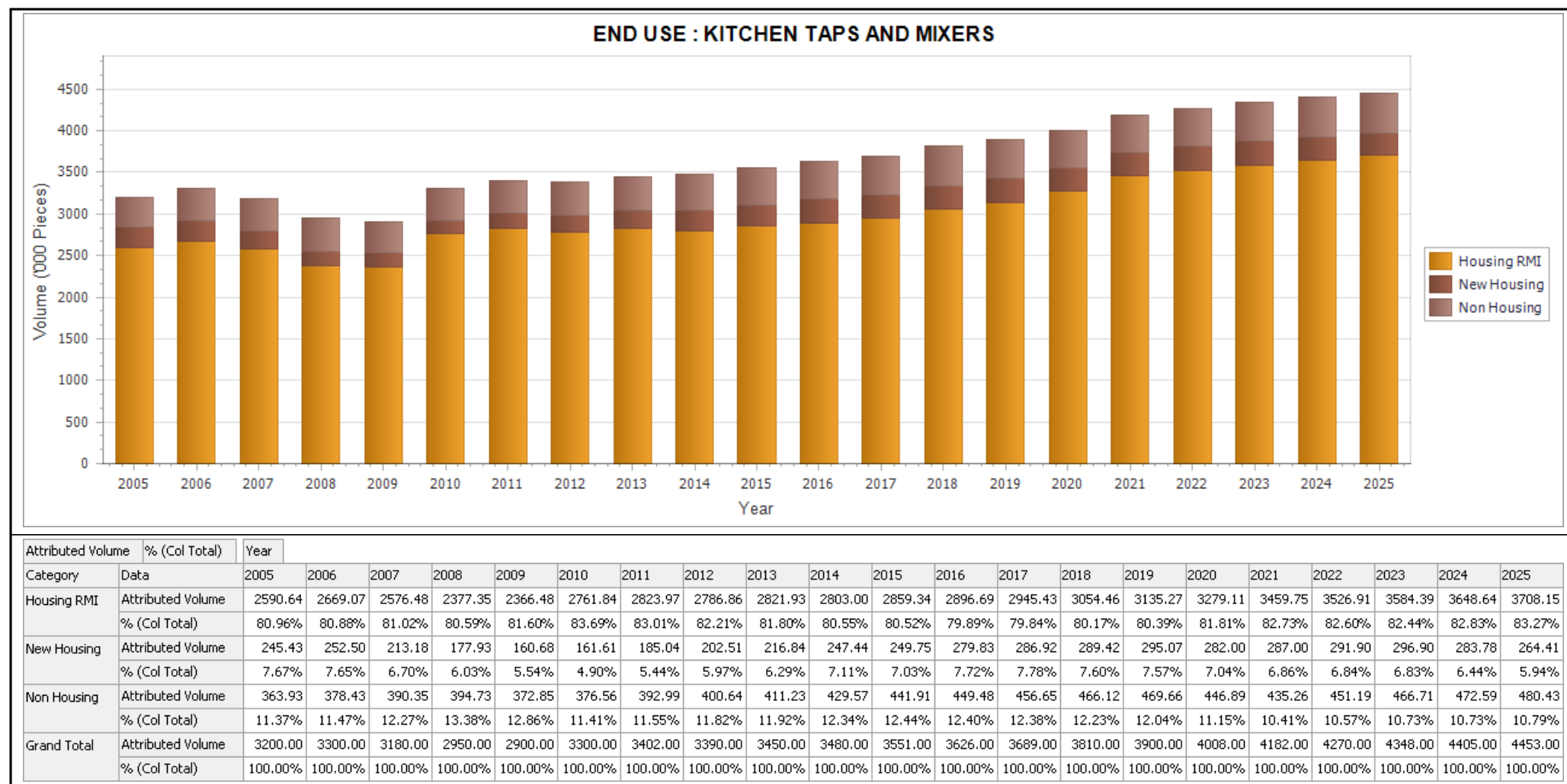
DE - SALES OF KITCHEN TAPS & MIXERS BY END USE SEGMENT ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

9.4.2 End Use : Historical and Forecasts

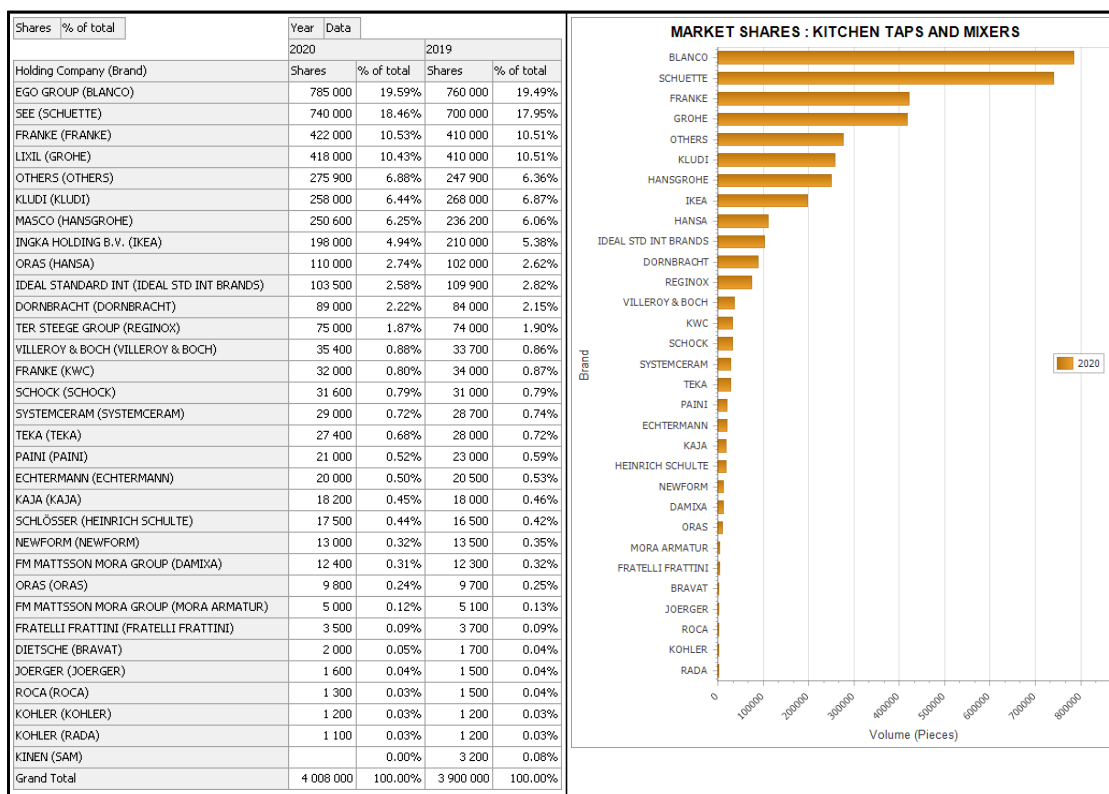
DE - SALES OF KITCHEN TAPS AND MIXERS BY END USE SEGMENT ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

9.5 Market Shares : Kitchen Taps & Mixers

DE - ESTIMATED MARKET SHARES FOR KITCHEN TAPS AND MIXERS



Source: BRG based on industry interviews.

9.6.1

Route to Market: Kitchen Taps & Mixers / Faucets
DE - ESTIMATED DISTRIBUTION FOR KITCHEN TAPS & MIXERS / FAUCETS: 2020

Manufacturer / Importer	Manufacturers / Importers
	100.0%

Wholesale Level	48.7%
	Wholesalers / Merchants
	48.7%

Retail Level	Manufacturer	Wholesaler	Manufacturer	Wholesaler	Manufacturer	Wholesaler
	3.9%	7.5%	17.1%	8.0%	24.5%	15.0%
	Internet Retailers		DIY Retailers		Specialist Retailers	
	11.4%		25.1%		39.5%	

Manufacturer	Wholesaler	Internet	DIY	Specialist
4.9%	18.0%	2.2%	1.0%	4.5%
Installers				
30.6%				

End Use Level	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer
	0.1%	0.1%	9.2%	24.1%	31.0%	24.3%	0.8%	0.1%	0.0%	0.0%	4.0%	6.3%
	Residential						Non-Residential					
	88.9%						11.1%					
	New Build			Residential RMI			New Build			Non-Residential RMI		
	7.0%			81.8%			4.0%			7.1%		
	Project	Non-Project		Project	Non-Project		Project	Non-Project		Project	Non-Project	
	2.8%	4.2%		2.0%	79.8%		1.9%	2.1%		2.0%	5.1%	

9.6.2

Route to Market: Kitchen Taps & Mixers / Faucets
DE - ESTIMATED DISTRIBUTION FOR KITCHEN TAPS & MIXERS / FAUCETS: 2025

Manufacturer / Importer	Manufacturers / Importers
	100.0%

Wholesale Level	47.0%
	Wholesalers / Merchants
	47.0%

Retail Level	Manufacturer	Wholesaler	Manufacturer	Wholesaler	Manufacturer	Wholesaler
	5.0%	11.0%	18.0%	7.0%	24.0%	16.0%
	Internet Retailers		DIY Retailers		Specialist Retailers	
	16.0%		25.0%		40.0%	

Manufacturer	Wholesaler	Internet	DIY	Specialist
4.8%	12.8%	5.0%	1.2%	5.2%
Installers				
29.0%				

End Use Level	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer
	0.2%	0.1%	11.0%	23.8%	31.0%	23.1%	1.0%	0.1%	0.0%	0.0%	3.8%	5.9%
	Residential						Non-Residential					
	89.2%						10.8%					
	New Build			Residential RMI			New Build			Non-Residential RMI		
	5.9%			83.3%			3.9%			6.9%		
	Project		Non-Project	Project		Non-Project	Project		Non-Project	Project		Non-Project
	2.8%		3.1%	2.0%		81.3%	1.8%		2.1%	2.1%		4.8%

Source: BRG Based on Industry Interviews. - Totals may vary due to roundings

9.6.3 Route to Market - Consolidated View : Kitchen Taps and Mixers / Faucets

DE - ESTIMATED DISTRIBUTION FOR KITCHEN TAPS AND MIXERS / FAUCETS : 2020

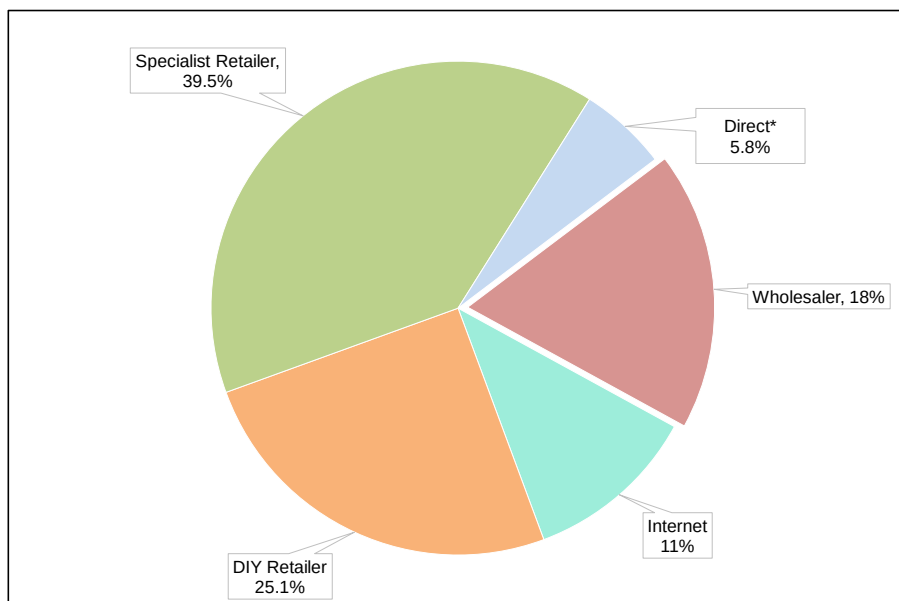
Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
18.2%	11.4%	25.1%	39.5%	5.8%	4.8%	3.9%	84.1%	7.2%
100%					8.7%		91.3%	

9.6.4 Route to Market - Consolidated View : Kitchen Taps and Mixers / Faucets

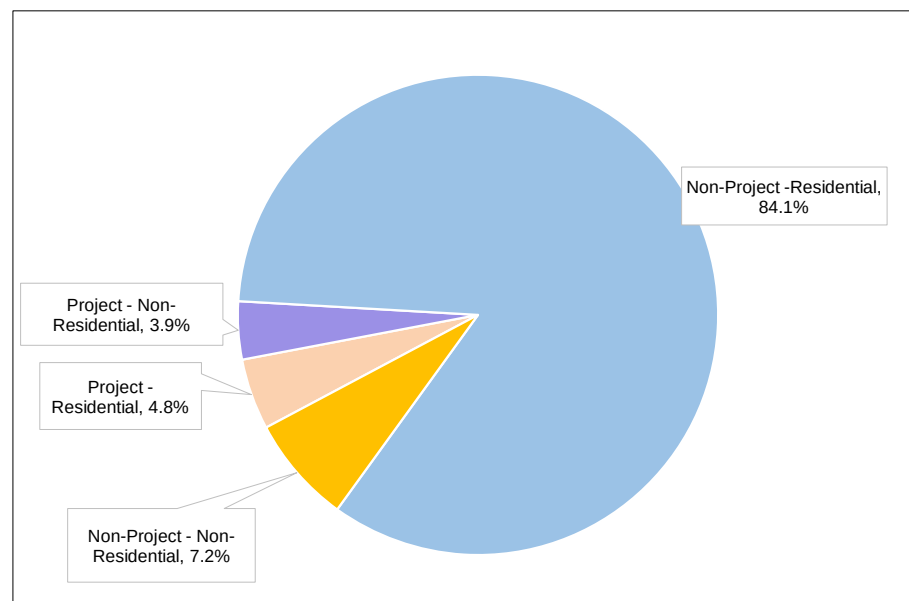
DE - ESTIMATED DISTRIBUTION FOR KITCHEN TAPS AND MIXERS / FAUCETS : 2025

Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
13.0%	16.0%	25.0%	40.0%	6.0%	4.8%	3.9%	84.4%	6.9%
100%					8.7%		91.3%	

DE - SALES OF KITCHEN TAPS AND MIXERS / FAUCETS BY DISTRIBUTION CHANNEL: 2020



DE - SALES OF KITCHEN TAPS AND MIXERS / FAUCETS BY POINT OF DECISION AND END-USE SECTOR: 2020



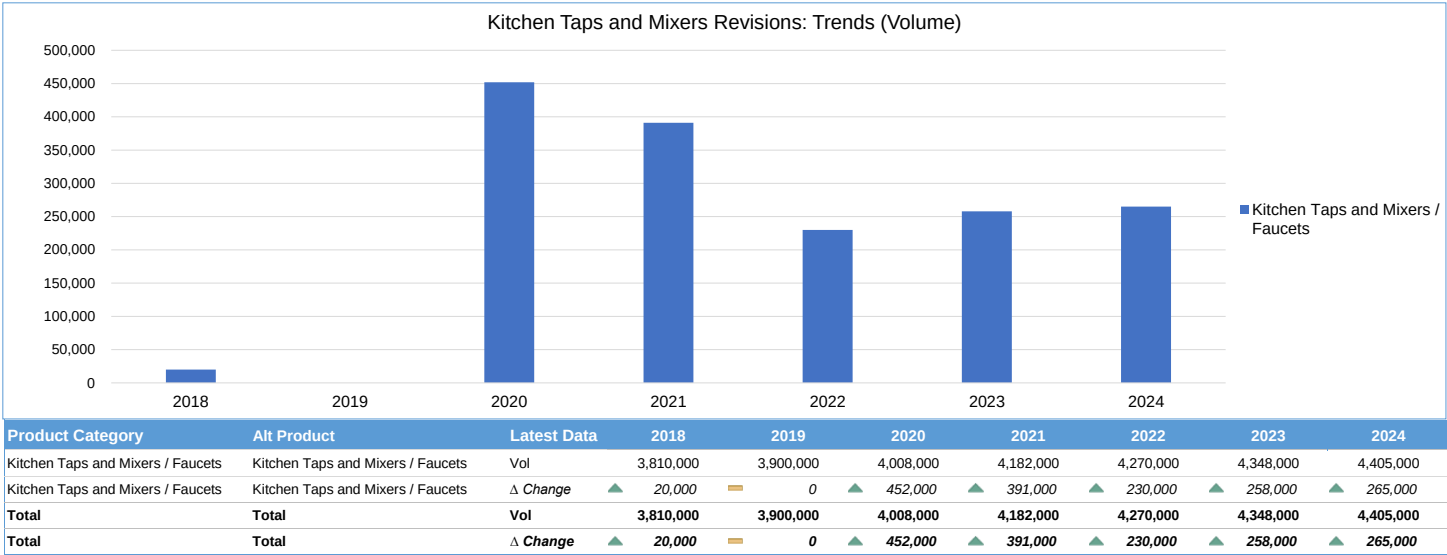
Source: BRG Based on Industry Interviews.

*Direct includes sales going directly from manufacturers to installers, contractors and private customers.

9.7 Revisions in the German Kitchen Taps and Mixers Market (April 2021)

DE - REVISIONS IN THE GERMAN KITCHEN TAPS AND MIXERS MARKET

Data Comparison: October 2020



No revisions have been made to historical volumes for this market.

Following a thorough detailed price analysis, by comparing domestically produced taps and mixers in contrast to imported taps and mixers, significant downward revisions were made to the MSP for all types of taps and mixers. Taps and mixers produced in Germany are very high priced, which is because Germany nowadays mostly produces high quality, high priced, bathroom and kitchen fittings, whereas more economy priced products are imported from abroad. This includes German manufacturers, which often produce their premium products in their domestic production site(s), whereas lower priced ranges are sourced from foreign factories, including Poland, Hungary and Portugal. Average prices for mixers, not including any type of taps, produced in Germany have an MSP of around €70 to €80 in recent years, whereas imported models, again only referring to mixers, are significantly lower prices, with MSPs between €35 to €40 in recent years. This includes models imported by German manufacturers from their foreign production plants but also all other imported brands, such as the ones imported from e.g., China, Italy, Spain, France and Poland.